Discrete Data Analysis and Lab

Lecture notes, Chapters 1–11

Based on A. Agresti, *Categorical Data Analysis*, 2nd ed. (Wiley, 2002)

Course map

The course develops methods for categorical responses, mainly from the biomedical and social sciences, and shows how to carry them out with statistical software in the weekly lab. A single generalized linear model framework links logistic regression and loglinear models for discrete data with normal regression for continuous data. The syllabus topics correspond to the chapters of these notes as follows.

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1.	Introduction	1
2.	Contingency table analysis: chi-squared tests, measures of association	2, 3
3.	Generalized linear models	4
4.	Logistic model	5, 6, 7
5.	Loglinear model: three-way and four-way contingency tables	8, 9
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Chapter 1. Distributions and Inference for Categorical Data

1.1 Categorical response data

Motivation. The measurement scale of a categorical response decides which models and summaries are meaningful.

A *categorical variable* has a measurement scale consisting of a set of categories (liberal / moderate / conservative; normal, benign, suspicious, malignant).

1.1.1 Response–explanatory variable distinction

These notes concern *categorical responses*; explanatory variables may be of any type.

1.1.2 Nominal–ordinal scale distinction

Definition (Nominal, ordinal, interval)

(a) A categorical variable is *nominal* if its categories have no natural ordering (religious affiliation, mode of transport).
(b) It is *ordinal* if the categories are ordered but the distances between them are unknown (patient condition good/fair/serious/critical).
(c) An *interval* variable has numerical distances between any two values (blood pressure, income); it is a *ratio* variable if ratios are also meaningful.

The scale depends on how a variable is measured. In the hierarchy interval > ordinal > nominal, methods for a lower level apply at a higher level (losing information), not conversely.

1.1.3 Continuous–discrete variable distinction

All measurement is discrete in practice; the subject here is variables taking few values.

1.1.4 Quantitative–qualitative variable distinction

Nominal is *qualitative*, interval *quantitative*; ordinal sits in between and is often handled by *scores* or a latent continuous variable.

1.1.5 Organization of this book

Binomial, multinomial and Poisson replace the normal; the central families are *logistic regression* and *loglinear models*.

1.2 Distributions for categorical data

Motivation. Binomial, multinomial and Poisson play the role for counts that the normal plays for continuous responses.

1.2.1 Binomial distribution

Let $Y_1, \ldots, Y_n$ be independent Bernoulli trials with $P(Y_i = 1) = \pi$. Then $Y = \sum_i Y_i \sim \text{bin}(n, \pi)$:

$$p(y) = \binom{n}{y} \pi^y (1-\pi)^{n-y}, \qquad y = 0, 1, \ldots, n. \tag{1.1}$$

Proposition 1.1 (Binomial moments)

If $Y \sim \text{bin}(n, \pi)$ then $\mu = E(Y) = n\pi$, $\sigma^2 = \text{Var}(Y) = n\pi(1-\pi)$, and

$$\frac{E(Y-\mu)^3}{\sigma^3} = \frac{1-2\pi}{\sqrt{n\pi(1-\pi)}}.$$

In particular the distribution becomes symmetric (and, by the CLT, normal) as $n \to \infty$ for fixed π.

Proof.

Step 1. For one trial $E(Y_i) = E(Y_i^2) = \pi$, so $\text{Var}(Y_i) = \pi(1-\pi)$; by linearity and independence $E(Y) = n\pi$, $\text{Var}(Y) = n\pi(1-\pi)$.

Step 2. Third central moments of independent variables add (cumulants are additive), and $E(Y_i - \pi)^3 = (1-\pi)^3\pi - \pi^3(1-\pi) = \pi(1-\pi)(1-2\pi)$. Hence $E(Y-\mu)^3 = n\pi(1-\pi)(1-2\pi)$; dividing by $\sigma^3 = [n\pi(1-\pi)]^{3/2}$ gives the formula, which tends to 0. ■

Without independent, identical trials the binomial fails (e.g. sampling without replacement gives the hypergeometric, Section 3.5).

1.2.2 Multinomial distribution

Let each of n independent, identical trials fall in one of c categories, $y_{ij} = 1$ if trial i falls in category j, $n_j = \sum_i y_{ij}$ and $\pi_j = P(y_{ij} = 1)$. Then $(n_1, \ldots, n_c)$ is *multinomial*:

$$p(n_1, \ldots, n_{c-1}) = \frac{n!}{n_1!\, n_2! \cdots n_c!} \pi_1^{n_1} \pi_2^{n_2} \cdots \pi_c^{n_c}, \tag{1.2}$$

a $(c-1)$-dimensional distribution since $\sum_j n_j = n$; $c = 2$ gives the binomial.

Proposition 1.2 (Multinomial moments)

For the multinomial $(n, \boldsymbol{\pi})$ distribution,

$$E(n_j) = n\pi_j, \qquad \text{Var}(n_j) = n\pi_j(1-\pi_j), \qquad \text{Cov}(n_j, n_k) = -n\pi_j\pi_k \ (j \neq k),$$

and each n_j is marginally $\text{bin}(n, \pi_j)$.

Proof. [Supplement]

Step 1. n_j is a sum of n independent Bernoulli(π_j) indicators y_{ij}, so $n_j \sim \text{bin}(n, \pi_j)$ (Proposition 1.1).

Step 2. For $j \neq k$, $y_{ij}y_{ik} = 0$, so $\text{Cov}(y_{ij}, y_{ik}) = -\pi_j\pi_k$; summing over independent trials gives $\text{Cov}(n_j, n_k) = -n\pi_j\pi_k$. ■

In matrix form, $\text{Cov}(\mathbf{n}) = n[\text{diag}(\boldsymbol{\pi}) - \boldsymbol{\pi}\boldsymbol{\pi}^\top]$.

1.2.3 Poisson distribution

Counts without a fixed upper limit are often modelled as *Poisson* with mean μ:

$$p(y) = \frac{e^{-\mu}\mu^y}{y!}, \qquad y = 0, 1, 2, \ldots \tag{1.3}$$

Its skewness is $1/\sqrt{\mu}$, so it approaches normality as $\mu \to \infty$.

Proposition 1.3 (Poisson moments and binomial limit)

(a) If $Y \sim \text{Poisson}(\mu)$ then $E(Y) = \text{Var}(Y) = \mu$.
(b) If $n \to \infty$ and $\pi \to 0$ with $n\pi \to \mu$, then $\text{bin}(n, \pi)$ probabilities converge to $\text{Poisson}(\mu)$ probabilities.

Proof. [Supplement]

Step (a). $E(Y) = \sum_{y \geq 1} e^{-\mu}\mu^y/(y-1)! = \mu$ and $E[Y(Y-1)] = \sum_{y \geq 2} e^{-\mu}\mu^y/(y-2)! = \mu^2$, so $\text{Var}(Y) = \mu^2 + \mu - \mu^2 = \mu$.

Step (b). $\binom{n}{y}\pi^y(1-\pi)^{n-y} = \frac{(n\pi)^y}{y!} \cdot \frac{n(n-1)\cdots(n-y+1)}{n^y} \cdot (1-\pi)^{n-y}$. For fixed y the middle factor $\to 1$, and $(1-\pi)^{n-y} = \exp\{-n\pi + O(n\pi^2)\}(1-\pi)^{-y} \to e^{-\mu}$. ■

1.2.4 Overdispersion

Counts often vary more than the binomial or Poisson allows (*overdispersion*) because the probability or rate varies with unmeasured factors.

Proposition 1.4 (Mixing creates overdispersion)

Let Y given λ have mean $E(Y \mid \lambda)$ and variance $\text{Var}(Y \mid \lambda)$, where λ is random. Then

$$E(Y) = E[E(Y \mid \lambda)], \qquad \text{Var}(Y) = E[\text{Var}(Y \mid \lambda)] + \text{Var}[E(Y \mid \lambda)].$$

If $Y \mid \lambda \sim \text{Poisson}(\lambda)$ with $E(\lambda) = \mu$, then $E(Y) = \mu$ and $\text{Var}(Y) = \mu + \text{Var}(\lambda) > \mu$ whenever λ is nondegenerate.

Proof. [Supplement]

Step 1. The first identity is the tower property. For the second, $\text{Var}(Y) = E[\text{Var}(Y \mid \lambda) + E(Y \mid \lambda)^2] - (E[E(Y \mid \lambda)])^2 = E[\text{Var}(Y \mid \lambda)] + \text{Var}[E(Y \mid \lambda)]$.

Step 2. In the Poisson case $E(Y \mid \lambda) = \text{Var}(Y \mid \lambda) = \lambda$. ■

Mixtures of binomials likewise give extra-binomial variation (negative binomial, Section 4.3.4; random effects, Chapters 12–13).

1.2.5 Connection between Poisson and multinomial distributions

Theorem 1.5 (Poisson conditioned on the total is multinomial)

Let $Y_1, \ldots, Y_c$ be independent Poisson with means $\mu_1, \ldots, \mu_c$. Then $n = \sum_i Y_i \sim \text{Poisson}(\sum_i \mu_i)$, and conditional on n,

$$(Y_1, \ldots, Y_c) \mid \textstyle\sum_i Y_i = n \;\sim\; \text{multinomial}(n, \{\pi_i\}), \qquad \pi_i = \mu_i / \textstyle\sum_j \mu_j.$$

Proof.

Step 1. The product of the pgfs $e^{\mu_i(s-1)}$ is $e^{(\sum \mu_i)(s-1)}$, the pgf of Poisson$(\sum \mu_i)$.

Step 2. For $\sum n_i = n$, the conditional probability is

$$\frac{\prod_i e^{-\mu_i}\mu_i^{n_i}/n_i!}{e^{-\sum_j \mu_j}(\sum_j \mu_j)^n/n!} = \frac{n!}{\prod_i n_i!}\prod_i\Big(\frac{\mu_i}{\sum_j \mu_j}\Big)^{n_i},$$

using $\sum_i n_i = n$ to distribute $(\sum_j \mu_j)^n$; this is (1.2). ■

The Poisson and multinomial likelihoods share the kernel in $\{\pi_i\}$, so they usually give the same estimates.

1.3 Statistical inference for categorical data

Motivation. ML estimation and the Wald, LR and score tests, with their inverted intervals, are the toolkit for the course.

1.3.1 Likelihood functions and maximum likelihood estimation

Definition (Likelihood, ML estimate, information)

The *likelihood* $\ell(\beta)$ is the probability of the observed data as a function of β, and $L(\beta) = \log \ell(\beta)$. The *ML estimate* $\hat\beta$ maximizes ℓ, typically solving $\partial L/\partial\beta = 0$. The *information matrix* has (j,k) element

$$-E\Big(\frac{\partial^2 L(\boldsymbol\beta)}{\partial\beta_j\,\partial\beta_k}\Big). \tag{1.4}$$

Under regularity conditions $\mathrm{Cov}(\hat{\boldsymbol\beta})$ is asymptotically the inverse information; greater curvature of L means smaller SEs.

Lemma 1.6 (Score identities)

Let $f(y;\beta)$ be a density or pmf such that differentiation under $\int dy$ (or $\sum_y$) is allowed twice. For one observation let $u(\beta) = \partial \log f(Y;\beta)/\partial\beta$. Then

$$E_\beta[u(\beta)] = 0, \qquad \mathrm{Var}_\beta[u(\beta)] = -E_\beta\Big[\frac{\partial^2 \log f(Y;\beta)}{\partial\beta^2}\Big] =: i(\beta).$$

Proof. [Supplement]

Step 1. Differentiate $\int f\,dy = 1$: $0 = \int(\partial_\beta \log f)\,f\,dy = E[u]$.

Step 2. Differentiate again: $0 = \int[(\partial_\beta^2 \log f)f + (\partial_\beta \log f)^2 f]\,dy$, so $\mathrm{Var}(u) = E[u^2] = -E[\partial_\beta^2 \log f]$. ■

Theorem 1.7 (Asymptotic normality of the ML estimator)

Let $Y_1,\dots,Y_n$ be i.i.d. $f(y;\beta_0)$ with β_0 interior to the parameter space, satisfying Lemma 1.6, $0 < i(\beta_0) < \infty$, and $|\partial_\beta^3 \log f(y;\beta)| \le M(y)$ near β_0 with $E\,M(Y) < \infty$. If $\hat\beta$ is a consistent root of the likelihood equation, then

$$\sqrt{n}(\hat\beta - \beta_0) \xrightarrow{d} N\big(0,\, 1/i(\beta_0)\big).$$

For a vector $\boldsymbol{\beta}$ the same holds with $i(\boldsymbol{\beta}_0)$ the per-observation information matrix: $\sqrt{n}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0) \xrightarrow{d} N(\mathbf{0}, i(\boldsymbol{\beta}_0)^{-1})$, and in fact $\sqrt{n}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0) = i(\boldsymbol{\beta}_0)^{-1}\, n^{-1/2} \nabla L(\boldsymbol{\beta}_0) + o_p(1)$.

Proof. [Supplement]

Step 1. Expanding $0 = L'(\hat{\beta}) = L'(\beta_0) + (\hat{\beta} - \beta_0)L''(\beta_0) + \frac{1}{2}(\hat{\beta} - \beta_0)^2 L'''(\beta^*)$, with β^* between $\hat{\beta}$ and β_0,

$$\sqrt{n}(\hat{\beta} - \beta_0) = \frac{n^{-1/2} L'(\beta_0)}{-n^{-1} L''(\beta_0) - \frac{1}{2}(\hat{\beta} - \beta_0)\, n^{-1} L'''(\beta^*)}.$$

Step 2. $L'(\beta_0)$ is a sum of i.i.d. terms with mean 0 and variance $i(\beta_0)$ (Lemma 1.6), so $n^{-1/2} L'(\beta_0) \xrightarrow{d} N(0, i(\beta_0))$.

Step 3. $-n^{-1} L''(\beta_0) \xrightarrow{p} i(\beta_0)$ (WLLN), and with probability $\to 1$, $|n^{-1} L'''(\beta^*)| \le n^{-1} \sum_i M(Y_i) = O_p(1)$, which times $\hat{\beta} - \beta_0 = o_p(1)$ vanishes. So the denominator $\xrightarrow{p} i(\beta_0)$.

Step 4. Slutsky gives $N(0, i(\beta_0))/i(\beta_0) = N(0, 1/i(\beta_0))$ and the stated representation. The vector case replaces L'' by the Hessian and division by matrix inversion. ■

1.3.2 Likelihood function and ML estimate for binomial parameter

Constants such as $\binom{n}{y}$ do not affect the maximizer; only the *kernel* matters.

Proposition 1.8 (Binomial ML estimate and information)

For y successes in n trials, the log likelihood, score and information are

$$L(\pi) = y \log \pi + (n - y) \log(1 - \pi), \tag{1.5}$$

$$\frac{\partial L}{\partial \pi} = \frac{y}{\pi} - \frac{n - y}{1 - \pi} = \frac{y - n\pi}{\pi(1 - \pi)}, \tag{1.6}$$

$$-E\Big(\frac{\partial^2 L}{\partial \pi^2}\Big) = \frac{n}{\pi(1 - \pi)}. \tag{1.7}$$

The ML estimate is $\hat{\pi} = y/n$, with $E(\hat{\pi}) = \pi$ and $\sigma(\hat{\pi}) = \sqrt{\pi(1 - \pi)/n}$, the inverse square root of the information.

Proof.

Step 1. (1.6) vanishes iff $\pi = y/n$, and $\partial^2 L/\partial \pi^2 = -y/\pi^2 - (n - y)/(1 - \pi)^2 < 0$, so y/n is the global maximizer (for $y = 0$ or n, L is monotone with maximum at the boundary, again y/n).

Step 2. With $E(Y) = n\pi$: $E[Y/\pi^2 + (n - Y)/(1 - \pi)^2] = n/[\pi(1 - \pi)]$; and $E(\hat{\pi}) = \pi$, $\mathrm{Var}(\hat{\pi}) = \pi(1 - \pi)/n$. ■

1.3.3 Wald–likelihood ratio–score test triad

For $H_0\colon \beta = \beta_0$ write $L_0 = L(\beta_0)$ and $L_1 = L(\hat{\beta})$.

Definition (The three likelihood-based statistics)

(a) *Wald*: $z_W = (\hat{\beta} - \beta_0)/\mathrm{SE}$ with SE evaluated at $\hat{\beta}$; for a vector, $W = (\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0)^\top [\widehat{\mathrm{Cov}}(\hat{\boldsymbol{\beta}})]^{-1} (\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0)$.
(b) *Likelihood ratio (LR)*: $\Lambda = \ell_0/\ell_1 \le 1$ and $-2 \log \Lambda = -2(L_0 - L_1)$.

(c) *Score*: with $u(\beta_0) = \partial L/\partial\beta|_{\beta_0}$ and $\iota(\beta_0) = -E[\partial^2 L/\partial\beta^2]|_{\beta_0}$, $z_S = u(\beta_0)/\iota(\beta_0)^{1/2}$; in the vector case $\mathbf{u}^\top \iota^{-1}\mathbf{u}$ at $\boldsymbol{\beta}_0$.

Wald uses L at $\hat\beta$, score its slope and curvature at β_0, and LR both values (Figure 1).

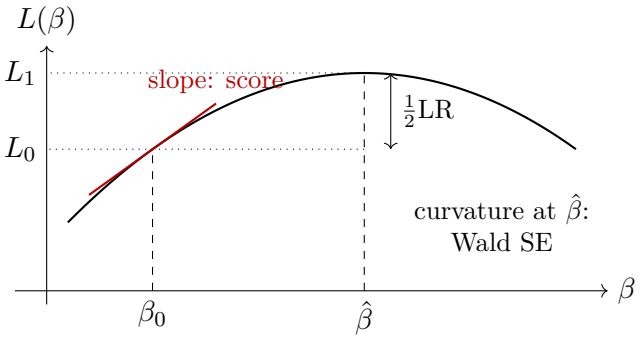


Figure 1: Information used by the three tests of $H_0\colon \beta = \beta_0$.

Lemma 1.9 (Normal quadratic forms)

(a) If $\mathbf{z} \sim N_q(\mathbf{0}, \Sigma)$ with Σ nonsingular, then $\mathbf{z}^\top\Sigma^{-1}\mathbf{z} \sim \chi^2_q$.
(b) If $\mathbf{z} \sim N_c(\mathbf{0}, P)$ with P symmetric idempotent of rank r, then $\mathbf{z}^\top\mathbf{z} \sim \chi^2_r$.

Proof. [Supplement]

Step (a). $\mathbf{w} = \Sigma^{-1/2}\mathbf{z} \sim N_q(\mathbf{0}, I)$ and $\mathbf{z}^\top\Sigma^{-1}\mathbf{z} = \sum_{k=1}^q w_k^2$.

Step (b). $P = Q\Delta Q^\top$ with Q orthogonal and Δ diagonal with r ones and zeros elsewhere (eigenvalues of an idempotent are 0 or 1). Then $\mathbf{w} = Q^\top\mathbf{z} \sim N(\mathbf{0}, \Delta)$, so $\mathbf{z}^\top\mathbf{z} = \mathbf{w}^\top\mathbf{w} = \sum_{k\le r} w_k^2 \sim \chi^2_r$. ■

Theorem 1.10 (Asymptotic null distributions of the triad)

Under the conditions of Theorem 1.7 and $H_0\colon \boldsymbol{\beta} = \boldsymbol{\beta}_0$ ($\boldsymbol{\beta} \in \mathbb{R}^q$), the Wald, LR and score statistics each converge in distribution to χ^2_q, and the difference between any two of them converges in probability to 0. For $q = 1$, z_W and z_S are asymptotically $N(0,1)$.

Proof. [Supplement] Write $\iota_1 = i(\boldsymbol{\beta}_0)$ and $\mathbf{s}_n = n^{-1/2}\nabla L(\boldsymbol{\beta}_0) \xrightarrow{d} N(\mathbf{0}, \iota_1)$ (CLT, Lemma 1.6).

Step 1 (score). The score statistic is $\nabla L^\top (n\iota_1)^{-1}\nabla L = \mathbf{s}_n^\top \iota_1^{-1}\mathbf{s}_n \xrightarrow{d} \chi^2_q$ by Lemma 1.9(a).

Step 2 (Wald). By Theorem 1.7, $\mathbf{z}_n = \sqrt{n}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0) = \iota_1^{-1}\mathbf{s}_n + o_p(1)$, and the Wald statistic is $\mathbf{z}_n^\top \hat\iota_1 \mathbf{z}_n$ with $\hat\iota_1 \xrightarrow{p} \iota_1$; so it equals $\mathbf{s}_n^\top\iota_1^{-1}\mathbf{s}_n + o_p(1)$.

Step 3 (LR). Expanding about $\hat{\boldsymbol{\beta}}$, where $\nabla L = \mathbf{0}$, $L(\boldsymbol{\beta}_0) = L(\hat{\boldsymbol{\beta}}) + \frac{1}{2}(\boldsymbol{\beta}_0 - \hat{\boldsymbol{\beta}})^\top \nabla^2 L(\tilde{\boldsymbol{\beta}})(\boldsymbol{\beta}_0 - \hat{\boldsymbol{\beta}})$ with $\tilde{\boldsymbol{\beta}}$ on the segment. Hence $-2(L_0 - L_1) = \mathbf{z}_n^\top[-n^{-1}\nabla^2 L(\tilde{\boldsymbol{\beta}})]\mathbf{z}_n = \mathbf{z}_n^\top \iota_1 \mathbf{z}_n + o_p(1)$ (WLLN and the third-derivative bound), the Wald statistic plus $o_p(1)$.

Step 4. All three are $\mathbf{s}_n^\top\iota_1^{-1}\mathbf{s}_n + o_p(1) \xrightarrow{d} \chi^2_q$. For $q = 1$, $z_S = \mathbf{s}_n/\sqrt{\iota_1} \xrightarrow{d} N(0,1)$ and $z_W = z_S + o_p(1)$. ■

For composite H_0, Wilks' theorem gives $-2\log\Lambda \xrightarrow{d} \chi^2_{\mathrm{df}}$, df = difference in parameter-space dimensions (Theorem 1.18 for the multinomial). For small to moderate n, LR is more reliable than Wald.

1.3.4 Constructing confidence intervals

Inverting a test gives an interval: the 95% interval is the set of β_0 whose test has $P > 0.05$.

(a) *Wald interval*: $\hat{\beta} \pm z_{\alpha/2}\text{SE}$.

(b) *LR interval*: $\{\beta_0 : -2[L(\beta_0) - L(\hat{\beta})] < \chi^2_1(\alpha)\}$, where $\chi^2_1(\alpha) = z^2_{\alpha/2}$.

(c) *Score interval*: $\{\beta_0 : |z_S(\beta_0)| < z_{\alpha/2}\}$.

If L is exactly parabolic the three coincide; with small categorical samples they can differ markedly. LR intervals are preferable for small to moderate n; Wald is the most common.

1.4 Statistical inference for binomial parameters

Motivation. Near the boundary the three methods behave very differently for the binomial.

1.4.1 Tests about a binomial parameter

Proposition 1.11 (Binomial test statistics)

For $H_0\colon \pi = \pi_0$ with $\hat{\pi} = y/n$:

$$z_W = \frac{\hat{\pi} - \pi_0}{\sqrt{\hat{\pi}(1-\hat{\pi})/n}}, \tag{1.8}$$

$$z_S = \frac{\hat{\pi} - \pi_0}{\sqrt{\pi_0(1-\pi_0)/n}}, \tag{1.9}$$

$$-2(L_0 - L_1) = 2\sum \text{observed} \log\frac{\text{observed}}{\text{fitted}} = 2\Big[y\log\frac{y}{n\pi_0} + (n-y)\log\frac{n-y}{n-n\pi_0}\Big]. \tag{1.10}$$

Each is asymptotically $N(0,1)$ or χ^2_1 under H_0.

Proof.

Step 1. Wald uses $\text{SE} = \sqrt{\hat{\pi}(1-\hat{\pi})/n}$ (Proposition 1.8 at $\pi = \hat{\pi}$).

Step 2. Score: by (1.6)–(1.7), $u(\pi_0)/\iota(\pi_0)^{1/2} = \frac{y-n\pi_0}{\pi_0(1-\pi_0)}\sqrt{\frac{\pi_0(1-\pi_0)}{n}} = \frac{y-n\pi_0}{\sqrt{n\pi_0(1-\pi_0)}}$, which is (1.9).

Step 3. LR: $-2(L_0 - L_1) = 2[y\log(\hat{\pi}/\pi_0) + (n-y)\log((1-\hat{\pi})/(1-\pi_0))]$, which is (1.10). The asymptotics are Theorem 1.10 with $q = 1$. ∎

The score statistic uses the null SE, and its null distribution is closer to $N(0,1)$.

1.4.2 Confidence intervals for a binomial parameter

Inverting the Wald test gives

$$\hat{\pi} \pm z_{\alpha/2}\sqrt{\hat{\pi}(1-\hat{\pi})/n}, \tag{1.11}$$

with coverage usually below nominal, much below for π near 0 or 1 (adding 2 successes and 2 failures helps greatly).

Proposition 1.12 (Wilson score interval)

The score interval $\{\pi_0 : |\hat\pi - \pi_0| < z\sqrt{\pi_0(1-\pi_0)/n}\}$, $z = z_{\alpha/2}$, has endpoints

$$\hat\pi\Big(\frac{n}{n+z^2}\Big) + \frac{1}{2}\Big(\frac{z^2}{n+z^2}\Big) \;\pm\; z\sqrt{\frac{1}{n+z^2}\Big[\hat\pi(1-\hat\pi)\Big(\frac{n}{n+z^2}\Big) + \frac{1}{2}\cdot\frac{1}{2}\Big(\frac{z^2}{n+z^2}\Big)\Big]}.$$

Its midpoint is $(y + z^2/2)/(n + z^2)$, the sample proportion after adding $z^2/2$ successes and $z^2/2$ failures.

Proof. [Supplement]

Step 1. The set $|z_S| < z$ is where $(\hat\pi - \pi_0)^2 - z^2\pi_0(1-\pi_0)/n < 0$, i.e. between the roots of $(1 + z^2/n)\pi_0^2 - (2\hat\pi + z^2/n)\pi_0 + \hat\pi^2 = 0$.

Step 2. Multiplying by n, the quadratic formula gives $\pi_0 = \big[n\hat\pi + z^2/2 \pm z\sqrt{n\hat\pi(1-\hat\pi) + z^2/4}\big]/(n + z^2)$; dividing numerator terms by $n + z^2$ gives the stated centre and half-width. ■

The coefficient of z^2 averages the variance of a proportion at $\hat\pi$ and at $1/2$, with sample size $n + z^2$. Wilson performs much better than Wald.

1.4.3 Proportion of vegetarians example

Example 1.13 (Proportion of vegetarians)

Of $n = 25$ students, $y = 0$ were vegetarian, so $\hat\pi = 0$.
(a) Wald: $(0, 0)$, useless at the boundary.
(b) Score: $(0.0,\ 0.133)$; e.g. $\pi_0 = 0.10$ gives $z_S = -1.67$ (inside).
(c) LR: $L(\pi) = 25\log(1-\pi)$, $L(\hat\pi) = 0$; the upper bound solves $-50\log(1-\pi_0) = 3.84$, giving $(0.0,\ 0.074)$.
The methods differ greatly because $\hat\pi$ is highly skewed near 0.

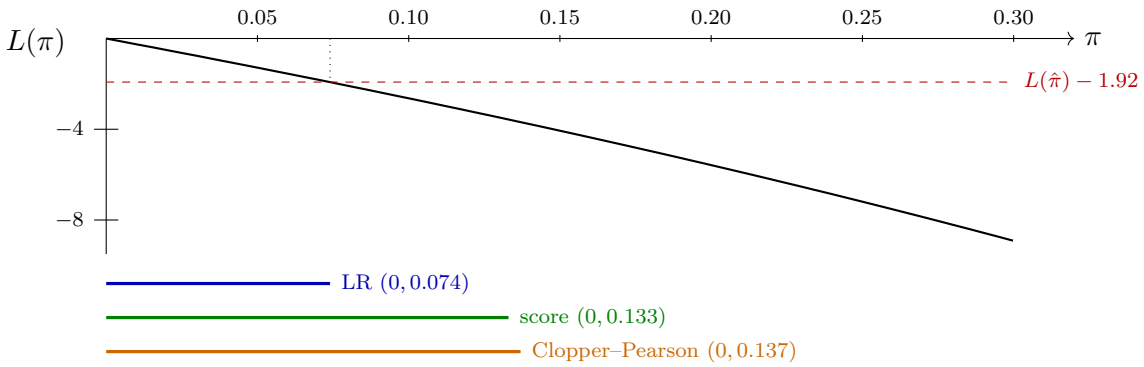


Figure 2: Binomial log likelihood for $y = 0$, $n = 25$; the LR interval is where L is within $3.84/2$ of its maximum.

1.5 Statistical inference for multinomial parameters

Motivation. Multinomial goodness of fit leads to Pearson's X^2 and the LR statistic G^2.

1.5.1 Estimation of multinomial parameters

Proposition 1.14 (Multinomial ML estimates)

The multinomial kernel

$$\prod_j \pi_j^{n_j}, \qquad \pi_j \geq 0, \quad \sum_j \pi_j = 1, \tag{1.12}$$

is maximized uniquely by the sample proportions $\hat{\pi}_j = p_j = n_j/n$.

Proof.

Step 1 (likelihood equations). With $\pi_c = 1 - \sum_{j<c} \pi_j$, $L = \sum_j n_j \log \pi_j$ has $\partial L/\partial \pi_j = n_j/\pi_j - n_c/\pi_c = 0$, $j < c$. So $\hat{\pi}_j/\hat{\pi}_c = n_j/n_c$; summing, $\hat{\pi}_c = n_c/n$ and $\hat{\pi}_j = n_j/n$.

Step 2 (global maximum). [Supplement] For any admissible $\boldsymbol{\pi}$ (sum over j with $n_j > 0$), Jensen's inequality gives

$$L(\boldsymbol{\pi}) - L(\mathbf{p}) = n \sum_j p_j \log \frac{\pi_j}{p_j} \leq n \log \sum_j p_j \frac{\pi_j}{p_j} \leq n \log 1 = 0,$$

with equality iff $\pi_j = p_j$ for all j. ■

1.5.2 Pearson statistic for testing a specified multinomial

For $H_0 \colon \pi_j = \pi_{j0}$ (Pearson, 1900), with *expected frequencies* $\mu_j = n\pi_{j0}$,

$$X^2 = \sum_j \frac{(n_j - \mu_j)^2}{\mu_j}, \tag{1.13}$$

and for large n, $X^2 \approx \chi^2_{c-1}$ (Theorem 1.16).

1.5.3 Example: testing Mendel's theories

Example 1.15 (Mendel's peas)

Mendel predicted 75% yellow seeds; 6022 of $n = 8023$ were yellow: $\mu_1 = 6017.25$, $X^2 = 0.015$ (df $= 1$), $P = 0.90$. Fisher (1936), summing over Mendel's experiments, got $X^2 = 42$ on df $= 84$ (mean 84, SD 13): a tiny left-tail P, a fit *too good*.

1.5.4 Chi-squared theoretical justification

Theorem 1.16 (Asymptotic null distribution of Pearson's X^2)

Under $H_0 \colon \boldsymbol{\pi} = \boldsymbol{\pi}_0$ with all $\pi_{j0} > 0$, $X^2 \xrightarrow{d} \chi^2_{c-1}$ as $n \to \infty$. Moreover X^2 is the score statistic for H_0.

Proof. [Supplement] Let $\hat{\boldsymbol{\pi}} = (p_1, \ldots, p_{c-1})^\top$, $\boldsymbol{\pi}_0 = (\pi_{10}, \ldots, \pi_{c-1,0})^\top$ and $\Sigma_0 = \operatorname{diag}(\boldsymbol{\pi}_0) - \boldsymbol{\pi}_0 \boldsymbol{\pi}_0^\top$.

Step 1 (CLT). $\hat{\boldsymbol{\pi}}$ is a mean of i.i.d. indicator vectors with covariance Σ_0 (Proposition 1.2), so $\sqrt{n}(\hat{\boldsymbol{\pi}} - \boldsymbol{\pi}_0) \xrightarrow{d} N_{c-1}(\mathbf{0}, \Sigma_0)$.

Step 2 (inverse). $\Sigma_0^{-1} = \operatorname{diag}(1/\pi_{j0}) + \mathbf{1}\mathbf{1}^\top/\pi_{c0}$: using $\boldsymbol{\pi}_0^\top \mathbf{1} = 1 - \pi_{c0}$, the product is $I + \boldsymbol{\pi}_0 \mathbf{1}^\top [1 - \pi_{c0} - (1 - \pi_{c0})]/\pi_{c0} = I$.

Step 3 (algebra). With $d_j = p_j - \pi_{j0}$, $j = 1, \dots, c$, and $d_c = -\sum_{j<c} d_j$,

$$n\,\mathbf{d}^\top \Sigma_0^{-1}\mathbf{d} = n\Big[\sum_{j<c}\frac{d_j^2}{\pi_{j0}} + \frac{(\sum_{j<c} d_j)^2}{\pi_{c0}}\Big] = n\sum_{j=1}^{c}\frac{d_j^2}{\pi_{j0}} = X^2. \tag{1.14}$$

Step 4 (limit). By continuous mapping and Lemma 1.9(a), $X^2 \xrightarrow{d} \chi^2_{c-1}$.

Step 5 (score test). By Proposition 1.14, Step 1, the score at $\boldsymbol{\pi}_0$ is $u_j = n(d_j/\pi_{j0} - d_c/\pi_{c0}) = n(\Sigma_0^{-1}\mathbf{d})_j$, and the information is $n\Sigma_0^{-1}$. So the score statistic is $\mathbf{u}^\top(n\Sigma_0^{-1})^{-1}\mathbf{u} = n\mathbf{d}^\top\Sigma_0^{-1}\mathbf{d} = X^2$. ■

Heuristically (Fisher), independent Poisson n_j give $\sum_j (n_j - \mu_j)^2/\mu_j \approx \chi^2_c$, and conditioning on $\sum n_j = n$ (Theorem 1.5) loses one df. For $c = 2$, $X^2 = z_S^2$ with z_S from (1.9).

1.5.5 Likelihood-ratio chi-squared

The kernel (1.12) is maximized at $\boldsymbol{\pi}_0$ under H_0 and at p_j in general, so

$$G^2 = -2\log\Lambda = 2\sum_j n_j \log\frac{n_j}{n\pi_{j0}}, \tag{1.15}$$

of the form (1.10), with $G^2 \xrightarrow{d} \chi^2_{c-1}$ under H_0 (Theorem 1.10, $q = c - 1$).

Theorem 1.17 (Asymptotic equivalence of X^2 and G^2)

Let μ_j be (possibly estimated) expected frequencies with $\sum_j \mu_j = n$ and $(n_j - \mu_j)/\mu_j = O_p(n^{-1/2})$ for each j, as holds under H_0. Then $G^2 - X^2 \xrightarrow{p} 0$; in particular $G^2 \xrightarrow{d} \chi^2_{c-1}$ under $H_0\colon \boldsymbol{\pi} = \boldsymbol{\pi}_0$.

Proof. [Supplement]

Step 1. Write $n_j = \mu_j(1 + x_j)$, $x_j = O_p(n^{-1/2})$. Since $(1+x)\log(1+x) = x + x^2/2 + R(x)$ with $|R(x)| \le C|x|^3$ for $|x| \le 1/2$,

$$G^2 = 2\sum_j \mu_j x_j + \sum_j \mu_j x_j^2 + 2\sum_j \mu_j R(x_j).$$

Step 2. $\sum_j \mu_j x_j = \sum_j (n_j - \mu_j) = 0$, $\sum_j \mu_j x_j^2 = X^2$, and $\mu_j R(x_j) = O(n)O_p(n^{-3/2}) \xrightarrow{p} 0$. Slutsky with Theorem 1.16 gives the limit. ■

Under H_a both grow like n but need not be close. X^2 approaches chi-squared faster; G^2 is often poor when $n/c < 5$.

1.5.6 Testing with estimated expected frequencies

Often $\pi_{j0} = \pi_j(\boldsymbol{\theta})$ with $\dim\boldsymbol{\theta} = p$ unknown; then $\hat\mu_j = n\pi_j(\hat{\boldsymbol{\theta}})$, and estimation reduces the df.

Theorem 1.18 (Degrees of freedom with estimated parameters)

Assume $\boldsymbol{\pi}(\boldsymbol{\theta})$ has continuous second derivatives, $\pi_j(\boldsymbol{\theta}_0) > 0$, the $c \times p$ Jacobian $\partial\boldsymbol{\pi}/\partial\boldsymbol{\theta}^\top$ has rank p at $\boldsymbol{\theta}_0$, and $\hat{\boldsymbol{\theta}}$ is a consistent ML estimate. If $H_0\colon \boldsymbol{\pi} = \boldsymbol{\pi}(\boldsymbol{\theta})$ holds, then

$$X^2 = \sum_j \frac{(n_j - \hat{\mu}_j)^2}{\hat{\mu}_j} \xrightarrow{d} \chi^2_{c-1-p}, \qquad \text{and } G^2 - X^2 \xrightarrow{p} 0.$$

Proof. [Supplement] Let $\boldsymbol{\pi}_0 = \boldsymbol{\pi}(\boldsymbol{\theta}_0)$, $D = \operatorname{diag}(\boldsymbol{\pi}_0)$, $\mathbf{s} = (\sqrt{\pi_{10}}, \ldots, \sqrt{\pi_{c0}})^\top$ (a unit vector), and $A = D^{-1/2}\,\partial\boldsymbol{\pi}/\partial\boldsymbol{\theta}^\top$ at $\boldsymbol{\theta}_0$ ($c \times p$, rank p).

Step 1. Since $\sum_j \pi_j(\boldsymbol{\theta}) = 1$, $\mathbf{s}^\top A = \sum_j \partial\pi_j/\partial\boldsymbol{\theta}^\top = \mathbf{0}$.

Step 2. With $\mathbf{p} = (p_1, \ldots, p_c)^\top$ and $\operatorname{Cov}(\sqrt{n}\,\mathbf{p}) = D - \boldsymbol{\pi}_0\boldsymbol{\pi}_0^\top$, the CLT gives $\mathbf{z}_n := \sqrt{n}\,D^{-1/2}(\mathbf{p} - \boldsymbol{\pi}_0) \xrightarrow{d} \mathbf{z} \sim N_c(\mathbf{0},\, I - \mathbf{s}\mathbf{s}^\top)$.

Step 3. $L(\boldsymbol{\theta}) = n\sum_j p_j \log \pi_j(\boldsymbol{\theta})$ has score at $\boldsymbol{\theta}_0$

$$\nabla L = n\,A^\top D^{-1/2}\mathbf{p} = n\,A^\top D^{-1/2}(\mathbf{p} - \boldsymbol{\pi}_0) = \sqrt{n}\,A^\top \mathbf{z}_n,$$

using $A^\top D^{-1/2}\boldsymbol{\pi}_0 = A^\top \mathbf{s} = \mathbf{0}$. The per-observation information is $\sum_j (\partial\pi_j/\partial\boldsymbol{\theta})(\partial\pi_j/\partial\boldsymbol{\theta})^\top/\pi_j = A^\top A$, so by Theorem 1.7 $\sqrt{n}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0) = (A^\top A)^{-1}A^\top \mathbf{z}_n + o_p(1)$.

Step 4. By the delta method, $\sqrt{n}\,D^{-1/2}(\boldsymbol{\pi}(\hat{\boldsymbol{\theta}}) - \boldsymbol{\pi}_0) = A\sqrt{n}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0) + o_p(1)$. So with $H = A(A^\top A)^{-1}A^\top$,

$$\mathbf{e}_n := \sqrt{n}\,D^{-1/2}(\mathbf{p} - \boldsymbol{\pi}(\hat{\boldsymbol{\theta}})) = (I - H)\mathbf{z}_n + o_p(1) \xrightarrow{d} (I - H)\mathbf{z}.$$

Step 5. X^2 differs from $\mathbf{e}_n^\top \mathbf{e}_n$ only by $\pi_j(\hat{\boldsymbol{\theta}})$ in place of π_{j0} in the denominators, a factor $\xrightarrow{p} 1$; so $X^2 = \mathbf{e}_n^\top \mathbf{e}_n + o_p(1)$.

Step 6. $(I - H)\mathbf{z}$ has covariance $(I - H)(I - \mathbf{s}\mathbf{s}^\top)(I - H) = I - H - \mathbf{s}\mathbf{s}^\top$, as $H\mathbf{s} = \mathbf{0}$. H and $\mathbf{s}\mathbf{s}^\top$ project onto orthogonal subspaces of dimensions p and 1, so $I - H - \mathbf{s}\mathbf{s}^\top$ is idempotent of rank $c - p - 1$; Lemma 1.9(b) gives $X^2 \xrightarrow{d} \chi^2_{c-1-p}$.

Step 7. Step 4 gives $(n_j - \hat{\mu}_j)/\hat{\mu}_j = O_p(n^{-1/2})$ and $\sum_j \hat{\mu}_j = n$, so Theorem 1.17 gives $G^2 - X^2 \xrightarrow{p} 0$. ■

Example 1.19 (Pneumonia infections in calves)

156 calves were classified by primary infection and, if infected, secondary infection; (no primary, yes secondary) is a *structural zero*. Counts (fitted): (yes,yes) 30 (38.1), (yes,no) 63 (39.0), (no,no) 63 (78.9). If $P(\text{primary}) = \pi = P(\text{secondary} \mid \text{primary})$, the cell probabilities are $(\pi^2, \pi(1-\pi), 1-\pi)$, the kernel is $\pi^{2n_{11}+n_{12}}(1-\pi)^{n_{12}+n_{22}}$, and

$$\hat{\pi} = \frac{2n_{11} + n_{12}}{2n_{11} + 2n_{12} + n_{22}} = \frac{123}{249} = 0.494.$$

$X^2 = 19.7$, df $= 3 - 1 - 1 = 1$ ($P < 0.00001$): primary infection seems to have an immunizing effect.

Chapter 2. Describing Contingency Tables

2.1 Probability structure for contingency tables

Motivation. The joint distribution contains the whole relationship; the sampling design decides which distributions a table can estimate.

2.1.1 Contingency tables and their distributions

Let X have I and Y have J categories. An $I \times J$ table of cell counts is a *contingency table* (Pearson, 1904).

Definition (Joint, marginal and conditional distributions)

(a) The *joint distribution* is $\{\pi_{ij}\}$, $\pi_{ij} = P(X = i, Y = j)$.
(b) The *marginal distributions* are $\pi_{i+} = \sum_j \pi_{ij}$ and $\pi_{+j} = \sum_i \pi_{ij}$, with $\sum_i \pi_{i+} = \sum_j \pi_{+j} = \sum_i \sum_j \pi_{ij} = 1$ (a "+" denotes summation over that index).
(c) When Y is a response, the *conditional distribution* of Y in row i is $\pi_{j|i} = P(Y = j \mid X = i) = \pi_{ij}/\pi_{i+}$, with $\sum_j \pi_{j|i} = 1$.

If X is fixed by design, only the conditional distributions are meaningful; studies typically compare those of Y across levels of X.

Example 2.1 (Aspirin and heart attacks)

The Physicians' Health Study: a five-year randomized blind trial of aspirin versus placebo (every other day) for cardiovascular mortality.

	Myocardial infarction			
Group	Fatal	Nonfatal	No attack	Total
Placebo	18	171	10,845	11,034
Aspirin	5	99	10,933	11,037

Of 11,034 placebo takers 18 had fatal attacks, against 5 of 11,037 aspirin takers.

2.1.2 Sensitivity and specificity

For a diagnostic test let X = true disease status and Y = diagnosis (positive/negative).

Definition (Sensitivity, specificity)

Sensitivity $= P(Y = \text{positive} \mid X = \text{diseased}) = \pi_{1|1}$; *specificity* $= P(Y = \text{negative} \mid X = \text{not diseased}) = \pi_{2|2}$.

Mammography with clinical examination: sensitivity 0.82, specificity 0.99 (literature survey).

2.1.3 Independence of categorical variables

Definition (Independence)

X and Y are *independent* if $\pi_{ij} = \pi_{i+}\pi_{+j}$ for all i, j.

Proposition 2.2 (Independence as homogeneity)

If all $\pi_{i+} > 0$, then X and Y are independent if and only if $\pi_{j|1} = \pi_{j|2} = \cdots = \pi_{j|I}$ for every j; the common value is then π_{+j}.

Proof.

Step 1. If $\pi_{ij} = \pi_{i+}\pi_{+j}$ then $\pi_{j|i} = \pi_{ij}/\pi_{i+} = \pi_{+j}$ for every i.

Step 2. Conversely, if $\pi_{j|i} = c_j$ for all i, then $\pi_{+j} = \sum_i \pi_{i+}\pi_{j|i} = c_j \sum_i \pi_{i+} = c_j$, so $\pi_{ij} = \pi_{i+}\pi_{j|i} = \pi_{i+}\pi_{+j}$. ■

For a response Y this *homogeneity of conditional distributions* is the natural definition. Sample analogues: $p_{ij} = n_{ij}/n$, $p_{j|i} = n_{ij}/n_{i+}$, $n = \sum_i \sum_j n_{ij}$ (also written $\hat{\pi}_{ij}$).

2.1.4 Poisson, binomial, and multinomial sampling

(a) *Poisson sampling*: the IJ counts Y_{ij} are independent Poisson(μ_{ij}), with joint pmf $\prod_i \prod_j e^{-\mu_{ij}} \mu_{ij}^{n_{ij}}/n_{ij}!$; nothing is fixed.

(b) *Multinomial sampling*: n is fixed and $\{n_{ij}\}$ is multinomial over the IJ cells, $\frac{n!}{\prod\prod n_{ij}!} \prod\prod \pi_{ij}^{n_{ij}}$.

(c) *Independent (product) multinomial sampling*: the row totals $n_i = n_{i+}$ are fixed and each row is an independent multinomial sample with probabilities $\{\pi_{j|i}\}$:

$$\prod_i \Big[\frac{n_i!}{\prod_j n_{ij}!} \prod_j \pi_{j|i}^{n_{ij}} \Big]. \tag{2.1}$$

(d) When both margins are fixed the distribution is hypergeometric (Section 3.5).

Theorem 2.3 (Conditioning on row totals gives product multinomial)

If $\{n_{ij}\}$ arise from independent Poisson sampling with means $\{\mu_{ij}\}$, or from multinomial sampling with probabilities $\pi_{ij} = \mu_{ij}/n$, then conditional on the row totals $\{n_{i+}\}$ the rows are independent multinomial samples with probabilities $\pi_{j|i} = \mu_{ij}/\mu_{i+}$.

Proof. [Supplement]

Step 1 (Poisson). By Theorem 1.5 applied to row i, given n_{i+} the counts $(n_{i1}, \ldots, n_{iJ})$ are multinomial$(n_{i+}, \{\mu_{ij}/\mu_{i+}\})$. Each row's counts and total involve only that row's Poisson variables, so the rows stay independent given all row totals.

Step 2 (multinomial). The row totals $(n_{1+}, \ldots, n_{I+})$ are multinomial$(n, \{\pi_{i+}\})$ (collapsing categories of a multinomial). Hence for tables with these row totals,

$$P(\{n_{ij}\} \mid \{n_{i+}\}) = \frac{\frac{n!}{\prod_i \prod_j n_{ij}!} \prod_i \prod_j \pi_{ij}^{n_{ij}}}{\frac{n!}{\prod_i n_{i+}!} \prod_i \pi_{i+}^{n_{i+}}} = \prod_i \Big[\frac{n_{i+}!}{\prod_j n_{ij}!} \prod_j \Big(\frac{\pi_{ij}}{\pi_{i+}} \Big)^{n_{ij}} \Big],$$

using $\pi_{i+}^{n_{i+}} = \prod_j \pi_{i+}^{n_{ij}}$. This is (2.1) with $\pi_{j|i} = \pi_{ij}/\pi_{i+} = \mu_{ij}/\mu_{i+}$. ■

So for explanatory X one may analyse conditional on the row totals even when the design did not fix them.

2.1.5 Seat belt example

Example 2.4 (Seat belts and crash outcome)

Seat-belt use (yes/no) by crash outcome (fatal/nonfatal), Massachusetts Turnpike:
(a) recording all accidents in the coming year: n random, the four counts independent Poisson;
(b) sampling 200 police records from last year: $n = 200$ fixed, multinomial over four cells;
(c) sampling 100 fatal and 100 nonfatal records: column totals fixed, each column an independent binomial;
(d) the (unethical) experiment randomizing 100 of 200 subjects to wear belts and forcing crashes: row totals fixed, independent binomial rows.

2.1.6 Types of studies

Example 2.5 (Smoking and lung cancer: a case–control study)

Doll and Hill: 709 lung cancer patients in 20 London hospitals, each with a matched noncancer control (hospital, gender, 5-year age group), asked about smoking.

	Lung cancer	
Smoker	Cases	Controls
Yes	688	650
No	21	59
Total	709	709

The cancer margin is fixed: $P(\text{smoker} \mid \text{case}) = 688/709 = 0.970$, $P(\text{smoker} \mid \text{control}) = 650/709 = 0.917$; $P(\text{cancer} \mid \text{smoking})$ needs the prevalence (Bayes).

Definition (Study designs)

(a) *Retrospective* (case–control): samples at fixed levels of Y, X looked up in the past; each column is multinomial on X.
(b) *Prospective*: groups defined by X are followed up (*clinical trials* randomize X, *cohort studies* do not); each row is multinomial on Y.
(c) *Cross-sectional*: subjects classified on both variables at once; one multinomial sample of IJ counts.

Case–control, cohort and cross-sectional studies are *observational* (more prone to bias); a clinical trial is *experimental*, randomization balancing other variables.

2.2 Comparing two proportions

Motivation. Difference, ratio and odds ratio compare π_1, π_2 and behave very differently under change of sampling design.

In a 2×2 table write $\pi_i = \pi_{1|i}$ for the success probability in row i, so $\pi_{2|i} = 1 - \pi_i$.

2.2.1 Difference of proportions

The *difference of proportions* $\pi_1 - \pi_2 \in [-1, 1]$ is the same for successes or failures up to sign, and is 0 iff Y is independent of the row classification. The column difference is in general different.

2.2.2 Relative risk

A difference of fixed size may matter more near 0 or 1: $0.010 - 0.001$ and $0.410 - 0.401$ are both 0.009, but the ratios are 10.0 and 1.02.

Definition (Relative risk)

The *relative risk* is $\pi_1/\pi_2 \in [0, \infty)$; it equals 1 under independence. Comparing the second response category gives a different value $(1 - \pi_1)/(1 - \pi_2)$.

2.2.3 Odds ratio

Definition (Odds, odds ratio)

For success probability π the *odds* are $\Omega = \pi/(1 - \pi)$, inverted by $\pi = \Omega/(\Omega + 1)$. With $\Omega_i = \pi_i/(1 - \pi_i) = \pi_{i1}/\pi_{i2}$, the *odds ratio* is

$$\theta = \frac{\Omega_1}{\Omega_2} = \frac{\pi_1/(1 - \pi_1)}{\pi_2/(1 - \pi_2)} = \frac{\pi_{11}\pi_{22}}{\pi_{12}\pi_{21}}, \tag{2.2}$$

also called the *cross-product ratio* (Yule). The sample odds ratio is $\hat{\theta} = n_{11}n_{22}/(n_{12}n_{21})$.

E.g. $\pi = 0.75$ gives $\Omega = 3$: a success is three times as likely as a failure; $\Omega = 1/3$ gives $\pi = 0.25$.

2.2.4 Properties of the odds ratio

Proposition 2.6 (Properties of the odds ratio)

Assume all $\pi_{ij} > 0$.
(a) $\theta = 1$ iff $\Omega_1 = \Omega_2$ iff $\pi_1 = \pi_2$ (independence); $\theta > 1$ iff $\pi_1 > \pi_2$.
(b) Reversing the order of the rows (or of the columns) replaces θ by $1/\theta$, i.e. $\log\theta$ by $-\log\theta$; transposing the table leaves θ unchanged.
(c) θ is the same whichever conditional direction is used:

$$\theta = \frac{P(Y = 1 \mid X = 1)/P(Y = 2 \mid X = 1)}{P(Y = 1 \mid X = 2)/P(Y = 2 \mid X = 2)} = \frac{P(X = 1 \mid Y = 1)/P(X = 2 \mid Y = 1)}{P(X = 1 \mid Y = 2)/P(X = 2 \mid Y = 2)}. \tag{2.3}$$

(d) $\hat{\theta}$ is unchanged when both counts in a row, or both counts in a column, are multiplied by a positive constant.

Proof.

Step (a). $x \mapsto x/(1 - x)$ is strictly increasing on $(0, 1)$, so $\Omega_1 \gtreqless \Omega_2$ iff $\pi_1 \gtreqless \pi_2$.

Step (b). Swapping rows maps $\pi_{11}\pi_{22}/(\pi_{12}\pi_{21})$ to $\pi_{21}\pi_{12}/(\pi_{22}\pi_{11}) = 1/\theta$; similarly for columns. Transposing swaps π_{12} and π_{21}, which leaves the cross-product ratio unchanged.

Step (c). The first ratio is $(\pi_{11}/\pi_{12})/(\pi_{21}/\pi_{22})$ after cancelling π_{1+} and π_{2+}; the second is $(\pi_{11}/\pi_{21})/(\pi_{12}/\pi_{22})$ after cancelling π_{+1}, π_{+2}. Both equal $\pi_{11}\pi_{22}/(\pi_{12}\pi_{21})$.

Step (d). Multiplying row 1 by c multiplies numerator $n_{11}n_{22}$ and denominator $n_{12}n_{21}$ each by c; columns likewise. ■

$\theta = 4$ means the odds (not the probability) in row 1 are four times those in row 2; $\log 4 = 1.39$ and $\log 0.25 = -1.39$ are equally strong. By (c)–(d), $\hat{\theta}$ estimates the same parameter in prospective, retrospective and cross-sectional designs (e.g. $100 + 100$ or $40 + 160$ in a case–control study). The sample difference and relative risk are invariant to scaling rows only, not columns or transposition.

2.2.5 Aspirin and heart attacks revisited

Example 2.7 (Three measures for the aspirin study)

Combining fatal and nonfatal attacks in Example 2.1: placebo $189/11{,}034 = 0.0171$, aspirin $104/11{,}037 = 0.0094$.
(a) Difference: $0.0171 - 0.0094 = 0.0077$.
(b) Relative risk: $0.0171/0.0094 = 1.82$.
(c) Odds ratio: $\hat{\theta} = \dfrac{189 \times 10{,}933}{10{,}845 \times 104} = 1.83$; the odds of heart attack were 83% higher with placebo.

2.2.6 Case–control studies and the odds ratio

A case–control study estimates $P(X = i \mid Y = j)$, not $P(Y = j \mid X = i)$, hence neither the difference nor the relative risk; by (2.3) the odds ratio is the same in either direction.

Example 2.8 (Odds ratio from Doll and Hill's data)

From Example 2.5,

$$\hat{\theta} = \frac{(688/709)/(21/709)}{(650/709)/(59/709)} = \frac{688 \times 59}{650 \times 21} = 3.0.$$

Estimated odds of lung cancer for smokers: 3.0 times those for nonsmokers.

2.2.7 Relationship between odds ratio and relative risk

Proposition 2.9 (Odds ratio versus relative risk)

$$\theta = \frac{\pi_1}{\pi_2} \times \frac{1 - \pi_2}{1 - \pi_1}.$$

So $\theta \approx$ relative risk when π_1, π_2 are both near 0, and θ is farther from 1 (same side).

Proof.

Step 1. Directly from (2.2): $\theta = \frac{\pi_1}{1-\pi_1} \cdot \frac{1-\pi_2}{\pi_2}$.

Step 2. If $\pi_1 > \pi_2$ then $(1 - \pi_2)/(1 - \pi_1) > 1$ so $\theta > \pi_1/\pi_2 > 1$; if $\pi_1 < \pi_2$ the factor is < 1 and $\theta < \pi_1/\pi_2 < 1$. As $\pi_1, \pi_2 \to 0$ the factor tends to 1. ■

Aspirin study ($\pi_i < 0.02$): relative risk 1.82, odds ratio 1.83. So for rare diseases $\hat{\theta}$ estimates the relative risk (Cornfield, 1951): smokers had about 3 times the lung cancer rate.

2.3 Partial association in stratified 2×2 tables

Motivation. A third variable Z affecting both X and Y can change, even reverse, an apparent association once held fixed.

E.g. a passive-smoking study must control age and socioeconomic status: spouses of nonsmokers may be younger, and the young get lung cancer less often.

2.3.1 Partial tables

Definition (Partial and marginal tables)

For fixed Z, the two-way cross-sections of an $X \times Y \times Z$ table are *partial tables* with *conditional associations*; summing them gives the XY *marginal table*, which ignores Z.

2.3.2 Death penalty example

Example 2.10 (Death penalty verdict by defendant's and victims' race)

674 defendants in Florida multiple-murder indictments, 1976–1987. Y = death penalty, X = defendant's race, Z = victims' race.

Victims' race	Defendant's race	Yes	No	Percent yes
White	White	53	414	11.3
	Black	11	37	22.9
Black	White	0	16	0.0
	Black	4	139	2.8
Total	White	53	430	11.0
	Black	15	176	7.9

(a) Controlling for victims' race, black defendants received the death penalty more often: by $22.9 - 11.3 = 11.6$ points with white victims and 2.8 points with black victims.
(b) Ignoring victims' race, white defendants received it more often (11.0% vs. 7.9%): the association reverses.
(c) Victims' race is strongly associated with defendant's race (marginal odds ratio $467 \times 143/(48 \times 16) = 87.0$), and the death penalty is much more likely for white victims.

Definition (Simpson's paradox)

A marginal association whose direction differs from that of each conditional association is an instance of *Simpson's paradox* (Yule 1903, Simpson 1951).

It warns against causal readings of an X–Y association. Against Fisher's genetic-factor objection, Cornfield showed that a confounder explaining a very strong association must itself be very strongly associated with X and Y.

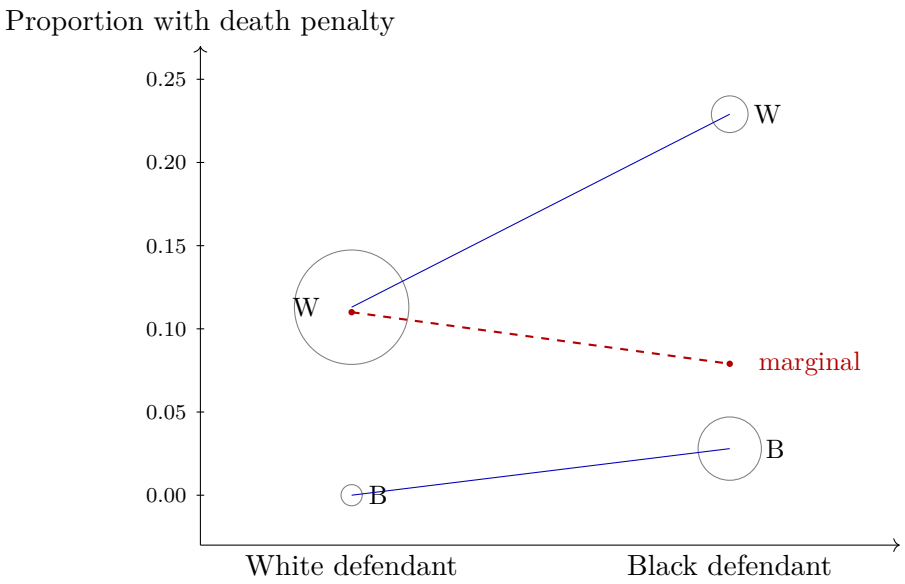


Figure 3: Simpson's paradox in the death penalty data (letters: victims' race; area $\propto$ cases). Conditional lines rise, the marginal line falls.

2.3.3 Conditional and marginal odds ratios

For a $2 \times 2 \times K$ table with expected frequencies μ_{ijk}:

$$\theta_{XY(k)} = \frac{\mu_{11k}\mu_{22k}}{\mu_{12k}\mu_{21k}} \quad \text{(conditional)}, \qquad \theta_{XY} = \frac{\mu_{11+}\mu_{22+}}{\mu_{12+}\mu_{21+}} \quad \text{(marginal)},$$

with $\mu_{ij+} = \sum_k \mu_{ijk}$; sample versions use counts.

Example 2.11 (Odds ratios for the death penalty data)

(a) White victims: $\hat{\theta}_{XY(1)} = \dfrac{53 \times 37}{414 \times 11} = 0.43$; the odds of the death penalty for white defendants were 43% of those for black defendants.
(b) Black victims: $\hat{\theta}_{XY(2)} = \dfrac{0 \times 139}{16 \times 4} = 0.0$ (no white defendant with black victims got the death penalty).
(c) Marginal: $\hat{\theta}_{XY} = \dfrac{53 \times 176}{430 \times 15} = 1.45$; the odds were 45% higher for white defendants.

2.3.4 Marginal versus conditional independence

Definition (Conditional independence)

X and Y are *conditionally independent at level* k of Z if they are independent in partial table k, i.e. $P(Y = j \mid X = i, Z = k) = P(Y = j \mid Z = k)$ for all i, j. They are *conditionally independent given* Z if this holds for every k.

Proposition 2.12 (Factorization under conditional independence)

For a single multinomial over the three-way table with $\pi_{ijk} = P(X = i, Y = j, Z = k)$ and all $\pi_{++k} > 0$, X and Y are conditionally independent given Z iff

$$\pi_{ijk} = \frac{\pi_{i+k}\,\pi_{+jk}}{\pi_{++k}} \qquad \text{for all } i, j, k.$$

Conditional independence does not imply marginal independence: in general $\pi_{ij+} = \sum_k \pi_{i+k}\pi_{+jk}/\pi_{++k} \neq \pi_{i++}\pi_{+j+}$.

Proof.

Step 1. Always $\pi_{ijk} = P(X = i, Z = k)\,P(Y = j \mid X = i, Z = k) = \pi_{i+k}\,P(Y = j \mid X = i, Z = k)$.

Step 2. Under conditional independence the last factor equals $P(Y = j \mid Z = k) = \pi_{+jk}/\pi_{++k}$, giving the factorization. Conversely, the factorization gives $P(Y = j \mid X = i, Z = k) = \pi_{ijk}/\pi_{i+k} = \pi_{+jk}/\pi_{++k}$ whenever $\pi_{i+k} > 0$.

Step 3. Summing over k gives $\pi_{ij+} = \sum_k \pi_{i+k}\pi_{+jk}/\pi_{++k}$, which involves k in every factor and does not reduce to $\pi_{i++}\pi_{+j+}$; Example 2.13 is an explicit counterexample. ■

For $2 \times 2 \times K$ tables, conditional independence means $\theta_{XY(k)} = 1$ for all k.

Example 2.13 (Conditional but not marginal independence)

Expected frequencies for Y = response, X = treatment, Z = clinic:

Clinic	Treatment	Success	Failure
1	A	18	12
	B	12	8
2	A	2	8
	B	8	32
Total	A	20	20
	B	20	40

(a) $\theta_{XY(1)} = \dfrac{18 \times 8}{12 \times 12} = 1.0$ and $\theta_{XY(2)} = \dfrac{2 \times 32}{8 \times 8} = 1.0$: conditional independence.

(b) Marginal: $\theta_{XY} = \dfrac{20 \times 40}{20 \times 20} = 2.0$.

(c) All XZ and YZ conditional odds ratios are 6.0, e.g. $\dfrac{18 \times 8}{12 \times 2}$: clinic 1 uses A more and has more successes, so the marginal table misleads.

2.3.5 Homogeneous association

Definition (Homogeneous association)

A $2 \times 2 \times K$ table has *homogeneous XY association* if $\theta_{XY(1)} = \cdots = \theta_{XY(K)}$; conditional independence is the case of common value 1. Otherwise there is *interaction* and Z is an *effect modifier*.

Proposition 2.14 (Homogeneous association is symmetric)

For a $2 \times 2 \times 2$ table with positive probabilities,

$$\frac{\theta_{XY(1)}}{\theta_{XY(2)}} = \frac{\theta_{XZ(1)}}{\theta_{XZ(2)}} = \frac{\theta_{YZ(1)}}{\theta_{YZ(2)}},$$

where $\theta_{XZ(j)}$ is the XZ odds ratio at level j of Y and $\theta_{YZ(i)}$ the YZ odds ratio at level i of X. So, for $2 \times 2 \times K$, homogeneous XY association holds iff for all k, k' the XZ (and YZ) odds ratio formed with k, k' is the same at both levels of Y (of X).

Proof. [Supplement]

Step 1. Directly,

$$\frac{\theta_{XY(1)}}{\theta_{XY(2)}} = \frac{\pi_{111}\pi_{221}}{\pi_{121}\pi_{211}} \cdot \frac{\pi_{122}\pi_{212}}{\pi_{112}\pi_{222}}, \qquad \frac{\theta_{XZ(1)}}{\theta_{XZ(2)}} = \frac{\pi_{111}\pi_{212}}{\pi_{112}\pi_{211}} \cdot \frac{\pi_{122}\pi_{221}}{\pi_{121}\pi_{222}}.$$

Both equal $\dfrac{\pi_{111}\pi_{122}\pi_{212}\pi_{221}}{\pi_{112}\pi_{121}\pi_{211}\pi_{222}}$ (odd index sums over even), symmetric in the three indices, so it also equals $\theta_{YZ(1)}/\theta_{YZ(2)}$.

Step 2. For $2 \times 2 \times K$, apply Step 1 to the $2 \times 2 \times 2$ subtable formed by levels k, k' of Z: $\theta_{XY(k)} = \theta_{XY(k')}$ iff the XZ odds ratio for (k, k') is the same at $Y = 1$ and $Y = 2$, and likewise for YZ. ■

Homogeneity is thus a joint property ("no three-factor interaction"). Odds ratios 1.2, 3.9, 8.8 in age groups < 45, 45–65, > 65 would make age an effect modifier. Death penalty: $\hat{\theta}_{XY(1)} = 0.43$, $\hat{\theta}_{XY(2)} = 0.0$ (zero count; 0.94 after adding 1/2), not contradicting homogeneity (tests in Section 6.3).

2.4 Extensions for $I \times J$ tables

Motivation. Beyond 2×2, a minimal set of odds ratios describes association completely; summary indices trade completeness for interpretability.

2.4.1 Odds ratios in $I \times J$ tables

Any pair of rows a, b and columns c, d gives an odds ratio $\pi_{ac}\pi_{bd}/(\pi_{bc}\pi_{ad})$; there are $\binom{I}{2}\binom{J}{2}$ of them, highly redundant.

Definition (Local and baseline odds ratios)

The *local odds ratios* are

$$\theta_{ij} = \frac{\pi_{ij}\,\pi_{i+1,j+1}}{\pi_{i,j+1}\,\pi_{i+1,j}}, \qquad i = 1, \ldots, I-1,\ j = 1, \ldots, J-1. \tag{2.4}$$

An alternative basic set uses the last row and column as baseline:

$$\alpha_{ij} = \frac{\pi_{ij}\,\pi_{IJ}}{\pi_{Ij}\,\pi_{iJ}}, \qquad i = 1, \ldots, I-1,\ j = 1, \ldots, J-1. \tag{2.5}$$

Proposition 2.15 ($(I-1)(J-1)$ odds ratios describe the association)

Assume all $\pi_{ij} > 0$.
(a) Every odds ratio $\pi_{ac}\pi_{bd}/(\pi_{bc}\pi_{ad})$ with $a < b$, $c < d$ is the product $\prod_{i=a}^{b-1}\prod_{j=c}^{d-1}\theta_{ij}$.
(b) Given the marginal distributions $\{\pi_{i+}\}, \{\pi_{+j}\}$, the local odds ratios determine the cell probabilities uniquely.
(c) Independence holds iff all $(I-1)(J-1)$ local odds ratios equal 1.

Proof. [Supplement]

Step (a). Let $\lambda_{ij} = \log \pi_{ij}$ and $\log \theta_{ij} = \lambda_{ij} - \lambda_{i,j+1} - \lambda_{i+1,j} + \lambda_{i+1,j+1}$. Summing over $j = c, \ldots, d-1$ telescopes to $\lambda_{ic} - \lambda_{id} - \lambda_{i+1,c} + \lambda_{i+1,d}$; summing that over $i = a, \ldots, b-1$ telescopes to $\lambda_{ac} - \lambda_{ad} - \lambda_{bc} + \lambda_{bd}$, the log of the stated odds ratio.

Step (b). Suppose $\{\pi_{ij}\}$ and $\{\pi'_{ij}\}$ have the same margins and the same local odds ratios. Let $\delta_{ij} = \log(\pi'_{ij}/\pi_{ij})$. Its local contrasts $\delta_{ij} - \delta_{i,j+1} - \delta_{i+1,j} + \delta_{i+1,j+1}$ vanish, so by the telescoping in (a) so does $\delta_{ij} - \delta_{iJ} - \delta_{Ij} + \delta_{IJ}$. With $r_i = \delta_{iJ} - \delta_{IJ}$ and $c_j = \delta_{Ij}$ this gives $\delta_{ij} = r_i + c_j$. Hence

$$\sum_i \sum_j (\pi'_{ij} - \pi_{ij}) \log \frac{\pi'_{ij}}{\pi_{ij}} = \sum_i r_i(\pi'_{i+} - \pi_{i+}) + \sum_j c_j(\pi'_{+j} - \pi_{+j}) = 0.$$

Each term $(\pi' - \pi)\log(\pi'/\pi)$ is ≥ 0, with equality iff $\pi' = \pi$ (log is increasing). Hence $\pi'_{ij} = \pi_{ij}$ for all i, j.

Step (c). If $\pi_{ij} = \pi_{i+}\pi_{+j}$ then every $\theta_{ij} = 1$. Conversely, the independence table $\{\pi_{i+}\pi_{+j}\}$ has the same margins and all local odds ratios 1; by (b) it is the only such table. ■

So $(I-1)(J-1)$ parameters describe any $I \times J$ association. Example 2.1: local odds ratios $\frac{18\times 99}{171\times 5} = 2.08$ (fatal vs. nonfatal) and $\frac{171\times 10{,}933}{99\times 10{,}845} = 1.74$ (nonfatal vs. none), product 3.63 (first vs. third column); the more serious outcome was more prevalent under placebo. For $I \times J \times K$, homogeneous XY association means each conditional XY odds ratio is constant in Z.

2.4.2 Summary measures of association

A *proportional reduction in variation* measure, with $V(Y)$ a variation measure for the marginal distribution $\{\pi_{+j}\}$ and $V(Y \mid i)$ that for $\{\pi_{j|i}\}$, is

$$\frac{V(Y) - E[V(Y \mid X)]}{V(Y)}, \qquad E[V(Y \mid X)] = \sum_i \pi_{i+} V(Y \mid i), \tag{2.6}$$

generalizing R^2. For a nominal response Theil used the *entropy* $V(Y) = -\sum_j \pi_{+j} \log \pi_{+j}$, giving the *uncertainty coefficient*

$$U = -\frac{\sum_i \sum_j \pi_{ij} \log(\pi_{ij}/(\pi_{i+}\pi_{+j}))}{\sum_j \pi_{+j} \log \pi_{+j}}. \tag{2.7}$$

Proposition 2.16 (Range of the uncertainty coefficient)

If more than one $\pi_{+j} > 0$, then $0 \leq U \leq 1$; $U = 0$ iff X and Y are independent, and $U = 1$ iff for each i some $\pi_{j|i} = 1$.

Proof. [Supplement] Write

$$H(Y) = -\sum_j \pi_{+j} \log \pi_{+j} > 0, \qquad H(Y \mid X) = -\sum_i \pi_{i+} \sum_j \pi_{j|i} \log \pi_{j|i} \geq 0.$$

Step 1. (2.7) is (2.6) with entropy: since $\log \pi_{j|i} = \log \pi_{ij} - \log \pi_{i+}$, $H(Y) - H(Y \mid X) = \sum_i \sum_j \pi_{ij} \log \frac{\pi_{ij}}{\pi_{i+}\pi_{+j}}$, so $U = [H(Y) - H(Y \mid X)]/H(Y)$.

Step 2. By Jensen, $-\sum\sum \pi_{ij} \log \frac{\pi_{i+}\pi_{+j}}{\pi_{ij}} \geq -\log \sum\sum \pi_{i+}\pi_{+j} \geq 0$, with equality iff $\pi_{ij} = \pi_{i+}\pi_{+j}$ on all cells (then the sum is 1). So $U \geq 0$, with equality iff independence.

Step 3. $U \leq 1$ iff $H(Y \mid X) \geq 0$, true since each $-\pi \log \pi \geq 0$; equality iff every conditional distribution is degenerate. ■

Such measures are hard to interpret (a "30% reduction in entropy"?); ordinal summaries are easier.

2.4.3 Ordinal trends: concordant and discordant pairs

For ordinal X, Y, a pair is *concordant* if the subject higher on X is also higher on Y, *discordant* if lower, and *tied* if they agree on X and/or Y.

Example 2.17 (Job satisfaction and income)

Black males, 1996 General Social Survey ($n = 96$); VD/LD/MS/VS = very / a little dissatisfied, moderately / very satisfied.

Income ($)	VD	LD	MS	VS
< 15,000	1	3	10	6
15,000–25,000	2	3	10	7
25,000–40,000	1	6	14	12
> 40,000	0	1	9	11

Each cell's count times the total count below and to the right gives concordant pairs:

$$\begin{aligned} C = {} & 1(3+10+7+6+14+12+1+9+11) + 3(10+7+14+12+9+11) \\ & + 10(7+12+11) + 2(6+14+12+1+9+11) + 3(14+12+9+11) + 10(12+11) \\ & + 1(1+9+11) + 6(9+11) + 14(11) = 1331, \end{aligned}$$

and similarly (below and to the left) $D = 3(2{+}1{+}0){+}10(2{+}3{+}1{+}6{+}0{+}1){+}\cdots{+}12(0{+}1{+}9) = 849$. Since $C > D$, low income tends to occur with low satisfaction.

For two independent observations from $\{\pi_{ij}\}$ the probabilities of concordance and discordance are

$$\Pi_c = 2\sum_i \sum_j \pi_{ij}\Big(\sum_{h>i}\sum_{k>j} \pi_{hk}\Big), \qquad \Pi_d = 2\sum_i \sum_j \pi_{ij}\Big(\sum_{h>i}\sum_{k<j} \pi_{hk}\Big), \tag{2.8}$$

the factor 2 because either observation may be the one in cell (i, j).

2.4.4 Ordinal measure of association: gamma

Definition (Goodman–Kruskal gamma)

Given that a pair is untied on both variables, $\Pi_c/(\Pi_c + \Pi_d)$ is the probability of concordance. *Gamma* is

$$\gamma = \frac{\Pi_c - \Pi_d}{\Pi_c + \Pi_d}, \qquad \text{sample version } \hat{\gamma} = \frac{C - D}{C + D}. \tag{2.9}$$

Proposition 2.18 (Properties of gamma)

(a) $-1 \leq \gamma \leq 1$; $\gamma = 1$ iff $\Pi_d = 0$ and $\gamma = -1$ iff $\Pi_c = 0$ (perfect monotonicity, not linearity).
(b) Reversing the category order of one variable changes the sign of γ; γ is symmetric in X

and Y.
(c) Independence implies $\gamma = 0$; the converse is false.

Proof. [Supplement]

Step (a). $\Pi_c, \Pi_d \geq 0$, so $|\Pi_c - \Pi_d| \leq \Pi_c + \Pi_d$, with the extreme cases as stated.

Step (b). Reversing one ordering turns concordant pairs into discordant ones and vice versa, swapping Π_c and Π_d. Interchanging the roles of X and Y does not change whether a pair is concordant.

Step (c). Let (X_1, Y_1), (X_2, Y_2) be independent draws. Then $\Pi_c = 2P(X_1 > X_2, Y_1 > Y_2)$ and $\Pi_d = 2P(X_1 > X_2, Y_1 < Y_2)$. Under independence of X and Y, (X_1, X_2) is independent of (Y_1, Y_2), so $\Pi_c = 2P(X_1 > X_2)P(Y_1 > Y_2)$ and $\Pi_d = 2P(X_1 > X_2)P(Y_1 < Y_2)$; since Y_1, Y_2 are i.i.d., $P(Y_1 > Y_2) = P(Y_1 < Y_2)$, so $\Pi_c = \Pi_d$ and $\gamma = 0$. For the converse, a U-shaped joint distribution can have $\Pi_c = \Pi_d$ without independence, e.g. the 3×3 table with $\pi_{11} = \pi_{31} = \pi_{22} = 1/3$ has $\Pi_c = 2\pi_{11}\pi_{22} = 2/9 = 2\pi_{22}\pi_{31} = \Pi_d$. ■

2.4.5 Gamma for job satisfaction example

Example 2.19 (Gamma for income and job satisfaction)

Example 2.17: $\hat{\gamma} = (1331 - 849)/(1331 + 849) = 0.221$, a weak tendency for satisfaction to increase with income.

Chapter 3. Inference for Contingency Tables

3.1 Confidence intervals for association parameters

Motivation. Standard errors for association measures come from the delta method; better intervals come from inverting score or LR tests.

3.1.1 Interval estimation of odds ratios

$\hat{\theta} = n_{11}n_{22}/(n_{12}n_{21})$ is 0 or ∞ when some $n_{ij} = 0$, so $E(\hat{\theta})$ and $\mathrm{Var}(\log\hat{\theta})$ do not exist; $\tilde{\theta} = \frac{(n_{11}+0.5)(n_{22}+0.5)}{(n_{12}+0.5)(n_{21}+0.5)}$ has the same limit and behaves better. $\log\hat{\theta}$ is far less skewed than $\hat{\theta}$, with estimated standard error (Corollary 3.7)

$$\hat{\sigma}(\log\hat{\theta}) = \Big(\frac{1}{n_{11}} + \frac{1}{n_{12}} + \frac{1}{n_{21}} + \frac{1}{n_{22}}\Big)^{1/2}. \tag{3.1}$$

Definition (Woolf interval)

$\log\hat{\theta} \pm z_{\alpha/2}\,\hat{\sigma}(\log\hat{\theta})$ is a Wald interval for $\log\theta$; exponentiating its endpoints gives the Woolf interval for θ.

It is slightly conservative. If $\hat{\theta} = 0$ (or ∞), take 0 (or ∞) as one limit and compute the other with $n_{ij} + 0.5$.

3.1.2 Aspirin and myocardial infarction example

Example 3.1 (Swedish aspirin study: odds ratio)

MI deaths: 28 of 684 (placebo), 18 of 676 (aspirin). $\hat{\theta} = \frac{28\times658}{656\times18} = 1.56$, $\log\hat{\theta} = 0.445$, $\hat{\sigma}(\log\hat{\theta}) = 0.307$, 95% interval $(e^{-0.157}, e^{1.047}) = (0.85, 2.85)$: equal odds are plausible.

3.1.3 Interval estimation of difference of proportions

Treat the rows as independent binomials $y_i \sim \mathrm{bin}(n_i, \pi_i)$, $\hat{\pi}_i = y_i/n_i$.

Proposition 3.2 (Difference of proportions)

$E(\hat{\pi}_1 - \hat{\pi}_2) = \pi_1 - \pi_2$ and

$$\sigma(\hat{\pi}_1 - \hat{\pi}_2) = \Big[\frac{\pi_1(1-\pi_1)}{n_1} + \frac{\pi_2(1-\pi_2)}{n_2}\Big]^{1/2}. \tag{3.2}$$

As $n_1, n_2 \to \infty$, $(\hat{\pi}_1 - \hat{\pi}_2 - (\pi_1 - \pi_2))/\hat{\sigma} \xrightarrow{d} N(0,1)$, where $\hat{\sigma}$ replaces π_i by $\hat{\pi}_i$; so $(\hat{\pi}_1 - \hat{\pi}_2) \pm z_{\alpha/2}\hat{\sigma}$ is a Wald interval.

Proof.

Step 1. Linearity and independence give the mean and variance, using $\mathrm{Var}(\hat{\pi}_i) = \pi_i(1-\pi_i)/n_i$ (Proposition 1.8).

Step 2. [Supplement] The two standardized proportions are independent and asymptotically $N(0,1)$ (CLT), so is the standardized difference; $\hat{\sigma}/\sigma \xrightarrow{p} 1$ and Slutsky completes the proof. ■

As with (1.11), coverage is usually below nominal near 0 or 1.

3.1.4 Interval estimation of relative risk

The sample relative risk $r = \hat{\pi}_1/\hat{\pi}_2$ is also better handled on the log scale, with

$$\sigma(\log r) = \left[\frac{1-\pi_1}{\pi_1 n_1} + \frac{1-\pi_2}{\pi_2 n_2}\right]^{1/2} \tag{3.3}$$

(Corollary 3.5(b)); exponentiate $\log r \pm z_{\alpha/2}\hat{\sigma}(\log r)$.

Example 3.3 (Swedish aspirin study: relative risk and difference)

Proportions 0.0409 (placebo), 0.0266 (aspirin).
(a) $r = 1.54$, interval (0.86, 2.75).
(b) Difference 0.014, interval $(-0.005, 0.033)$.
No effect is plausible; the larger study of Example 2.7 shows a benefit.

3.1.5 Deriving standard errors with the delta method*

Theorem 3.4 (Delta method)

Suppose $\sqrt{n}(T_n - \theta) \xrightarrow{d} N(0, \sigma^2)$ and g is differentiable at θ. Then

$$\sqrt{n}\,[g(T_n) - g(\theta)] \xrightarrow{d} N\big(0,\, [g'(\theta)]^2\sigma^2\big).$$

If moreover g' is continuous at θ, $g'(\theta) \neq 0$, and $\hat{\sigma} \xrightarrow{p} \sigma > 0$, then $\sqrt{n}[g(T_n) - g(\theta)]/(|g'(T_n)|\hat{\sigma}) \xrightarrow{d} N(0,1)$.

Proof. [Supplement]

Step 1. Let $h(t) = \frac{g(t)-g(\theta)}{t-\theta} - g'(\theta)$ for $t \neq \theta$, $h(\theta) = 0$; h is continuous at θ. Since $T_n \xrightarrow{p} \theta$, $h(T_n) \xrightarrow{p} 0$.

Step 2. $\sqrt{n}[g(T_n) - g(\theta)] = [g'(\theta) + h(T_n)]\sqrt{n}(T_n - \theta) \xrightarrow{d} g'(\theta)N(0,\sigma^2)$ by Slutsky.

Step 3. $|g'(T_n)|\hat{\sigma} \xrightarrow{p} |g'(\theta)|\sigma > 0$, and Slutsky again gives the standardized version. ∎

Locally g has slope $g'(\theta)$, so $g(T_n)$ spreads $|g'(\theta)|$ times as much as T_n (Figure 4).

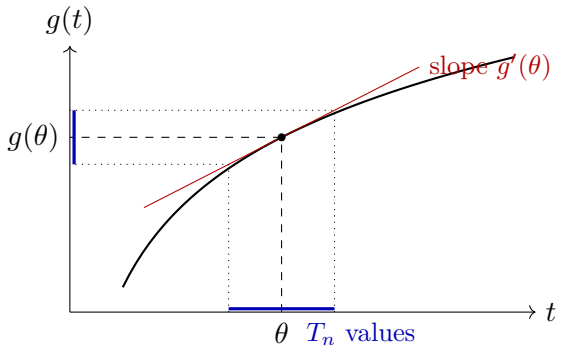


Figure 4: The delta method: near θ, g is nearly linear, so a band of T_n maps to a band about $|g'(\theta)|$ times as wide.

3.1.6 Delta method applied to sample logit*

Corollary 3.5 (Sample logit and log proportion)

Let $\hat{\pi} = y/n$ with $y \sim \text{bin}(n, \pi)$, $0 < \pi < 1$.
(a) The sample logit satisfies

$$\sqrt{n}\Big[\log\frac{\hat{\pi}}{1-\hat{\pi}} - \log\frac{\pi}{1-\pi}\Big] \xrightarrow{d} N\Big(0, \frac{1}{\pi(1-\pi)}\Big).$$

(b) $\sqrt{n}(\log\hat{\pi} - \log\pi) \xrightarrow{d} N(0, (1-\pi)/\pi)$; hence for independent samples the log relative risk has asymptotic variance (3.3).

Proof.

Step (a). $\sqrt{n}(\hat{\pi} - \pi) \xrightarrow{d} N(0, \pi(1-\pi))$, and $g(\pi) = \log[\pi/(1-\pi)]$ has $g'(\pi) = 1/[\pi(1-\pi)]$; Theorem 3.4 gives variance $\pi(1-\pi)[\pi(1-\pi)]^{-2}$.

Step (b). With $g = \log$, $g'(\pi) = 1/\pi$ and the variance is $(1-\pi)/\pi$. Dividing by n_i and adding over the two independent samples gives (3.3). ■

This is the variance of the limiting normal; the true variance does not exist ($\hat{\pi} \in \{0, 1\}$ with positive probability).

3.1.7 Delta method for log odds ratio*

Theorem 3.6 (Multivariate delta method for multinomial proportions)

Let $(n_1, \ldots, n_c)$ be multinomial$(n, \boldsymbol{\pi})$, $\hat{\boldsymbol{\pi}} = \mathbf{n}/n$, and let g be differentiable at $\boldsymbol{\pi}$ with $\phi_i = \partial g/\partial\pi_i$. Define

$$\sigma^2 = \sum_i \pi_i\phi_i^2 - \Big(\sum_i \pi_i\phi_i\Big)^2. \tag{3.4}$$

If $\sigma > 0$ then $\sqrt{n}[g(\hat{\boldsymbol{\pi}}) - g(\boldsymbol{\pi})]/\sigma \xrightarrow{d} N(0, 1)$, and the same holds with σ replaced by its ML estimate $\hat{\sigma}$ (substituting $\hat{\pi}_i$ for π_i in (3.4)).

Proof. [Supplement]

Step 1. By the multivariate CLT, $\sqrt{n}(\hat{\boldsymbol{\pi}} - \boldsymbol{\pi}) \xrightarrow{d} \mathbf{z} \sim N_c(\mathbf{0}, \Sigma)$, $\Sigma = \text{diag}(\boldsymbol{\pi}) - \boldsymbol{\pi}\boldsymbol{\pi}^\top$ (Proposition 1.2).

Step 2. Differentiability gives $\sqrt{n}[g(\hat{\boldsymbol{\pi}}) - g(\boldsymbol{\pi})] = \boldsymbol{\phi}^\top\sqrt{n}(\hat{\boldsymbol{\pi}} - \boldsymbol{\pi}) + o_p(1) \xrightarrow{d} \boldsymbol{\phi}^\top\mathbf{z}$, since $\|\hat{\boldsymbol{\pi}} - \boldsymbol{\pi}\| = O_p(n^{-1/2})$.

Step 3. $\boldsymbol{\phi}^\top\mathbf{z} \sim N(0, \boldsymbol{\phi}^\top\Sigma\boldsymbol{\phi})$, and $\boldsymbol{\phi}^\top\Sigma\boldsymbol{\phi} = \sum_i \pi_i\phi_i^2 - (\sum_i \pi_i\phi_i)^2 = \sigma^2$.

Step 4. If the ϕ_i are continuous, $\hat{\sigma} \xrightarrow{p} \sigma$ (WLLN), and Slutsky applied to the product of the standardized statistic and $\sigma/\hat{\sigma} \xrightarrow{p} 1$ gives the limit $N(0, 1)$. ■

Corollary 3.7 (Standard error of the log odds ratio)

For a multinomial 2×2 table, $\log\hat{\theta}$ has asymptotic standard error $\big(\sum_i\sum_j 1/(n\pi_{ij})\big)^{1/2}$, estimated by (3.1).

Proof.

Step 1. $g(\boldsymbol{\pi}) = \log\pi_{11} + \log\pi_{22} - \log\pi_{12} - \log\pi_{21}$ has $\phi_{ij} = \pm 1/\pi_{ij}$ (+ on the diagonal).

Step 2. $\sum\sum \pi_{ij}\phi_{ij} = 1-1-1+1 = 0$ and $\sum\sum \pi_{ij}\phi_{ij}^2 = \sum\sum 1/\pi_{ij}$; Theorem 3.6 gives $\sigma/\sqrt{n} = (\sum\sum 1/(n\pi_{ij}))^{1/2}$, estimated by $(\sum\sum 1/n_{ij})^{1/2}$. ■

The same holds for independent binomial rows (Corollary 3.5(a)). A Wald interval for θ itself is not recommended.

3.2 Testing independence in two-way contingency tables

Motivation. Independence, $\pi_{ij} = \pi_{i+}\pi_{+j}$, is the starting point of contingency-table analysis, tested by X^2 and G^2.

Under independent multinomial rows the hypothesis is homogeneity; the same tests apply.

3.2.1 Pearson and likelihood-ratio chi-squared tests

Proposition 3.8 (ML estimates under independence)

Under H_0, the ML estimates are $\hat\pi_{i+} = n_{i+}/n$, $\hat\pi_{+j} = n_{+j}/n$, so the estimated expected frequencies are

$$\hat\mu_{ij} = n\hat\pi_{i+}\hat\pi_{+j} = \frac{n_{i+}n_{+j}}{n}.$$

Proof. The kernel $\prod_i\prod_j(\pi_{i+}\pi_{+j})^{n_{ij}} = \prod_i \pi_{i+}^{n_{i+}} \cdot \prod_j \pi_{+j}^{n_{+j}}$ is a product of two multinomial kernels in separate parameters, each maximized by sample proportions (Proposition 1.14). ■

$$X^2 = \sum_i\sum_j \frac{(n_{ij}-\hat\mu_{ij})^2}{\hat\mu_{ij}}, \qquad G^2 = 2\sum_i\sum_j n_{ij}\log\frac{n_{ij}}{\hat\mu_{ij}}. \tag{3.5}$$

Theorem 3.9 (Null distribution of X^2 and G^2 for independence)

Under H_0 with all $\pi_{i+}, \pi_{+j} > 0$, X^2 and G^2 both converge in distribution to $\chi^2_{(I-1)(J-1)}$, and $X^2 - G^2 \xrightarrow{p} 0$.

Proof. [Supplement] Apply Theorem 1.18 to the $c = IJ$ cells with the $p = (I-1)+(J-1)$ parameters $\pi_{1+},\ldots,\pi_{I-1,+},\pi_{+1},\ldots,\pi_{+,J-1}$.

Step 1 (rank). With $a_i = \pi_{i+}$, $b_j = \pi_{+j}$, a tangent direction is $d\pi_{ij} = b_j\,da_i + a_i\,db_j$ with $\sum_i da_i = \sum_j db_j = 0$. If all $d\pi_{ij} = 0$, summing over j gives $da_i = 0$, and then $db_j = 0$; so the Jacobian has rank p.

Step 2 (df). Hence $X^2 \xrightarrow{d} \chi^2_{IJ-1-p}$ with $IJ-1-(I-1)-(J-1) = (I-1)(J-1)$, and $G^2 - X^2 \xrightarrow{p} 0$. ■

Fisher (1922) corrected Pearson's df $= IJ-1$. The limits also hold for Poisson and product multinomial sampling; X^2 converges faster than G^2 (Section 3.5).

3.2.2 Education and religious fundamentalism example

Example 3.10 (Education and religious beliefs)

1996 GSS: count, $\hat{\mu}_{ij}$, and standardized residual (3.6).

Highest degree	Fundamentalist		Moderate		Liberal		Total
< High school	178 (137.8)	4.5	138 (161.5)	−2.6	108 (124.7)	−1.9	424
HS / junior college	570 (539.5)	2.6	648 (632.1)	1.3	442 (488.4)	−4.0	1660
Bachelor / graduate	138 (208.7)	−6.8	252 (244.5)	0.7	252 (188.9)	6.3	642
Total	886		1038		802		2726

$X^2 = 69.2$, $G^2 = 69.8$, df $= 4$, $P < 0.0001$.

3.3 Following-up chi-squared tests

Motivation. Residuals, partitions of G^2 and odds ratios reveal the nature of an association that a test only detects.

3.3.1 Pearson and standardized residuals

Definition (Pearson and standardized residuals)

The *Pearson residual* and *standardized Pearson residual* for independence are

$$e_{ij} = \frac{n_{ij} - \hat{\mu}_{ij}}{\hat{\mu}_{ij}^{1/2}}, \qquad r_{ij} = \frac{n_{ij} - \hat{\mu}_{ij}}{[\hat{\mu}_{ij}(1 - p_{i+})(1 - p_{+j})]^{1/2}}, \tag{3.6}$$

with $p_{i+} = n_{i+}/n$, $p_{+j} = n_{+j}/n$. Note $\sum_i \sum_j e_{ij}^2 = X^2$.

Theorem 3.11 (Asymptotic variance of Pearson residuals)

Under independence, $e_{ij} \xrightarrow{d} N(0, (1 - \pi_{i+})(1 - \pi_{+j}))$; hence $r_{ij} \xrightarrow{d} N(0, 1)$. The average asymptotic variance of the e_{ij} is $(I - 1)(J - 1)/(IJ) < 1$.

Proof. [Supplement] Use the notation of the proof of Theorem 1.18 with cells (i, j), $a_i = \pi_{i+}$, $b_j = \pi_{+j}$, and unit vectors $\mathbf{r} = (\sqrt{a_i})_i$, $\mathbf{s}_c = (\sqrt{b_j})_j$, so $\mathbf{s} = \mathbf{r} \otimes \mathbf{s}_c$.

Step 1. By that proof the Pearson residuals are $(I - H)\mathbf{z}_n + o_p(1)$ with limiting covariance $I - H - \mathbf{s}\mathbf{s}^\top$, H the projection onto the column space of A.

Step 2. A tangent direction $d\pi_{ij} = b_j\, da_i + a_i\, db_j$ gives $D^{-1/2} d\boldsymbol{\pi}$ with entries $\sqrt{b_j}\, x_i + \sqrt{a_i}\, y_j$, $x_i = da_i/\sqrt{a_i}$, $y_j = db_j/\sqrt{b_j}$, where $\mathbf{x} \perp \mathbf{r}$, $\mathbf{y} \perp \mathbf{s}_c$. So the column space is $(\mathbf{r}^\perp \otimes \mathbf{s}_c) \oplus (\mathbf{r} \otimes \mathbf{s}_c^\perp)$, two orthogonal subspaces, and

$$H = (I_I - \mathbf{r}\mathbf{r}^\top) \otimes \mathbf{s}_c\mathbf{s}_c^\top + \mathbf{r}\mathbf{r}^\top \otimes (I_J - \mathbf{s}_c\mathbf{s}_c^\top).$$

Step 3. The (ij, ij) entries of H and $\mathbf{s}\mathbf{s}^\top$ are $(1 - a_i)b_j + a_i(1 - b_j)$ and $a_i b_j$, so e_{ij} has asymptotic variance $1 - a_i - b_j + a_i b_j = (1 - a_i)(1 - b_j)$.

Step 4. Slutsky gives $r_{ij} \xrightarrow{d} N(0, 1)$, and $\frac{1}{IJ} \sum_i \sum_j (1 - a_i)(1 - b_j) = \frac{(I-1)(J-1)}{IJ}$. ■

Comparing e_{ij} with $N(0, 1)$ is conservative; $|r_{ij}|$ above about 2 or 3 indicates lack of fit in that cell.

3.3.2 Education and religious fundamentalism revisited

Example 3.12 (Residuals for education and religion)

$r_{11} = (178 - 137.8)/[137.8(1 - 0.156)(1 - 0.325)]^{1/2} = 4.5$. Large positive residuals at (less than HS, fundamentalist) and (bachelor+, liberal), large negative at the other corners; corner odds ratio $\frac{178\times252}{108\times138} = 3.0$.

3.3.3 Partitioning chi-squared

Independent $\chi^2_{\nu_1}$ and $\chi^2_{\nu_2}$ sum to $\chi^2_{\nu_1+\nu_2}$, so a statistic can be split into components showing which categories drive an association.

Theorem 3.13 (Exact partition of G^2 by successive columns)

For an $I \times J$ table, for $j = 1, \ldots, J-1$ let G^2_j be the G^2 statistic for independence in the $I \times 2$ table whose first column merges columns $1, \ldots, j$ and whose second column is column $j+1$. Then

$$G^2 = \sum_{j=1}^{J-1} G^2_j,$$

and each G^2_j has df $= I - 1$, so the df also add up to $(I-1)(J-1)$.

Proof. [Supplement]

Step 1. With $\Phi(x) = x \log x$, for any table with counts m_{ij}, margins m_{i+}, m_{+j} and total m, (3.5) gives

$$G^2/2 = \sum_i \sum_j \Phi(m_{ij}) - \sum_i \Phi(m_{i+}) - \sum_j \Phi(m_{+j}) + \Phi(m),$$

since $\log(m_{ij}/\hat{\mu}_{ij}) = \log m_{ij} + \log m - \log m_{i+} - \log m_{+j}$.

Step 2. Let $S_{ij} = \sum_{l \le j} n_{il}$ and $S_{+j} = \sum_i S_{ij}$. The jth subtable has cells S_{ij}, $n_{i,j+1}$, row totals $S_{i,j+1}$, column totals S_{+j}, $n_{+,j+1}$ and total $S_{+,j+1}$, so $G^2_j/2$ equals

$$\sum_i [\Phi(S_{ij}) + \Phi(n_{i,j+1}) - \Phi(S_{i,j+1})] - \Phi(S_{+j}) - \Phi(n_{+,j+1}) + \Phi(S_{+,j+1}).$$

Step 3. Summing over j, $\Phi(S_{ij}) - \Phi(S_{i,j+1})$ telescopes to $\Phi(n_{i1}) - \Phi(n_{i+})$, which with $\sum_{j \ge 2} \Phi(n_{ij})$ gives $\sum_j \Phi(n_{ij}) - \Phi(n_{i+})$; likewise the column terms give $\Phi(n) - \sum_j \Phi(n_{+j})$. The total is $G^2/2$ by Step 1. ■

Repeating this on rows gives Lancaster's partition into $(I-1)(J-1)$ components with df $= 1$, from the 2×2 tables

$$\begin{matrix} \sum_{a<i} \sum_{b<j} n_{ab} & \sum_{a<i} n_{aj} \\ \sum_{b<j} n_{ib} & n_{ij} \end{matrix} \qquad i = 2, \ldots, I, \ j = 2, \ldots, J. \tag{3.7}$$

Comparing each column with the last does *not* give an additive partition (Section 9.2.4 views partitions as nested models).

3.3.4 Origin of schizophrenia example

Example 3.14 (School of psychiatric thought and origin of schizophrenia)

Schools (eclectic, medical, psychoanalytic) by ascribed origin (biogenic, environmental, combination): $G^2 = 23.04$, df $= 4$. By (3.7): Ecl vs. Med 0.29, 1.36; (Ecl+Med) vs. Psy 12.95 (Bio vs. Env), 8.43 ((Bio+Env) vs. Com); total 23.04. Psychoanalysts more often ascribe a combined or environmental origin.

3.3.5 Rules for partitioning

Necessary conditions for independent components (Goodman; Lancaster):

(a) the subtable df sum to the df of the full table;

(b) each cell count of the full table is a cell count in exactly one subtable;

(c) each marginal total of the full table is a marginal total of exactly one subtable.

Exact partitions hold for G^2 but not X^2.

3.3.6 Limitations of chi-squared tests

They only measure evidence, need large samples, and ignore ordinal information (invariant to reordering; Section 3.4).

3.3.7 Why consider independence?

If independence approximately holds, $p_{i+}p_{+j}$ can beat p_{ij} in MSE: biased, but with fewer parameters.

Proposition 3.15 (MSE of the two estimators)

For multinomial sampling,

$$\sum_i \sum_j E(p_{ij} - \pi_{ij})^2 = \frac{1 - \sum_i \sum_j \pi_{ij}^2}{n}, \qquad \sum_i \sum_j E(p_{i+}p_{+j} - \pi_{ij})^2 \to \sum_i \sum_j (\pi_{i+}\pi_{+j} - \pi_{ij})^2.$$

Under independence, $n \sum \sum E(p_{i+}p_{+j} - \pi_{ij})^2 \to \sum_i \sum_j [\pi_{+j}^2 \pi_{i+}(1 - \pi_{i+}) + \pi_{i+}^2 \pi_{+j}(1 - \pi_{+j})]$.

Proof. [Supplement]

Step 1. p_{ij} is unbiased with $\mathrm{Var}(p_{ij}) = \pi_{ij}(1 - \pi_{ij})/n$; sum over cells.

Step 2. With $a = p_{i+} - \pi_{i+}$, $b = p_{+j} - \pi_{+j}$, $p_{i+}p_{+j} - \pi_{ij} = (\pi_{i+}\pi_{+j} - \pi_{ij}) + \pi_{+j}a + \pi_{i+}b + ab$. Since $E(a^2), E(b^2) = O(n^{-1})$ and $E(a^2b^2) \le [E(a^4)E(b^4)]^{1/2} = O(n^{-2})$, all but the first term vanish in mean square.

Step 3. Under independence the first term is 0, and $nE(\pi_{+j}a + \pi_{i+}b)^2 = \pi_{+j}^2 \pi_{i+}(1 - \pi_{i+}) + \pi_{i+}^2 \pi_{+j}(1 - \pi_{+j}) + 2\pi_{i+}\pi_{+j}\, n\,\mathrm{Cov}(a, b)$, where $n\,\mathrm{Cov}(p_{i+}, p_{+j}) = \pi_{ij} - \pi_{i+}\pi_{+j} = 0$; the terms involving ab are $o(n^{-1})$ by Cauchy–Schwarz. ∎

Example 3.16 (A 3×3 illustration)

Let $\pi_{i+} = \pi_{+j} = 1/3$ and $\pi_{ij} = \frac{1}{9}[1 + \delta(i - 2)(j - 2)]$.
(a) At $\delta = 0$ the total MSE is $8/(9n)$ for p_{ij} and $\approx 4/(9n)$ for $p_{i+}p_{+j}$.

(b) For $\delta \neq 0$ the latter tends to $4\delta^2/81$, yet still wins for moderate n (e.g. $n = 500$, $\delta = 0.1$).

3.4 Two-way tables with ordered classifications

Motivation. Ordinal trend statistics give df $= 1$ tests with much more power against monotone alternatives.

3.4.1 Linear trend alternative to independence

With row scores $u_1 \leq \cdots \leq u_I$ and column scores $v_1 \leq \cdots \leq v_J$, let r be the sample correlation of the scores over the n subjects:

$$r = \frac{\sum\sum u_i v_j n_{ij} - (\sum_i u_i n_{i+})(\sum_j v_j n_{+j})/n}{\left[\sum_i u_i^2 n_{i+} - \frac{(\sum_i u_i n_{i+})^2}{n}\right]^{1/2}\left[\sum_j v_j^2 n_{+j} - \frac{(\sum_j v_j n_{+j})^2}{n}\right]^{1/2}}.$$

Definition (Linear trend statistic)

$M^2 = (n-1)r^2$ (Mantel, 1963), referred to χ^2_1; for one-sided alternatives use $M = \sqrt{n-1}\, r$ with the normal tail.

Theorem 3.17 (Null distribution of M^2)

Under independence of X and Y (multinomial sampling, each score variable nondegenerate), $\sqrt{n}\, r \xrightarrow{d} N(0,1)$; hence $M \xrightarrow{d} N(0,1)$ and $M^2 \xrightarrow{d} \chi^2_1$.

Proof. [Supplement] Let (U_k, V_k), $k \leq n$, be the subjects' score pairs: i.i.d., U independent of V, with means μ, ν and variances $\sigma^2, \tau^2 > 0$.

Step 1. $S_{UV} = \frac{1}{n}\sum_k W_k - (\bar{U} - \mu)(\bar{V} - \nu)$ with $W_k = (U_k - \mu)(V_k - \nu)$ i.i.d., $E(W_k) = 0$ and $\mathrm{Var}(W_k) = \sigma^2\tau^2$ by independence; so $n^{-1/2}\sum_k W_k \xrightarrow{d} N(0, \sigma^2\tau^2)$.

Step 2. $\sqrt{n}(\bar{U} - \mu)(\bar{V} - \nu) = O_p(1)o_p(1) \xrightarrow{p} 0$, so $\sqrt{n}S_{UV} \xrightarrow{d} N(0, \sigma^2\tau^2)$.

Step 3. $S_{UU} \xrightarrow{p} \sigma^2$, $S_{VV} \xrightarrow{p} \tau^2$, so by Slutsky $\sqrt{n}\, r \xrightarrow{d} N(0,1)$; as $\sqrt{(n-1)/n} \to 1$, $M \xrightarrow{d} N(0,1)$ and $M^2 \xrightarrow{d} \chi^2_1$. ■

3.4.2 Job satisfaction example revisited

Example 3.18 (Linear trend test for income and job satisfaction)

For Example 2.17 ($n = 96$), $X^2 = 6.0$, $G^2 = 6.8$, df $= 9$ ($P \approx 0.7$). With scores $1, 2, 3, 4$ and income midpoints $7.5, 20, 32.5, 60$, $r = 0.200$, $M^2 = 3.81$ ($P = 0.051$) and $M = 1.95$ (one-sided $P = 0.026$): much stronger evidence.

3.4.3 Monotone trend alternatives to independence

Ordinal measures such as gamma (Section 2.4.4) give $z = \hat{\gamma}/\mathrm{SE}$; for Example 2.17, $\hat{\gamma} = 0.221$, SE $= 0.117$, $z = 1.89$, 95% interval $(-0.01, 0.45)$.

3.4.4 Extra power with ordinal tests

X^2, G^2 spread power over $(I-1)(J-1)$ parameters; M^2 and $(\hat{\gamma}/\mathrm{SE})^2$ concentrate it on one (Section 6.4).

3.4.5 Choice of scores

Scores fixed before seeing the data give a valid test, poor scores an insensitive one; M^2 is invariant to linear transformations.

Example 3.19 (Maternal drinking and congenital malformations)

Malformation (present/absent) by drinks per day (0, < 1, 1–2, 3–5, ≥ 6), with present counts $48, 38, 5, 1, 1$ among 17,114, 14,502, 793, 127, 38 subjects.
(a) Scores $0, 0.5, 1.5, 4.0, 7.0$: $M^2 = 6.57$, $P = 0.010$.
(b) Scores $1, 2, 3, 4, 5$: $M^2 = 1.83$, $P = 0.18$.
(c) Midranks: $M^2 = 0.35$, $P = 0.55$; they treat the sparse last three categories as nearly equal, which is inappropriate.

Use scores reflecting distances and check sensitivity. With midranks for both variables, r is *Spearman's rho*.

3.4.6 Trend tests for $I \times 2$ and $2 \times J$ tables

For $2 \times J$ with $u = (0, 1)$, M^2 compares the two row means of the Y scores; with midranks it is the *Wilcoxon–Mann–Whitney* test. For $I \times 2$ it is the *Cochran–Armitage trend test* (Section 5.3.5).

3.4.7 Nominal–ordinal tables

For a nominal variable with more than two categories, compare ordinal mean scores across categories (Section 7.5.3).

3.5 Small-sample tests of independence

Motivation. Conditioning on the margins eliminates nuisance parameters and gives the exact hypergeometric null distribution.

3.5.1 Fisher's exact test for 2×2 tables

Theorem 3.20 (Hypergeometric null distribution)

Under independence, for Poisson, multinomial, or independent binomial sampling, conditional on both sets of marginal totals,

$$p(t) = P(n_{11} = t) = \frac{\binom{n_{1+}}{t}\binom{n_{2+}}{n_{+1}-t}}{\binom{n}{n_{+1}}}, \qquad m_- \le t \le m_+, \tag{3.8}$$

where $m_- = \max(0,\, n_{1+} + n_{+1} - n)$ and $m_+ = \min(n_{1+},\, n_{+1})$.

Proof.

Step 1. By Theorem 2.3, given row totals, $n_{11} \sim \text{bin}(n_{1+}, \pi)$ and $n_{21} \sim \text{bin}(n_{2+}, \pi)$ independently under H_0, and $n_{+1} \sim \text{bin}(n, \pi)$.

Step 2. So $P(n_{11} = t \mid n_{+1})$ is $\binom{n_{1+}}{t}\binom{n_{2+}}{n_{+1}-t}\pi^{n_{+1}}(1-\pi)^{n-n_{+1}}$ divided by $\binom{n}{n_{+1}}\pi^{n_{+1}}(1-\pi)^{n-n_{+1}}$; the powers cancel, leaving (3.8).

Step 3. $0 \le t \le n_{1+}$ and $0 \le n_{+1} - t \le n_{2+}$ give $m_- \le t \le m_+$. ■

For $H_a\colon \theta > 1$, $P = P(n_{11} \ge t_o)$.

Proposition 3.21 (Orderings of 2×2 tables with fixed margins)

With all margins fixed, the sample odds ratio and the difference of proportions $n_{11}/n_{1+} - n_{21}/n_{2+}$ are increasing in n_{11}, and

$$X^2 = \frac{n^3\big(n_{11} - E(n_{11})\big)^2}{n_{1+}n_{2+}n_{+1}n_{+2}}, \qquad E(n_{11}) = \frac{n_{1+}n_{+1}}{n}.$$

Hence one-sided P-values agree for all three orderings, and $P(X^2 \ge X_o^2) = P(|n_{11} - E(n_{11})| \ge |t_o - E(n_{11})|)$.

Proof. [Supplement]

Step 1. With $n_{12} = n_{1+} - n_{11}$, $n_{21} = n_{+1} - n_{11}$, $n_{22} = n - n_{1+} - n_{+1} + n_{11}$, the numerator of $\hat\theta$ increases and its denominator decreases in n_{11}, and $n_{11}/n_{1+} - (n_{+1} - n_{11})/n_{2+}$ increases.

Step 2. $n_{11}n_{22} - n_{12}n_{21} = n\,n_{11} - n_{1+}n_{+1} = n[n_{11} - E(n_{11})]$ by expanding.

Step 3. In (3.5) all four $n_{ij} - \hat\mu_{ij}$ equal $\pm(n_{11}n_{22} - n_{12}n_{21})/n$ and $\sum_{i,j} 1/\hat\mu_{ij} = n^3/(n_{1+}n_{2+}n_{+1}n_{+2})$, so $X^2 = n(n_{11}n_{22} - n_{12}n_{21})^2/(n_{1+}n_{2+}n_{+1}n_{+2})$; Step 2 gives the display. ■

3.5.2 Fisher's tea drinker

Example 3.22 (Fisher's tea-tasting experiment)

A lady tasted 8 cups, 4 milk-first, named 4 as milk-first and got 3 right ($n_{11} = 3$, all margins 4 by design). For $H_a\colon \theta > 1$,

$$P(n_{11} = 3) = \frac{\binom{4}{3}\binom{4}{1}}{\binom{8}{4}} = \frac{16}{70} = 0.229, \qquad P(n_{11} = 4) = \frac{1}{70} = 0.014,$$

so $P = 0.243$: too small an experiment to establish her ability.

3.5.3 Two-sided P-values for Fisher's exact test

Two-sided criteria include:

(a) sum $p(t)$ over t with $p(t) \le p(t_o)$ (the most popular);

(b) $P(|n_{11} - E(n_{11})| \ge |t_o - E(n_{11})|)$, which equals $P(X^2 \ge X_o^2)$ by Proposition 3.21;

(c) $2\min[P(n_{11} \ge t_o), P(n_{11} \le t_o)]$, which can exceed 1;

(d) the smaller tail plus the largest attainable probability in the other tail not exceeding it (Blaker).

3.5.4 Discreteness and conservatism issues

Here one-sided P-values can only be 0.014, 0.243, 0.757, 0.986, 1, so rejecting at $P \leq 0.05$ has size 0.014: the test is *conservative*.

Example 3.23 (Randomized test of exact size 0.05)

Reject if $n_{11} = 4$, and with probability 0.157 if $n_{11} = 3$: $P(\text{reject}) = 0.014 + 0.157(0.229) = 0.05$.

Instead of randomization, report the *mid-P-value* $\frac{1}{2}P(n_{11} = t_o) + P(n_{11} > t_o)$ (0.129 for the tea data), rarely much above nominal size.

3.6 Extensions for multiway tables and nontabulated responses

Motivation. Models extend these tests to conditional independence and continuous predictors.

3.6.1 Categorical data need not be contingency tables

GLMs (next chapter) contain this chapter's methods as special cases.

Chapter 4. Introduction to Generalized Linear Models

GLMs contain logistic regression, Poisson loglinear models and ordinary regression.

4.1 Generalized linear model

Motivation. A GLM keeps the linear predictor but chooses the response distribution and the linearizing scale separately.

4.1.1 Components of generalized linear models

Definition (Natural exponential family)

A family of the form

$$f(y_i;\theta_i) = a(\theta_i)\, b(y_i) \exp\{y_i\, Q(\theta_i)\} \tag{4.1}$$

is a *natural exponential family*; $Q(\theta)$ is the *natural parameter*.

Definition (Generalized linear model)

(i) *Random component*: independent $y_1, \ldots, y_N$, each from (4.1) (more generally, the exponential dispersion family of Section 4.4).
(ii) *Systematic component*: a *linear predictor* $\eta_i = \sum_{j=1}^{p} \beta_j x_{ij}$ (usually $x_{i1} \equiv 1$, an intercept α).
(iii) *Link function*: a monotone differentiable g with

$$g(\mu_i) = \eta_i = \sum_j \beta_j x_{ij}, \qquad \mu_i = E(Y_i). \tag{4.2}$$

The link with $g(\mu_i) = Q(\theta_i)$ is the *canonical link*; $g(\mu) = \mu$ is the *identity link*.

4.1.2 Binomial logit models for binary data

For $Y \in \{0, 1\}$ with $P(Y = 1) = \pi$, $f(y;\pi) = (1-\pi)\exp\{y \log[\pi/(1-\pi)]\}$, which is (4.1) with $a(\pi) = 1 - \pi$, $b(y) = 1$, $Q(\pi) = \operatorname{logit} \pi$. The logit is the canonical link; GLMs with it are *logit models*.

4.1.3 Poisson loglinear models for count data

For a Poisson count, $f(y;\mu) = e^{-\mu}(1/y!)\exp(y \log \mu)$, so $Q(\mu) = \log \mu$ and the canonical link is the log link:

$$\log \mu_i = \sum_j \beta_j x_{ij}, \qquad i = 1, \ldots, N, \tag{4.3}$$

the *Poisson loglinear model*.

4.1.4 Generalized linear models for continuous responses

The normal (with dispersion parameter) has the mean as natural parameter, so regression, ANOVA and ANCOVA are identity-link GLMs.

Random component	Link	Systematic	Model
Normal	identity	cont./categ./mixed	regression / ANOVA / ANCOVA
Binomial	logit	mixed	logistic regression (Ch. 5–6)
Poisson	log	mixed	loglinear (Ch. 8–9)
Multinomial	generalized logit	mixed	multinomial response (Ch. 7)

Unlike transforming Y, the link need not stabilize variance or produce normality: fitting maximizes the chosen likelihood.

4.1.5 Deviance

Let $L(\boldsymbol{\mu}; \mathbf{y})$ be the log likelihood in terms of the means. The *saturated model* ($\hat{\mu}_i = y_i$, one parameter per observation) attains the largest value $L(\mathbf{y}; \mathbf{y})$.

Definition (Deviance)

For a Poisson or binomial GLM, $D(\mathbf{y}; \hat{\boldsymbol{\mu}}) = -2[L(\hat{\boldsymbol{\mu}}; \mathbf{y}) - L(\mathbf{y}; \mathbf{y})]$, the LR statistic for the model against the saturated alternative.

With N fixed and growing counts it is asymptotically χ^2_{N-p} (Theorem 4.19).

Proposition 4.1 (Deviance of the homogeneity model)

Let $y_i \sim \text{bin}(n_i, \pi_i)$ independently, $i = 1, \ldots, N$, and consider the model $\pi_i = \pi$ for all i ($p = 1$). Its deviance equals the statistic G^2 for testing independence in the $N \times 2$ table of successes and failures, and has df $= N - 1$.

Proof.

Step 1. Let $n = \sum n_i$, $s = \sum y_i$. With $L = \sum_i [y_i \log \pi_i + (n_i - y_i) \log(1 - \pi_i)]$, the MLEs are $\hat{\pi} = s/n$ (model) and $\tilde{\pi}_i = y_i/n_i$ (saturated).

Step 2. Hence

$$D = 2 \sum_i \Big[y_i \log \frac{y_i}{n_i s/n} + (n_i - y_i) \log \frac{n_i - y_i}{n_i(n-s)/n} \Big].$$

Step 3. The independence fitted values of the $N \times 2$ table are $n_i s/n$ and $n_i(n-s)/n$, so $D = 2 \sum \text{obs} \log(\text{obs}/\text{fit}) = G^2$, with df $= N - 1$ (N saturated parameters minus one). ■

4.1.6 Advantages of the GLM formulation

One theory and one algorithm (IRLS, Section 4.6) cover every exponential-family response and link; deviances generalize ANOVA (Section 4.5).

4.2 Generalized linear models for binary data

Motivation. For binary Y, the choice of which function of $\pi(x)$ is linear in x determines how effects are interpreted.

4.2.1 Linear probability model

The *linear probability model* $\pi(x) = \alpha + \beta x$ is the identity-link binomial GLM; β is the change in probability per unit x, but $\alpha + \beta x$ eventually leaves $[0, 1]$. Fit by binomial ML, not least squares (the variance depends on x); they agree only for fitted values in about 0.3–0.7.

4.2.2 Snoring and heart disease example

Example 4.2 (Snoring and heart disease)

2484 subjects, snoring scores $x = (0, 2, 4, 5)$, independent binomial rows.

Snoring	Yes	No	Proportion	Linear fit	Logit fit
Never ($x = 0$)	24	1355	0.017	0.017	0.021
Occasionally ($x = 2$)	35	603	0.055	0.057	0.044
Nearly every night ($x = 4$)	21	192	0.099	0.096	0.093
Every night ($x = 5$)	30	224	0.118	0.116	0.132

ML linear fit $\hat{\pi}(x) = 0.0172 + 0.0198x$ ($\widehat{\text{se}} = 0.0028$): risk 0.02 for nonsnorers, +0.04 per two score units. Logistic fit $\operatorname{logit} \hat{\pi}(x) = -3.87 + 0.40x$; over this range the fits are similar (Figure 5).

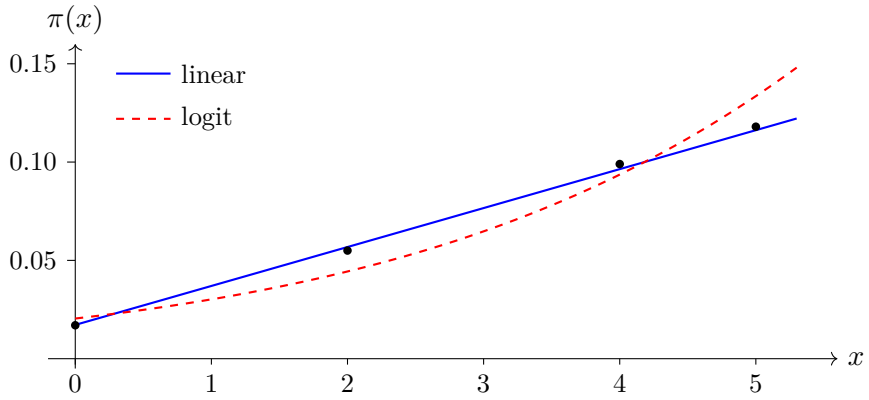


Figure 5: Snoring data: sample proportions with the linear probability and logistic fits.

4.2.3 Logistic regression model

Usually a fixed change in x matters less when $\pi(x)$ is near 0 or 1, giving S-shaped curves such as

$$\pi(x) = \frac{\exp(\alpha + \beta x)}{1 + \exp(\alpha + \beta x)}. \tag{4.4}$$

Proposition 4.3 (Logistic regression is the binomial GLM with logit link)

Model (4.4) holds if and only if

$$\log \frac{\pi(x)}{1 - \pi(x)} = \alpha + \beta x.$$

Hence logistic regression is the binomial GLM with the canonical (logit) link.

Proof. From (4.4), $1-\pi(x) = 1/[1+\exp(\alpha+\beta x)]$, so the odds are $\exp(\alpha+\beta x)$; take logs. Conversely, solving $\log[\pi/(1-\pi)] = \eta$ gives $\pi = e^\eta/(1+e^\eta)$. The logit is the binomial natural parameter (Section 4.1.2). ■

The logit ranges over $\mathbb{R}$, removing the structural problem; for the snoring data $\hat\beta = 0.40 > 0$.

4.2.4 Binomial GLM for 2×2 contingency tables

Proposition 4.4 (Effect parameters for a binary predictor)

Let $x \in \{0,1\}$ and $\text{link}[\pi(x)] = \alpha+\beta x$. Then $\beta = \text{link}[\pi(1)] - \text{link}[\pi(0)]$; in particular
(a) identity link: $\beta = \pi(1)-\pi(0)$, the difference of proportions;
(b) log link: $\beta = \log[\pi(1)/\pi(0)]$, the log relative risk;
(c) logit link: $\beta = \log\dfrac{\pi(1)/[1-\pi(1)]}{\pi(0)/[1-\pi(0)]}$, the log odds ratio.

Proof. Subtract $\text{link}[\pi(0)] = \alpha$ from $\text{link}[\pi(1)] = \alpha+\beta$ and specialize the link; for (c), a difference of log odds is a log odds ratio. ■

4.2.5 Probit and inverse cdf link functions

A monotone curve has the shape of a cdf: for a standard cdf Φ of a location–scale class,

$$\pi(x) = \Phi(\alpha+\beta x) \quad\Longleftrightarrow\quad \Phi^{-1}[\pi(x)] = \alpha+\beta x, \tag{4.5}$$

with link Φ^{-1}. Φ standard normal gives the *probit model* (Section 6.6); the standard logistic cdf $e^x/(1+e^x)$ (sd $\pi/\sqrt{3}$, slightly heavier tails) has inverse logit π.

4.3 Generalized linear models for counts

Motivation. Counts have variance growing with the mean: the Poisson log-linear GLM is basic, overdispersion leads to the negative binomial.

4.3.1 Poisson loglinear models

With one predictor,

$$\log\mu = \alpha+\beta x, \qquad \mu = \exp(\alpha+\beta x) = e^\alpha (e^\beta)^x, \tag{4.6}$$

so a one-unit increase in x multiplies the mean by e^β.

4.3.2 Horseshoe crab mating example

Example 4.5 (Horseshoe crab satellites)

173 female crabs; Y = number of satellite males; predictors color, spine condition, weight, width x (mean 26.3 cm). Grouped by width:

Width (cm)	<23.25	-24.25	-25.25	-26.25	-27.25	-28.25	-29.25	>29.25
Cases	14	14	28	39	22	24	18	14
Mean	1.00	1.43	2.39	2.69	2.86	3.87	3.94	5.14
Variance	2.77	8.88	6.54	11.38	6.88	8.81	16.88	8.29

(a) Log link: $\log\hat{\mu} = -3.305 + 0.164x$ ($\widehat{\text{se}} = 0.020$); $\hat{\mu} = 2.74$ at $x = 26.3$, and $e^{0.164} = 1.18$: each cm raises the mean by 18%.
(b) Identity link: $\hat{\mu} = -11.53 + 0.55x$ ($\widehat{\text{se}} = 0.06$). The fits agree where most data lie.

4.3.3 Overdispersion for Poisson GLMs

Above, the variances far exceed the means: *overdispersion*, typically from unobserved heterogeneity.

Proposition 4.6 (Mixtures of Poissons are overdispersed)

If $Y \mid \lambda \sim \text{Poisson}(\lambda)$ and λ is random with $E(\lambda) = \mu$, then $E(Y) = \mu$ and $\text{Var}(Y) = \mu + \text{Var}(\lambda) \geq \mu$, with equality only when λ is degenerate.

Proof. [Supplement] $E(Y) = E[E(Y \mid \lambda)] = \mu$ and $\text{Var}(Y) = E[\text{Var}(Y \mid \lambda)] + \text{Var}[E(Y \mid \lambda)] = \mu + \text{Var}(\lambda)$. ■

If the mean is right but the distribution is not Poisson, the ML estimates stay consistent but the standard errors are wrong.

4.3.4 Negative binomial GLMs

The negative binomial mass function is

$$f(y; k, \mu) = \frac{\Gamma(y+k)}{\Gamma(k)\Gamma(y+1)}\Big(\frac{k}{\mu+k}\Big)^k\Big(1-\frac{k}{\mu+k}\Big)^y, \qquad y = 0, 1, 2, \ldots \tag{4.7}$$

Proposition 4.7 (Negative binomial as a gamma mixture)

If $Y \mid \lambda \sim \text{Poisson}(\lambda)$ and λ has the gamma density with shape k and mean μ, then Y has the mass function (4.7), with $E(Y) = \mu$ and $\text{Var}(Y) = \mu + \mu^2/k$. For fixed k, (4.7) is a natural exponential family with natural parameter $\log[\mu/(\mu+k)]$, so negative binomial regression is a GLM; as $k^{-1} \to 0$ it tends to the Poisson.

Proof. [Supplement]

Step 1. With $h(\lambda) = (k/\mu)^k\lambda^{k-1}e^{-k\lambda/\mu}/\Gamma(k)$,

$$P(Y=y) = \frac{(k/\mu)^k}{\Gamma(k)\,y!}\int_0^\infty \lambda^{y+k-1}e^{-\lambda(1+k/\mu)}d\lambda = \frac{\Gamma(y+k)}{\Gamma(k)\,y!}\Big(\frac{k}{\mu+k}\Big)^k\Big(\frac{\mu}{\mu+k}\Big)^y,$$

which is (4.7).

Step 2. The gamma law has mean μ, variance μ^2/k; Proposition 4.6 gives the moments.

Step 3. With $Q = \log[\mu/(\mu+k)]$, (4.7) is $(1-e^Q)^k\,b(y)\exp(yQ)$, $b(y) = \Gamma(y+k)/[\Gamma(k)y!]$: the form (4.1).

Step 4. As $k \to \infty$, $\Gamma(y+k)/[\Gamma(k)(\mu+k)^y] \to 1$ and $(1+\mu/k)^{-k} \to e^{-\mu}$, so $f(y) \to e^{-\mu}\mu^y/y!$. ■

k^{-1} is a *dispersion parameter* (Section 13.4).

Example 4.8 (Crab satellites: negative binomial fit)

Identity link: Poisson $\hat{\mu} = -11.53 + 0.55x$ ($\widehat{\text{se}} = 0.06$); negative binomial $\hat{\mu} = -11.15 + 0.53x$ ($\widehat{\text{se}} = 0.11$), $\hat{k}^{-1} = 0.98$. Similar fits, but the honest SE is nearly doubled.

4.3.5 Poisson regression for rates

For a rate μ_i/t_i with exposure t_i, $\log(\mu_i/t_i) = \sum_j \beta_j x_{ij}$ is the Poisson GLM $\log \mu_i = \log t_i + \sum_j \beta_j x_{ij}$ with *offset* $\log t_i$ (Section 9.7).

4.3.6 Poisson GLM of independence in $I \times J$ contingency tables

Proposition 4.9 (Poisson form of independence)

Let Y_{ij} be independent Poisson with means $\mu_{ij} = \mu\alpha_i\beta_j$, where $\alpha_i, \beta_j > 0$ and $\sum_i \alpha_i = \sum_j \beta_j = 1$; equivalently

$$\log \mu_{ij} = \lambda + \lambda_i^X + \lambda_j^Y, \qquad \lambda = \log \mu,\ \lambda_i^X = \log \alpha_i,\ \lambda_j^Y = \log \beta_j. \tag{4.8}$$

Conditional on $n = \sum_{i,j} Y_{ij}$, the cell counts are multinomial with probabilities $\pi_{ij} = \alpha_i\beta_j$, which satisfy $\pi_{i+} = \alpha_i$, $\pi_{+j} = \beta_j$ and $\pi_{ij} = \pi_{i+}\pi_{+j}$.

Proof.

Step 1. $n \sim \text{Poisson}(\sum \mu_{ij}) = \text{Poisson}(\mu)$.

Step 2. For counts $\{n_{ij}\}$ with total n,

$$P(\{Y_{ij} = n_{ij}\} \mid n) = \frac{\prod_{ij} e^{-\mu_{ij}} \mu_{ij}^{n_{ij}}/n_{ij}!}{e^{-\mu}\mu^n/n!} = \frac{n!}{\prod_{ij} n_{ij}!} \prod_{ij} \Big(\frac{\mu_{ij}}{\mu}\Big)^{n_{ij}},$$

multinomial with $\pi_{ij} = \alpha_i\beta_j$.

Step 3. $\pi_{i+} = \alpha_i \sum_j \beta_j = \alpha_i$ and $\pi_{+j} = \beta_j$, so $\pi_{ij} = \pi_{i+}\pi_{+j}$. ∎

So Chapter 3 independence inferences are inferences for (4.8); Chapters 8–9 extend them.

4.4 Moments and likelihood for generalized linear models

Motivation. Mean, variance, likelihood equations and information all come from derivatives of $b(\theta)$.

Definition (Exponential dispersion family)

Independent $y_1, \ldots, y_N$ with

$$f(y_i; \theta_i, \phi) = \exp\Big\{\frac{y_i\theta_i - b(\theta_i)}{a(\phi)} + c(y_i, \phi)\Big\}; \tag{4.9}$$

θ_i is the *natural* and ϕ the *dispersion parameter*; usually $a(\phi) = \phi/\omega_i$ with known weight ω_i. For known ϕ this is (4.1) with $Q(\theta) = \theta/a(\phi)$.

4.4.1 Mean and variance functions for the random component

With $L_i = \log f(y_i; \theta_i, \phi)$ and $L = \sum_i L_i$,

$$L_i = \frac{y_i\theta_i - b(\theta_i)}{a(\phi)} + c(y_i, \phi). \tag{4.10}$$

Theorem 4.10 (Moments of the exponential dispersion family)

For (4.9), with θ_i interior to the natural parameter space,

$$\mu_i = E(Y_i) = b'(\theta_i), \qquad \mathrm{Var}(Y_i) = b''(\theta_i)\, a(\phi). \tag{4.11}$$

Proof. [Supplement]

Step 1. (Bartlett identities.) Let $\ell = \log f$. For θ interior, differentiation under $\int f\, dy = 1$ is permitted (support free of θ, locally dominated derivatives), giving $E(\partial\ell/\partial\theta) = 0$; differentiating again with $\partial f/\partial\theta = f\, \partial\ell/\partial\theta$ gives $-E(\partial^2\ell/\partial\theta^2) = E(\partial\ell/\partial\theta)^2$ (sums for discrete Y).

Step 2. From (4.10), $\partial L_i/\partial\theta_i = [y_i - b'(\theta_i)]/a(\phi)$ and $\partial^2 L_i/\partial\theta_i^2 = -b''(\theta_i)/a(\phi)$.

Step 3. The first identity gives $\mu_i = b'(\theta_i)$; the second gives $b''(\theta_i)/a(\phi) = \mathrm{Var}(Y_i)/a(\phi)^2$, i.e. $\mathrm{Var}(Y_i) = b''(\theta_i)a(\phi)$. ■

Since $b'' > 0$, b' is strictly increasing, so θ_i and μ_i determine each other.

4.4.2 Mean and variance functions for Poisson and binomial

Proposition 4.11 (Poisson and binomial in dispersion form)

(a) Poisson: $\theta_i = \log\mu_i$, $b(\theta_i) = e^{\theta_i}$, $a(\phi) = 1$, $c = -\log y_i!$; then $E(Y_i) = \mathrm{Var}(Y_i) = \mu_i$.
(b) Binomial proportion: if $n_i Y_i \sim \mathrm{bin}(n_i, \pi_i)$, then $\theta_i = \log[\pi_i/(1-\pi_i)]$, $b(\theta_i) = \log(1 + e^{\theta_i})$, $a(\phi) = 1/n_i$, $c = \log\binom{n_i}{n_i y_i}$; then $E(Y_i) = \pi_i$ and $\mathrm{Var}(Y_i) = \pi_i(1-\pi_i)/n_i$.

Proof.

(a) $e^{-\mu_i}\mu_i^{y_i}/y_i! = \exp[y_i\theta_i - e^{\theta_i} - \log y_i!]$, and $b'(\theta_i) = b''(\theta_i) = e^{\theta_i} = \mu_i$ (Theorem 4.10).

(b) Since $\log(1-\pi_i) = -\log(1+e^{\theta_i})$,

$$\binom{n_i}{n_i y_i}\pi_i^{n_i y_i}(1-\pi_i)^{n_i - n_i y_i} = \exp\Big[\frac{y_i\theta_i - \log(1+e^{\theta_i})}{1/n_i} + \log\binom{n_i}{n_i y_i}\Big].$$

Then $b'(\theta_i) = e^{\theta_i}/(1+e^{\theta_i}) = \pi_i$ and $b''(\theta_i)a(\phi) = \pi_i(1-\pi_i)/n_i$. ■

For $N(\mu_i, \sigma^2)$: $\theta_i = \mu_i$, $b(\theta) = \theta^2/2$, $a(\phi) = \sigma^2$.

4.4.3 Systematic component and link function

In matrix form $\boldsymbol{\eta} = X\boldsymbol{\beta}$ with $N \times p$ *model matrix* X, and $\eta_i = g(\mu_i)$.

Proposition 4.12 (Canonical link)

The canonical link ($g(\mu_i) = \theta_i$) is $g = (b')^{-1}$: log for the Poisson, logit for the binomial.

Proof. By Theorem 4.10, $\mu_i = b'(\theta_i)$ with b' strictly increasing, so $\theta_i = (b')^{-1}(\mu_i)$. Poisson: $b' = \exp$, inverse log. Binomial: $b'(\theta) = e^\theta/(1+e^\theta)$, inverse logit. ■

4.4.4 Likelihood equations for a GLM

The log likelihood

$$L(\boldsymbol{\beta}) = \sum_i \frac{y_i\theta_i - b(\theta_i)}{a(\phi)} + \sum_i c(y_i, \phi) \tag{4.12}$$

depends on $\boldsymbol{\beta}$ through $\theta_i(\mu_i)$, $\mu_i = g^{-1}(\eta_i)$, $\eta_i = \sum_j \beta_j x_{ij}$.

Theorem 4.13 (GLM likelihood equations)

For a GLM with exponential dispersion random component,

$$\frac{\partial L_i}{\partial \beta_j} = \frac{(y_i - \mu_i)x_{ij}}{\mathrm{Var}(Y_i)} \frac{\partial \mu_i}{\partial \eta_i}, \tag{4.13}$$

and the likelihood equations are

$$\sum_{i=1}^{N} \frac{(y_i - \mu_i)x_{ij}}{\mathrm{Var}(Y_i)} \frac{\partial \mu_i}{\partial \eta_i} = 0, \qquad j = 1, \ldots, p. \tag{4.14}$$

Proof.

Step 1. By the chain rule, $\partial L_i/\partial \beta_j = (\partial L_i/\partial \theta_i)(\partial \theta_i/\partial \mu_i)(\partial \mu_i/\partial \eta_i)(\partial \eta_i/\partial \beta_j)$.

Step 2. $\partial L_i/\partial \theta_i = (y_i - \mu_i)/a(\phi)$ (proof of Theorem 4.10); $\partial \mu_i/\partial \theta_i = b''(\theta_i) = \mathrm{Var}(Y_i)/a(\phi)$; $\partial \eta_i/\partial \beta_j = x_{ij}$.

Step 3. Multiplying, $a(\phi)$ cancels, giving (4.13); summing over i gives (4.14). ■

The equations depend on the distribution only through μ_i and the *variance function* $v(\mu_i) = \mathrm{Var}(Y_i)$: μ (Poisson), $\mu(1-\mu)$ (Bernoulli), σ^2 (normal).

4.4.5 Likelihood equations for binomial GLMs

Corollary 4.14 (Binomial likelihood equations)

Let $n_i Y_i \sim \mathrm{bin}(n_i, \pi_i)$ and $\pi_i = \Phi(\eta_i)$, $\eta_i = \sum_j \beta_j x_{ij}$, for a continuous cdf Φ with density ϕ. The likelihood equations are

$$\sum_i \frac{n_i(y_i - \pi_i)\, x_{ij}}{\pi_i(1-\pi_i)} \phi(\eta_i) = 0, \qquad j = 1, \ldots, p. \tag{4.15}$$

For the logit link they reduce to

$$\sum_i n_i(y_i - \pi_i)x_{ij} = 0, \qquad j = 1, \ldots, p. \tag{4.16}$$

Proof. Substitute $\partial \pi_i/\partial \eta_i = \phi(\eta_i)$ and $\mathrm{Var}(Y_i) = \pi_i(1-\pi_i)/n_i$ in (4.14). For the logit, $\partial \eta_i/\partial \pi_i = 1/[\pi_i(1-\pi_i)]$, so $\phi(\eta_i) = \pi_i(1-\pi_i)$ cancels the denominator. ■

4.4.6 Asymptotic covariance matrix of model parameter estimators

Theorem 4.15 (Information matrix of a GLM)

The expected information matrix $\mathcal{I}$ with entries $E[-\partial^2 L(\boldsymbol{\beta})/\partial \beta_h \partial \beta_j]$ is

$$\mathcal{I} = X^\top W X, \qquad W = \mathrm{diag}(w_1, \ldots, w_N), \quad w_i = \frac{(\partial \mu_i/\partial \eta_i)^2}{\mathrm{Var}(Y_i)}. \tag{4.17}$$

Proof.

Step 1. By the second Bartlett identity applied to $\boldsymbol{\beta}$, $E(\partial^2 L_i/\partial \beta_h \partial \beta_j) = -E(\partial L_i/\partial \beta_h \cdot \partial L_i/\partial \beta_j)$.

Step 2. By (4.13) and $E(Y_i-\mu_i)^2 = \mathrm{Var}(Y_i)$, the latter equals $x_{ih}x_{ij}(\partial\mu_i/\partial\eta_i)^2/\mathrm{Var}(Y_i) = x_{ih}w_ix_{ij}$.

Step 3. Summing over independent observations gives the (h,j) entry of $X^\top WX$. ■

Hence

$$\widehat{\mathrm{Cov}}(\hat{\boldsymbol{\beta}}) = \hat{\mathcal{I}}^{-1} = (X^\top \hat{W} X)^{-1}, \tag{4.18}$$

justified by the next theorem.

Theorem 4.16 (Asymptotic normality of the ML estimator)

Let $\boldsymbol{\beta}_0$ be the true value, interior to the parameter space, and write $\boldsymbol{u}(\boldsymbol{\beta}) = \partial L/\partial\boldsymbol{\beta}$, $H(\boldsymbol{\beta}) = \partial^2 L/\partial\boldsymbol{\beta}\partial\boldsymbol{\beta}^\top$, $\mathcal{I}_N(\boldsymbol{\beta}) = X^\top W(\boldsymbol{\beta})X$. Assume:
(R1) $N^{-1}\mathcal{I}_N(\boldsymbol{\beta}_0) \to \Sigma$, positive definite;
(R2) Lindeberg condition: for each $\boldsymbol{c} \in \mathbb{R}^p$, the terms $\boldsymbol{c}^\top \partial L_i/\partial\boldsymbol{\beta}$ $(i \le N)$, normalized by $(N\boldsymbol{c}^\top\Sigma\boldsymbol{c})^{1/2}$, satisfy Lindeberg's condition;
(R3) $\sup_{\|\boldsymbol{\beta}-\boldsymbol{\beta}_0\|\le\delta} \|N^{-1}H(\boldsymbol{\beta}) + \Sigma\| \xrightarrow{p} 0$ as $N\to\infty$ and then $\delta \to 0$.
Then with probability tending to one the likelihood equations have a root $\hat{\boldsymbol{\beta}}$ with $\hat{\boldsymbol{\beta}} \xrightarrow{p} \boldsymbol{\beta}_0$, and

$$\sqrt{N}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0) \xrightarrow{d} N_p(\mathbf{0}, \Sigma^{-1}), \qquad \text{i.e.} \qquad \hat{\boldsymbol{\beta}} \approx N_p(\boldsymbol{\beta}_0, (X^\top W X)^{-1}).$$

Proof. [Supplement]

Step 1. (Score CLT.) The summands $\partial L_i/\partial\boldsymbol{\beta}$ are independent, mean $\mathbf{0}$, with covariances summing to $\mathcal{I}_N(\boldsymbol{\beta}_0)$. By (R1), (R2), Lindeberg–Feller and Cramér–Wold, $N^{-1/2}\boldsymbol{u}(\boldsymbol{\beta}_0) \xrightarrow{d} N_p(\mathbf{0},\Sigma)$; so $N^{-1}\boldsymbol{u}(\boldsymbol{\beta}_0) \xrightarrow{p} \mathbf{0}$.

Step 2. (Consistent root.) Let $\lambda > 0$ be the smallest eigenvalue of Σ, and fix small δ so that, by (R3), with probability $\to 1$ all eigenvalues of $-N^{-1}H$ are $\ge \lambda/2$ on $B_\delta = \{\|\boldsymbol{\beta} - \boldsymbol{\beta}_0\| \le \delta\}$. For $\boldsymbol{\beta} \in \partial B_\delta$, Taylor's theorem gives

$$L(\boldsymbol{\beta}) - L(\boldsymbol{\beta}_0) = \boldsymbol{u}(\boldsymbol{\beta}_0)^\top(\boldsymbol{\beta}-\boldsymbol{\beta}_0) + \tfrac{1}{2}(\boldsymbol{\beta}-\boldsymbol{\beta}_0)^\top H(\bar{\boldsymbol{\beta}})(\boldsymbol{\beta}-\boldsymbol{\beta}_0) \le N\delta\Big(\frac{\|\boldsymbol{u}(\boldsymbol{\beta}_0)\|}{N} - \frac{\lambda\delta}{4}\Big),$$

negative with probability $\to 1$ by Step 1. So the maximum of L on the compact B_δ is interior, where $\boldsymbol{u} = \mathbf{0}$; as δ is arbitrary, $\hat{\boldsymbol{\beta}} \xrightarrow{p} \boldsymbol{\beta}_0$.

Step 3. (Expansion.) Coordinatewise mean value theorem: $\mathbf{0} = \boldsymbol{u}(\hat{\boldsymbol{\beta}}) = \boldsymbol{u}(\boldsymbol{\beta}_0) + \tilde{H}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0)$, rows of $\tilde{H}$ evaluated at intermediate points; by consistency and (R3), $-N^{-1}\tilde{H} \xrightarrow{p} \Sigma$.

Step 4. So $\sqrt{N}(\hat{\boldsymbol{\beta}}-\boldsymbol{\beta}_0) = (-N^{-1}\tilde{H})^{-1}N^{-1/2}\boldsymbol{u}(\boldsymbol{\beta}_0) \xrightarrow{d} N_p(\mathbf{0},\Sigma^{-1})$ by Slutsky; and $N^{-1}X^\top\hat{W}X \xrightarrow{p} \Sigma$ by continuity, justifying (4.18). ■

For grouped binomial or Poisson data the same holds with N fixed and $n_i \to \infty$.

4.4.7 Likelihood equations and covariance matrix for Poisson loglinear model

Corollary 4.17 (Poisson loglinear model)

For $\log\boldsymbol{\mu} = X\boldsymbol{\beta}$ with Poisson responses, the likelihood equations are

$$\sum_i (y_i - \mu_i)x_{ij} = 0, \qquad j = 1,\dots,p, \tag{4.19}$$

and $\widehat{\mathrm{Cov}}(\hat{\boldsymbol{\beta}}) = (X^\top\hat{W}X)^{-1}$ with $\hat{W} = \mathrm{diag}(\hat{\mu}_1,\dots,\hat{\mu}_N)$.

Proof. For the log link $\partial\mu_i/\partial\eta_i = \mu_i = \mathrm{Var}(Y_i)$, so (4.14) becomes (4.19) and $w_i = \mu_i^2/\mu_i = \mu_i$. ■

4.5 Inference for generalized linear models

Motivation. Wald, score and LR methods apply to any GLM; LR inference runs through the deviance, the analogue of the residual sum of squares.

Theorem 4.18 (Wald, likelihood-ratio and score tests)

Under the conditions of Theorem 4.16, for $H_0 : \boldsymbol{\beta} = \boldsymbol{\beta}_0$ each of

$$\begin{aligned} W &= (\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0)^\top \hat{\mathcal{I}}\, (\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0) && \text{(Wald)}, \\ \Lambda &= -2[L(\boldsymbol{\beta}_0) - L(\hat{\boldsymbol{\beta}})] && \text{(likelihood ratio)}, \\ S &= \boldsymbol{u}(\boldsymbol{\beta}_0)^\top \mathcal{I}(\boldsymbol{\beta}_0)^{-1} \boldsymbol{u}(\boldsymbol{\beta}_0) && \text{(score)} \end{aligned}$$

converges in distribution to χ^2_p, and the three differ by $o_p(1)$. For nested models $M_0 \subset M_1$, where M_0 imposes q linear restrictions on $\boldsymbol{\beta}$, $\Lambda = -2[L(\hat{\boldsymbol{\beta}}_0) - L(\hat{\boldsymbol{\beta}}_1)] \xrightarrow{d} \chi^2_q$.

Proof. [Supplement] Let $\boldsymbol{Z}_N = N^{-1/2}\boldsymbol{u}(\boldsymbol{\beta}_0) \xrightarrow{d} \boldsymbol{Z} \sim N_p(\mathbf{0}, \Sigma)$; by Theorem 4.16, $\sqrt{N}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0) = \Sigma^{-1}\boldsymbol{Z}_N + o_p(1)$.

Step 1. (Score.) $S = \boldsymbol{Z}_N^\top (N^{-1}\mathcal{I}_N)^{-1} \boldsymbol{Z}_N \xrightarrow{d} \boldsymbol{Z}^\top \Sigma^{-1} \boldsymbol{Z} \sim \chi^2_p$.

Step 2. (Wald.) $W = [\sqrt{N}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0)]^\top (N^{-1}\hat{\mathcal{I}})[\sqrt{N}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0)] = \boldsymbol{Z}_N^\top \Sigma^{-1} \boldsymbol{Z}_N + o_p(1)$.

Step 3. (LR.) Expanding about $\hat{\boldsymbol{\beta}}$, where $\boldsymbol{u} = \mathbf{0}$, $L(\boldsymbol{\beta}_0) = L(\hat{\boldsymbol{\beta}}) + \frac{1}{2}(\boldsymbol{\beta}_0 - \hat{\boldsymbol{\beta}})^\top H(\bar{\boldsymbol{\beta}})(\boldsymbol{\beta}_0 - \hat{\boldsymbol{\beta}})$, so by (R3) $\Lambda = \boldsymbol{Z}_N^\top \Sigma^{-1} \boldsymbol{Z}_N + o_p(1)$.

Step 4. (Nested.) Write M_0 as $\boldsymbol{\beta} = \boldsymbol{\beta}_0 + A\boldsymbol{\gamma}$, A of full rank $p-q$; M_0 has score $A^\top \boldsymbol{u}$ and information $A^\top \mathcal{I} A$. Step 3 for both models gives

$$\Lambda = \boldsymbol{Z}_N^\top [\Sigma^{-1} - A(A^\top \Sigma A)^{-1} A^\top] \boldsymbol{Z}_N + o_p(1).$$

With $\boldsymbol{Z} = \Sigma^{1/2}\boldsymbol{\varepsilon}$, $\boldsymbol{\varepsilon} \sim N_p(\mathbf{0}, I)$, the limit is $\boldsymbol{\varepsilon}^\top (I - P)\boldsymbol{\varepsilon}$ with $P = \Sigma^{1/2} A(A^\top \Sigma A)^{-1} A^\top \Sigma^{1/2}$ a projection of rank $p - q$; so it is χ^2_q. ∎

Inverting the tests gives the Wald interval $\hat{\beta}_j \pm z_{\alpha/2}\widehat{\mathrm{se}}(\hat{\beta}_j)$ and the profile-likelihood interval $\{\beta_j : \Lambda(\beta_j) \le \chi^2_1(\alpha)\}$.

4.5.1 Deviance and goodness of fit

Let $\tilde{\theta}_i$ ($\tilde{\mu}_i = y_i$) and $\hat{\theta}_i$ be the saturated and fitted natural parameters. By (4.12) with $a(\phi) = \phi/\omega_i$, $-2[L(\hat{\boldsymbol{\mu}}; \mathbf{y}) - L(\mathbf{y}; \mathbf{y})]$ equals the following divided by ϕ.

Definition (Scaled deviance and deviance)

$$D(\mathbf{y}; \hat{\boldsymbol{\mu}}) = 2 \sum_i \omega_i [y_i(\tilde{\theta}_i - \hat{\theta}_i) - b(\tilde{\theta}_i) + b(\hat{\theta}_i)] \tag{4.20}$$

is the *deviance*, and D/ϕ the *scaled deviance*; they coincide for Poisson and binomial ($\phi = 1$). For the normal linear model $D = \sum_i (y_i - \hat{\mu}_i)^2$.

Theorem 4.19 (Asymptotic distribution of the deviance)

For a binomial or Poisson GLM with a fixed number N of observations whose counts grow ($n_i \to \infty$, or $\mu_i \to \infty$), if the model with p parameters holds then $D(\mathbf{y}; \hat{\boldsymbol{\mu}}) \xrightarrow{d} \chi^2_{N-p}$.

Proof. [Supplement] The saturated model is a GLM with N free θ_i, and the working model is the submodel given by $N - p$ linear restrictions $\boldsymbol{\theta} \in \mathrm{col}(X)$ (canonical scale; otherwise reparametrize smoothly). D is the LR statistic for these nested models, so Theorem 4.18 (growing counts as sample size) gives χ^2_{N-p}. ■

4.5.2 Deviance for Poisson models

Proposition 4.20 (Poisson deviance)

For a Poisson GLM,

$$D(\mathbf{y}; \hat{\boldsymbol{\mu}}) = 2 \sum_i \big[y_i \log(y_i/\hat{\mu}_i) - y_i + \hat{\mu}_i \big]. \tag{4.21}$$

If the model has log link and an intercept, $\sum_i y_i = \sum_i \hat{\mu}_i$ and $D(\mathbf{y}; \hat{\boldsymbol{\mu}}) = 2 \sum_i y_i \log(y_i/\hat{\mu}_i)$.

Proof. Put $\hat{\theta}_i = \log \hat{\mu}_i$, $b(\hat{\theta}_i) = \hat{\mu}_i$, $\tilde{\theta}_i = \log y_i$, $b(\tilde{\theta}_i) = y_i$, $\omega_i = 1$ in (4.20). With an intercept the first equation of (4.19) is $\sum_i (y_i - \hat{\mu}_i) = 0$, so the linear terms cancel ($0 \log 0 = 0$). ■

For a two-way table under independence this is G^2 of Chapter 3.

4.5.3 Deviance for binomial models: grouped and ungrouped data

Proposition 4.21 (Binomial deviance)

For binomial proportions y_i based on n_i trials,

$$D(\mathbf{y}; \hat{\boldsymbol{\mu}}) = 2 \sum_i n_i y_i \log \frac{n_i y_i}{n_i \hat{\pi}_i} + 2 \sum_i (n_i - n_i y_i) \log \frac{n_i - n_i y_i}{n_i - n_i \hat{\pi}_i}, \tag{4.22}$$

that is, $D = 2 \sum \text{observed} \times \log(\text{observed}/\text{fitted})$ over the $2N$ cells of successes and failures.

Proof. With $\hat{\theta}_i = \operatorname{logit} \hat{\pi}_i$, $b(\hat{\theta}_i) = -\log(1 - \hat{\pi}_i)$, $\tilde{\theta}_i = \operatorname{logit} y_i$, $b(\tilde{\theta}_i) = -\log(1 - y_i)$, $\omega_i = n_i$, (4.20) gives

$$D = 2 \sum_i n_i \Big[y_i \log \frac{y_i}{\hat{\pi}_i} + (1 - y_i) \log \frac{1 - y_i}{1 - \hat{\pi}_i} \Big],$$

which is (4.22). ■

Grouped and ungrouped (0/1) data give the same ML estimates but different saturated models and deviances; the χ^2 approximation needs grouped data with $n_i \to \infty$.

4.5.4 Likelihood-ratio model comparison using the deviance

Proposition 4.22 (Deviance difference)

Let $M_0 \subset M_1$ be nested Poisson or binomial GLMs with fitted values $\hat{\boldsymbol{\mu}}_0, \hat{\boldsymbol{\mu}}_1$. Then $D(\mathbf{y}; \hat{\boldsymbol{\mu}}_1) \leq D(\mathbf{y}; \hat{\boldsymbol{\mu}}_0)$ and the likelihood-ratio statistic for $H_0 : M_0$ given M_1 is

$$-2\big[L(\hat{\boldsymbol{\mu}}_0; \mathbf{y}) - L(\hat{\boldsymbol{\mu}}_1; \mathbf{y})\big] = D(\mathbf{y}; \hat{\boldsymbol{\mu}}_0) - D(\mathbf{y}; \hat{\boldsymbol{\mu}}_1),$$

asymptotically χ^2 with df equal to the difference in numbers of parameters. With log link and intercept (Poisson) or for binomial data, it equals $2 \sum \text{observed} \times \log(\text{fitted}_1/\text{fitted}_0)$.

Proof. Maximizing over a smaller space gives $L(\hat{\boldsymbol{\mu}}_0) \leq L(\hat{\boldsymbol{\mu}}_1)$; the identity follows by adding and subtracting $L(\mathbf{y}; \mathbf{y})$; the distribution is Theorem 4.18. Subtracting (4.22) (or (4.21) without linear terms) for the two models cancels the log observed terms. ■

The saturated part cancels, so the comparison is valid for ungrouped data too.

4.5.5 Residuals for GLMs

Definition (Deviance and Pearson residuals)

With $D = \sum_i d_i$, $d_i = 2\omega_i[y_i(\tilde{\theta}_i - \hat{\theta}_i) - b(\tilde{\theta}_i) + b(\hat{\theta}_i)]$, the *deviance residual* is $\sqrt{d_i}\ \operatorname{sgn}(y_i - \hat{\mu}_i)$ and the *Pearson residual* is $e_i = (y_i - \hat{\mu}_i)/[\widehat{\operatorname{Var}}(Y_i)]^{1/2}$.

For independence in a two-way table $\sum e_{ij}^2 = X^2$, $\sum d_{ij} = G^2$. These are less variable than $N(0,1)$; standardizing uses the *hat matrix*

$$\text{HatM} = W^{1/2} X (X^\top W X)^{-1} X^\top W^{1/2}. \tag{4.23}$$

Proposition 4.23 (Leverages and residual covariance)

(a) HatM is symmetric and idempotent with trace p; its diagonal elements (*leverages*) satisfy $0 \leq h_i \leq 1$ and $\sum_i h_i = p$.
(b) To first order, with $V = \operatorname{Cov}(\boldsymbol{Y}) = \operatorname{diag}\{\operatorname{Var}(Y_i)\}$,

$$\operatorname{Cov}(\boldsymbol{Y} - \hat{\boldsymbol{\mu}}) \approx V^{1/2}(I - \text{HatM})V^{1/2}, \qquad \operatorname{Var}(Y_i - \hat{\mu}_i) \approx \operatorname{Var}(Y_i)(1 - h_i).$$

Hence the *standardized Pearson residual* is

$$r_i = \frac{y_i - \hat{\mu}_i}{[\widehat{\operatorname{Var}}(Y_i)(1 - \hat{h}_i)]^{1/2}} = \frac{e_i}{\sqrt{1 - \hat{h}_i}} \qquad \left(\text{Poisson: } \frac{y_i - \hat{\mu}_i}{\sqrt{\hat{\mu}_i(1 - \hat{h}_i)}}\right). \tag{4.24}$$

Proof. [Supplement] (a) With $A = W^{1/2}X$, $\text{HatM} = A(A^\top A)^{-1}A^\top$ is symmetric, $\text{HatM}^2 = \text{HatM}$, $\operatorname{tr}\text{HatM} = \operatorname{tr} I_p = p$, and $h_i = \sum_j h_{ij}^2 \geq h_i^2$ gives $0 \leq h_i \leq 1$.

(b)

Step 1. Let $G = \operatorname{diag}(\partial\eta_i/\partial\mu_i)$; by (4.13), $\boldsymbol{u} = X^\top W G(\boldsymbol{y} - \boldsymbol{\mu})$.

Step 2. By Theorem 4.16, $\hat{\boldsymbol{\beta}} - \boldsymbol{\beta} \approx (X^\top W X)^{-1} X^\top W G(\boldsymbol{y} - \boldsymbol{\mu})$ and $\hat{\boldsymbol{\mu}} - \boldsymbol{\mu} \approx G^{-1}X(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})$, so $\boldsymbol{y} - \hat{\boldsymbol{\mu}} \approx [I - G^{-1}X(X^\top W X)^{-1}X^\top W G](\boldsymbol{y} - \boldsymbol{\mu})$.

Step 3. Since $w_i = 1/(G_{ii}^2 \operatorname{Var}(Y_i))$, $V^{1/2} = G^{-1}W^{-1/2} =: M$, and the bracket is $M(I - \text{HatM})M^{-1}$.

Step 4. With $\operatorname{Cov}(\boldsymbol{y} - \boldsymbol{\mu}) = M^2$ and HatM symmetric idempotent, $\operatorname{Cov}(\boldsymbol{y} - \hat{\boldsymbol{\mu}}) \approx M(I - \text{HatM})M^{-1}M^2M^{-1}(I - \text{HatM})M = M(I - \text{HatM})M$, with ith diagonal entry $\operatorname{Var}(Y_i)(1 - h_i)$.

■

Leverages depend on the fit through W, so extreme predictor values need not have high leverage.

4.6 Fitting generalized linear models

Motivation. Newton–Raphson and Fisher scoring (that is, IRLS) solve the nonlinear likelihood equations.

4.6.1 Newton–Raphson method

With $\boldsymbol{u}^{(t)}, H^{(t)}$ the score and Hessian at $\boldsymbol{\beta}^{(t)}$, maximizing the quadratic approximation $L(\boldsymbol{\beta}^{(t)}) + \boldsymbol{u}^{(t)\top}(\boldsymbol{\beta}-\boldsymbol{\beta}^{(t)}) + \tfrac{1}{2}(\boldsymbol{\beta}-\boldsymbol{\beta}^{(t)})^\top H^{(t)}(\boldsymbol{\beta}-\boldsymbol{\beta}^{(t)})$ gives the *Newton–Raphson* step (Figure 6)

$$\boldsymbol{\beta}^{(t+1)} = \boldsymbol{\beta}^{(t)} - \left(H^{(t)}\right)^{-1}\boldsymbol{u}^{(t)}. \tag{4.25}$$

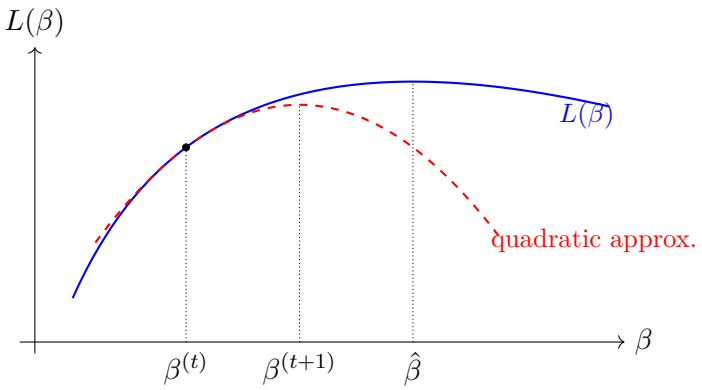


Figure 6: One Newton–Raphson cycle: the next guess maximizes the quadratic approximation at $\beta^{(t)}$.

Example 4.24 (Newton–Raphson for a binomial parameter)

For $y \sim \mathrm{bin}(n,\pi)$, $u = (y-n\pi)/[\pi(1-\pi)]$, $H = -[y/\pi^2 + (n-y)/(1-\pi)^2]$. From $\pi^{(0)} = \tfrac{1}{2}$ one step gives y/n; otherwise four or five iterations.

Theorem 4.25 (Quadratic convergence of Newton–Raphson)

Suppose $\boldsymbol{u}(\hat{\boldsymbol{\beta}}) = \mathbf{0}$, H is Lipschitz on a ball B around $\hat{\boldsymbol{\beta}}$ with constant γ ($\|H(\boldsymbol{a}) - H(\boldsymbol{b})\| \le \gamma\|\boldsymbol{a}-\boldsymbol{b}\|$), and $\|H(\boldsymbol{\beta})^{-1}\| \le M$ on B. If $\boldsymbol{\beta}^{(t)} \in B$, then

$$\|\boldsymbol{\beta}^{(t+1)} - \hat{\boldsymbol{\beta}}\| \le \frac{M\gamma}{2}\,\|\boldsymbol{\beta}^{(t)} - \hat{\boldsymbol{\beta}}\|^2.$$

In particular, if $\boldsymbol{\beta}^{(0)} \in B$ and $\|\boldsymbol{\beta}^{(0)} - \hat{\boldsymbol{\beta}}\| < 2/(M\gamma)$, the iterates converge to $\hat{\boldsymbol{\beta}}$ and the number of correct digits roughly doubles each step.

Proof. [Supplement]

Step 1. With $\boldsymbol{e}_t = \boldsymbol{\beta}^{(t)} - \hat{\boldsymbol{\beta}}$, $\boldsymbol{u}^{(t)} = \boldsymbol{u}^{(t)} - \boldsymbol{u}(\hat{\boldsymbol{\beta}}) = \int_0^1 H(\hat{\boldsymbol{\beta}} + s\boldsymbol{e}_t)\,ds\;\boldsymbol{e}_t$.

Step 2. By (4.25),

$$\boldsymbol{e}_{t+1} = \boldsymbol{e}_t - (H^{(t)})^{-1}\boldsymbol{u}^{(t)} = (H^{(t)})^{-1}\int_0^1 [H(\hat{\boldsymbol{\beta}} + \boldsymbol{e}_t) - H(\hat{\boldsymbol{\beta}} + s\boldsymbol{e}_t)]ds\, \boldsymbol{e}_t.$$

Step 3. The integrand has norm $\leq \gamma(1-s)\|\boldsymbol{e}_t\|$ and $\int_0^1 (1-s)ds = \frac{1}{2}$, so $\|\boldsymbol{e}_{t+1}\| \leq \frac{M\gamma}{2}\|\boldsymbol{e}_t\|^2$.

Step 4. If $\rho = \frac{M\gamma}{2}\|\boldsymbol{e}_0\| < 1$, induction gives $\|\boldsymbol{e}_{t+1}\| \leq \rho\|\boldsymbol{e}_t\|$, the iterates stay in B and $\boldsymbol{e}_t \to \mathbf{0}$, the error being roughly squared each step. ■

4.6.2 Fisher scoring method

Fisher scoring replaces $-H$ (observed information) by the expected information $\mathcal{I}$:

$$\boldsymbol{\beta}^{(t+1)} = \boldsymbol{\beta}^{(t)} + (\mathcal{I}^{(t)})^{-1}\boldsymbol{u}^{(t)} \quad \Longleftrightarrow \quad \mathcal{I}^{(t)}\boldsymbol{\beta}^{(t+1)} = \mathcal{I}^{(t)}\boldsymbol{\beta}^{(t)} + \boldsymbol{u}^{(t)}. \tag{4.26}$$

For the binomial parameter $\mathcal{I} = n/[\pi(1-\pi)]$ and one step gives y/n from any start. For a GLM, $\mathcal{I}^{(t)} = X^\top W^{(t)} X$ and $\boldsymbol{u}^{(t)}$ has elements

$$u_j = \sum_{i=1}^{N} \frac{(y_i - \mu_i)x_{ij}}{\mathrm{Var}(Y_i)}\frac{\partial \mu_i}{\partial \eta_i}; \tag{4.27}$$

at convergence $(\mathcal{I}^{(t)})^{-1}$ is the covariance estimate (4.18). For canonical links the two methods coincide (Theorem 4.27); otherwise Fisher scoring has nonnegative definite $\mathcal{I}$ but need not converge quadratically.

4.6.3 ML as iterative reweighted least squares

For $\boldsymbol{z} = X\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ with $\mathrm{Cov}(\boldsymbol{\varepsilon}) = V$, the WLS estimator is $(X^\top V^{-1}X)^{-1}X^\top V^{-1}\boldsymbol{z}$.

Theorem 4.26 (Fisher scoring is IRLS)

For a GLM, the Fisher scoring step (4.26) is

$$\boldsymbol{\beta}^{(t+1)} = (X^\top W^{(t)} X)^{-1} X^\top W^{(t)} \boldsymbol{z}^{(t)},$$

the WLS fit of the *working response*

$$z_i^{(t)} = \eta_i^{(t)} + (y_i - \mu_i^{(t)})\frac{\partial \eta_i^{(t)}}{\partial \mu_i^{(t)}} \tag{4.28}$$

on X with weight matrix (inverse covariance) $W^{(t)}$.

Proof.

Step 1. $\mathcal{I}^{(t)}\boldsymbol{\beta}^{(t)} = X^\top W^{(t)} X\boldsymbol{\beta}^{(t)} = X^\top W^{(t)}\boldsymbol{\eta}^{(t)}$.

Step 2. Since $w_i = (\partial\mu_i/\partial\eta_i)^2/\mathrm{Var}(Y_i)$, each term of (4.27) is $x_{ij}w_i(y_i - \mu_i)\,\partial\eta_i/\partial\mu_i$, so $\boldsymbol{u}^{(t)} = X^\top W^{(t)}\boldsymbol{d}^{(t)}$ with $d_i^{(t)} = (y_i - \mu_i^{(t)})\,\partial\eta_i^{(t)}/\partial\mu_i^{(t)}$.

Step 3. The right side of (4.26) is $X^\top W^{(t)}(\boldsymbol{\eta}^{(t)} + \boldsymbol{d}^{(t)}) = X^\top W^{(t)}\boldsymbol{z}^{(t)}$: the WLS normal equations. ■

z_i linearizes the link, $g(y_i) \approx \eta_i + (y_i - \mu_i)\,\partial\eta_i/\partial\mu_i$, and W^{-1} approximates its covariance. Iterating is *iteratively reweighted least squares* (IRLS); a start is $\hat{\boldsymbol{\mu}}^{(0)} = \mathbf{y}$, adjusted where $g(y_i)$ is infinite.

4.6.4 Simplifications for canonical links

Theorem 4.27 (Canonical link properties)

Consider a GLM with the canonical link, $\theta_i = \eta_i = \sum_j \beta_j x_{ij}$, and a common $a(\phi)$ for all i (e.g. Poisson; binomial with all $n_i = 1$). Then:
(a) $\sum_i y_i x_{ij}$, $j = 1, \ldots, p$, are sufficient statistics for $\boldsymbol{\beta}$;
(b) $\partial L_i / \partial \beta_j = (y_i - \mu_i) x_{ij} / a(\phi)$, so the likelihood equations are

$$\sum_i x_{ij} y_i = \sum_i x_{ij} \mu_i, \qquad j = 1, \ldots, p, \qquad \text{i.e.} \quad X^\top \mathbf{y} = X^\top \boldsymbol{\mu}; \tag{4.29}$$

(c) $\partial^2 L / \partial \beta_h \partial \beta_j = -\sum_i x_{ih} x_{ij} b''(\theta_i)/a(\phi)$ does not involve $\mathbf{y}$; hence observed and expected information coincide, $H = -\mathcal{I}$, and Newton–Raphson equals Fisher scoring;
(d) L is concave in $\boldsymbol{\beta}$, strictly if X has full column rank; so a solution of (4.29), if one exists, is the unique ML estimate.
With weights $a(\phi) = \phi/\omega_i$ the same holds with y_i, μ_i replaced by $\omega_i y_i, \omega_i \mu_i$.

Proof.

Step 1. By (4.12) the only term involving both data and parameters is $a(\phi)^{-1} \sum_i y_i \theta_i = a(\phi)^{-1} \sum_j \beta_j (\sum_i y_i x_{ij})$; the factorization theorem gives (a).

Step 2. Here $\partial \mu_i / \partial \eta_i = b''(\theta_i) = \mathrm{Var}(Y_i)/a(\phi)$; inserting into (4.13) gives (b).

Step 3. Since $\partial \mu_i / \partial \beta_h = b''(\theta_i) x_{ih}$, $\partial^2 L_i / \partial \beta_h \partial \beta_j = -x_{ij} x_{ih} b''(\theta_i)/a(\phi)$, free of y_i and equal to its expectation: (c).

Step 4. For $\boldsymbol{c} \in \mathbb{R}^p$, $\boldsymbol{c}^\top H \boldsymbol{c} = -\sum_i b''(\theta_i)(\boldsymbol{x}_i^\top \boldsymbol{c})^2/a(\phi) \le 0$, with equality only if $X\boldsymbol{c} = \mathbf{0}$, impossible for $\boldsymbol{c} \neq \mathbf{0}$ under full rank. A stationary point of a strictly concave function is its unique maximum: (d). ■

So (4.29) equates sufficient statistics to their expectations (cf. the normal equations, (4.19), (4.16) with $n_i = 1$).

Chapter 5. Logistic Regression

Logistic regression is the central model for binary responses.

5.1 Interpreting parameters in logistic regression

Motivation. People think in probabilities, so β needs several readings: odds ratio, local slope, change in π over x.

For binary Y and predictor X let $\pi(x) = P(Y = 1 \mid X = x)$. The logistic regression model is

$$\pi(x) = \frac{\exp(\alpha + \beta x)}{1 + \exp(\alpha + \beta x)}, \qquad \text{equivalently} \qquad \operatorname{logit} \pi(x) = \log \frac{\pi(x)}{1 - \pi(x)} = \alpha + \beta x. \tag{5.1}$$

5.1.1 Interpreting β: odds, probabilities, and linear approximations

The sign of β gives the direction; the curve steepens as $|\beta|$ grows and is flat at $\beta = 0$ (independence). For $\beta > 0$ it is a logistic cdf, approaching 1 and 0 at the same rate.

Proposition 5.1 (Odds-ratio interpretation)

Under (5.1) the odds are $\pi(x)/[1 - \pi(x)] = e^{\alpha}(e^{\beta})^{x}$; hence for every x,

$$\frac{\pi(x+1)/[1 - \pi(x+1)]}{\pi(x)/[1 - \pi(x)]} = e^{\beta} :$$

the odds multiply by e^{β} for each unit increase in x, and e^{β} is an odds ratio.

Proof. Exponentiate the logit form: odds $= \exp(\alpha + \beta x)$. The ratio at $x+1$ and x is $\exp(\alpha + \beta x + \beta)/\exp(\alpha + \beta x) = e^{\beta}$. ■

Proposition 5.2 (Linear approximation (Berkson 1951))

Under (5.1),

$$\frac{d\pi(x)}{dx} = \beta\, \pi(x)[1 - \pi(x)].$$

(a) The slope is greatest, $\beta/4$, where $\pi(x) = \frac{1}{2}$, i.e. at $x = -\alpha/\beta$ (the median effective level EL_{50}; in toxicology the LD_{50}). At $\pi(x) = 0.9$ or 0.1 it is 0.09β, and it tends to 0 as $\pi(x) \to 0$ or 1.
(b) $\pi(-\alpha/\beta \pm 1/\beta) = e^{\pm 1}/(1 + e^{\pm 1})$, i.e. 0.73 and 0.27: a change of $1/\beta$ in x from the EL_{50} changes π by about 0.23, close to the value $\frac{1}{4}$ suggested by the tangent line.

Proof. [Supplement]

Step 1. With $\eta = \alpha + \beta x$ and $\pi = e^{\eta}/(1 + e^{\eta})$,

$$\frac{d\pi}{d\eta} = \frac{e^{\eta}(1 + e^{\eta}) - e^{\eta} e^{\eta}}{(1 + e^{\eta})^2} = \frac{e^{\eta}}{(1 + e^{\eta})^2} = \pi(1 - \pi),$$

and $d\eta/dx = \beta$, so $d\pi/dx = \beta\pi(1 - \pi)$.

Step 2. The function $\pi(1-\pi)$ on $(0, 1)$ is maximized at $\pi = \frac{1}{2}$ with value $\frac{1}{4}$; $\pi(x) = \frac{1}{2}$ iff $\alpha + \beta x = 0$ iff $x = -\alpha/\beta$. Also $0.9 \times 0.1 = 0.09$.

Step 3. At $x = -\alpha/\beta \pm 1/\beta$, $\eta = \pm 1$, so $\pi = e^{\pm 1}/(1 + e^{\pm 1}) = 0.731$ or 0.269. ■

So a unit change in x changes π by about $\beta\pi(1-\pi)$ (Figure 7). Alternatively report $\pi(x)$ at the quartiles of x, comparable across units. After centering x, the intercept is the logit at $\bar{x}$, and correlations among estimates with quadratic or interaction terms drop.

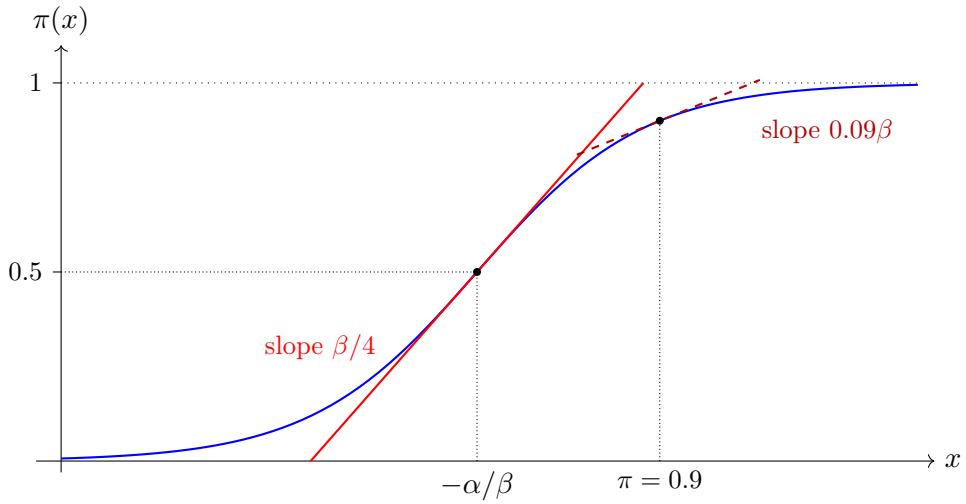


Figure 7: Logistic curve ($\alpha = -3$, $\beta = 1$) with tangent lines: slope $\beta\pi(1-\pi)$ equals $\beta/4$ at $\pi = \frac{1}{2}$ and 0.09β at $\pi = 0.9$.

5.1.2 Looking at the data

Before fitting, plot sample proportions or logits, not 0/1 responses. The sample logit $\log[y_i/(n_i - y_i)]$ is infinite when $y_i = 0$ or n_i; the adjusted version

$$\log \frac{y_i + \frac{1}{2}}{n_i - y_i + \frac{1}{2}}$$

is least biased of this form (bias $O(n_i^{-2})$, Haldane 1956). The plot should be roughly linear; with continuous x and small n_i, group nearby x or smooth and look for departures from an S-shape.

5.1.3 Horseshoe crabs revisited

Example 5.3 (Horseshoe crabs: width and presence of satellites)

Crabs (Example 4.5), $Y = 1$ if at least one satellite. The GAM fit and the proportions in the eight width categories show an increasing S-shape. With x = width (cm),

$$\hat{\pi}(x) = \frac{\exp(-12.351 + 0.497x)}{1 + \exp(-12.351 + 0.497x)}.$$

Parameter	Estimate	SE	LR 95% limits	Wald χ^2
Intercept	−12.3508	2.6287	(−17.8097, −7.4573)	22.07
width	0.4972	0.1017	(0.3084, 0.7090)	23.89

Deviance 194.45 and Pearson $X^2 = 165.14$ on 171 df; log likelihood −97.23.

(a) At the mean width 26.3, $\hat{\pi} = 0.674$; $\hat{\pi} = \frac{1}{2}$ at $x = 12.351/0.497 = 24.8$.
(b) The odds multiply by $e^{0.497} = 1.64$ per cm (a 64% increase).

(c) At $\bar{x}$ the probability rises by about $0.497(0.674)(0.326) = 0.11$ per cm.
(d) At the width quartiles 24.9, 26.1, 27.7, $\hat{\pi} = 0.51,\ 0.65,\ 0.81$: an increase of 0.30 over the middle half.
(e) With weight (kg) instead, $\operatorname{logit}\hat{\pi} = -3.695 + 1.815x$; $\hat{\beta}$'s are not comparable across units, but at the weight quartiles 2.00, 2.35, 2.85 we get $\hat{\pi} = 0.48,\ 0.64,\ 0.81$, an increase of 0.33, similar to width.

5.1.4 Logistic regression with retrospective studies

In case–control studies Y is fixed by design and X observed; logistic effects are odds ratios, which are estimable retrospectively.

Theorem 5.4 (Logistic regression under case–control sampling)

Let $Z = 1$ indicate that a subject is sampled, and suppose the sampling probabilities depend only on y: $P(Z = 1 \mid y, x) = \rho_y$ for $y = 0, 1$. If $\operatorname{logit} P(Y = 1 \mid x) = \alpha + \beta x$, then

$$\operatorname{logit} P(Y = 1 \mid Z = 1, x) = \alpha^* + \beta x, \qquad \alpha^* = \alpha + \log(\rho_1/\rho_0).$$

So the slope β is the same as in the prospective model; only the intercept changes.

Proof.

Step 1. By Bayes' theorem,

$$P(Y = 1 \mid Z = 1, x) = \frac{P(Z = 1 \mid Y = 1, x)P(Y = 1 \mid x)}{\sum_{j=0}^{1} P(Z = 1 \mid Y = j, x)P(Y = j \mid x)} = \frac{\rho_1 P(Y = 1 \mid x)}{\rho_0 P(Y = 0 \mid x) + \rho_1 P(Y = 1 \mid x)}.$$

Step 2. Divide numerator and denominator by $P(Y = 0 \mid x)$ and use $P(Y = 1 \mid x)/P(Y = 0 \mid x) = \exp(\alpha + \beta x)$:

$$P(Y = 1 \mid Z = 1, x) = \frac{\rho_1 \exp(\alpha + \beta x)}{\rho_0 + \rho_1 \exp(\alpha + \beta x)} = \frac{\exp[\alpha + \log(\rho_1/\rho_0) + \beta x]}{1 + \exp[\alpha + \log(\rho_1/\rho_0) + \beta x]}.$$

■

Sampling cases at 10 times the rate of controls raises the intercept by $\log 10 = 2.3$. Other binary links are not recoverable from case–control data, a main reason for the logit's popularity in biomedicine. Matched studies need conditional methods (Section 10.2.5).

Proposition 5.5 (Normal discriminant model implies logistic regression (Cornfield 1962))

Suppose $X \mid Y = i \sim N(\mu_i, \sigma^2)$, $i = 0, 1$, and $P(Y = 1) = \rho$. Then $P(Y = 1 \mid X = x)$ satisfies (5.1) with

$$\beta = \frac{\mu_1 - \mu_0}{\sigma^2}, \qquad \alpha = \log\frac{\rho}{1-\rho} - \frac{\mu_1^2 - \mu_0^2}{2\sigma^2}.$$

If the two variances differ, the logit is quadratic in x.

Proof. [Supplement] Let φ_i be the $N(\mu_i, \sigma_i^2)$ density. By Bayes' theorem,

$$\operatorname{logit} P(Y = 1 \mid x) = \log\frac{\rho\varphi_1(x)}{(1-\rho)\varphi_0(x)} = \log\frac{\rho}{1-\rho} + \log\frac{\sigma_0}{\sigma_1} - \frac{(x-\mu_1)^2}{2\sigma_1^2} + \frac{(x-\mu_0)^2}{2\sigma_0^2}.$$

With $\sigma_0 = \sigma_1 = \sigma$ the x^2 terms cancel and

$$-\frac{(x-\mu_1)^2-(x-\mu_0)^2}{2\sigma^2} = \frac{(\mu_1-\mu_0)x}{\sigma^2} - \frac{\mu_1^2-\mu_0^2}{2\sigma^2},$$

giving the stated α, β. If $\sigma_0 \neq \sigma_1$ the coefficient of x^2 is $\frac{1}{2}(\sigma_0^{-2} - \sigma_1^{-2}) \neq 0$. ■

So two roughly normal groups with similar spread give a good logistic fit; unequal spreads make $\pi(x)$ nonmonotone.

5.2 Inference for logistic regression

Motivation. The Wald–LR–score triad of Theorem 4.18 applies; LR is most reliable, and goodness of fit with continuous predictors needs care.

5.2.1 Types of inference

For logit $\pi(x) = \alpha + \beta x$, the test of independence $H_0 : \beta = 0$ can use

(i) Wald: $z = \hat{\beta}/\widehat{\text{se}}$, or $z^2 \sim \chi^2_1$;

(ii) likelihood ratio: $-2(L_0 - L_1)$, twice the difference of maximized log likelihoods under H_0 and the full model, χ^2_1;

(iii) score: the derivative of the log likelihood at $\beta = 0$, comparing the sufficient statistic with its null expectation (Section 5.3.5).

They agree in large samples. LR is preferred; for large $|\beta|$ the Wald test loses power (Hauck and Donner 1977). Wald intervals are $\hat{\beta} \pm z_{\alpha/2}\widehat{\text{se}}$; inverting LR gives the profile-likelihood interval.

Proposition 5.6 (Interval for $\pi(x_0)$)

The estimated variance of the fitted logit at x_0 is

$$\widehat{\text{Var}}(\hat{\alpha} + \hat{\beta}x_0) = \widehat{\text{Var}}(\hat{\alpha}) + x_0^2\,\widehat{\text{Var}}(\hat{\beta}) + 2x_0\,\widehat{\text{Cov}}(\hat{\alpha}, \hat{\beta}),$$

and if (l, u) is a $1-\alpha$ interval for logit $\pi(x_0)$, then $\left(e^l/(1+e^l),\ e^u/(1+e^u)\right)$ is a $1-\alpha$ interval for $\pi(x_0)$.

Proof. The first formula is $\text{Var}[(1, x_0)(\hat{\alpha}, \hat{\beta})^\top] = (1, x_0)\widehat{\text{Cov}}(\hat{\alpha}, \hat{\beta})(1, x_0)^\top$. As $t \mapsto e^t/(1+e^t)$ is strictly increasing, the coverage is preserved. ■

Equivalently, refit with $x^* = x - x_0$: the intercept and its SE are the fitted logit at x_0 and its SE.

5.2.2 Inference for horseshoe crab data

Example 5.7 (Horseshoe crabs: tests and intervals)

(a) Wald: $z = 0.497/0.102 = 4.9$, $z^2 = 23.9$ ($P < 0.0001$). LR: $-2(-112.88 + 97.23) = 31.3$ (df $= 1$), stronger still.
(b) Wald interval for β: $0.497 \pm 1.96(0.102) = (0.298, 0.697)$; profile: $(0.308, 0.709)$. Odds ratio per cm: $(1.36, 2.03)$.
(c) At $x = 26.5$: logit $= -12.351 + 0.497(26.5) = 0.825$, $\hat{\pi} = 0.695$. With $\widehat{\text{Var}}(\hat{\alpha}) = 6.910$,

$\widehat{\mathrm{Var}}(\hat{\beta}) = 0.01035$, $\widehat{\mathrm{Cov}} = -0.2668$,

$$\widehat{\mathrm{Var}} = 6.910 + (26.5)^2(0.01035) + 2(26.5)(-0.2668) = 0.038,$$

so the logit interval is $0.825 \pm 1.96\sqrt{0.038} = (0.44, 1.21)$, i.e. $(0.61, 0.77)$ for $\pi(26.5)$.
(d) At $x = 26.5$ (6 crabs, 4 with satellites) the sample proportion 0.67 has SE $= \sqrt{0.67(0.33)/6} = 0.19$ and score interval (0.30, 0.90), versus model SE = 0.04: the model borrows strength from all 173 observations (Section 6.4.5).

5.2.3 Checking goodness of fit: ungrouped and grouped data

Lack of fit is detected by an LR test against a more complex model (quadratic term, interaction), or globally by X^2, G^2 on observed versus fitted counts $n_i\hat{\pi}_i$, $n_i(1-\hat{\pi}_i)$: χ^2 with df = (settings) − (parameters) for a fixed number of settings with growing counts. This needs categorical predictors (Section 4.5.3).

5.2.4 Goodness of fit of model for horseshoe crabs

Example 5.8 (Horseshoe crabs: goodness of fit)

(a) Quadratic in centered width: $\operatorname{logit}\hat{\pi} = 0.618 + 0.533x + 0.040x^2$, $\widehat{\mathrm{se}} = 0.046$ for x^2; LR = 0.83 (df = 1): not needed.
(b) Width has 66 distinct values; a 66×2 table violates the chi-squared theory. Using the eight width categories of Example 4.5, fitted values summed over crabs in each category:

Width (cm)	Yes	No	Fitted yes	Fitted no
< 23.25	5	9	3.64	10.36
23.25–24.25	4	10	5.31	8.69
24.25–25.25	17	11	13.78	14.22
25.25–26.25	21	18	24.23	14.77
26.25–27.25	15	7	15.94	6.06
27.25–28.25	20	4	19.38	4.62
28.25–29.25	15	3	15.65	2.35
> 29.25	14	0	13.08	0.92

$X^2 = 5.3$, $G^2 = 6.2$, df $= 8 - 2 = 6$ ($P > 0.4$): no lack of fit (approximate, as $\pi(x)$ varies within a category).

5.2.5 Checking goodness of fit with ungrouped data by grouping

With many predictors, grouping each variable gives a sparse table; group instead by fitted probabilities.

Definition (Hosmer–Lemeshow statistic)

Order observations by $\hat{\pi}$ (ungrouped fit) into g groups of about equal size (usually $g = 10$). With y_{ij}, $\hat{\pi}_{ij}$ the outcome and fitted probability of observation j in group i ($j = 1, \ldots, n_i$),

$$\mathrm{HL} = \sum_{i=1}^{g} \frac{\left(\sum_j y_{ij} - \sum_j \hat{\pi}_{ij}\right)^2}{\left(\sum_j \hat{\pi}_{ij}\right)\left[1 - \left(\sum_j \hat{\pi}_{ij}\right)/n_i\right]}.$$

Proposition 5.9 (HL as a Pearson statistic)

HL equals the Pearson statistic $\sum(\text{observed} - \text{fitted})^2/\text{fitted}$ for the $g \times 2$ table of observed and fitted counts of successes and failures in the groups.

Proof. [Supplement] In group i let $O_i = \sum_j y_{ij}$, $E_i = \sum_j \hat{\pi}_{ij}$. The failure cell has observed $n_i - O_i$ and fitted $n_i - E_i$, so its deviation is $-(O_i - E_i)$. The two Pearson terms of row i add to

$$(O_i - E_i)^2\Big(\frac{1}{E_i} + \frac{1}{n_i - E_i}\Big) = \frac{(O_i - E_i)^2\, n_i}{E_i(n_i - E_i)} = \frac{(O_i - E_i)^2}{E_i(1 - E_i/n_i)}.$$

Summing over i gives HL. ■

Reference distribution and limitations

HL is not chi-squared in the usual sense (probabilities vary within a group); by simulation, with n distinct covariate patterns its null distribution is about χ^2_{g-2} (Hosmer–Lemeshow). Ties in $\hat{\pi}$ make software values differ slightly. Crab width model, $g = 10$: HL $= 3.5$ on 8 df, a decent fit. HL has low power and does not say what is wrong; comparisons with richer models and diagnostics (Section 6.2) are more informative.

5.3 Logit models with categorical predictors

Motivation. Codings of dummy variables change parameters, not fit or odds ratios; ordered factors with scores give the linear logit model and the Cochran–Armitage test.

5.3.1 ANOVA-type representation of factors

For a single factor X with I categories, let $y_i \sim \text{bin}(n_i, \pi_i)$ in row i of an $I \times 2$ table. The model

$$\log \frac{\pi_i}{1 - \pi_i} = \alpha + \beta_i, \qquad i = 1, \ldots, I, \tag{5.2}$$

resembles one-way ANOVA and has one redundant parameter, so a constraint is imposed, e.g. $\beta_I = 0$.

Proposition 5.10 (Coding invariance and saturation)

(a) Model (5.2) is saturated: for any $\{\pi_i\} \subset (0, 1)$ there are parameters satisfying it, and its fitted values are the sample proportions.
(b) Under any constraint, $\alpha + \beta_i = \text{logit}\,\pi_i$ and $\beta_a - \beta_b$ is the log odds ratio for rows a, b; these and the fit are the same for every coding. With $\beta_I = 0$, $\alpha = \text{logit}\,\pi_I$ and β_i is the log odds ratio of row i versus row I.
(c) No effect, $\beta_1 = \cdots = \beta_I$, is statistical independence of X and Y.

Proof.

(a) Take $\alpha = \text{logit}\,\pi_I$ and $\beta_i = \text{logit}\,\pi_i - \text{logit}\,\pi_I$. The model has I free parameters for I binomials, so it is saturated and $\hat{\pi}_i = y_i/n_i$.

(b) If $(\alpha, \boldsymbol{\beta})$ satisfies the model, so does $(\alpha - c, \beta_i + c)$ for any c, and every constraint selects one such c; $\alpha + \beta_i$ and $\beta_a - \beta_b$ do not depend on c. In particular $\beta_a - \beta_b = \text{logit}\,\pi_a - \text{logit}\,\pi_b$, a log odds ratio.

(c) Equal β_i means equal π_i across rows. ■

5.3.2 Dummy variables in logit models

Equivalently, with dummy variables $x_i = 1$ for row i and 0 otherwise ($i = 1, \ldots, I-1$),

$$\operatorname{logit} \pi = \alpha + \beta_1 x_1 + \cdots + \beta_{I-1} x_{I-1};$$

no dummy for category I means $\beta_I = 0$ (some software sets $\beta_1 = 0$). *Effect coding* imposes $\sum_i \beta_i = 0$; for $I = 2$, $x = \pm 1$ and the log odds ratio is $2\beta_1$ (versus β_1 for a 0/1 dummy). A single $\hat{\beta}_i$ is meaningful only relative to another category.

5.3.3 Alcohol and infant malformation example revisited

Example 5.11 (Maternal alcohol consumption and malformation)

Alcohol (drinks/day)	Present	Absent	Logit	Observed prop.	Fitted (linear logit)
0	48	17066	−5.87	0.0028	0.0026
< 1	38	14464	−5.94	0.0026	0.0030
1–2	5	788	−5.06	0.0063	0.0041
3–5	1	126	−4.84	0.0079	0.0091
≥ 6	1	37	−3.61	0.0263	0.0231

For (5.2), $\hat{\alpha} + \hat{\beta}_i$ are the sample logits, e.g. $\log(48/17066) = -5.87$. With $\beta_5 = 0$: $\hat{\alpha} = -3.61$, $\hat{\beta}_1 = -2.26$; with $\beta_1 = 0$: $\hat{\alpha} = -5.87$. Apart from the first two rows, logits increase with alcohol. Independence ($\hat{\alpha} = \log(93/32481) = -5.86$, df = 4): $X^2 = 12.1$ ($P = 0.02$), $G^2 = 6.2$ ($P = 0.19$), mixed because the counts are very unbalanced (exact P: 0.03, 0.13). Both ignore the ordering, and one case more or less in the last rows changes those proportions a lot.

5.3.4 Linear logit model for $I \times 2$ tables

For ordered categories with scores $x_1, \ldots, x_I$ the *linear logit model*

$$\operatorname{logit} \pi_i = \alpha + \beta x_i \tag{5.3}$$

lies between independence ($\beta = 0$) and the saturated model (5.2).

Example 5.12 (Alcohol: linear logit fit)

Scores (0, 0.5, 1.5, 4.0, 7.0): $\hat{\alpha} = -5.9605$ (0.1154), $\hat{\beta} = 0.3166$ (0.1254), profile interval (0.0187, 0.5236); odds multiply by $e^{0.317} = 1.37$ per drink per day. $G^2 = 1.95$, $X^2 = 2.05$, df = 3; log likelihood −635.60. Fitted proportions: Example 5.11.

5.3.5 Cochran–Armitage trend test

For an $I \times 2$ table with ordered rows and independent $y_i \sim \text{bin}(n_i, \pi_i)$, Cochran (1954) and Armitage (1955) partitioned X^2 via $\pi_i = \alpha + \beta x_i$ fitted by weighted least squares. With $n = \sum n_i$, $p_i = y_i/n_i$, $p = \sum y_i/n$, $\bar{x} = \sum n_i x_i/n$, the fit is $\hat{\pi}_i = p + b(x_i - \bar{x})$ with

$$b = \frac{\sum_i n_i (p_i - p)(x_i - \bar{x})}{\sum_i n_i (x_i - \bar{x})^2}.$$

Theorem 5.13 (Partition of the Pearson statistic)

Let $X^2(I) = \sum_i n_i(p_i - p)^2/[p(1-p)]$ be the Pearson statistic for independence in the $I \times 2$ table. Then

$$X^2(I) = z^2 + X^2(L),$$

where

$$X^2(L) = \frac{\sum_i n_i(p_i - \hat{\pi}_i)^2}{p(1-p)}, \qquad z^2 = \frac{b^2 \sum_i n_i(x_i - \bar{x})^2}{p(1-p)} = \frac{\left[\sum_i (x_i - \bar{x})y_i\right]^2}{p(1-p)\sum_i n_i(x_i - \bar{x})^2}.$$

Under independence z^2 is asymptotically χ^2_1 (the *Cochran–Armitage trend test* of $\beta = 0$); when the linear probability model holds, $X^2(L)$ is asymptotically χ^2_{I-2} (a test of linearity).

Proof. [Supplement]

Step 1. (The Pearson form.) The cells of row i have observed y_i, $n_i - y_i$ and fitted $n_i p$, $n_i(1-p)$, with deviations $\pm(y_i - n_i p)$. As in Proposition 5.9 the two terms add to $(y_i - n_i p)^2/[n_i p(1-p)] = n_i(p_i - p)^2/[p(1-p)]$.

Step 2. (Decomposition.) Write $p_i - p = (p_i - \hat{\pi}_i) + (\hat{\pi}_i - p)$. Then

$$\sum n_i(p_i - p)^2 = \sum n_i(p_i - \hat{\pi}_i)^2 + \sum n_i(\hat{\pi}_i - p)^2 + 2\sum n_i(p_i - \hat{\pi}_i)(\hat{\pi}_i - p).$$

Step 3. (Cross term vanishes.) Since $\hat{\pi}_i - p = b(x_i - \bar{x})$,

$$\sum n_i(p_i - \hat{\pi}_i)(x_i - \bar{x}) = \sum n_i(p_i - p)(x_i - \bar{x}) - b\sum n_i(x_i - \bar{x})^2 = 0$$

by the definition of b.

Step 4. (Middle term.) $\sum n_i(\hat{\pi}_i - p)^2 = b^2 \sum n_i(x_i - \bar{x})^2$. Since $\sum n_i(x_i - \bar{x}) = 0$, $\sum n_i(p_i - p)(x_i - \bar{x}) = \sum (y_i - n_i p)(x_i - \bar{x}) = \sum (x_i - \bar{x})y_i$, which gives the second form of z^2. Dividing Step 2 by $p(1-p)$ yields the partition.

Step 5. (Null distribution of z^2.) Under independence $E\sum(x_i - \bar{x})Y_i = \pi \sum n_i(x_i - \bar{x}) = 0$ and $\operatorname{Var}\sum(x_i - \bar{x})Y_i = \pi(1-\pi)\sum n_i(x_i - \bar{x})^2$. By the CLT and $p \xrightarrow{p} \pi$ (Slutsky), $z \xrightarrow{d} N(0,1)$. The limit of $X^2(L)$ follows from the chi-squared theory for the fit of a 2-parameter model to I binomials. ■

Proposition 5.14 (Cochran–Armitage = score test)

The score statistic for $H_0 : \beta = 0$ in the linear logit model (5.3) equals z^2 of Theorem 5.13.

Proof. [Supplement]

Step 1. Under H_0 the MLE is $\tilde{\alpha} = \operatorname{logit} p$, so $\tilde{\pi}_i = p$. By (4.16) in count form the score components at $(\tilde{\alpha}, 0)$ are

$$u_\alpha = \sum_i (y_i - n_i p) = 0, \qquad u_\beta = \sum_i x_i(y_i - n_i p) = \sum_i (x_i - \bar{x})y_i.$$

Step 2. The information at $(\tilde{\alpha}, 0)$ is $X^\top \operatorname{diag}\{n_i p(1-p)\} X$:

$$\mathcal{I} = p(1-p)\begin{pmatrix} n & \sum n_i x_i \\ \sum n_i x_i & \sum n_i x_i^2 \end{pmatrix},$$

and by the formula for the inverse of a 2×2 matrix,

$$[\mathcal{I}^{-1}]_{\beta\beta} = \frac{n}{p(1-p)\left[n\sum n_i x_i^2 - (\sum n_i x_i)^2\right]} = \frac{1}{p(1-p)\sum n_i(x_i - \bar{x})^2}.$$

Step 3. The score statistic $\boldsymbol{u}^\top \mathcal{I}^{-1}\boldsymbol{u}$ with $u_\alpha = 0$ is $u_\beta^2[\mathcal{I}^{-1}]_{\beta\beta} = z^2$. ■

The statistic also equals $M^2 = (n-1)r^2$ of Section 3.4 for $J = 2$, with $n-1$ replaced by n; for $I = 2$, $X^2(L) = 0$ and $z^2 = X^2(I)$.

Example 5.15 (Alcohol: trend test)

Here $z^2 = 6.6$ ($P = 0.010$) and $X^2(I) = 12.1 = 6.6 + 5.5$, $X^2(L) = 5.5$ (df $= 3$): slight nonlinearity. $r = 0.014$ ($n = 32{,}574$) is a poor descriptive measure for such unbalanced tables. Wald $(0.317/0.125)^2 = 6.4$ ($P = 0.012$), LR 4.25 ($P = 0.039$); LR is safest here, and the profile interval (0.02, 0.52) is preferable to Wald (0.07, 0.56). Exact methods (Section 6.7.4) matter even for huge n.

5.4 Multiple logistic regression

Motivation. Each coefficient is a conditional effect; deviance differences of nested models decide which terms are needed.

For predictors $\mathbf{x} = (x_1, \ldots, x_p)$,

$$\operatorname{logit}\pi(\mathbf{x}) = \alpha + \beta_1 x_1 + \cdots + \beta_p x_p, \qquad \pi(\mathbf{x}) = \frac{\exp(\alpha + \beta_1 x_1 + \cdots + \beta_p x_p)}{1 + \exp(\alpha + \beta_1 x_1 + \cdots + \beta_p x_p)}. \tag{5.4}$$

e^{β_j} is the multiplicative effect on the odds of a unit increase in x_j, at fixed levels of the other predictors.

5.4.1 Logit models for multiway contingency tables

With binary X and Z coded by dummies ($x_1 = z_1 = 1$, $x_2 = z_2 = 0$) and independent binomials at each (i, k), consider

$$\operatorname{logit} P(Y = 1) = \alpha + \beta_1 x_i + \beta_2 z_k. \tag{5.5}$$

Proposition 5.16 (Main-effects model and homogeneous association)

In (5.5), at each level z_k of Z,

$$\operatorname{logit} P(Y = 1 \mid x = 1, z_k) - \operatorname{logit} P(Y = 1 \mid x = 0, z_k) = \beta_1,$$

so e^{β_1} is the conditional XY odds ratio, the same at each level of Z (homogeneous association). $\beta_1 = 0$ is conditional independence of X and Y given Z.

Proof. $(\alpha + \beta_1 + \beta_2 z_k) - (\alpha + \beta_2 z_k) = \beta_1$, free of z_k; a difference of logits is a log odds ratio. If $\beta_1 = 0$ the conditional odds ratios all equal 1. ■

No interaction means additivity on the logit scale; on another scale interaction may appear or disappear. For factors with any numbers of categories:

$$\operatorname{logit} P(Y = 1) = \alpha + \beta_i^X + \beta_k^Z, \tag{5.6}$$

with one redundant parameter per factor (e.g. $\beta_I^X = \beta_K^Z = 0$; superscripts are labels). Conditional independence of X and Y is $\beta_1^X = \cdots = \beta_I^X$. For two categories each, (5.6) with $\beta_2^X = \beta_2^Z = 0$ is (5.5) with $\beta_1^X = \beta_1$, $\beta_1^Z = \beta_2$.

5.4.2 AIDS and AZT example

Example 5.17 (AIDS symptoms, AZT use and race)

338 veterans with faltering immune systems, randomized to immediate or deferred AZT; response: AIDS symptoms within 3 years.

		Symptoms	
Race	AZT use	Yes	No
White	Yes	14	93
	No	32	81
Black	Yes	11	52
	No	12	43

In (5.5) let $x = 1$ for immediate AZT and $z = 1$ for whites. Fit:

Parameter	Estimate	SE	Wald χ^2
Intercept	-1.0736	0.2629	16.67
AZT	-0.7195	0.2790	6.65
race	0.0555	0.2886	0.04

(a) Conditional odds ratio for AZT $e^{-0.7195} = 0.487$: odds of symptoms about halved, for each race. Wald $\exp[-0.720 \pm 1.96(0.279)] = (0.28, 0.84)$; profile (0.279, 0.835). Race: 1.057, (0.60, 1.86).
(b) $H_0 : \beta_1 = 0$ (conditional independence of AZT and symptoms given race): LR statistic 6.9 (df $= 1$, $P = 0.01$); Wald $(-0.720/0.279)^2 = 6.65$.
(c) Codings (last $= 0$, first $= 0$, sum $= 0$): intercept $(-1.074, -1.738, -1.406)$, AZT-yes $(-0.720, 0, -0.360)$, white $(0.055, 0, 0.028)$. Always intercept + AZT-yes + white $= -1.738$ ($\hat{\pi} = 0.15$, whites with immediate AZT) and $\hat{\beta}_1^X - \hat{\beta}_2^X = -0.720$.
(d) Fitted probabilities (95% limits): white/AZT 0.150 (0.099, 0.220); white/no 0.265 (0.197, 0.348); black/AZT 0.143 (0.087, 0.225); black/no 0.255 (0.170, 0.364).

5.4.3 Goodness of fit as a likelihood-ratio test

Let $G^2(M_0 \mid M_1) = -2(L_0 - L_1)$. The fit statistic $G^2(M)$ is the case M_1 = saturated, df = (binomials) − (parameters). By Proposition 4.22, with L_S the saturated log likelihood,

$$G^2(M_0 \mid M_1) = -2(L_0 - L_S) - [-2(L_1 - L_S)] = G^2(M_0) - G^2(M_1).$$

Example 5.18 (AIDS: goodness of fit)

Whites with immediate AZT: $\hat{\pi} = 0.150$, fitted counts $107(0.150) = 16.0$ and 91.0; likewise for all 8 cells. $G^2 = 1.38$, $X^2 = 1.39$, df $= 4 - 3 = 1$ ($P > 0.2$), also a test of homogeneous odds ratios. No race effect: $G^2(M_0) - G^2(M_1) = 1.42 - 1.38 = 0.04$.

The difference is often nearly chi-squared even when the separate $G^2(M_i)$ are not, if the models differ by few parameters.

5.4.4 Horseshoe crab example revisited

Example 5.19 (Horseshoe crabs: width and color)

Color (medium light, medium, medium dark, dark; a surrogate for age), dummies c_1, c_2, c_3 with dark as reference:

$$\operatorname{logit}\pi = \alpha + \beta_1 c_1 + \beta_2 c_2 + \beta_3 c_3 + \beta_4 x. \tag{5.7}$$

Parameter	Estimate	SE	LR 95% limits	χ^2
intercept	−12.7151	2.7618	(−18.4564, −7.5788)	21.20
c_1	1.3299	0.8525	(−0.2738, 3.1354)	2.43
c_2	1.4023	0.5484	(0.3527, 2.5260)	6.54
c_3	1.1061	0.5921	(−0.0279, 2.3138)	3.49
width	0.4680	0.1055	(0.2713, 0.6870)	19.66

Deviance 187.46, Pearson 168.66 on 168 df; log likelihood −93.73.

Dark: $\operatorname{logit}\hat{\pi} = -12.715 + 0.468x$; medium light: $-11.385 + 0.468x$. At $x = 26.3$, $\hat{\pi} = 0.399$ and 0.715. Without interaction the curves are horizontal shifts and never cross; 1 cm multiplies the odds by $e^{0.468} = 1.60$ for every color, and medium light vs. dark by $e^{1.330} = 3.8$ at any width.

5.4.5 Model comparison

Example 5.20 (Horseshoe crabs: testing color and interaction)

No color effect given width: LR = 7.0, df = 3, $P = 0.07$. Adding color×width (separate fits per color): LR = 4.4, df = 3, $P = 0.22$, weak evidence of interaction.

5.4.6 Quantitative treatment of ordinal predictor

Example 5.21 (Horseshoe crabs: color scores)

With scores $c = 1, 2, 3, 4$,

$$\operatorname{logit}\pi = \alpha + \beta_1 c + \beta_2 x, \qquad \hat{\beta}_1 = -0.509\ (0.224), \quad \hat{\beta}_2 = 0.458\ (0.104).$$

Each step darker multiplies the odds by $e^{-0.509} = 0.60$ at fixed width; versus (5.7), LR = 1.7 (df = 2, $P = 0.44$). The estimates (1.33, 1.40, 1.11, 0) suggest dark (0) versus other (1); versus (5.7), LR = 0.5 (df = 2), with fit

$$\operatorname{logit}\hat{\pi} = -12.980 + 1.300c + 0.478x \qquad (\widehat{\text{se}}:\ 0.526,\ 0.104). \tag{5.8}$$

Non-dark crabs have $e^{1.300} = 3.7$ times the odds of dark crabs at a given width. All three models make dark crabs least likely to have satellites. A quantitative treatment of an ordinal predictor, when it fits, is simpler and gives more powerful one-df tests (Section 6.4).

5.4.7 Standardized and probability-based interpretations

To compare predictors with different units, multiply $\hat{\beta}_j$ by s_{x_j} (a one-SD effect). On the probability scale a unit increase in x_j changes π by about $\hat{\beta}_j\hat{\pi}(1-\hat{\pi})$ (Proposition 5.2); or report $\hat{\pi}$ at the quartiles of x_j with others at their means (at both values for a dummy).

Example 5.22 (Horseshoe crabs: effects on the probability scale)

For (5.8): at $\hat{\pi} = 0.5$, 1 cm raises π by about $0.478(0.5)(0.5) = 0.12$ (1 cm $< \frac{1}{2}$ SD). With $\bar{c} = 0.873$, $\hat{\pi} = 0.51$ and 0.80 at the width quartiles 24.9, 27.7. At $x = 26.3$: 0.40 (dark), 0.71 (other).

Variable	Estimate	SE	Comparison	Change in probability
No interaction model				
Intercept	−12.980	2.727		
Color (0 dark, 1 other)	1.300	0.526	$(1,0)$ at $\bar{x}$	$0.31 = 0.71 - 0.40$
Width x (cm)	0.478	0.104	(UQ, LQ) at $\bar{c}$	$0.29 = 0.80 - 0.51$
Interaction model				
Intercept	−5.854	6.694		
Color	−6.958	7.318		
Width x	0.200	0.262	(UQ, LQ) at $c = 0$	$0.13 = 0.43 - 0.30$
Width × color	0.322	0.286	(UQ, LQ) at $c = 1$	$0.29 = 0.84 - 0.55$

The width effect appears larger for lighter crabs, but the interaction is not significant.

5.5 Fitting logistic regression models

Motivation. As a canonical-link GLM, logistic regression has a concave likelihood, Newton–Raphson = Fisher scoring = IRLS, and finite estimates exactly under overlap.

Let $\mathbf{x}_i = (x_{i1}, \ldots, x_{ip})$ be setting i ($i = 1, \ldots, N$), $x_{i1} = 1$, with y_i successes in n_i trials (continuous predictors: typically $N = n$, $n_i = 1$). The model is

$$\pi(\mathbf{x}_i) = \frac{\exp\left(\sum_{j=1}^{p} \beta_j x_{ij}\right)}{1 + \exp\left(\sum_{j=1}^{p} \beta_j x_{ij}\right)}. \tag{5.9}$$

5.5.1 Likelihood equations

Theorem 5.23 (Logistic log likelihood and likelihood equations)

For independent $y_i \sim \text{bin}(n_i, \pi(\mathbf{x}_i))$ under (5.9),

$$L(\boldsymbol{\beta}) = \sum_j \Big(\sum_i y_i x_{ij}\Big)\beta_j - \sum_i n_i \log\Big[1 + \exp\Big(\sum_j \beta_j x_{ij}\Big)\Big], \tag{5.10}$$

so $\{\sum_i y_i x_{ij}\}$ are sufficient, and the likelihood equations are

$$\sum_i y_i x_{ij} - \sum_i n_i \hat{\pi}_i x_{ij} = 0, \quad j = 1, \ldots, p, \qquad \text{i.e.} \qquad X^\top \mathbf{y} = X^\top \hat{\boldsymbol{\mu}}, \tag{5.11}$$

where $\hat{\mu}_i = n_i \hat{\pi}_i$.

Proof.

Step 1. The joint mass function is proportional to

$$\prod_i \pi_i^{y_i}(1-\pi_i)^{n_i-y_i} = \Big\{\prod_i (1-\pi_i)^{n_i}\Big\} \exp\Big[\sum_i y_i \log\frac{\pi_i}{1-\pi_i}\Big].$$

Step 2. With $\operatorname{logit}\pi_i = \sum_j \beta_j x_{ij}$, the exponent is $\sum_i y_i \sum_j \beta_j x_{ij} = \sum_j \beta_j \sum_i y_i x_{ij}$, and $1-\pi_i = [1+\exp(\sum_j \beta_j x_{ij})]^{-1}$. Taking logs gives (5.10); the factorization theorem gives sufficiency.

Step 3. Differentiating,

$$\frac{\partial L}{\partial \beta_j} = \sum_i y_i x_{ij} - \sum_i n_i x_{ij} \frac{\exp(\sum_k \beta_k x_{ik})}{1+\exp(\sum_k \beta_k x_{ik})} = \sum_i (y_i - n_i\pi_i) x_{ij}.$$

■

These are (4.16) with counts: by Theorem 4.27 they equate sufficient statistics to their expectations. They are nonlinear and need iteration.

5.5.2 Asymptotic covariance matrix of parameter estimators

Theorem 5.24 (Information matrix for logistic regression)

$$-\frac{\partial^2 L}{\partial\beta_a \partial\beta_b} = \sum_i x_{ia}x_{ib}\, n_i\pi_i(1-\pi_i),$$

which does not involve $\mathbf{y}$; so observed and expected information coincide, and

$$\widehat{\operatorname{Cov}}(\hat{\boldsymbol{\beta}}) = \{X^\top \operatorname{diag}[n_i\hat\pi_i(1-\hat\pi_i)]X\}^{-1}. \tag{5.12}$$

Proof. From Step 3 above, $\partial L/\partial\beta_a = \sum_i (y_i - n_i\pi_i)x_{ia}$, and by Proposition 5.2, Step 1, $\partial\pi_i/\partial\beta_b = \pi_i(1-\pi_i)x_{ib}$. Hence $\partial^2 L/\partial\beta_a\partial\beta_b = -\sum_i n_i\pi_i(1-\pi_i)x_{ia}x_{ib}$. This is free of y_i, so it equals its expectation, and Theorem 4.15 gives (5.12): the GLM form $(X^\top \hat W X)^{-1}$ with $w_i = n_i\pi_i(1-\pi_i)$. ■

The square roots of the diagonal of (5.12) are the standard errors of $\hat{\boldsymbol{\beta}}$.

5.5.3 Distribution of probability estimators

For a setting $\mathbf{x}$, the fitted logit $\mathbf{x}^\top\hat{\boldsymbol{\beta}}$ has estimated variance $\mathbf{x}^\top\widehat{\operatorname{Cov}}(\hat{\boldsymbol{\beta}})\mathbf{x}$, so

$$\mathbf{x}^\top\hat{\boldsymbol{\beta}} \pm z_{\alpha/2}\sqrt{\mathbf{x}^\top\widehat{\operatorname{Cov}}(\hat{\boldsymbol{\beta}})\mathbf{x}}$$

is a large-sample interval for $\operatorname{logit}\pi(\mathbf{x})$; applying $t\mapsto e^t/(1+e^t)$ to the endpoints gives one for $\pi(\mathbf{x})$ (Proposition 5.6).

5.5.4 Newton–Raphson method applied to logistic regression

Proposition 5.25 (Newton–Raphson / IRLS for logistic regression)

With $\pi_i^{(t)} = \exp(\mathbf{x}_i^\top\boldsymbol{\beta}^{(t)})/[1+\exp(\mathbf{x}_i^\top\boldsymbol{\beta}^{(t)})]$ and $\mu_i^{(t)} = n_i\pi_i^{(t)}$, the Newton–Raphson step is

$$\boldsymbol{\beta}^{(t+1)} = \boldsymbol{\beta}^{(t)} + \{X^\top\operatorname{diag}[n_i\pi_i^{(t)}(1-\pi_i^{(t)})]X\}^{-1}X^\top(\mathbf{y}-\boldsymbol{\mu}^{(t)}). \tag{5.13}$$

Equivalently $\boldsymbol{\beta}^{(t+1)} = (X^\top V_t^{-1} X)^{-1} X^\top V_t^{-1} \boldsymbol{z}^{(t)}$, with $V_t = \text{diag}\{1/[n_i \pi_i^{(t)}(1-\pi_i^{(t)})]\}$ and

$$z_i^{(t)} = \log \frac{\pi_i^{(t)}}{1-\pi_i^{(t)}} + \frac{y_i - n_i \pi_i^{(t)}}{n_i \pi_i^{(t)}(1-\pi_i^{(t)})}. \tag{5.14}$$

At convergence, $-H^{-1}$ is the estimated covariance (5.12).

Proof. By Theorems 5.23 and 5.24, $\boldsymbol{u}^{(t)} = X^\top(\mathbf{y} - \boldsymbol{\mu}^{(t)})$ and $H^{(t)} = -X^\top \text{diag}[n_i \pi_i^{(t)}(1-\pi_i^{(t)})] X$; substitute in (4.25). Since $H = -\mathcal{I}$ (canonical link), this is also Fisher scoring, hence IRLS by Theorem 4.26. For the logit, $\partial \eta_i / \partial \mu_i = 1/[n_i \pi_i (1-\pi_i)]$ when $\mu_i = n_i \pi_i$, so the working response (4.28) is (5.14), and $W = V_t^{-1}$. ■

V_t holds the estimated variances of sample logits (Section 3.1.6) and $\boldsymbol{z}^{(t)}$ is the linearized sample logit. From $\boldsymbol{\beta}^{(0)}$ (e.g. $\mathbf{0}$) the iterates converge to $\hat{\boldsymbol{\beta}}$ (Walker and Duncan 1967).

5.5.5 Convergence and existence of finite estimates

Theorem 5.26 (Concavity and uniqueness)

If X has full column rank, $L(\boldsymbol{\beta})$ in (5.10) is strictly concave; hence the ML estimate, when it exists, is unique, and any root of (5.11) is it.

Proof. By Theorem 5.24, $\boldsymbol{c}^\top H \boldsymbol{c} = -\sum_i n_i \pi_i (1-\pi_i)(\mathbf{x}_i^\top \boldsymbol{c})^2$ with every $n_i \pi_i (1-\pi_i) > 0$. It is < 0 unless $X\boldsymbol{c} = \mathbf{0}$, i.e. $\boldsymbol{c} = \mathbf{0}$. A strictly concave function has at most one maximizer, and a stationary point of a concave function is a global maximum. ■

Theorem 5.27 (Existence of the ML estimate: overlap versus separation)

Write the data as n individual binary observations $(\mathbf{x}_i, y_i)$ and let $s_i = 2y_i - 1 \in \{-1, 1\}$. Assume X has full column rank. Call the data *separated* if some $\boldsymbol{d} \neq \mathbf{0}$ satisfies $s_i \mathbf{x}_i^\top \boldsymbol{d} \geq 0$ for all i, with strict inequality for at least one i (a hyperplane with all $y = 1$ on one side and all $y = 0$ on the other, ties on it allowed). Otherwise the data *overlap*.
(a) If the data are separated, no ML estimate exists: L increases indefinitely along $\boldsymbol{d}$ and some $|\hat{\beta}_j|$ is "infinite".
(b) If the data overlap, a finite ML estimate exists, and it is unique.

Proof. [Supplement]

Step 1. (A convenient form.) For one binary observation, $y\eta - \log(1+e^\eta) = -\log(1+e^{-s\eta})$ (check $y = 1$ and $y = 0$). Hence

$$L(\boldsymbol{\beta}) = -\sum_i \log[1 + \exp(-s_i \mathbf{x}_i^\top \boldsymbol{\beta})] < 0.$$

Each term is a strictly increasing function of $s_i \mathbf{x}_i^\top \boldsymbol{\beta}$.

Step 2. (a) Suppose $\boldsymbol{d}$ separates and $\hat{\boldsymbol{\beta}}$ were a maximizer. For $t > 0$, $s_i \mathbf{x}_i^\top (\hat{\boldsymbol{\beta}} + t\boldsymbol{d}) \geq s_i \mathbf{x}_i^\top \hat{\boldsymbol{\beta}}$ for all i, with strict inequality for some i. By Step 1, $L(\hat{\boldsymbol{\beta}} + t\boldsymbol{d}) > L(\hat{\boldsymbol{\beta}})$, a contradiction. So the supremum is not attained; along $\boldsymbol{\beta} + t\boldsymbol{d}$ the likelihood keeps increasing as $t \to \infty$ (under complete separation, $s_i \mathbf{x}_i^\top \boldsymbol{d} > 0$ for all i, it tends to the perfect-fit value $L = 0$).

Step 3. (b) Overlap and full rank imply: for every unit vector $\boldsymbol{d}$, $m(\boldsymbol{d}) = \min_i s_i \mathbf{x}_i^\top \boldsymbol{d} < 0$. Indeed, otherwise $s_i \mathbf{x}_i^\top \boldsymbol{d} \geq 0$ for all i; not separated forces all equalities, so $X\boldsymbol{d} = \mathbf{0}$, contradicting full rank.

Step 4. m is continuous on the compact unit sphere, so $\max_{\|\boldsymbol{d}\|=1} m(\boldsymbol{d}) = -c < 0$. For $\boldsymbol{\beta} = r\boldsymbol{d}$

with $r > 0$, $\|\boldsymbol{d}\| = 1$, choose i with $s_i \mathbf{x}_i^\top \boldsymbol{d} \le -c$; then by Step 1

$$L(r\boldsymbol{d}) \le -\log[1 + \exp(rc)] \le -rc.$$

Step 5. Hence $\{\boldsymbol{\beta} : L(\boldsymbol{\beta}) \ge L(\mathbf{0})\} \subset \{\|\boldsymbol{\beta}\| \le -L(\mathbf{0})/c\}$ is bounded; it is closed by continuity, hence compact and nonempty. The continuous L attains its maximum on it, which is a global maximum. Uniqueness is Theorem 5.26. ■

Similar results hold for the probit and some other links (Silvapulle 1981). Software rarely detects $\hat{\beta} = \pm\infty$: the log likelihood flattens, convergence criteria are met, and enormous standard errors are reported.

Example 5.28 (Perfect discrimination)

$y = 0$ at $x = 10, 20, 30, 40$, $y = 1$ at $x = 60, 70, 80, 90$. With $\alpha = -50\beta$ ($\hat{\pi}(50) = 0.5$) and $\beta \to \infty$ the likelihood increases toward a perfect fit (Figure 8). PROC GENMOD reports $\operatorname{logit} \hat{\pi} = -192.2 + 3.8x$ with standard errors 8.0×10^8 and 1.5×10^7, artifacts of $\hat{\beta} = \infty$.

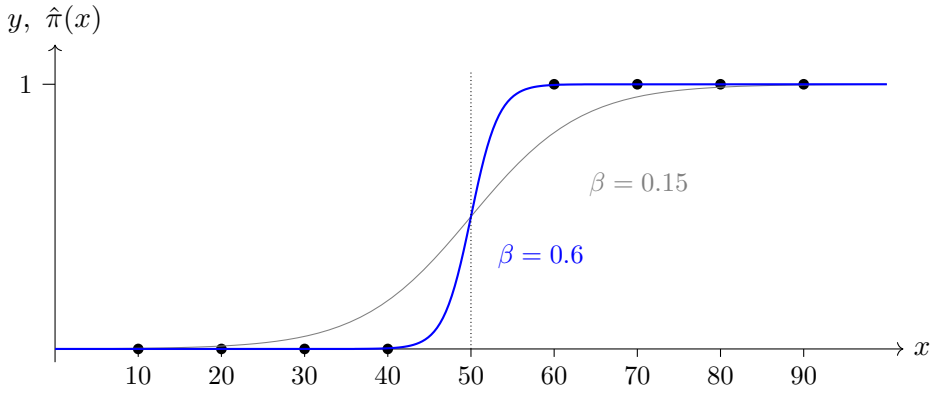


Figure 8: Complete separation: with $\alpha = -50\beta$ the curve sharpens to a step at $x = 50$ as $\beta \to \infty$; the likelihood has no maximum.

Chapter 6. Building and Applying Logistic Regression Models

6.1 Strategies in model selection

Motivation. We want a model complex enough to fit but simple enough to interpret.

Confirmatory studies compare a few prespecified models; exploratory searches only generate hypotheses. Look first at each predictor's marginal effect. Guidelines: at least 10 outcomes of each type per predictor; beware multicollinearity.

6.1.1 Horseshoe crab example revisited

Example 6.1 (Horseshoe crabs: all main effects)

Crabs: $y = 1$ if a satellite; predictors color C (4 categories), spine S (3), weight, width W. The main-effects model (dummies c_1, c_2, c_3, s_1, s_2) has LR 40.6 against the null ($df = 7$, $P < 0.0001$), yet no single effect is clearly significant:

Parameter	Estimate	SE	Parameter	Estimate	SE
Intercept	−9.273	3.838	color 2	1.506	0.567
weight	0.826	0.704	color 3	1.120	0.593
width	0.263	0.195	spine 1	−0.400	0.503
color 1	1.609	0.936	spine 2	−0.496	0.629

Multicollinearity (corr(weight, width) = 0.887): keep W, C, S. Models are named by highest-order terms ($C + S + W$, $C + S{*}W$, ...), main effects included.

6.1.2 Stepwise procedures

Forward selection adds the term with the smallest P-value (*stepwise* also re-tests earlier terms); *backward elimination* removes the term with the largest P-value until every deletion significantly worsens the fit. Add or drop factors as a whole. Backward is preferred (forward may stop early through one low-power test). Selected effects are overestimated; a variable central to the study may be kept anyway.

6.1.3 Backward elimination for the horseshoe crab example

With a continuous predictor G^2 is not χ^2, but deviance differences of nested models are approximately χ^2.

Example 6.2 (Horseshoe crabs: backward elimination)

Selected rows of the book's Table 6.2:

	Model	G^2	df	AIC	Comparison	Difference
1	$C{*}S{*}W$	170.44	152	212.4	–	–
2	$C{*}S + C{*}W + S{*}W$	173.68	155	209.7	(2)–(1)	3.2 ($df = 3$)
3c	$C{*}S + C{*}W$	173.69	157	205.7	(3c)–(2)	0.0 ($df = 2$)
4b	$W + C{*}S$	177.61	160	203.6	(4b)–(3c)	3.9 ($df = 3$)
5	$C + S + W$	186.61	166	200.6	(5)–(4b)	9.0 ($df = 6$)
6c	$C + W$	187.46	168	197.5	(6c)–(5)	0.8 ($df = 2$)
7a	C	212.06	169	220.1	(7a)–(6c)	24.5 ($df = 1$)
7b	W	194.45	171	198.5	(7b)–(6c)	7.0 ($df = 3$)
8	$C = \text{dark} + W$	187.96	170	194.0	(8)–(6c)	0.5 ($df = 2$)
9	None	225.76	172	227.8	(9)–(8)	37.8 ($df = 2$)

All interactions drop out, then S; removing C from $C + W$ costs 7.0 ($df = 3$, $P = 0.07$) and removing W costs 24.5. A single dark/not-dark dummy loses almost nothing: final model width + dark.

6.1.4 AIC, model selection, and the "correct" model

No model is exactly true; a parsimonious one may estimate better. AIC measures the expected closeness of the fit to the truth.

Definition (AIC)

For a model with p parameters and maximized log likelihood L_M, AIC $= -2(L_M - p) = -2L_M + 2p$; smaller is better.

Proposition 6.3 (AIC and the adjusted deviance)

Fit models to the same data with N grouped observations (saturated model N parameters, $df = N - p$). Then

$$\text{AIC} = [G^2 - 2\,df] + \text{const},$$

where $-2L_S + 2N$ is common to all models; so AIC orders models as $G^2 - 2\,df$ does.

Proof. $G^2 = -2(L_M - L_S)$ gives $-2L_M = G^2 - 2L_S$; with $p = N - df$, AIC $= [G^2 - 2\,df] + (2N - 2L_S)$, and the last term depends only on the data. ■

Theorem 6.4 (Akaike's bias correction)

Let $Y_1, \ldots, Y_n$ be i.i.d. with density $f(\cdot;\theta_0)$ belonging to a regular parametric model with p-dimensional parameter, $\hat\theta$ the MLE and $\ell(\theta) = \sum_i \log f(Y_i;\theta)$. Let $Y_1^*, \ldots, Y_n^*$ be an independent copy of the data and $\ell^*(\theta)$ its log likelihood. Then, to first order,

$$E[\ell(\hat\theta)] - E[\ell^*(\hat\theta)] \approx p.$$

So AIC $= -2\ell(\hat\theta) + 2p$ estimates the expected out-of-sample deviance $-2E[\ell^*(\hat\theta)]$ (i.e. $2n$ times the Kullback–Leibler discrepancy, up to a constant).

Proof. [Supplement] Let I be the per-observation Fisher information, so $\sqrt{n}(\hat\theta - \theta_0) \xrightarrow{d} N(0, I^{-1})$.

Step 1. Expanding about $\hat\theta$ ($\ell'(\hat\theta) = 0$), $\ell(\theta_0) \approx \ell(\hat\theta) - \frac{n}{2}(\hat\theta - \theta_0)^\top I(\hat\theta - \theta_0)$; since $n(\hat\theta - \theta_0)^\top I(\hat\theta - \theta_0) \xrightarrow{d} \chi^2_p$, $E[\ell(\hat\theta)] \approx E[\ell(\theta_0)] + p/2$.

Step 2. Expanding ℓ^* about θ_0, and using that $\ell^{*\prime}(\theta_0)$ has mean zero and is independent of $\hat\theta$,

$$E[\ell^*(\hat\theta)] \approx E[\ell^*(\theta_0)] - \tfrac{n}{2} E\big[(\hat\theta-\theta_0)^\top I(\hat\theta-\theta_0)\big] = E[\ell^*(\theta_0)] - \tfrac{p}{2}.$$

Step 3. $\ell(\theta_0)$ and $\ell^*(\theta_0)$ are equal in distribution, so subtracting gives $E[\ell(\hat\theta)] - E[\ell^*(\hat\theta)] \approx p$, i.e. $E[-2\ell(\hat\theta)+2p] \approx -2E[\ell^*(\hat\theta)]$. Finally $-2E[\ell^*(\theta)]/n = 2\,\mathrm{KL}(f_{\theta_0}\|f_\theta) - 2E\log f(Y;\theta_0)$, whose last term is model-free. ■

Example 6.5 (Horseshoe crabs: AIC)

AIC is smallest for $C+W$ (197.5) among C, S, W models, smaller still for the dark-indicator model (194.0), itself suggested by the $C+W$ fit.

6.1.5 Using causal hypotheses to guide model building

A time ordering suggests a sequence of logit models, each response explained by earlier variables.

Example 6.6 (Divorce and pre/extramarital sex)

G = gender, P = premarital sex, E = extramarital sex, M = marital status:

	Women				Men			
PMS	Yes		No		Yes		No	
EMS	Yes	No	Yes	No	Yes	No	Yes	No
Divorced	17	54	36	214	28	60	17	68
Still married	4	25	4	322	11	42	4	130

For $G \to P \to E \to M$: P on none, $G^2 = 75.3$ ($df = 1$; G–P odds ratio 0.27); E on none / P / $G+P$, $G^2 = 48.9/2.9/0.0$ ($df = 3/2/1$), weak evidence of a direct G effect; M on $E+P$ / $E{*}P$ / $E{*}P+G$, $G^2 = 18.2/5.2/0.7$ ($df = 5/4/3$), so $E \times P$ is needed. The causal reasoning led close to a well-fitting model.

6.1.6 New model-building strategies for data mining

In huge data sets nearly every effect is significant; predictive-power summaries (Section 6.2) matter more.

6.2 Logistic regression diagnostics

Motivation. Residuals and influence measures show where a model fails.

6.2.1 Pearson, deviance, and standardized residuals

Let $y_i \sim \mathrm{bin}(n_i, \pi_i)$ at setting $i = 1, \ldots, N$, with fitted $\hat\pi_i$ and leverage $\hat h_i$, the ith diagonal element of $\hat W^{1/2} X (X^\top \hat W X)^{-1} X^\top \hat W^{1/2}$, $\hat W = \mathrm{diag}(n_i \hat\pi_i(1-\hat\pi_i))$.

Definition (Residuals for binomial GLMs)

(a) Pearson residual:

$$e_i = \frac{y_i - n_i\hat{\pi}_i}{\sqrt{n_i\hat{\pi}_i(1-\hat{\pi}_i)}}. \tag{6.1}$$

(b) Standardized (adjusted) Pearson residual: $r_i = e_i/\sqrt{1-\hat{h}_i}$.
(c) Deviance residual: $\sqrt{d_i}\ \mathrm{sgn}(y_i - n_i\hat{\pi}_i)$, with $d_i = 2[y_i \log \frac{y_i}{n_i\hat{\pi}_i} + (n_i - y_i)\log \frac{n_i - y_i}{n_i - n_i\hat{\pi}_i}]$.

Thus $X^2 = \sum_i e_i^2$ and $G^2 = \sum_i d_i$.

Theorem 6.7 (Approximate distribution of residuals)

Suppose the logistic model holds with p parameters and the n_i are large. Then, approximately,
(a) $\mathrm{Var}(y_i - n_i\hat{\pi}_i) \approx n_i\pi_i(1-\pi_i)(1-h_i)$, so $\mathrm{Var}(e_i) \approx 1 - h_i < 1$ while r_i is approximately $N(0,1)$;
(b) $E(\sum_i e_i^2) \approx N - p = df$, so $E(\sum_i e_i^2)/N \approx df/N < 1$.

Proof. [Supplement] Write $\mu = (n_i\pi_i)$, $W = \mathrm{diag}(n_i\pi_i(1-\pi_i)) = \mathrm{Cov}(y)$, $A = X^\top W X$.

Step 1. The logit likelihood equations are $X^\top(y - \hat{\mu}) = 0$, and $\partial\mu/\partial\beta^\top = WX$, so $\hat{\mu} - \mu \approx WX(\hat{\beta} - \beta)$ and $0 \approx X^\top(y-\mu) - A(\hat{\beta} - \beta)$, i.e. $\hat{\beta} - \beta \approx A^{-1}X^\top(y - \mu)$.

Step 2. Hence $y - \hat{\mu} \approx (I - WXA^{-1}X^\top)(y - \mu)$ and

$$\mathrm{Cov}(y - \hat{\mu}) \approx (I - WXA^{-1}X^\top)\, W\, (I - XA^{-1}X^\top W) = W - WXA^{-1}X^\top W.$$

Step 3. So $\mathrm{Cov}(W^{-1/2}(y - \hat{\mu})) \approx I - H$, $H = W^{1/2}XA^{-1}X^\top W^{1/2}$ with diagonal (h_i). The components of $W^{-1/2}(y - \hat{\mu})$ are the e_i (with π_i for $\hat{\pi}_i$, a negligible change), proving (a); normality follows from that of y, and $r_i = e_i/\sqrt{1-h_i}$ has unit variance.

Step 4. H is a projection of rank p, so $E\sum e_i^2 \approx \mathrm{tr}(I - H) = N - p$. ■

With $n_i = 1$ (continuous predictors) residuals take two values and are uninformative; group observations where possible.

6.2.2 Heart disease example

Example 6.8 (Blood pressure and heart disease)

Framingham men, eight blood-pressure categories (scores $111.5, \ldots, 191.5$), response: heart disease within six years.

Blood pressure	Sample size	Heart disease	Fitted		Std. residual	
			Indep.	Linear	Indep.	Linear
< 117	156	3	10.8	5.2	−2.62	−1.11
117–126	252	17	17.4	10.6	−0.12	2.37
127–136	284	12	19.7	15.1	−2.02	−0.95
137–146	271	16	18.8	18.1	−0.74	−0.57
147–156	139	12	9.6	11.6	0.84	0.13
157–166	85	8	5.9	8.9	0.93	−0.33
167–186	99	16	6.9	14.2	3.76	0.65
> 186	43	8	3.0	8.4	3.07	−0.18

Independence fits poorly ($G^2 = 30.0$, $df = 7$), residuals increasing in x; the linear logit model removes the trend ($G^2 = 5.9$, $df = 6$), leaving only $r_2 = 2.37$.

6.2.3 Graduate admissions example

Example 6.9 (Admissions at the University of Florida)

23 departments: no gender effect, $\operatorname{logit}\pi_{ik} = \alpha + \beta_k^D$, fits rather poorly ($G^2 = 44.7$, $df = 23$). Large standardized residuals for (female, admitted): astronomy 2.87 (6 of 6 females, 3 of 11 males), geography 2.17, psychology -2.27; without these three, $G^2 = 24.4$, $df = 20$. A gender effect barely helps ($G^2 = 42.4$, $df = 22$; deviations go both ways). Its conditional odds ratio 1.19 (favouring females) versus marginal 0.94 is Simpson's paradox.

Proposition 6.10 (Residuals under conditional independence)

In a $2 \times 2 \times K$ table with $y_{ik} \sim \text{bin}(n_{ik}, \pi_{ik})$, $\operatorname{logit}\pi_{ik} = \alpha + \beta_k$ has $\hat\pi_{1k} = \hat\pi_{2k} = y_{+k}/n_{+k}$ and $y_{1k} - n_{1k}\hat\pi_{1k} = -(y_{2k} - n_{2k}\hat\pi_{2k})$. So the two standardized residuals in stratum k are equal and opposite; the Pearson residuals need not be.

Proof.

Step 1. The likelihood equation for β_k is $\sum_i y_{ik} = \sum_i n_{ik}\hat\pi_{ik}$, and $\hat\pi_{1k} = \hat\pi_{2k}$, so $\hat\pi_{ik} = y_{+k}/n_{+k}$.

Step 2. Then $y_{1k} - n_{1k}\hat\pi_{1k} = (n_{2k}y_{1k} - n_{1k}y_{2k})/n_{+k} = -(y_{2k} - n_{2k}\hat\pi_{2k})$.

Step 3. Negatives of each other have the same standard error, so the standardized residuals are $\pm$ the same number; the Pearson residuals divide instead by $\sqrt{n_{ik}\hat\pi_{ik}(1-\hat\pi_{ik})}$, which differs for $i = 1, 2$ when $n_{1k} \neq n_{2k}$. ■

6.2.4 Influence diagnostics for logistic regression

Deleting an observation equals adding a parameter that fits it perfectly. Leverage depends on $\hat W$, so an extreme x_i with $\hat\pi_i$ near 0 or 1 can have small leverage. Influence measures: *Dfbeta* (change in $\hat\beta_j$ on deleting i, over its SE); *confidence interval displacement* c_i; the drops ΔX_i^2, ΔG_i^2.

Theorem 6.11 (One-step influence approximations)

Let $A = X^\top \hat W X$, $w_i = n_i\hat\pi_i(1-\hat\pi_i)$, $h_i = w_i x_i^\top A^{-1} x_i$, and let $\hat\beta_{(i)}$ be the one-step Newton approximation (started from $\hat\beta$) to the estimate without observation i. Then

$$\hat\beta - \hat\beta_{(i)} = \frac{A^{-1}x_i\,(y_i - n_i\hat\pi_i)}{1-h_i}, \qquad c_i = (\hat\beta - \hat\beta_{(i)})^\top A(\hat\beta - \hat\beta_{(i)}) = \frac{e_i^2 h_i}{(1-h_i)^2},$$

and the score statistic for the extra parameter that fits observation i perfectly equals $r_i^2 = e_i^2/(1-h_i)$, the usual approximation to ΔX_i^2.

Proof. [Supplement]

Step 1. The full score vanishes at $\hat\beta$, so the reduced-data score is $U_{(i)}(\hat\beta) = -x_i(y_i - n_i\hat\pi_i)$, with information $A - w_i x_i x_i^\top$; one Newton step gives $\hat\beta_{(i)} = \hat\beta + (A - w_i x_i x_i^\top)^{-1} U_{(i)}(\hat\beta)$.

Step 2. By Sherman–Morrison,

$$(A - w_i x_i x_i^\top)^{-1} x_i = A^{-1}x_i + \frac{w_i A^{-1}x_i x_i^\top A^{-1}x_i}{1 - w_i x_i^\top A^{-1} x_i} = \frac{A^{-1}x_i}{1-h_i},$$

giving $\hat\beta - \hat\beta_{(i)}$.

Step 3. Then $c_i = (y_i - n_i\hat\pi_i)^2 x_i^\top A^{-1}x_i/(1-h_i)^2 = (y_i - n_i\hat\pi_i)^2 h_i/\{w_i(1-h_i)^2\} = e_i^2 h_i/(1-h_i)^2$.

Step 4. Add γ with covariate $\mathbf{e}_i$. At $(\hat\beta, 0)$, $U_\gamma = y_i - n_i\hat\pi_i$, $I_{\gamma\gamma} = w_i$, $I_{\gamma\beta} = w_i x_i^\top$, $I_{\beta\beta} = A$; the efficient information is $w_i - w_i^2 x_i^\top A^{-1} x_i = w_i(1-h_i)$, so the score statistic is $(y_i - n_i\hat\pi_i)^2/\{w_i(1-h_i)\} = r_i^2$. Fitting γ equals deleting i, so this approximates the drop in X^2 (and G^2). ■

Example 6.12 (Heart disease: influence)

Linear logit model of Example 6.8: category 2 has by far the largest Dfbeta (-1.14), c (2.26), ΔX^2 (5.64), ΔG^2 (5.04); all others are below 1.4.

6.2.5 Summarizing predictive power: R and R-squared measures

No R^2 analog is universally useful. Proposals:

(a) the correlation $r(y, \hat\pi)$ (crabs: 0.402 for W, 0.452 for $C+W$, 0.526 for $C{*}S{*}W$, the last inflated by its many parameters);

(b) $1 - \sum(y_i - \hat\pi_i)^2 / \sum(y_i - \bar y)^2$;

(c) likelihood-based measures.

Proposition 6.13 (Likelihood R^2 measures)

Let $L_0 \le L_M \le L_S$ be the maximized log likelihoods of the null (intercept-only), the given and the saturated models. Then $(L_M - L_0)/(L_S - L_0) \in [0,1]$, and:
(a) for n ungrouped Bernoulli observations, $L_S = 0$ and the measure reduces to McFadden's $D = (L_0 - L_M)/L_0$, where $L_0 = n[\bar y \log \bar y + (1-\bar y)\log(1-\bar y)]$;
(b) for N grouped binomial counts it equals $D^* = \{G^2(0) - G^2(M)\}/G^2(0)$.

Proof. The models are nested, so $L_0 \le L_M \le L_S \le 0$, giving the range.

(a) The saturated fit $\hat\pi_i = y_i \in \{0,1\}$ makes every term $y_i \log\hat\pi_i + (1-y_i)\log(1-\hat\pi_i)$ zero, so $L_S = 0$; the null fit $\hat\pi_i = \bar y$ gives L_0.

(b) $G^2(M) = -2(L_M - L_S)$ and $G^2(0) = -2(L_0 - L_S)$; take the ratio. ■

D^* is misleading: it is 1 for any perfect fit and tends to 1 whenever the model holds; use D on ungrouped data.

6.2.6 Summarizing predictive power: classification tables and ROC curves

A *classification table* cross-classifies y with $\hat y = \mathbb{1}\{\hat\pi_i > \pi_0\}$ (sensitivity, specificity); it discards information and depends on the arbitrary π_0.

Definition (ROC curve and concordance index)

The *ROC curve* plots sensitivity against 1−specificity as π_0 ranges over $[0,1]$. With n_1, n_0 the numbers of 1's and 0's, the *concordance index*

$$c = \frac{1}{n_1 n_0} \sum_{i:\, y_i=1} \sum_{j:\, y_j=0} \Big[\mathbb{1}\{\hat\pi_i > \hat\pi_j\} + \tfrac{1}{2}\mathbb{1}\{\hat\pi_i = \hat\pi_j\} \Big]$$

is the proportion of (case, non-case) pairs in which the case has the larger fitted probability.

Proposition 6.14 (Area under the ROC curve)

The area under the empirical ROC curve (with linear interpolation across ties) equals the concordance index c.

Proof. [Supplement]

Step 1. Lower π_0 from 1 to 0. Passing a distinct fitted value v moves the curve from (a, b) by $(m_0(v)/n_0, m_1(v)/n_1)$, where $m_y(v)$ counts observations with response y and $\hat{\pi} = v$, and $b = \#\{i : y_i = 1, \hat{\pi}_i > v\}/n_1$.

Step 2. The trapezoid added has area

$$\frac{m_0(v)}{n_0}\Big(b + \frac{m_1(v)}{2n_1}\Big) = \frac{1}{n_0 n_1} \sum_{j:\, y_j=0,\ \hat{\pi}_j=v} \Big[\#\{i : y_i = 1, \hat{\pi}_i > v\} + \tfrac{1}{2}\#\{i : y_i = 1, \hat{\pi}_i = v\}\Big].$$

Step 3. Summing over v runs over every non-case j once, giving c. ■

$c = 0.5$ is random guessing. Crabs: $c = 0.639$ (color), 0.742 (width), 0.771 (width and color).

6.3 Inference about conditional associations in $2 \times 2 \times K$ tables

Motivation. Logit models, the CMH test and the Mantel–Haenszel estimator compare two treatments across strata.

Example 6.15 (Eight-center clinical trial)

Drug vs. control cream for curing an infection (μ_{11k}, $\mathrm{Var}(n_{11k})$: null hypergeometric moments of drug successes):

	Drug		Control				
Center	Succ.	Fail.	Succ.	Fail.	Odds ratio	μ_{11k}	$\mathrm{Var}(n_{11k})$
1	11	25	10	27	1.19	10.36	3.79
2	16	4	22	10	1.82	14.62	2.47
3	14	5	7	12	4.80	10.50	2.41
4	2	14	1	16	2.29	1.45	0.70
5	6	11	0	12	∞	3.52	1.20
6	1	10	0	10	∞	0.52	0.25
7	1	4	1	8	2.0	0.71	0.42
8	4	2	6	1	0.33	4.62	0.62

6.3.1 Using logit models to test conditional independence

For binary Y, binary X ($x_1 = 1$, $x_2 = 0$) and control Z, let $\pi_{ik} = P(Y = 1 \mid X = i, Z = k)$ and

$$\operatorname{logit} \pi_{ik} = \alpha + \beta x_i + \beta_k^Z, \qquad i = 1, 2,\ k = 1, \ldots, K, \tag{6.2}$$

with common conditional odds ratio e^{β}. Conditional independence $H_0 : \beta = 0$ is tested by Wald or by LR against

$$\operatorname{logit} \pi_{ik} = \alpha + \beta_k^Z, \tag{6.3}$$

both $df = 1$. The goodness-of-fit test of (6.3) ($df = K$) is less powerful when the association is similar across strata.

6.3.2 Cochran–Mantel–Haenszel test of conditional independence

Given both margins of each partial table, n_{11k} determines the table.

Proposition 6.16 (Null hypergeometric moments)

Under conditional independence, given the margins of partial table k, n_{11k} is hypergeometric with

$$\mu_{11k} = E(n_{11k}) = \frac{n_{1+k}n_{+1k}}{n_{++k}}, \qquad \mathrm{Var}(n_{11k}) = \frac{n_{1+k}n_{2+k}n_{+1k}n_{+2k}}{n_{++k}^2(n_{++k}-1)}.$$

Proof. [Supplement] Drop k, let $n = n_{++}$, $q = n_{+1}/n$. Under H_0 the n_{+1} successes are a random subset of the n subjects. With I_s the success indicator of row-1 subject s, $n_{11} = \sum_s I_s$, $P(I_s = 1) = q$ and $P(I_s = I_t = 1) = n_{+1}(n_{+1}-1)/\{n(n-1)\}$ for $s \neq t$.

Step 1. $E(n_{11}) = n_{1+}q = n_{1+}n_{+1}/n$.

Step 2. $\mathrm{Var}(I_s) = q(1-q)$ and $\mathrm{Cov}(I_s, I_t) = -q(1-q)/(n-1)$, so

$$\mathrm{Var}(n_{11}) = n_{1+}q(1-q)\Big[1 - \frac{n_{1+}-1}{n-1}\Big] = n_{1+}q(1-q)\frac{n-n_{1+}}{n-1} = \frac{n_{1+}n_{2+}n_{+1}n_{+2}}{n^2(n-1)}.$$

■

Definition (CMH statistic)

$$\mathrm{CMH} = \frac{\left[\sum_k (n_{11k} - \mu_{11k})\right]^2}{\sum_k \mathrm{Var}(n_{11k})}, \tag{6.4}$$

approximately χ^2_1 under H_0. Cochran's version uses the variance $n_{1+k}n_{2+k}n_{+1k}n_{+2k}/n_{++k}^3$.

Deviations accumulate when the association has the same direction in every stratum. The hypergeometric form also covers retrospective studies and randomized trials.

Theorem 6.17 (CMH as a score test for the logit model)

For model (6.2) with independent binomial rows:
(a) the (unconditional) score statistic for $H_0 : \beta = 0$ is Cochran's statistic;
(b) the score statistic for β based on the conditional likelihood given all row and column margins is exactly CMH. So CMH and Cochran differ only by the factors $(n_{++k}-1)/n_{++k}$ in the variance, and both are asymptotically equivalent to the Wald and LR tests of $\beta = 0$.

Proof. [Supplement] Absorb α: $\mathrm{logit}\,\pi_{ik} = \beta x_i + \gamma_k$.

Step 1. (Score.) From $L = \sum_{ik}[y_{ik}(\beta x_i + \gamma_k) - n_{ik}\log(1 + e^{\beta x_i + \gamma_k})]$, $\partial L/\partial\beta = \sum_k (y_{1k} - n_{1k}\pi_{1k})$ and $\partial L/\partial\gamma_k = \sum_i (y_{ik} - n_{ik}\pi_{ik})$. Under H_0, $\hat\pi_{1k} = \hat\pi_{2k} = n_{+1k}/n_{++k}$ ($y_{ik} = n_{i1k}$), so $U = \sum_k (n_{11k} - \mu_{11k})$.

Step 2. (Information.) With $v_k = \pi_k(1-\pi_k)$: $I_{\beta\beta} = \sum_k n_{1+k}v_k$, $I_{\beta\gamma_k} = n_{1+k}v_k$, $I_{\gamma_k\gamma_k} = n_{++k}v_k$, $I_{\gamma_k\gamma_l} = 0$ ($k \neq l$). The efficient information is

$$I_{\beta\beta} - \sum_k \frac{I^2_{\beta\gamma_k}}{I_{\gamma_k\gamma_k}} = \sum_k v_k \frac{n_{1+k}n_{2+k}}{n_{++k}};$$

with $\hat v_k = n_{+1k}n_{+2k}/n_{++k}^2$, U^2 over this is Cochran's statistic: (a).

Step 3. (Conditional score.) Given all margins,

$$P(n_{11k} = t_k,\ k = 1, \ldots, K) = \prod_k \frac{c_k(t_k)e^{\beta t_k}}{\sum_u c_k(u)e^{\beta u}},$$

a one-parameter exponential family with statistic $T = \sum_k n_{11k}$, score $T - E_\beta(T)$ and information $\mathrm{Var}_\beta(T)$. At $\beta = 0$ these are $\sum_k (n_{11k} - \mu_{11k})$ and $\sum_k \mathrm{Var}(n_{11k})$ (Proposition 6.16), so the statistic is CMH: (b). The last claim is standard score/Wald/LR asymptotics. ■

6.3.3 Multicenter clinical trial example

Example 6.18 (Clinical trial: tests)

All centers but the last show a positive association: CMH = 6.38 ($P = 0.012$); model (6.2) gives $\hat{\beta} = 0.777$ (SE 0.307), Wald 6.42, LR 6.67.

6.3.4 Estimation of common odds ratio

Under (6.2) the ML estimate of $\theta = e^\beta$ is $e^{\hat{\beta}}$; non-model alternatives are Woolf's estimator and the Mantel–Haenszel estimator.

Definition (Mantel–Haenszel estimator)

$$\hat{\theta}_{MH} = \frac{\sum_k n_{11k} n_{22k}/n_{++k}}{\sum_k n_{12k} n_{21k}/n_{++k}} = \frac{\sum_k R_k}{\sum_k S_k} = \frac{R}{S}. \tag{6.5}$$

Theorem 6.19 (Unbiased estimating equation for $\hat{\theta}_{MH}$)

If the rows of each partial table are independent binomials and the odds ratio equals θ in every stratum, then $E(R - \theta S) = 0$. Consequently $\hat{\theta}_{MH}$ is consistent both as $n \to \infty$ with K fixed and as $K \to \infty$ with bounded strata sizes.

Proof. [Supplement]

Step 1. With independent $n_{11k} \sim \mathrm{bin}(n_{1+k}, \pi_{1k})$, $n_{21k} \sim \mathrm{bin}(n_{2+k}, \pi_{2k})$: $E(n_{11k} n_{22k}) = n_{1+k} n_{2+k} \pi_{1k}(1 - \pi_{2k})$ and $E(n_{12k} n_{21k}) = n_{1+k} n_{2+k}(1 - \pi_{1k})\pi_{2k}$.

Step 2. Since $\theta = \pi_{1k}(1 - \pi_{2k})/\{(1 - \pi_{1k})\pi_{2k}\}$, $E(R_k - \theta S_k) = 0$; sum over k.

Step 3. Under either scheme, with $m = n$ or $m = K$, R/m and S/m are averages of independent terms with bounded variances, so $(R - \theta S)/m \xrightarrow{p} 0$ while S/m stays bounded away from 0 (nondegenerate strata). Hence $\hat{\theta}_{MH} - \theta = (R - \theta S)/S \xrightarrow{p} 0$. ■

Theorem 6.20 (Unconditional ML breaks down for matched pairs)

In model (6.2) with one matched pair per stratum, let b (c) be the number of pairs in which only the first (only the second) member has $y = 1$. Then the unconditional ML estimator is $\hat{\beta} = 2\log(b/c)$, whereas $\hat{\theta}_{MH} = b/c$. As $K \to \infty$, $\log(b/c) \xrightarrow{p} \beta$, so $\hat{\beta} \xrightarrow{p} 2\beta$.

Proof. [Supplement]

Step 1. A concordant pair's likelihood is maximized at $\gamma_k = \pm\infty$ with value 1: no information on β.

Step 2. With $u = e^{\gamma_k}$, $\psi = e^\beta$, the two discordant types have likelihoods $\frac{u\psi}{(1+u\psi)(1+u)}$ and $\frac{u}{(1+u\psi)(1+u)}$, both maximized at $u = \psi^{-1/2}$ (log-derivative $1/u - \psi/(1 + u\psi) - 1/(1 + u) = 0 \iff u^2\psi = 1$), with values $\psi/(1 + \psi^{1/2})^2$ and $1/(1 + \psi^{1/2})^2$.

Step 3. The profile log likelihood $b\beta - 2(b+c)\log(1+e^{\beta/2})$ has derivative $b-(b+c)e^{\beta/2}/(1+e^{\beta/2})$, zero at $e^{\beta/2} = b/c$: $\hat{\beta} = 2\log(b/c)$.

Step 4. $R_k = \frac{1}{2}$ in a b-type pair, $S_k = \frac{1}{2}$ in a c-type pair, both 0 otherwise, so $\hat{\theta}_{MH} = b/c$.

Step 5. Given discordance, a pair is b-type with probability $\pi_1(1-\pi_2)/\{\pi_1(1-\pi_2)+(1-\pi_1)\pi_2\} = \psi/(1+\psi)$, so $b/c \xrightarrow{p} \psi$. ■

The Robins–Breslow–Greenland variance estimate, valid under both asymptotic schemes, is

$$\hat{\sigma}^2(\log\hat{\theta}_{MH}) = \frac{\sum_k (n_{11k}+n_{22k})R_k/n_{++k}}{2R^2} + \frac{\sum_k (n_{12k}+n_{21k})S_k/n_{++k}}{2S^2} + \frac{\sum_k [(n_{11k}+n_{22k})S_k + (n_{12k}+n_{21k})R_k]/n_{++k}}{2RS}.$$

Example 6.21 (Clinical trial: common odds ratio)

For Example 6.15, $\hat{\theta}_{MH} = 2.13$, $\log\hat{\theta}_{MH} = 0.758$ ($\hat{\sigma} = 0.303$), 95% CI $(1.18, 3.87)$; the Wald and LR intervals for e^{β} are $(1.19, 3.97)$ and $(1.20, 4.02)$.

Neither $\hat{\theta}_{MH}$ nor CMH requires exactly equal odds ratios; both work well when the effects mostly share a direction.

6.3.5 Testing homogeneity of odds ratios

Homogeneity $\theta_{XY(1)} = \cdots = \theta_{XY(K)}$ is model (6.2), tested by its G^2 or X^2 with $df = K-1$.

Proposition 6.22 (Fitted counts under a common odds ratio)

Fix the margins of stratum k and a value $\theta > 0$. There is a unique m_k with $\max(0, n_{1+k}+n_{+1k}-n_{++k}) < m_k < \min(n_{1+k}, n_{+1k})$ such that the table with $(1,1)$ entry m_k and the given margins has odds ratio θ:

$$\frac{m_k\,(n_{2+k}-n_{+1k}+m_k)}{(n_{1+k}-m_k)(n_{+1k}-m_k)} = \theta.$$

Proof. [Supplement] The other cells $n_{1+k}-m$, $n_{+1k}-m$, $n_{2+k}-n_{+1k}+m$ are all positive exactly on the stated open interval. There the left side is continuous, strictly increasing in m, tends to 0 at the lower end and to $+\infty$ at the upper end; by the intermediate value theorem there is exactly one root (of a quadratic in m). ■

Definition (Breslow–Day statistic)

With m_k from Proposition 6.22 at $\theta = \hat{\theta}_{MH}$ and $v_k \approx [1/m_k + 1/(n_{1+k}-m_k) + 1/(n_{+1k}-m_k) + 1/(n_{2+k}-n_{+1k}+m_k)]^{-1}$, $\mathrm{BD} = \sum_k (n_{11k}-m_k)^2/v_k$ is approximately χ^2_{K-1} under homogeneity.

Example 6.23 (Clinical trial: homogeneity)

$G^2 = 9.7$, $X^2 = 8.0$ ($df = 7$): no evidence against a common odds ratio.

6.3.6 Summarizing heterogeneity in odds ratios

Exactly equal odds ratios are unrealistic; a *random-effects* logit model takes the stratum log odds ratios from $N(\mu, \sigma^2)$ (Chapter 12).

Chapter 7. Logit Models for Multinomial Responses

7.1 Nominal responses: baseline-category logit models

Motivation. Pairing each of J unordered categories with a baseline gives $J-1$ logit equations that determine all pairwise odds.

Let Y have J categories, $\pi_j(\mathbf{x}) = P(Y = j \mid \mathbf{x})$, $\sum_j \pi_j(\mathbf{x}) = 1$; at fixed $\mathbf{x}$ the counts are multinomial. Any suitable $J-1$ of the $\binom{J}{2}$ pairwise logits determine the rest.

7.1.1 Baseline-category logits

Definition (Baseline-category logit model)

With category J as baseline,

$$\log \frac{\pi_j(\mathbf{x})}{\pi_J(\mathbf{x})} = \alpha_j + \boldsymbol{\beta}_j^\top \mathbf{x}, \qquad j = 1, \ldots, J-1. \tag{7.1}$$

The effects $\boldsymbol{\beta}_j$ depend on which category is paired with the baseline.

Proposition 7.1 (All pairwise logits)

Under (7.1), for any categories a, b,

$$\log \frac{\pi_a(\mathbf{x})}{\pi_b(\mathbf{x})} = (\alpha_a - \alpha_b) + (\boldsymbol{\beta}_a - \boldsymbol{\beta}_b)^\top \mathbf{x},$$

with $\alpha_J = 0$, $\boldsymbol{\beta}_J = \mathbf{0}$. So the baseline is arbitrary: changing it reparametrizes, not changing the model or fit.

Proof. $\log(\pi_a/\pi_b) = \log(\pi_a/\pi_J) - \log(\pi_b/\pi_J)$; substitute (7.1) (the convention covers a or $b = J$). Each parametrization is a linear reparametrization of the others, so the set of probability vectors is the same. ■

With categorical predictors and nonsparse data, X^2 and G^2 check fit; otherwise they still compare nested models differing by few terms.

7.1.2 Alligator food choice example

Example 7.2 (Alligator food choice)

219 alligators, four Florida lakes; response: primary food type (fish, invertebrate, reptile, bird, other); predictors lake L, gender G, size S (≤ 2.3 m vs. > 2.3 m). Sparse (219 in 80 cells): G^2 suits model comparison better than fit testing. $G^2[(G) \mid (\cdot)] = 2.1$ and $G^2[(L+S) \mid (G+L+S)] = 2.2$ ($df = 4$ each): collapse over gender. Then $(L+S)$ has $G^2 = 17.1$, $X^2 = 15.0$ ($df = 12$), versus 38.2 for (L) and 66.2 for (S); both matter. Counts (fitted for $L+S$):

Lake	Size	Fish	Invert.	Reptile	Bird	Other
Hancock	≤ 2.3	23 (20.9)	4 (3.6)	2 (1.9)	2 (2.7)	8 (9.9)
	> 2.3	7 (9.1)	0 (0.4)	1 (1.1)	3 (2.3)	5 (3.1)
Oklawaha	≤ 2.3	5 (5.2)	11 (12.0)	1 (1.5)	0 (0.2)	3 (1.1)
	> 2.3	13 (12.8)	8 (7.0)	6 (5.5)	1 (0.8)	0 (1.9)
Trafford	≤ 2.3	5 (4.4)	11 (12.4)	2 (2.1)	1 (0.9)	5 (4.2)
	> 2.3	8 (8.6)	7 (5.6)	6 (5.9)	3 (3.1)	5 (5.8)
George	≤ 2.3	16 (18.5)	19 (16.9)	1 (0.5)	2 (1.2)	3 (3.8)
	> 2.3	17 (14.5)	1 (3.1)	0 (0.5)	1 (1.8)	3 (2.2)

Two of 40 standardized residuals exceed 2. Fish baseline; $s = 1$ for small; z_H, z_O, z_T for Hancock, Oklawaha, Trafford (George reference). Estimates (SE):

Logit	Intercept	Size ≤ 2.3	Hancock	Oklawaha	Trafford
$\log(\pi_I/\pi_F)$	-1.55	1.46 (0.40)	-1.66 (0.61)	0.94 (0.47)	1.12 (0.49)
$\log(\pi_R/\pi_F)$	-3.31	-0.35 (0.58)	1.24 (1.19)	2.46 (1.12)	2.94 (1.12)
$\log(\pi_B/\pi_F)$	-2.09	-0.63 (0.64)	0.70 (0.78)	-0.65 (1.20)	1.09 (0.84)
$\log(\pi_O/\pi_F)$	-1.90	0.33 (0.45)	0.83 (0.56)	0.01 (0.78)	1.52 (0.62)

Given lake, small alligators have odds of invertebrates rather than fish $e^{1.46} = 4.3$ times those of large ones (Wald CI $e^{1.46 \pm 1.96(0.396)} = (2.0, 9.3)$); these odds are higher in Trafford and Oklawaha, lower in Hancock, than in George. Other pairs follow by subtraction (Proposition 7.1), e.g.

$$\begin{aligned}\log(\hat{\pi}_I/\hat{\pi}_O) &= (-1.55 + 1.46s - 1.66z_H + 0.94z_O + 1.12z_T) \\ &\quad - (-1.90 + 0.33s + 0.83z_H + 0.01z_O + 1.52z_T) \\ &= 0.35 + 1.13s - 2.48z_H + 0.93z_O - 0.39z_T.\end{aligned}$$

7.1.3 Estimating response probabilities

Proposition 7.3 (Response probabilities)

Model (7.1) is equivalent to

$$\pi_j(\mathbf{x}) = \frac{\exp(\alpha_j + \boldsymbol{\beta}_j^\top \mathbf{x})}{1 + \sum_{h=1}^{J-1} \exp(\alpha_h + \boldsymbol{\beta}_h^\top \mathbf{x})}, \qquad j = 1, \ldots, J, \tag{7.2}$$

with $\alpha_J = 0$, $\boldsymbol{\beta}_J = \mathbf{0}$. For $J = 2$ this is binary logistic regression.

Proof. From (7.1), $\pi_j = \pi_J \exp(\eta_j)$ with $\eta_j = \alpha_j + \boldsymbol{\beta}_j^\top \mathbf{x}$, valid also for $j = J$ with $\eta_J = 0$. Summing over j, $1 = \pi_J \sum_{h=1}^{J} e^{\eta_h} = \pi_J(1 + \sum_{h<J} e^{\eta_h})$. Solve for π_J and substitute. Conversely (7.2) gives $\pi_j/\pi_J = e^{\eta_j}$. ■

Baseline parameters are zero for identifiability: adding a common vector to all $(\alpha_j, \boldsymbol{\beta}_j)$ leaves (7.2) unchanged.

Example 7.4 (Alligators: fitted probabilities)

For a large alligator in Lake Hancock ($s = 0$, $z_H = 1$),

$$\hat{\pi}_I = \frac{e^{-1.55-1.66}}{1 + e^{-1.55-1.66} + e^{-3.31+1.24} + e^{-2.09+0.70} + e^{-1.90+0.83}} = 0.023,$$

and $\hat{\pi}_R = 0.072$, $\hat{\pi}_B = 0.141$, $\hat{\pi}_O = 0.194$, $\hat{\pi}_F = 0.570$. For quantitative x, plot $\hat{\pi}_j(x)$; with $J > 2$ a single $\hat{\pi}_j(x)$ need not be monotone.

7.2 Ordinal responses: cumulative logit models

Motivation. Using the ordering gives parsimony and power; cumulative logits with a common effect, from a latent continuous response, are the standard choice.

7.2.1 Cumulative logits

Definition (Cumulative logits)

With $P(Y \le j \mid \mathbf{x}) = \pi_1(\mathbf{x}) + \cdots + \pi_j(\mathbf{x})$,

$$\operatorname{logit} P(Y \le j \mid \mathbf{x}) = \log \frac{P(Y \le j \mid \mathbf{x})}{1 - P(Y \le j \mid \mathbf{x})} = \log \frac{\pi_1(\mathbf{x}) + \cdots + \pi_j(\mathbf{x})}{\pi_{j+1}(\mathbf{x}) + \cdots + \pi_J(\mathbf{x})}, \quad j = 1, \ldots, J-1.$$

Each cumulative logit uses all J categories; alone it is a binary logit for $\{1, \ldots, j\}$ vs. $\{j+1, \ldots, J\}$.

7.2.2 Proportional odds model

Definition (Proportional odds model)

$$\operatorname{logit} P(Y \le j \mid \mathbf{x}) = \alpha_j + \boldsymbol{\beta}^\top \mathbf{x}, \qquad j = 1, \ldots, J-1, \tag{7.3}$$

with intercepts $\alpha_1 < \alpha_2 < \cdots < \alpha_{J-1}$ (since $P(Y \le j \mid \mathbf{x})$ increases in j) and a single effect vector $\boldsymbol{\beta}$ common to all cutpoints.

Proposition 7.5 (Parallel curves and cumulative odds ratios)

Under (7.3):
(a) for a single predictor and $\beta \neq 0$, the curve of $P(Y \le k \mid x)$ is the curve of $P(Y \le j \mid x)$ shifted horizontally: $P(Y \le k \mid X = x) = P(Y \le j \mid X = x + (\alpha_k - \alpha_j)/\beta)$;
(b) for any $\mathbf{x}_1, \mathbf{x}_2$ and every j, the cumulative odds ratio is

$$\frac{P(Y \le j \mid \mathbf{x}_1)/P(Y > j \mid \mathbf{x}_1)}{P(Y \le j \mid \mathbf{x}_2)/P(Y > j \mid \mathbf{x}_2)} = \exp[\boldsymbol{\beta}^\top(\mathbf{x}_1 - \mathbf{x}_2)],$$

the same for all j and proportional (on the log scale) to the distance between $\mathbf{x}_1$ and $\mathbf{x}_2$;
(c) the conditional distributions are stochastically ordered: for each pair $\mathbf{x}_1, \mathbf{x}_2$, either $P(Y \le j \mid \mathbf{x}_1) \le P(Y \le j \mid \mathbf{x}_2)$ for all j, or $\ge$ for all j.

Proof. Let $F(z) = e^z/(1+e^z)$, so $P(Y \le j \mid \mathbf{x}) = F(\alpha_j + \boldsymbol{\beta}^\top \mathbf{x})$.

(a) $F(\alpha_j + \beta[x + (\alpha_k - \alpha_j)/\beta]) = F(\alpha_k + \beta x)$.

(b) The log of the cumulative odds ratio is the difference of logits, $(\alpha_j + \boldsymbol{\beta}^\top \mathbf{x}_1) - (\alpha_j + \boldsymbol{\beta}^\top \mathbf{x}_2) = \boldsymbol{\beta}^\top(\mathbf{x}_1 - \mathbf{x}_2)$.

(c) F is increasing, so $P(Y \le j \mid \mathbf{x}_1) \le P(Y \le j \mid \mathbf{x}_2)$ iff $\boldsymbol{\beta}^\top \mathbf{x}_1 \le \boldsymbol{\beta}^\top \mathbf{x}_2$, a condition not involving j. ■

With one predictor, e^{β} is the cumulative odds ratio for $x_1 - x_2 = 1$, the same for every cutpoint: McCullagh's *proportional odds*. The curves for $j = 1, 2, 3$ ($J = 4$) have the same shape (Figure 9).

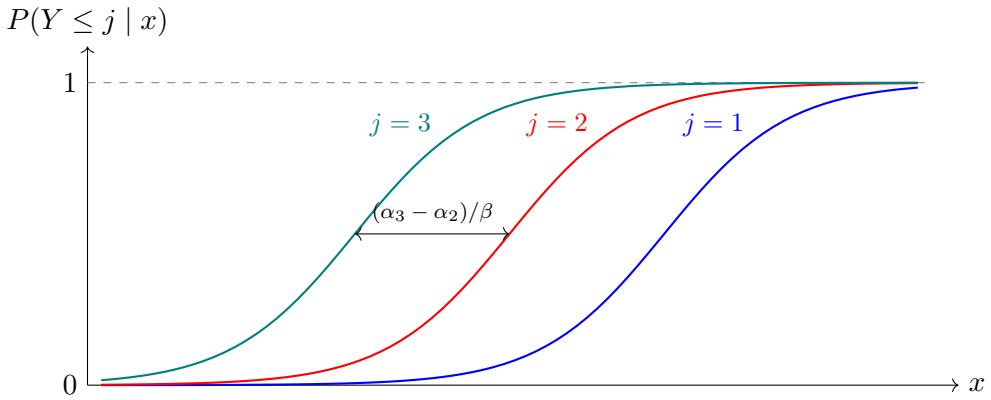


Figure 9: Proportional odds, $J = 4$, $\beta > 0$: three logistic curves with a common slope, horizontally shifted.

The shared shape makes the fit differ from separate binary fits. With y_{ij} indicators of the response of subject i, the likelihood is

$$\prod_{i=1}^{n}\prod_{j=1}^{J}\Big[\frac{\exp(\alpha_j + \boldsymbol{\beta}^{\top}\mathbf{x}_i)}{1+\exp(\alpha_j + \boldsymbol{\beta}^{\top}\mathbf{x}_i)} - \frac{\exp(\alpha_{j-1} + \boldsymbol{\beta}^{\top}\mathbf{x}_i)}{1+\exp(\alpha_{j-1} + \boldsymbol{\beta}^{\top}\mathbf{x}_i)}\Big]^{y_{ij}} \tag{7.4}$$

with $\alpha_0 = -\infty$, $\alpha_J = +\infty$; it is maximized by Fisher scoring.

7.2.3 Mental impairment example

Example 7.6 (Mental impairment)

40 adults, Alachua County, Florida. Y = mental impairment (1 well, 2 mild, 3 moderate symptom formation, 4 impaired); x_1 = life events index (past three years); x_2 = SES (1 high, 0 low). (x_2, x_1) by response category:

Well	(1,1) (1,9) (1,4) (1,3) (0,2) (1,0) (0,1) (1,3) (1,3) (1,7) (0,1) (0,2)
Mild	(1,5) (0,6) (1,3) (0,1) (1,8) (1,2) (0,5) (1,5) (1,9) (0,3) (1,3) (1,1)
Moderate	(0,0) (1,4) (0,3) (0,9) (1,6) (0,4) (0,3)
Impaired	(1,8) (1,2) (1,7) (0,5) (0,4) (0,4) (1,8) (0,8) (0,9)

The fit of logit $P(Y \le j \mid \mathbf{x}) = \alpha_j + \beta_1 x_1 + \beta_2 x_2$:

Parameter	Estimate	SE	LR 95% CI
α_1	−0.282	0.642	(−1.56, 0.98)
α_2	1.213	0.661	(−0.05, 2.57)
α_3	2.209	0.721	(0.86, 3.71)
life (β_1)	−0.319	0.121	(−0.57, −0.09)
SES (β_2)	1.111	0.611	(−0.06, 2.35)

At the mean life events $\bar{x}_1 = 4.275$ and low SES,

$$\hat{P}(Y=1) = \frac{\exp[-0.282 - 0.319(4.275)]}{1+\exp[-0.282 - 0.319(4.275)]} = 0.16.$$

$P(Y \le j)$ decreases with life events and is higher at high SES: given life events, the cumulative odds at high SES are $e^{1.111} = 3.0$ times those at low SES. At $\bar{x}_1$, $\hat{P}(Y=1) = 0.37$ (high) vs. 0.16 (low SES); between the life-events quartiles 2.0 and 6.5 it falls from 0.55 to 0.22 (high) and 0.28 to 0.09 (low). Score test of proportional odds: 2.33, $df = 4$, $P = 0.68$.

7.2.4 More complex models

Interactions and factors enter as in binary logistic regression, with $J-1$ cutpoint intercepts. Adding the interaction:

$$\operatorname{logit} \hat{P}(Y \le j \mid \mathbf{x}) = \hat{\alpha}_j - 0.420x_1 + 0.371x_2 + 0.181x_1x_2,$$

interaction SE 0.238: the life-events effect is -0.420 at low SES and -0.239 at high SES, not significantly different.

When proportional odds fails

Replacing $\boldsymbol{\beta}$ by $\boldsymbol{\beta}_j$ gives nonparallel curves, which cross for some $\mathbf{x}$ (misordered cumulative probabilities). A small P-value in the proportional odds test may reflect large n rather than important differences, and the simpler model may have smaller MSE. If the fit is poor in practical terms, options include:
(1) a link with a nonsymmetric response curve (complementary log-log);
(2) extra terms such as interactions;
(3) dispersion parameters;
(4) separate effects for some predictors only (partial proportional odds);
(5) a baseline-category logit model, using the ordering informally in interpretation.

7.3 Ordinal responses: cumulative link models

Motivation. The latent-variable argument allows any inverse-cdf link, such as probit or complementary log-log.

Definition (Cumulative link model)

For a continuous cdf G,

$$G^{-1}[P(Y \le j \mid \mathbf{x})] = \alpha_j + \boldsymbol{\beta}^\top \mathbf{x}, \qquad j = 1, \ldots, J-1. \tag{7.5}$$

Model (7.5) arises from crude measurement of a latent Y^* with cdf $G(y^* - \boldsymbol{\beta}^\top \mathbf{x})$; the α_j are cutpoints on the latent scale ($\pm\boldsymbol{\beta}$ only flips signs).

7.3.1 Types of cumulative links

(a) *Cumulative probit* ($G = \Phi$, normal Y^*): in $\Phi^{-1}[P(Y \le j)] = \alpha_j - \boldsymbol{\beta}^\top \mathbf{x}$, a unit increase in x_k raises $E(Y^*)$ by β_k SDs of ε. Logit and probit fits are similar; logit parameters are easier to interpret.

(b) *Complementary log-log* ($G(z) = 1 - \exp(-e^z)$, extreme-value latent): $\log\{-\log[1 - P(Y \le j \mid \mathbf{x})]\} = \alpha_j + \boldsymbol{\beta}^\top \mathbf{x}$, the grouped-time *proportional hazards* model; $P(Y \le j)$ approaches 1 faster than 0. The log-log link is cloglog for reversed categories ($G \mapsto 1 - G(-z)$).

Proposition 7.7 (Power interpretation of the cumulative cloglog model)

Under the cumulative complementary log-log model, for all j,

$$P(Y > j \mid \mathbf{x}_1) = P(Y > j \mid \mathbf{x}_2)^{\exp[\boldsymbol{\beta}^\top(\mathbf{x}_1 - \mathbf{x}_2)]}.$$

Proof. $P(Y > j \mid \mathbf{x}) = 1 - G(\alpha_j + \boldsymbol{\beta}^\top \mathbf{x}) = \exp[-e^{\alpha_j} e^{\boldsymbol{\beta}^\top \mathbf{x}}]$. Hence

$$\log P(Y > j \mid \mathbf{x}_1) = -e^{\alpha_j} e^{\boldsymbol{\beta}^\top \mathbf{x}_2} e^{\boldsymbol{\beta}^\top (\mathbf{x}_1 - \mathbf{x}_2)} = e^{\boldsymbol{\beta}^\top (\mathbf{x}_1 - \mathbf{x}_2)} \log P(Y > j \mid \mathbf{x}_2);$$

exponentiate. ■

7.3.2 Estimation for cumulative link models

Cumulative link models are multivariate GLMs, fitted by McCullagh's Fisher scoring on (7.4) with G for the logistic cdf. For large n the maximum is unique; the log likelihood is concave for logit, probit and cloglog links, so iterations converge rapidly.

7.3.3 Life table example

Example 7.8 (U.S. life lengths, 1981)

Percent distribution of life length by gender and race (fitted cloglog values in parentheses):

	Males		Females	
Life length	White	Black	White	Black
0–20	2.4 (2.4)	3.6 (4.4)	1.6 (1.2)	2.7 (2.3)
20–40	3.4 (3.5)	7.5 (6.4)	1.4 (1.9)	2.9 (3.4)
40–50	3.8 (4.4)	8.3 (7.7)	2.2 (2.4)	4.4 (4.3)
50–65	17.5 (16.7)	25.0 (26.1)	9.9 (9.6)	16.3 (16.3)
Over 65	72.9 (73.0)	55.6 (55.4)	84.9 (84.9)	73.7 (73.7)

The life-length cdf rises slowly at young ages and sharply at old ones, suggesting cloglog (also implied by a Gompertz hazard). For G (1 female) and R (1 black),

$$\log\{-\log[1 - P(Y \le j \mid G = g, R = r)]\} = \alpha_j + \beta_1 g + \beta_2 r, \qquad \hat\beta_1 = -0.658,\ \hat\beta_2 = 0.626.$$

(These are population values.) By Proposition 7.7, $P(Y > j \mid G = 0, r) = P(Y > j \mid G = 1, r)^{\exp(0.658)}$: given race, men's survival beyond any age is women's to the power 1.93; given gender, blacks' is whites' to the power $e^{0.626} = 1.87$. So if white women survive with probability π, white men and black women do so with about π^2, black men with about π^4.

7.4 Selected exercises

Motivation. Theory problems from Agresti's chapter end, filling in structural facts used above.

Exercise 7.24 (Multinomial trial as an exponential family)

For one multinomial trial with probabilities $\pi_1, \dots, \pi_J > 0$, let $\mathbf{y} = (y_1, \dots, y_{J-1})^\top$, $y_j = \mathbb{1}\{\text{category } j\}$. Show that $\mathbf{y}$ has the form

$$f(\mathbf{y}; \boldsymbol{\theta}) = \exp[\mathbf{y}^\top \boldsymbol{\theta} - b(\boldsymbol{\theta})],$$

a $(J-1)$-parameter exponential family whose natural parameters are the baseline-category logits $\theta_j = \log(\pi_j / \pi_J)$. Find b and check that $\partial b / \partial \theta_j = \mathbb{E}(y_j)$.

Solution.

Step 1. Put $y_J = 1 - \sum_{j<J} y_j$. Exactly one of $y_1, \dots, y_J$ equals 1, so $P(\mathbf{y}) = \prod_{j=1}^{J} \pi_j^{y_j}$.

Step 2. Factor out π_J:

$$\prod_{j=1}^{J} \pi_j^{y_j} = \pi_J^{1-\sum_{j<J} y_j} \prod_{j<J} \pi_j^{y_j} = \pi_J \prod_{j<J} \Big(\frac{\pi_j}{\pi_J}\Big)^{y_j} = \exp\Big[\sum_{j<J} y_j \theta_j + \log \pi_J\Big].$$

Step 3. By Proposition 7.3 (with no predictors), $\pi_J = 1/(1 + \sum_{h<J} e^{\theta_h})$, so $-\log \pi_J = b(\boldsymbol{\theta}) := \log(1 + \sum_{h<J} e^{\theta_h})$, a function of $\boldsymbol{\theta}$ alone. This gives the stated form with dispersion $a(\phi) = 1$ and $c(\mathbf{y}, \phi) = 0$; the map $(\pi_1, \dots, \pi_{J-1}) \mapsto \boldsymbol{\theta}$ is one-to-one onto $\mathbb{R}^{J-1}$.

Step 4. $\partial b / \partial \theta_j = e^{\theta_j}/(1 + \sum_h e^{\theta_h}) = \pi_j = \mathbb{E}(y_j)$, as for any exponential family.

So (7.1) uses the canonical link of this multivariate family, the analogue of logistic regression as a canonical GLM, with sufficient statistics $\sum_i y_{ij}$ and $\sum_i x_{ik} y_{ij}$. ■

Exercise 7.27 (A nonmonotone category probability)

Let $J = 3$ and

$$\pi_3(x) = \frac{1}{1 + \exp(\alpha_1 + \beta_1 x) + \exp(\alpha_2 + \beta_2 x)}.$$

Show that $\pi_3(x)$ is
(a) decreasing in x if $\beta_1 > 0$ and $\beta_2 > 0$;
(b) increasing in x if $\beta_1 < 0$ and $\beta_2 < 0$;
(c) nonmonotone when β_1 and β_2 have different signs.

Solution. Let $D(x) = 1 + e^{\alpha_1 + \beta_1 x} + e^{\alpha_2 + \beta_2 x} > 1$, so $\pi_3 = 1/D$ and $\pi_3' = -D'/D^2$ with

$$D'(x) = \beta_1 e^{\alpha_1 + \beta_1 x} + \beta_2 e^{\alpha_2 + \beta_2 x}.$$

(a) If $\beta_1, \beta_2 > 0$ then $D' > 0$, so $\pi_3' < 0$ everywhere.

(b) If $\beta_1, \beta_2 < 0$ then $D' < 0$, so $\pi_3' > 0$ everywhere.

(c) Say $\beta_1 > 0 > \beta_2$ (the other case is symmetric).

Step 1. $D''(x) = \beta_1^2 e^{\alpha_1 + \beta_1 x} + \beta_2^2 e^{\alpha_2 + \beta_2 x} > 0$, so D' is strictly increasing.

Step 2. As $x \to +\infty$ the first term of D' tends to $+\infty$ and the second to 0; as $x \to -\infty$ the first tends to 0 and the second to $-\infty$. Hence D' changes sign exactly once, at some x_0.

Step 3. Therefore $\pi_3' > 0$ for $x < x_0$ and $\pi_3' < 0$ for $x > x_0$: π_3 rises to a maximum at x_0 and then falls (with $\pi_3 \to 0$ at both ends). In particular it is not monotone.

This is the nonmonotonicity noted after the alligator example. ■

Exercise 7.31 (Cumulative logits need a common effect)

Consider logit $P(Y \le j \mid x) = \alpha_j + \beta_j x$, $j = 1, \dots, J-1$, with x ranging over $\mathbb{R}$ and the β_j not all equal. Show that the cumulative probabilities are misordered for some x, so the model does not define a probability distribution for all x.

Solution.

Step 1. Since the β_j are not all equal, there is a j with $\beta_j \neq \beta_{j+1}$ (otherwise all consecutive slopes, hence all slopes, would coincide).

Step 2. The difference of the two logits,

$$g(x) = (\alpha_j + \beta_j x) - (\alpha_{j+1} + \beta_{j+1} x) = (\alpha_j - \alpha_{j+1}) + (\beta_j - \beta_{j+1}) x,$$

is a nonconstant linear function, so $g(x) > 0$ for all x on one side of the root $x^* = (\alpha_{j+1} - \alpha_j)/(\beta_j - \beta_{j+1})$.

Step 3. The logistic cdf F is strictly increasing, so for such x, $P(Y \le j \mid x) = F(\alpha_j + \beta_j x) > F(\alpha_{j+1} + \beta_{j+1} x) = P(Y \le j+1 \mid x)$. This would make $\pi_{j+1}(x) = P(Y \le j+1 \mid x) - P(Y \le j \mid x) < 0$, which is impossible.

The two logistic curves have different slopes, so they cross (the remark in Section 7.2.4). On a bounded range of x the model can still be valid. ■

Exercise 7.32 (Uniform association in cumulative odds ratios)

For an $I \times J$ table with ordinal Y, independent multinomial rows, and scores $x_i = i$ for the rows, consider

$$\text{logit}\, P(Y \le j \mid X = x_i) = \alpha_j + \beta x_i.$$

(a) Show that $\text{logit}\, P(Y \le j \mid X = x_{i+1}) - \text{logit}\, P(Y \le j \mid X = x_i) = \beta$, and that this is the log cumulative odds ratio of the 2×2 table formed by rows $i, i+1$ and the binary response $(Y \le j,\ Y > j)$.
(b) Show that the residual df equal $IJ - I - J$.
(c) Show that independence of X and Y is the special case $\beta = 0$.
(d) If βx_i is replaced by unordered row parameters μ_i (treating X as nominal), show that for rows a, b the log cumulative odds ratio equals $\mu_a - \mu_b$ for all $J - 1$ cutpoints.

Solution.

(a) The difference is $(\alpha_j + \beta(i+1)) - (\alpha_j + \beta i) = \beta$. Writing $P_i = P(Y \le j \mid X = x_i)$, the logit difference is

$$\log \frac{P_{i+1}/(1 - P_{i+1})}{P_i/(1 - P_i)},$$

the log odds ratio of the 2×2 table with rows $i+1, i$ and columns $Y \le j$, $Y > j$. It equals β for all adjacent rows and cutpoints: uniform association in cumulative odds ratios.

(b) The saturated model has $I(J-1)$ parameters, the model J ($J-1$ intercepts and β); residual df $I(J-1) - J = IJ - I - J$.

(c) If $\beta = 0$, $P(Y \le j \mid X = x_i) = F(\alpha_j)$ is free of i: independence. Conversely, independence requires $\alpha_j + \beta x_i$ free of i, i.e. $\beta = 0$. So testing $\beta = 0$ is a one-df test of independence using both orderings.

(d) With $\text{logit}\, P(Y \le j \mid X = a) = \alpha_j + \mu_a$, the log cumulative odds ratio of rows a, b is $(\alpha_j + \mu_a) - (\alpha_j + \mu_b) = \mu_a - \mu_b$, free of j; the rows are stochastically ordered by μ_i (Proposition 7.5(c)). ■

Chapter 8. Loglinear Models for Contingency Tables

Loglinear models are log-link Poisson GLMs for cell counts that treat all variables symmetrically as joint responses; logit models with categorical predictors are special cases (Section 8.5). $\{n\}$, $\{\mu\}$, $\{\pi\}$ denote observed counts, expected counts and cell probabilities; "+" denotes summation.

8.1 Loglinear models for two-way tables

Motivation. Independence $\pi_{ij} = \pi_{i+}\pi_{+j}$ is additive on the log scale; adding "interaction" terms describes any association.

$\mu_{ij} = n\pi_{ij}$ (multinomial) or $\mu_{ij} = E(Y_{ij})$ for $N = IJ$ independent Poisson counts; formulas in μ_{ij} cover both. Rows are X, columns Y.

8.1.1 Independence model

Definition (Loglinear model of independence)

$$\log \mu_{ij} = \lambda + \lambda_i^X + \lambda_j^Y, \tag{8.1}$$

with identifiability constraints such as $\lambda_I^X = \lambda_J^Y = 0$ (baseline coding) or $\sum_i \lambda_i^X = \sum_j \lambda_j^Y = 0$.

Theorem 8.1 (Independence as a loglinear model)

For positive $\{\mu_{ij}\}$ with $\mu_{ij} = n\pi_{ij}$, the variables X and Y are independent if and only if (8.1) holds for some parameters. The ML fitted values are $\hat{\mu}_{ij} = n_{i+}n_{+j}/n$.

Proof.

Step 1. If $\pi_{ij} = \pi_{i+}\pi_{+j}$, then $\log \mu_{ij} = \log n + \log \pi_{i+} + \log \pi_{+j}$. Put $\lambda_i^X = \log \pi_{i+} - \log \pi_{I+}$, $\lambda_j^Y = \log \pi_{+j} - \log \pi_{+J}$ and $\lambda = \log n + \log \pi_{I+} + \log \pi_{+J}$; this is (8.1) with baseline constraints.

Step 2. Conversely, if $\mu_{ij} = a_i b_j$ with $a_i = e^{\lambda + \lambda_i^X}$, $b_j = e^{\lambda_j^Y}$, then $\mu_{i+} = a_i b_+$, $\mu_{+j} = a_+ b_j$, $\mu_{++} = a_+ b_+$, so

$$\pi_{ij} = \frac{a_i b_j}{a_+ b_+} = \frac{a_i b_+}{a_+ b_+} \cdot \frac{a_+ b_j}{a_+ b_+} = \pi_{i+}\pi_{+j}.$$

Step 3. The fitted values $n_{i+}n_{+j}/n$ satisfy the model and reproduce the row and column totals; by Birch's result (Theorem 8.19) they are the ML estimates. ■

Hence the Pearson X^2 and G^2 tests of independence are goodness-of-fit tests of the loglinear model (8.1), with $\mathrm{df} = (I-1)(J-1)$.

8.1.2 Interpretation of parameters

A loglinear GLM takes the N cell counts as observations and does not distinguish response from explanatory variables; forming a logit still helps interpretation.

Proposition 8.2 (Independence means a constant logit)

Let $J = 2$. Under (8.1),

$$\operatorname{logit} P(Y = 1 \mid X = i) = \lambda_1^Y - \lambda_2^Y \qquad \text{for every } i,$$

so the odds of column 1 equal $\exp(\lambda_1^Y - \lambda_2^Y)$ in every row.

Proof. $P(Y=1 \mid X=i)/P(Y=2 \mid X=i) = \mu_{i1}/\mu_{i2}$, hence the logit equals

$$\log \mu_{i1} - \log \mu_{i2} = (\lambda + \lambda_i^X + \lambda_1^Y) - (\lambda + \lambda_i^X + \lambda_2^Y) = \lambda_1^Y - \lambda_2^Y,$$

which does not depend on i. ■

Differences of main-effect parameters of one variable are log odds of one of its categories versus another.

8.1.3 Saturated model

Definition (Saturated two-way model)

$$\log \mu_{ij} = \lambda + \lambda_i^X + \lambda_j^Y + \lambda_{ij}^{XY}. \tag{8.2}$$

The association terms λ_{ij}^{XY} (coefficients of products of dummy variables) measure departures from independence; with baseline constraints $\lambda_{Ij}^{XY} = \lambda_{iJ}^{XY} = 0$.

Theorem 8.3 (Saturated model and odds ratios)

(a) Model (8.2) has IJ nonredundant parameters and describes every positive table $\{\mu_{ij}\}$ exactly.
(b) Every odds ratio is a contrast of the association parameters; for rows a, c and columns b, d,

$$\log \frac{\mu_{ab}\mu_{cd}}{\mu_{ad}\mu_{cb}} = \lambda_{ab}^{XY} + \lambda_{cd}^{XY} - \lambda_{ad}^{XY} - \lambda_{cb}^{XY}.$$

In particular, for a 2×2 table $\log \theta = \lambda_{11}^{XY} + \lambda_{22}^{XY} - \lambda_{12}^{XY} - \lambda_{21}^{XY}$.
(c) Independence holds iff all $\lambda_{ij}^{XY} = 0$; the test has df $= (I-1)(J-1)$.

Proof.

Step 1. Counting under baseline constraints: $1 + (I-1) + (J-1) + (I-1)(J-1) = IJ$.

Step 2. Given positive $\{\mu_{ij}\}$, set $h_{ij} = \log \mu_{ij}$ and define

$$\lambda = h_{IJ}, \quad \lambda_i^X = h_{iJ} - h_{IJ}, \quad \lambda_j^Y = h_{Ij} - h_{IJ}, \quad \lambda_{ij}^{XY} = h_{ij} - h_{iJ} - h_{Ij} + h_{IJ}.$$

These satisfy the baseline constraints and $\lambda + \lambda_i^X + \lambda_j^Y + \lambda_{ij}^{XY} = h_{ij}$, so every positive table is represented (uniquely, since the map is invertible).

Step 3. For (b), substitute (8.2): in $h_{ab} + h_{cd} - h_{ad} - h_{cb}$ the terms λ, $\lambda_a^X, \lambda_c^X, \lambda_b^Y, \lambda_d^Y$ each appear once with $+$ and once with $-$, leaving the stated contrast.

Step 4. For (c), by step 2 all $\lambda_{ij}^{XY} = 0$ iff $h_{ij} - h_{iJ} - h_{Ij} + h_{IJ} = 0$ for all i, j, i.e. all odds ratios with the last row and column are 1, which is equivalent to independence. The $(I-1)(J-1)$ free association parameters give the df. ■

Hierarchical models

A model is *hierarchical* if each term comes with all lower-order terms in its variables (e.g. λ_{ij}^{XY} requires λ_i^X, λ_j^Y). Nonhierarchical models such as $\log \mu_{ij} = \lambda + \lambda_i^X + \lambda_{ij}^{XY}$ depend on the coding (baseline coding forces $\log \mu_{Ij} = \lambda$) and are rarely sensible. Interpret the highest-order

terms, not main effects.

8.1.4 Alternative parameter constraints

Sum-to-zero constraints $\sum_i \lambda_{ij}^{XY} = \sum_j \lambda_{ij}^{XY} = 0$ are an alternative; only contrasts like Theorem 8.3(b) are invariant. For 2×2 with $\log\theta = 2.0$: baseline coding gives $\lambda_{11}^{XY} = 2.0$, others 0; sum-to-zero gives $\lambda_{11}^{XY} = \lambda_{22}^{XY} = 0.5$, $\lambda_{12}^{XY} = \lambda_{21}^{XY} = -0.5$. Baseline coding allows some estimates in a set to be infinite while others stay finite.

8.1.5 Multinomial models for cell probabilities

Proposition 8.4 (Poisson to multinomial)

If Y_{ij} are independent Poisson with $\log\mu_{ij}$ given by (8.2), then conditional on $\sum Y_{ij} = n$ the counts are multinomial$(n, \{\pi_{ij}\})$ with

$$\pi_{ij} = \frac{\exp(\lambda + \lambda_i^X + \lambda_j^Y + \lambda_{ij}^{XY})}{\sum_a \sum_b \exp(\lambda + \lambda_a^X + \lambda_b^Y + \lambda_{ab}^{XY})}, \tag{8.3}$$

in which λ cancels.

Proof. Let $\mu_{++} = \sum \mu_{ij}$; the total is Poisson(μ_{++}). For counts $\{n_{ij}\}$ summing to n,

$$P(\{Y_{ij} = n_{ij}\} \mid Y_{++} = n) = \frac{\prod_{ij} e^{-\mu_{ij}} \mu_{ij}^{n_{ij}}/n_{ij}!}{e^{-\mu_{++}}\mu_{++}^n/n!} = \frac{n!}{\prod n_{ij}!} \prod_{ij} \Big(\frac{\mu_{ij}}{\mu_{++}}\Big)^{n_{ij}},$$

since $\prod e^{-\mu_{ij}} = e^{-\mu_{++}}$ and $\sum n_{ij} = n$. This is multinomial with $\pi_{ij} = \mu_{ij}/\mu_{++}$, which is (8.3); the common factor e^{λ} cancels. ∎

The intercept λ only reflects the total, random under Poisson and fixed under multinomial sampling.

8.2 Loglinear models for independence and interaction in three-way tables

Motivation. Each kind of three-way independence or association is a loglinear model with certain terms absent, so the model symbol shows the structure.

The following lemma turns every independence statement into a statement about which terms appear.

Lemma 8.5 (Generating classes and factorization)

Let a hierarchical loglinear model have generating class $\{C_1, \ldots, C_m\}$ (the highest-order terms, i.e. the model symbol; e.g. (XZ, YZ) has $C_1 = \{X, Z\}$, $C_2 = \{Y, Z\}$). Then a positive table $\{\mu\}$ satisfies the model if and only if

$$\mu(\mathbf{x}) = \prod_{r=1}^{m} \psi_r(\mathbf{x}_{C_r})$$

for some positive functions ψ_r that depend on the cell $\mathbf{x}$ only through the coordinates in C_r.

Proof. [Supplement]

Step 1. Every term of the model involves a subset of some C_r, hence is a function of $\mathbf{x}_{C_r}$. Grouping terms by a generator containing them, $\log \mu = \sum_r g_r(\mathbf{x}_{C_r})$; exponentiate to get the product form.

Step 2. Conversely, by Theorem 8.3 (step 2, extended to more variables) any $g(\mathbf{x}_C)$ is a constant plus main effects plus interactions of every subset of C, all of which are model terms (hierarchy). Summing these decompositions of $\log \psi_r$ gives $\log \mu$ in the model's form. ■

8.2.1 Types of independence

Consider an $I \times J \times K$ table of three responses X, Y, Z with probabilities $\{\pi_{ijk}\}$ (or Poisson means $\{\mu_{ijk}\}$).

Definition (Three kinds of independence)

(i) *Mutual independence*: $\pi_{ijk} = \pi_{i++}\pi_{+j+}\pi_{++k}$ for all i, j, k.
(ii) *Joint independence* of Y from (X, Z): $\pi_{ijk} = \pi_{i+k}\pi_{+j+}$, i.e. two-way independence between Y and the IK combinations of (X, Z).
(iii) *Conditional independence* of X and Y given Z: $\pi_{ij|k} = \pi_{i+|k}\pi_{+j|k}$ for all k, where $\pi_{ij|k} = P(X = i, Y = j \mid Z = k)$; equivalently

$$\pi_{ijk} = \pi_{i+k}\pi_{+jk}/\pi_{++k} \qquad \text{for all } i, j, k. \tag{8.4}$$

Theorem 8.6 (Independence structures as loglinear models)

For positive tables:
(a) mutual independence $\iff$ (X, Y, Z): $\log \mu_{ijk} = \lambda + \lambda_i^X + \lambda_j^Y + \lambda_k^Z$;
(b) Y jointly independent of (X, Z) $\iff$ (XZ, Y): $\log \mu_{ijk} = \lambda + \lambda_i^X + \lambda_j^Y + \lambda_k^Z + \lambda_{ik}^{XZ}$;
(c) $X \perp Y \mid Z$ $\iff$ (XZ, YZ): $\log \mu_{ijk} = \lambda + \lambda_i^X + \lambda_j^Y + \lambda_k^Z + \lambda_{ik}^{XZ} + \lambda_{jk}^{YZ}$.
Moreover (a) $\Rightarrow$ (b) $\Rightarrow$ (c).

Proof. By Lemma 8.5 it suffices to match each independence statement with a product form. Write $\mu = n\pi$.

Step 1 (c). If (8.4) holds, $\mu_{ijk} = \psi_1(i,k)\psi_2(j,k)$ with $\psi_1 = n\pi_{i+k}$ and $\psi_2 = \pi_{+jk}/\pi_{++k}$. Conversely, if $\mu_{ijk} = A_{ik}B_{jk}$, then $\mu_{i+k} = A_{ik}B_{+k}$, $\mu_{+jk} = A_{+k}B_{jk}$ and $\mu_{++k} = A_{+k}B_{+k}$, whence

$$\frac{\mu_{i+k}\mu_{+jk}}{\mu_{++k}} = \frac{A_{ik}B_{+k}\, A_{+k}B_{jk}}{A_{+k}B_{+k}} = A_{ik}B_{jk} = \mu_{ijk},$$

which is (8.4) after dividing by n.

Step 2 (b). If $\pi_{ijk} = \pi_{i+k}\pi_{+j+}$ we have a product $\psi_1(i,k)\psi_2(j)$. Conversely if $\mu_{ijk} = A_{ik}B_j$ then $\mu_{i+k} = A_{ik}B_+$, $\mu_{+j+} = A_{++}B_j$, $\mu_{+++} = A_{++}B_+$, and $\mu_{i+k}\mu_{+j+}/\mu_{+++} = A_{ik}B_j$, i.e. $\pi_{ijk} = \pi_{i+k}\pi_{+j+}$.

Step 3 (a). Same computation with $\mu_{ijk} = A_iB_jC_k$: $\mu_{i++}\mu_{+j+}\mu_{++k}/\mu_{+++}^2 = A_iB_jC_k$.

Step 4. The implications follow because the model in (a) is a special case of (b) (set $\lambda^{XZ} = 0$), which is a special case of (c) (set $\lambda^{YZ} = 0$). ■

Conditional and marginal independence do not imply each other (Simpson's paradox); both hold under the stronger structures (Figure 10).

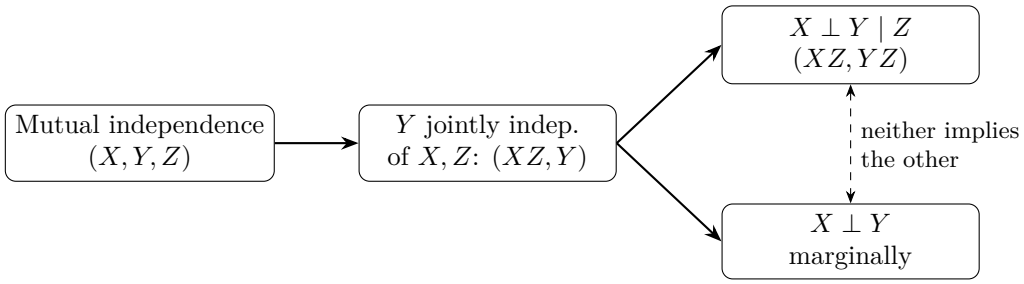


Figure 10: Relationships among types of XY independence (book Figure 8.1).

8.2.2 Homogeneous association and three-factor interaction

Definition (Homogeneous association and the saturated model)

The model of *homogeneous association* (no three-factor interaction) is

$$\log \mu_{ijk} = \lambda + \lambda_i^X + \lambda_j^Y + \lambda_k^Z + \lambda_{ij}^{XY} + \lambda_{ik}^{XZ} + \lambda_{jk}^{YZ}, \tag{8.5}$$

symbol (XY, XZ, YZ); equivalently $\pi_{ijk} = \psi_{ij}\phi_{jk}\omega_{ik}$. The general (saturated) model adds λ_{ijk}^{XYZ}, symbol (XYZ).

The saturated model has

$$1+(I-1)+(J-1)+(K-1)+(I-1)(J-1)+(I-1)(K-1)+(J-1)(K-1)+(I-1)(J-1)(K-1) = IJK$$

parameters, one per cell (expand $IJK = [(I-1)+1][(J-1)+1][(K-1)+1]$). Unlike the models of Theorem 8.6, (8.5) has no closed-form expression of ψ, ϕ, ω in terms of margins.

Model symbol	Association terms	Interpretation
(X, Y, Z)	none	mutual independence
(XY, Z)	λ^{XY}	Z jointly independent of X, Y
(XY, YZ)	$\lambda^{XY}, \lambda^{YZ}$	$X \perp Z \mid Y$
(XY, XZ, YZ)	all two-factor	homogeneous association
(XYZ)	all, incl. λ^{XYZ}	saturated

8.2.3 Interpreting model parameters

At level k of Z the XY conditional association is described by the $(I-1)(J-1)$ local odds ratios

$$\theta_{ij(k)} = \frac{\mu_{ijk}\,\mu_{i+1,j+1,k}}{\mu_{i,j+1,k}\,\mu_{i+1,j,k}}, \tag{8.6}$$

and similarly $\theta_{i(j)k}$ for XZ and $\theta_{(i)jk}$ for YZ. Conditional independence of X, Y given Z means all $\theta_{ij(k)} = 1$.

Theorem 8.7 (Homogeneous association)

(a) Under (XY, XZ, YZ),

$$\log \theta_{ij(k)} = \lambda_{ij}^{XY} + \lambda_{i+1,j+1}^{XY} - \lambda_{i,j+1}^{XY} - \lambda_{i+1,j}^{XY} \quad \text{for all } k, \tag{8.7}$$

and similarly for XZ and YZ odds ratios.

(b) Conversely, a positive table satisfies (XY, XZ, YZ) if and only if $\theta_{ij(1)} = \cdots = \theta_{ij(K)}$ for all i, j. Equivalently, iff the XZ odds ratios are the same at every j, or iff the YZ odds ratios are the same at every i.

Proof. [Supplement]

Step 1. For (a), substitute (8.5) into $\log \theta_{ij(k)}$. Each of $\lambda, \lambda^X, \lambda^Y, \lambda^Z_k, \lambda^{XZ}_{ik}, \lambda^{XZ}_{i+1,k}, \lambda^{YZ}_{jk}, \lambda^{YZ}_{j+1,k}$ appears once with each sign and cancels; only the XY contrast remains, and it does not involve k.

Step 2. For (b), suppose $\theta_{ij(k)} = \theta_{ij}$ does not depend on k. Let $h_{ijk} = \log \mu_{ijk}$. Telescoping the local log odds ratios over the rectangle $[1, i] \times [1, j]$,

$$h_{ijk} - h_{1jk} - h_{i1k} + h_{11k} = \sum_{a<i} \sum_{b<j} \log \theta_{ab} =: g(i, j),$$

which is free of k. Therefore

$$h_{ijk} = g(i, j) + [h_{1jk} - h_{11k}] + h_{i1k},$$

a sum of a function of (i, j), a function of (j, k) and a function of (i, k). By Lemma 8.5 the table satisfies (XY, XZ, YZ).

Step 3. The symmetric statements follow by permuting the variables; each is equivalent to (XY, XZ, YZ). ■

The three-factor term describes how an odds ratio changes across the third variable. For $2 \times 2 \times 2$, substituting the saturated model gives

$$\log \frac{\theta_{11(1)}}{\theta_{11(2)}} = (\lambda^{XYZ}_{111} + \lambda^{XYZ}_{221} - \lambda^{XYZ}_{121} - \lambda^{XYZ}_{211}) - (\lambda^{XYZ}_{112} + \lambda^{XYZ}_{222} - \lambda^{XYZ}_{122} - \lambda^{XYZ}_{212}),$$

which equals λ^{XYZ}_{111} when second-category parameters are set to 0. When X, Y are binary and second-level parameters are zero, $\log \theta_{11(k)} = \lambda^{XY}_{11}$.

8.2.4 Alcohol, cigarette, and marijuana use example

Example 8.8 (Alcohol, cigarette, marijuana use)

1992 survey of 2276 high-school seniors near Dayton, Ohio: ever used alcohol (A), cigarettes (C), marijuana (M).

		Observed: M		Fit of (AC, AM, CM)	
A	C	yes	no	yes	no
yes	yes	911	538	910.4	538.6
yes	no	44	456	44.6	455.4
no	yes	3	43	3.6	42.4
no	no	2	279	1.4	279.6

Estimated odds ratios under several models:

	Conditional			Marginal		
Model	AC	AM	CM	AC	AM	CM
(A, C, M)	1.0	1.0	1.0	1.0	1.0	1.0
(AC, M)	17.7	1.0	1.0	17.7	1.0	1.0
(AM, CM)	1.0	61.9	25.1	2.7	61.9	25.1
(AC, AM, CM)	7.8	19.8	17.3	17.7	61.9	25.1
(ACM), M=yes	13.8	24.3	17.5	17.7	61.9	25.1
(ACM), M=no	7.7	13.5	9.7			

A term absent from a model gives a conditional odds ratio 1.0; e.g. under (AM, CM),

$$\frac{909.24 \times 0.24}{45.76 \times 4.76} = \frac{438.84 \times 179.84}{555.16 \times 142.16} = 1.0,$$

yet the fitted marginal AC odds ratio is 2.7. Under (AC, AM, CM) the AC conditional odds ratio is 7.8 at both levels of M, $= \exp(\hat{\lambda}_{11}^{AC} + \hat{\lambda}_{22}^{AC} - \hat{\lambda}_{12}^{AC} - \hat{\lambda}_{21}^{AC})$. Estimates depend heavily on the model.

8.3 Inference for loglinear models

Motivation. Fit, needed terms and association sizes are judged by the usual GLM deviance tests and intervals.

8.3.1 Chi-squared goodness-of-fit tests

$X^2 = \sum(n - \hat{\mu})^2/\hat{\mu}$ and $G^2 = 2\sum n \log(n/\hat{\mu})$ test fit, df = cells − nonredundant parameters. AIC $= -2$(maximized log likelihood − number of parameters) amounts to minimizing $G^2 - 2\,\text{df}$.

Example 8.9 (Goodness of fit for the student survey)

Model	G^2	X^2	df	P-value
(A, C, M)	1286.0	1411.4	4	< 0.001
(A, CM)	534.2	505.6	3	< 0.001
(C, AM)	939.6	824.2	3	< 0.001
(M, AC)	843.8	704.9	3	< 0.001
(AC, AM)	497.4	443.8	2	< 0.001
(AC, CM)	92.0	80.8	2	< 0.001
(AM, CM)	187.8	177.6	2	< 0.001
(AC, AM, CM)	0.4	0.4	1	0.54
(ACM)	0.0	0.0	0	–

Only the homogeneous association model fits; it also minimizes AIC.

8.3.2 Inference about conditional associations

Tests about a term compare nested models: $-2(L_0 - L_1) = G^2(M_0) - G^2(M_1)$. For XY conditional independence within (XY, XZ, YZ), $H_0 : \lambda_{ij}^{XY} = 0$ for all $(I-1)(J-1)$ parameters, with statistic

$$G^2[(XZ, YZ) \mid (XY, XZ, YZ)] = G^2(XZ, YZ) - G^2(XY, XZ, YZ), \qquad \text{df} = (I-1)(J-1).$$

This is a model-based alternative to the generalized CMH test.

Example 8.10 (Conditional associations in the student survey)

AC conditional independence: $G^2[(AM, CM) \mid (AC, AM, CM)] = 187.8 - 0.4 = 187.4$, df $= 1$, $P < 0.001$; AM and CM are likewise needed (91.6, 497.0). Baseline coding (last category $= 0$), (AC, AM, CM):

$$\hat{\lambda}_{11}^{AC} = 2.054 \text{ (SE} = 0.174), \quad \hat{\lambda}_{11}^{AM} = 2.986 \text{ (0.465)}, \quad \hat{\lambda}_{11}^{CM} = 2.848 \text{ (0.164)}.$$

These are conditional log odds ratios. 95% Wald intervals: AC $\exp(2.054 \pm 1.96 \times 0.174) =$ (5.5, 11.0), AM (8.0, 49.2), CM (12.5, 23.8). All are strong, the marginal ones stronger still.

With large samples, significance alone is not informative; confidence intervals show whether associations are practically important.

8.4 Loglinear models for higher dimensions

Motivation. The three-way ideas extend to more variables, at the cost of many possible terms and sparse tables.

8.4.1 Four-way contingency tables

For variables W, X, Y, Z, the model with all two-factor terms,

$$\begin{aligned}\log \mu_{hijk} = \lambda &+ \lambda_h^W + \lambda_i^X + \lambda_j^Y + \lambda_k^Z \\ &+ \lambda_{hi}^{WX} + \lambda_{hj}^{WY} + \lambda_{hk}^{WZ} + \lambda_{ij}^{XY} + \lambda_{ik}^{XZ} + \lambda_{jk}^{YZ},\end{aligned}$$

symbol (WX, WY, WZ, XY, XZ, YZ), has each pair conditionally dependent with odds ratios constant over the other two variables; removing a two-factor term gives conditional independence of that pair. Under (WXY, WZ, XZ, YZ), the WX, WY, XY associations may vary over the third member of the triple, while all associations with Z are homogeneous. The saturated model adds all three-factor terms and λ^{WXYZ}.

Proposition 8.11 (Odds ratios in models without a variable in higher-order terms)

If a variable V appears in no term of order three or more, then for every other variable U the conditional UV odds ratios are the same at every combination of levels of the remaining variables, and they equal the exponentiated contrast of λ^{UV}.

Proof. [Supplement] Fix the other variables and form $\mu_{ab\cdot}\mu_{cd\cdot}/(\mu_{ad\cdot}\mu_{cb\cdot})$ (U levels a, c; V levels b, d). In the log, a term without V appears at $U = a$ once with each sign (in μ_{ab}, μ_{ad}), likewise at $U = c$, so it cancels; terms without U cancel the same way. By hypothesis the only term with both is λ^{UV}, free of the other variables, so the log odds ratio is $\lambda_{ab}^{UV} + \lambda_{cd}^{UV} - \lambda_{ad}^{UV} - \lambda_{cb}^{UV}$ at every level combination. ■

8.4.2 Automobile accident example

Example 8.12 (Automobile accidents and seat belts)

Maine, 1991: 68,694 auto-accident passengers by gender (G), location (L: urban, rural), seat-belt use (S), injury (I).

Gender	Location	Seat belt	Injury no	Injury yes
Female	Urban	No	7287	996
		Yes	11587	759
	Rural	No	3246	973
		Yes	6134	757
Male	Urban	No	10381	812
		Yes	10969	380
	Rural	No	6123	1084
		Yes	6693	513

Goodness of fit:

Model	G^2	df	P
(G, I, L, S)	2792.8	11	< 0.0001
(GI, GL, GS, IL, IS, LS)	23.4	5	< 0.001
(GIL, GIS, GLS, ILS)	1.3	1	0.25
(GIL, GS, IS, LS)	18.6	4	0.001
(GIS, GL, IL, LS)	22.8	4	< 0.001
(GLS, GI, IL, IS)	7.5	4	0.11
(ILS, GI, GL, GS)	20.6	4	< 0.001

Estimated conditional odds ratios:

Odds ratio	(GI, GL, GS, IL, IS, LS)	(GLS, GI, IL, IS)
GI	0.58	0.58
IL	2.13	2.13
IS	0.44	0.44
GL (S=no / yes)	1.23 / 1.23	1.33 / 1.17
GS (L=urban / rural)	0.63 / 0.63	0.66 / 0.58
LS (G=female / male)	1.09 / 1.09	1.17 / 1.03

Pairwise model: IS conditional log odds ratio -0.814 (SE 0.028), odds ratio interval $\exp(-0.814 \pm 1.96 \times 0.028) = (0.42, 0.47)$: seat belts more than halve the odds of injury in each gender–location group. Injury is more likely in rural accidents and for females; males have 0.63 times the odds of females of wearing belts. In (GLS, GI, IL, IS), I is in no three-factor term, so its odds ratios with G, L, S are homogeneous (Proposition 8.11); the GL, GS, LS odds ratios vary with the third variable, e.g. $0.66 = (7273.2 \times 10{,}959.2)/(11{,}632.6 \times 10{,}358.9)$ for urban accidents.

8.4.3 Large samples and statistical versus practical significance

(GLS, GI, IL, IS) fits better than the pairwise model ($23.4 - 7.5 = 15.9$, df $= 1$, $P = 0.0001$), yet the fitted GL, GS, LS odds ratios barely change across the third variable: the significance reflects the huge n. With large samples focus on estimation; the pairwise model is adequate for most purposes.

8.4.4 Dissimilarity index

Definition (Dissimilarity index)

For cell counts $n_i = np_i$ and fitted values $\hat{\mu}_i = n\hat{\pi}_i$,

$$\hat{\Delta} = \frac{\sum_i |n_i - \hat{\mu}_i|}{2n} = \frac{1}{2}\sum_i |p_i - \hat{\pi}_i|.$$

It estimates the population index $\Delta = \frac{1}{2}\sum |\pi_i - \pi_i^{\text{model}}|$.

Proposition 8.13 (Meaning of $\hat{\Delta}$)

If $\sum \hat{\mu}_i = n$ (true for any loglinear model with an intercept), then $0 \le \hat{\Delta} \le 1$, and $n\hat{\Delta}$ is the minimum number of observations that must be moved to other cells to make the observed table equal to the fitted one.

Proof. [Supplement]

Step 1. $\sum |p_i - \hat{\pi}_i| \le \sum p_i + \sum \hat{\pi}_i = 2$, so $\hat{\Delta} \le 1$; clearly $\hat{\Delta} \ge 0$.

Step 2. Let $d_i = n_i - \hat{\mu}_i$. Since $\sum d_i = 0$, $\sum d_i^+ = \sum d_i^- = \frac{1}{2}\sum |d_i| = n\hat{\Delta}$.

Step 3. One move changes $\sum d_i^+$ by at most 1, so at least $\sum d_i^+$ moves are needed; moving d_i^+ out of each surplus cell into the deficit cells (total $\sum d_i^-$, equal) suffices (fractional moves allowed). So the minimum is $n\hat{\Delta}$. ■

$\hat{\Delta} < 0.02$ or 0.03 means the model fits closely in practice even if rejected; $\hat{\Delta}$ overestimates a Δ near 0, especially for small n. Accident data: $\hat{\Delta} = 0.008$ (pairwise), 0.003 $((GLS, GI, IL, IS))$: moving under 1% of the data gives a perfect fit.

8.5 Loglinear–logit model connection

Motivation. Every logit model with categorical predictors is a loglinear model containing the full interaction among the predictors.

8.5.1 Using logit models to interpret loglinear models

Proposition 8.14 (Logit form of (XY, XZ, YZ))

If Y is binary and (XY, XZ, YZ) holds, then

$$\text{logit}\, P(Y = 1 \mid X = i, Z = k) = \alpha + \beta_i^X + \beta_k^Z, \tag{8.8}$$

with $\alpha = \lambda_1^Y - \lambda_2^Y$, $\beta_i^X = \lambda_{i1}^{XY} - \lambda_{i2}^{XY}$, $\beta_k^Z = \lambda_{1k}^{YZ} - \lambda_{2k}^{YZ}$.

Proof. The logit is $\log(\mu_{i1k}/\mu_{i2k})$. Subtracting (8.5) at $j = 2$ from (8.5) at $j = 1$, the terms $\lambda, \lambda_i^X, \lambda_k^Z, \lambda_{ik}^{XZ}$ cancel, leaving

$$(\lambda_1^Y - \lambda_2^Y) + (\lambda_{i1}^{XY} - \lambda_{i2}^{XY}) + (\lambda_{1k}^{YZ} - \lambda_{2k}^{YZ}).$$

■

The λ^{XZ} terms, which describe association among explanatory variables, cancel: the logit model $(X + Z)$ says nothing about it.

8.5.2 Auto accident example revisited

Example 8.15 (Injury as a response)

For (GLS, GI, IL, IS), the logit on injury cancels all terms without I, including λ^{GLS}, giving the logit model $(G + L + S)$:

$$\operatorname{logit} P(I = 1 \mid G = g, L = l, S = s) = \alpha + \beta_g^G + \beta_l^L + \beta_s^S, \qquad \beta_s^S = \lambda_{1s}^{IS} - \lambda_{2s}^{IS}.$$

Fitted values, G^2, df and standardized residuals are identical. The seat-belt log odds ratio is $\hat\beta_1^S - \hat\beta_2^S = -0.817$ in the logit model and $\hat\lambda_{11}^{IS} + \hat\lambda_{22}^{IS} - \hat\lambda_{12}^{IS} - \hat\lambda_{21}^{IS} = -0.817$ in the loglinear one. The loglinear model treats the 16 counts as Poisson; the logit model fixes the GLS margin and treats $\{n_{g1ls}\}$ as binomial.

8.5.3 Correspondence between loglinear and logit models

The logit on Y in (XY, YZ) is also (8.8), but (XY, YZ) additionally forces $X \perp Z \mid Y$, so it is *not* equivalent: the loglinear model must contain the full interaction among the explanatory variables.

Theorem 8.16 (Loglinear–logit equivalence)

Let Y be the response and $\mathbf{x}$ index the combinations of the explanatory variables. Consider a loglinear model containing the full interaction term among the explanatory variables, $\log \mu_{\mathbf{x}j} = \gamma_{\mathbf{x}} + \eta_{\mathbf{x}j}(\boldsymbol{\beta})$, where $\gamma_{\mathbf{x}}$ is unrestricted and $\eta_{\mathbf{x}j}$ collects the terms involving Y (with $\eta_{\mathbf{x}J} = 0$). Then the ML fitted values under Poisson sampling coincide with those of the baseline-category logit model $\log(\pi_{j|\mathbf{x}}/\pi_{J|\mathbf{x}}) = \eta_{\mathbf{x}j}(\boldsymbol{\beta})$ fitted with the margins $n_{\mathbf{x}+}$ fixed, and the ML estimates of $\boldsymbol{\beta}$ agree.

Proof. [Supplement]

Step 1. Reparametrize: $\mu_{\mathbf{x}+} = e^{\gamma_{\mathbf{x}}} \sum_j e^{\eta_{\mathbf{x}j}}$ and $\pi_{j|\mathbf{x}} = e^{\eta_{\mathbf{x}j}} / \sum_h e^{\eta_{\mathbf{x}h}}$, so $\mu_{\mathbf{x}j} = \mu_{\mathbf{x}+}\pi_{j|\mathbf{x}}$. Since $\gamma_{\mathbf{x}}$ is free, $(\{\mu_{\mathbf{x}+}\}, \boldsymbol{\beta})$ ranges over $(0,\infty)^{\#\mathbf{x}} \times \{\text{all } \boldsymbol{\beta}\}$, a product set (variation independence).

Step 2. The Poisson log likelihood splits:

$$\sum_{\mathbf{x},j} (n_{\mathbf{x}j} \log \mu_{\mathbf{x}j} - \mu_{\mathbf{x}j}) = \sum_{\mathbf{x}} (n_{\mathbf{x}+} \log \mu_{\mathbf{x}+} - \mu_{\mathbf{x}+}) + \sum_{\mathbf{x},j} n_{\mathbf{x}j} \log \pi_{j|\mathbf{x}}(\boldsymbol{\beta}).$$

Step 3. The two parts depend on separate parameters, so they are maximized separately: $\hat\mu_{\mathbf{x}+} = n_{\mathbf{x}+}$, and $\hat{\boldsymbol{\beta}}$ maximizes the second part, which is exactly the product-multinomial (binomial if $J = 2$) log likelihood of the logit model. Hence $\hat\mu_{\mathbf{x}j} = n_{\mathbf{x}+}\hat\pi_{j|\mathbf{x}}$ in both models. ■

Loglinear symbol	Logit model	Logit symbol
(Y, XZ)	α	$(-)$
(XY, XZ)	$\alpha + \beta_i^X$	(X)
(YZ, XZ)	$\alpha + \beta_k^Z$	(Z)
(XY, YZ, XZ)	$\alpha + \beta_i^X + \beta_k^Z$	$(X + Z)$
(XYZ)	$\alpha + \beta_i^X + \beta_k^Z + \beta_{ik}^{XZ}$	$(X*Z)$

With several responses the loglinear approach is more general; with one, logit models are more natural.

8.5.4 Generalized loglinear model

Write the counts as a vector $\mathbf{n} = (n_1, \ldots, n_N)'$ with means $\boldsymbol{\mu}$. An ordinary loglinear model is

$$\log \boldsymbol{\mu} = \mathbf{X}\boldsymbol{\beta}. \tag{8.9}$$

For instance, independence in a 2×2 table with $\lambda_2^X = \lambda_2^Y = 0$ is

$$\begin{pmatrix} \log \mu_{11} \\ \log \mu_{12} \\ \log \mu_{21} \\ \log \mu_{22} \end{pmatrix} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} \lambda \\ \lambda_1^X \\ \lambda_1^Y \end{pmatrix}.$$

The *generalized loglinear model* is $\mathbf{C}\log(\mathbf{A}\boldsymbol{\mu}) = \mathbf{X}\boldsymbol{\beta}$. With $\mathbf{A} = \mathbf{I}_4$, $\mathbf{C} = \begin{pmatrix} 1 & -1 & 0 & 0 \\ 0 & 0 & 1 & -1 \end{pmatrix}$, $\mathbf{X} = (1, 1)'$, $\boldsymbol{\beta} = \alpha$, both row logits equal α: independence. The class includes logit models and marginal models.

8.6 Loglinear model fitting: likelihood equations

Motivation. Fitted values match marginal tables because those margins are the sufficient statistics of a Poisson exponential family.

Derivations use Poisson sampling, which needs no constraint on parameters; Theorem 8.21 transfers them to multinomial sampling.

8.6.1 Minimal sufficient statistics

The kernel of the Poisson log likelihood for a three-way table is

$$L(\boldsymbol{\mu}) = \sum_{i,j,k} n_{ijk} \log \mu_{ijk} - \sum_{i,j,k} \mu_{ijk}. \tag{8.10}$$

Substituting the saturated model,

$$\begin{aligned} L = n\lambda + \sum_i n_{i++}\lambda_i^X + \sum_j n_{+j+}\lambda_j^Y + \sum_k n_{++k}\lambda_k^Z + \sum_{i,j} n_{ij+}\lambda_{ij}^{XY} + \sum_{i,k} n_{i+k}\lambda_{ik}^{XZ} \\ + \sum_{j,k} n_{+jk}\lambda_{jk}^{YZ} + \sum_{i,j,k} n_{ijk}\lambda_{ijk}^{XYZ} - \sum_{i,j,k} \exp(\lambda + \lambda_i^X + \cdots + \lambda_{ijk}^{XYZ}). \end{aligned} \tag{8.11}$$

Theorem 8.17 (Sufficient statistics are marginal tables)

For a hierarchical loglinear model, the minimal sufficient statistics are the marginal tables corresponding to the terms in the model symbol: e.g. $\{n_{i++}\}, \{n_{+j+}\}, \{n_{++k}\}$ for (X, Y, Z); $\{n_{ij+}\}, \{n_{++k}\}$ for (XY, Z); $\{n_{ij+}\}, \{n_{+jk}\}$ for (XY, YZ); $\{n_{ij+}\}, \{n_{i+k}\}, \{n_{+jk}\}$ for (XY, XZ, YZ).

Proof.

Step 1. With absent parameters zero in (8.11), the family is exponential with canonical statistics the marginal tables of the terms present; they are sufficient (factorization) and minimal (full rank after removing redundant parameters).

Step 2. A lower-order margin is a sum of a higher-order one (e.g. $n_{i++} = \sum_j n_{ij+}$), so only the margins of the highest-order terms (the generators) are needed. ■

For the saturated model the full table is sufficient: no data reduction.

8.6.2 Likelihood equations for loglinear models

In the matrix form (8.9), $\log \mu_i = \sum_j x_{ij}\beta_j$ and

$$L(\boldsymbol{\beta}) = \sum_i n_i\Big(\sum_j x_{ij}\beta_j\Big) - \sum_i \exp\Big(\sum_j x_{ij}\beta_j\Big). \tag{8.12}$$

Theorem 8.18 (Likelihood equations match margins)

(a) The likelihood equations are

$$\mathbf{X}'\mathbf{n} = \mathbf{X}'\hat{\boldsymbol{\mu}}\,: \tag{8.13}$$

sufficient statistics equal their fitted (expected) values.
(b) For a hierarchical model, (8.13) is equivalent to equality of the observed and fitted marginal tables in the model symbol. For (XZ, YZ):

$$\hat{\mu}_{i+k} = n_{i+k} \text{ for all } i,k, \qquad \hat{\mu}_{+jk} = n_{+jk} \text{ for all } j,k. \tag{8.14}$$

Proof.

Step 1. Since $\partial \exp(\sum_j x_{ij}\beta_j)/\partial\beta_j = x_{ij}\mu_i$,

$$\frac{\partial L}{\partial \beta_j} = \sum_i n_i x_{ij} - \sum_i \mu_i x_{ij}, \qquad j = 1,\ldots,p.$$

Setting these to 0 gives (8.13).

Step 2. (8.13) says $\mathbf{v}'\mathbf{n} = \mathbf{v}'\hat{\boldsymbol{\mu}}$ for all $\mathbf{v} \in \mathrm{col}(\mathbf{X})$. By Lemma 8.5 this space consists of the functions $\sum_r g_r(\mathbf{x}_{C_r})$, spanned by the indicators $\mathbf{v} = \mathbb{1}\{\mathbf{x}_{C_r} = \mathbf{c}\}$, for which $\mathbf{v}'\mathbf{n}$ is the marginal count at $\mathbf{c}$. So (8.13) is equality of all generator margins; for (XZ, YZ) this is (8.14). ■

So fitted values agree with the data on certain margins while their associations follow the model.

8.6.3 Birch's results for loglinear models

Theorem 8.19 (Birch)

There is at most one $\hat{\boldsymbol{\mu}}$ that both satisfies the loglinear model and matches the data in the minimal sufficient statistics, and if it exists it is the ML estimate. Conversely, any ML estimate has these two properties.

Proof. [Supplement]

Step 1. Write $\boldsymbol{\eta} = \log \boldsymbol{\mu} \in \mathcal{V} = \mathrm{col}(\mathbf{X})$. As a function of $\boldsymbol{\eta}$, $L = \mathbf{n}'\boldsymbol{\eta} - \sum_i e^{\eta_i}$ is strictly concave on $\mathbb{R}^N$: its Hessian is $-\mathrm{diag}(e^{\eta_i})$, negative definite. Its restriction to the subspace $\mathcal{V}$ is still strictly concave.

Step 2. A point $\boldsymbol{\eta} \in \mathcal{V}$ is stationary for $L|_{\mathcal{V}}$ iff the gradient $\mathbf{n} - \boldsymbol{\mu}$ is orthogonal to $\mathcal{V}$, i.e. $\mathbf{X}'(\mathbf{n} - \boldsymbol{\mu}) = \mathbf{0}$, which is margin matching by Theorem 8.18.

Step 3. A stationary point of a differentiable concave function is a global maximum, unique under strict concavity. So a model $\boldsymbol{\mu}$ matching the margins is the unique MLE; conversely the (interior) MLE is stationary, hence matches the margins. ■

E.g. for two-way independence, $\hat{\mu}_{ij} = n_{i+}n_{+j}/n$ satisfies the model and the margins, so it is the MLE.

8.6.4 Direct versus iterative calculation of fitted values

Theorem 8.20 (Closed-form (direct) estimates)

The ML fitted values for three-way models other than (XY, XZ, YZ) are:

Model	Probabilistic form	Fitted value $\hat{\mu}_{ijk}$
(X, Y, Z)	$\pi_{i++}\pi_{+j+}\pi_{++k}$	$n_{i++}n_{+j+}n_{++k}/n^2$
(XY, Z)	$\pi_{ij+}\pi_{++k}$	$n_{ij+}n_{++k}/n$
(XY, XZ)	$\pi_{ij+}\pi_{i+k}/\pi_{i++}$	$n_{ij+}n_{i+k}/n_{i++}$
(XY, XZ, YZ)	$\psi_{ij}\phi_{jk}\omega_{ik}$	iterative (IPF)
(XYZ)	none	n_{ijk}

(other models by symmetry, e.g. (XZ, YZ): $\hat{\mu}_{ijk} = n_{i+k}n_{+jk}/n_{++k}$).

Proof. Take (XZ, YZ); the others are identical in structure.

Step 1. By Theorem 8.6, under the model $\mu_{ijk} = \mu_{i+k}\mu_{+jk}/\mu_{++k}$.

Step 2. Define $\hat{\mu}_{ijk} = n_{i+k}n_{+jk}/n_{++k}$. It has the product form $A_{ik}B_{jk}$, so it satisfies the model.

Step 3. Summing over j: $\hat{\mu}_{i+k} = n_{i+k}n_{++k}/n_{++k} = n_{i+k}$; summing over i: $\hat{\mu}_{+jk} = n_{+jk}$. Thus it matches the sufficient margins (8.14).

Step 4. By Birch's theorem it is the MLE. For (X, Y, Z), (XY, Z), (XY, XZ) the displayed fits likewise have the factorized form and reproduce the sufficient margins (e.g. for (XY, XZ): $\sum_k n_{ij+}n_{i+k}/n_{i++} = n_{ij+}$ and $\sum_j n_{ij+}n_{i+k}/n_{i++} = n_{i+k}$). For (XYZ), $\hat{\mu} = \mathbf{n}$ trivially. ■

Models with closed-form estimates are *direct* (*decomposable*: products and ratios of fitted margins). Unsaturated models with all two-factor terms, such as (XY, XZ, YZ), are not; software iterates anyway.

8.6.5 Connection between multinomial and Poisson loglinear models

Theorem 8.21 (Poisson and multinomial give the same estimates)

Write the Poisson model as $\log \mu_i = \alpha + \mathbf{x}_i'\boldsymbol{\beta}$ (α the intercept). Then the Poisson log likelihood decomposes as

$$L(\alpha, \boldsymbol{\beta}) = \Big[\sum_i n_i \log \pi_i(\boldsymbol{\beta})\Big] + [n \log \tau - \tau], \qquad \pi_i(\boldsymbol{\beta}) = \frac{e^{\mathbf{x}_i'\boldsymbol{\beta}}}{\sum_h e^{\mathbf{x}_h'\boldsymbol{\beta}}}, \tag{8.15}$$

where $\tau = \sum_i \mu_i$. The first bracket is the multinomial log likelihood given n, the second the Poisson log likelihood of n. Consequently $\hat{\boldsymbol{\beta}}$ and its asymptotic covariance are the same under Poisson and multinomial sampling.

Proof.

Step 1. $L = \sum n_i \log \mu_i - \sum \mu_i = \sum n_i(\alpha + \mathbf{x}_i'\boldsymbol{\beta}) - \tau = n\alpha + \sum n_i \mathbf{x}_i'\boldsymbol{\beta} - \tau$.

Step 2. Since $\tau = e^{\alpha} \sum_h e^{\mathbf{x}_h'\boldsymbol{\beta}}$, $\alpha = \log \tau - \log \sum_h e^{\mathbf{x}_h'\boldsymbol{\beta}}$. Substituting,

$$L = \sum_i n_i \mathbf{x}_i'\boldsymbol{\beta} - n \log \sum_h e^{\mathbf{x}_h'\boldsymbol{\beta}} + n \log \tau - \tau = \sum_i n_i \log \pi_i(\boldsymbol{\beta}) + (n \log \tau - \tau).$$

Step 3. The map $(\alpha, \boldsymbol{\beta}) \mapsto (\tau, \boldsymbol{\beta})$ is one-to-one onto $(0, \infty) \times \mathbb{R}^p$, and $\boldsymbol{\beta}$ enters only the first bracket. Hence $\hat{\boldsymbol{\beta}}$ maximizes the multinomial log likelihood, $\hat{\tau} = n$, and the information matrix is block

diagonal in $(\tau, \boldsymbol{\beta})$, so the $\boldsymbol{\beta}$ block of the inverse information is the same for both sampling schemes. ■

The same holds for several independent multinomial samples if the model contains the term of the fixed margin (e.g. λ^{XZ} for fixed $\{n_{i+k}\}$), forcing $\hat{\mu}_{i+k} = n_{i+k}$ (Theorem 8.16). For one multinomial sample, $\widehat{\text{cov}}(\hat{\boldsymbol{\beta}}) = \{\mathbf{X}'[\text{diag}(\hat{\boldsymbol{\mu}}) - \hat{\boldsymbol{\mu}}\hat{\boldsymbol{\mu}}'/n]\mathbf{X}\}^{-1}$ ($\mathbf{X}$ without intercept), the Poisson covariance $\{\mathbf{X}' \text{diag}(\hat{\boldsymbol{\mu}})\mathbf{X}\}^{-1}$ with the α row and column deleted.

Chapter 9. Building and Extending Loglinear–Logit Models

Association graphs and collapsibility (Section 9.1), model selection (Section 9.2), residuals (Section 9.3), ordinal models (Section 9.4) and rate models (Section 9.5).

9.1 Association graphs and collapsibility

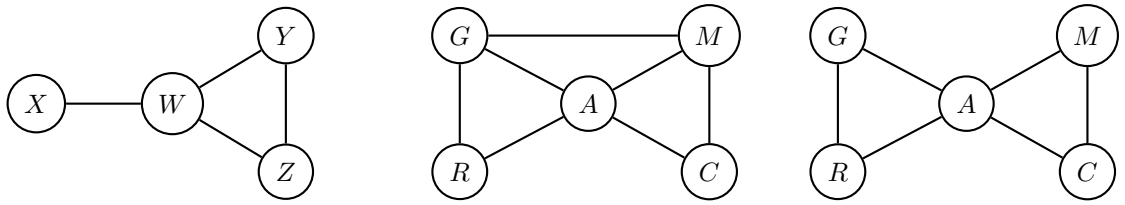
Motivation. Drawing a model as a graph turns conditional independence and collapsibility into questions about paths.

9.1.1 Association graphs

Definition (Association graph, path, separation)

The *association graph* of a loglinear model has a vertex per variable and an edge between two variables iff some term involves both; a *path* is a sequence of edges. A set S *separates* X and Y if every path between them passes through S.

Models with the same pairwise associations have the same graph; e.g. (WX, WY, WZ, YZ) and (WX, WYZ) share Figure 11.



(WX, WY, WZ, YZ) model 6: $(AC, AM, CM, AG, AR, GM, GR)$ model 7: (AC, AM, CM, AG, AR, GR)

Figure 11: Association graphs. Left: book Figure 9.1, W separates X from $\{Y, Z\}$. Middle, right: models for the five-way survey (Example 9.5); in model 7, A separates $\{G, R\}$ from $\{C, M\}$.

Theorem 9.1 (Separation implies conditional independence)

In a hierarchical loglinear model, if a set S of variables separates X and Y in the association graph, then X and Y are conditionally independent given S (after marginalizing over all other variables). More generally, if S separates sets A_0 and B_0, then $A_0 \perp B_0 \mid S$.

Proof. [Supplement]

Step 1. Let V be the set of all variables, A the set of vertices reachable from X by paths avoiding S, and $B = V \setminus (A \cup S)$. Since S separates X and Y, $Y \in B$.

Step 2. Every term T lies in $A \cup S$ or in $B \cup S$: if $T \ni a, b$ with $a \in A$, $b \in B$, hierarchy gives the term $\{a, b\}$, an edge, so $b \in A$, a contradiction.

Step 3. Grouping terms accordingly, $\mu(\mathbf{x}) = f(\mathbf{x}_A, \mathbf{x}_S)\, g(\mathbf{x}_B, \mathbf{x}_S)$ (assign terms inside S to f). Hence, for fixed $\mathbf{x}_S$,

$$P(\mathbf{x}_A, \mathbf{x}_B \mid \mathbf{x}_S) = \frac{f(\mathbf{x}_A, \mathbf{x}_S)\, g(\mathbf{x}_B, \mathbf{x}_S)}{\sum_{\mathbf{a}} f(\mathbf{a}, \mathbf{x}_S) \sum_{\mathbf{b}} g(\mathbf{b}, \mathbf{x}_S)},$$

which factors into a function of $\mathbf{x}_A$ times a function of $\mathbf{x}_B$: $\mathbf{x}_A \perp \mathbf{x}_B \mid \mathbf{x}_S$.

Step 4. Summing over the coordinates of A other than X and the coordinates of B other than Y preserves the product form, so $X \perp Y \mid S$. The statement for sets $A_0 \subseteq A$, $B_0 \subseteq B$ is the same computation. ■

In Figure 11 (left), W separates X and Y, so $X \perp Y \mid W$ (and also given $\{W, Z\}$); likewise $X \perp Z \mid W$.

9.1.2 Collapsibility in three-way contingency tables

Theorem 9.2 (Collapsibility conditions)

In a three-way table the XY marginal and conditional odds ratios are equal if $Z \perp X \mid Y$ or $Z \perp Y \mid X$, i.e. under (XY, YZ) or (XY, XZ) (or simpler models).

Proof. [Supplement]

Step 1. Under (XY, XZ), by Lemma 8.5, $\mu_{ijk} = A_{ij}C_{ik}$. The conditional odds ratio at level k is

$$\frac{A_{ij}C_{ik}\,A_{i'j'}C_{i'k}}{A_{ij'}C_{ik}\,A_{i'j}C_{i'k}} = \frac{A_{ij}A_{i'j'}}{A_{ij'}A_{i'j}},$$

free of k. The marginal table is $\mu_{ij+} = A_{ij}C_{i+}$, and the row factors $C_{i+}, C_{i'+}$ cancel in the odds ratio, giving the same value.

Step 2. Under (XY, YZ), $\mu_{ijk} = A_{ij}B_{jk}$ and $\mu_{ij+} = A_{ij}B_{j+}$; now column factors cancel, with the same conclusion. ■

Example 9.3 (Collapsibility in the student survey)

In (AM, CM) (graph A–M–C) the AM association is collapsible over C: conditional and marginal fitted odds ratios are both 61.9 (Example 8.8); likewise CM (25.1). AC is not (M is linked to both): conditional 1.0, marginal 2.7. In (AC, AM, CM) nothing is collapsible.

9.1.3 Collapsibility and logit models

Multicenter trial (treatment x, response Y, centers Z): $\operatorname{logit} P(Y = 1 \mid X = i, Z = k) = \alpha + \beta x_i + \beta_k^Z$ is (XY, XZ, YZ), so e^{β} typically differs from the marginal odds ratio. Without center effects the model is (XY, XZ), collapsible over Z; so with negligible center effects the treatment effect is about the marginal odds ratio.

9.1.4 Collapsibility and association graphs for multiway tables

Theorem 9.4 (Parametric collapsibility, Bishop et al.)

Partition the variables into disjoint A, B, C with B separating A and C. After collapsing over C, the parameters within A and between A and B are unchanged.

Proof. [Supplement]

Step 1. No term meets both A and C (an A–C edge would avoid B). Put terms meeting A into f, the rest into g; terms of f lie in $A \cup B$, those of g in $B \cup C$:

$$\mu(\mathbf{a}, \mathbf{b}, \mathbf{c}) = f(\mathbf{a}, \mathbf{b})\, g(\mathbf{b}, \mathbf{c}), \qquad \log f = \sum_{T \cap A \neq \emptyset} \lambda^T.$$

Step 2. Collapsing over C: $\mu(\mathbf{a},\mathbf{b}) = f(\mathbf{a},\mathbf{b})\,G(\mathbf{b})$ with $G(\mathbf{b}) = \sum_{\mathbf{c}} g(\mathbf{b},\mathbf{c})$. Thus

$$\log \mu(\mathbf{a},\mathbf{b}) = \sum_{T \cap A \neq \emptyset} \lambda^T + \log G(\mathbf{b}).$$

Step 3. $\log G(\mathbf{b})$ involves only B. By uniqueness of the decomposition of $\log \mu(\mathbf{a},\mathbf{b})$ under fixed constraints, every term with a variable of A in the collapsed table equals the corresponding λ^T. ■

For (WX, WY, WZ, YZ): $A = \{X\}$, $B = \{W\}$, $C = \{Y, Z\}$ keeps WX after collapsing over Y, Z; $A = \{Y, Z\}$, $B = \{W\}$, $C = \{X\}$ keeps the W, Y, Z associations after collapsing over X. A variable independent of all others can be collapsed over freely. With several variables in B the parameters are unchanged but ML estimates may differ slightly; they agree if the model contains the highest-order term among the variables of B.

9.2 Model selection and comparison

Motivation. Thousands of hierarchical models exist for five variables; inclusion principles and LR comparisons find a simple one that fits.

9.2.1 Considerations in model selection

A model should be complex enough to fit well yet simple to interpret, smoothing rather than overfitting. Guidelines:

(i) Distinguish responses from explanatory variables and concentrate on terms linking responses and linking explanatory variables to responses.

(ii) Include the full interaction among the explanatory variables and any design-fixed margin, so the fit reproduces these totals. For Table 8.8 (responses I, S, predictors G, L) the model should contain GL (e.g. 20,629 urban women, as observed): at least (GL, S, I).

(iii) With a single categorical response, use the equivalent logit model.

(iv) To explore, fit all single-factor, then all two-factor, then all three-factor terms, and so on; use automatic searches with care.

9.2.2 Model building for the Dayton student survey

Example 9.5 (Five-way student survey)

The survey of Example 8.8 also recorded gender (G) and race (R):

		Race = white				Race = other			
		Female		Male		Female		Male	
A	C	M=yes	no	yes	no	yes	no	yes	no
yes	yes	405	268	453	228	23	23	30	19
yes	no	13	218	28	201	2	19	1	18
no	yes	1	17	1	17	0	1	1	8
no	no	1	117	1	133	0	12	0	17

Treating A, C, M as responses and G, R as explanatory, every model contains GR.

Model	G^2	df
1. Mutual independence $+GR$	1325.1	25
2. Homogeneous association (all two-factor terms)	15.3	16
3. All three-factor terms	5.3	6
4g. (2) $-CR$	15.8	17
5. $(AC, AM, CM, AG, AR, GM, GR, MR)$	16.7	18
6. $(AC, AM, CM, AG, AR, GM, GR)$	19.9	19
7. (AC, AM, CM, AG, AR, GR)	28.8	20

Model 3 improves on model 2 by only $15.3 - 5.3 = 10.0$ on 10 df: drop three-factor terms. Backward elimination from model 2 (models 4a–4i; e.g. removing AC gives 201.2, CM 513.5) removes CR (+0.5, df 1), CG (+0.9), MR (+3.2); removing AG would add 5.3 ($P = 0.02$): stop at model 6. (The P-values are not literal, and many counts are small, but G^2 differences remain useful.)

In model 6 (Figure 11, middle) $\{A, M\}$ separates C from $\{G, R\}$: given alcohol and marijuana, cigarette use is independent of gender and race, and by Theorem 9.4 the AC, CM associations equal those of (AC, AM, CM) on the ACM table. Model 7 drops GM, so A separates $\{G, R\}$ from $\{C, M\}$ and all A, C, M associations can be read from the collapsed table. Model 7 fits reasonably ($G^2 = 28.8$, df $= 20$, $\hat{\Delta} = 0.036$).

9.2.3 Loglinear model comparison statistics

Let M_0 be a special case of M_1, with fitted values $\hat{\boldsymbol{\mu}}_0, \hat{\boldsymbol{\mu}}_1$.

Theorem 9.6 (Partitioning G^2)

$$G^2(M_0) = G^2(M_1) + G^2(M_0 \mid M_1), \tag{9.1}$$

where the likelihood-ratio statistic for M_0 against M_1 has the form

$$G^2(M_0 \mid M_1) = G^2(M_0) - G^2(M_1) = 2\sum_i \hat{\mu}_{1i} \log \frac{\hat{\mu}_{1i}}{\hat{\mu}_{0i}}, \tag{9.2}$$

which is the G^2 statistic for M_0 with $\hat{\boldsymbol{\mu}}_1$ playing the role of the data.

Proof.

Step 1. $G^2(M) = 2\sum n_i \log(n_i/\hat{\mu}_i)$, so

$$G^2(M_0) - G^2(M_1) = 2\sum_i n_i(\log \hat{\mu}_{1i} - \log \hat{\mu}_{0i}).$$

Step 2. Write $\log \hat{\boldsymbol{\mu}}_1 = \mathbf{X}_1\hat{\boldsymbol{\beta}}_1$ and, since $M_0 \subseteq M_1$, $\log \hat{\boldsymbol{\mu}}_0 = \mathbf{X}_0\hat{\boldsymbol{\beta}}_0 = \mathbf{X}_1\hat{\boldsymbol{\beta}}_0^*$, where $\hat{\boldsymbol{\beta}}_0^*$ is $\hat{\boldsymbol{\beta}}_0$ with zeros appended for the extra parameters. Then

$$2\mathbf{n}'(\log \hat{\boldsymbol{\mu}}_1 - \log \hat{\boldsymbol{\mu}}_0) = 2\,\mathbf{n}'\mathbf{X}_1(\hat{\boldsymbol{\beta}}_1 - \hat{\boldsymbol{\beta}}_0^*) = 2\,\hat{\boldsymbol{\mu}}_1'\mathbf{X}_1(\hat{\boldsymbol{\beta}}_1 - \hat{\boldsymbol{\beta}}_0^*),$$

using the likelihood equations $\mathbf{X}_1'\mathbf{n} = \mathbf{X}_1'\hat{\boldsymbol{\mu}}_1$ for M_1 (Theorem 8.18). This is $2\sum \hat{\mu}_{1i}\log(\hat{\mu}_{1i}/\hat{\mu}_{0i})$.

Step 3. $G^2(M_0 \mid M_1)$ has the G^2 form. (Note also $\sum \hat{\mu}_{0i} = \sum \hat{\mu}_{1i} = n$ when both models contain an intercept.) Equation (9.1) is the definition rearranged. ■

$G^2(M_0 \mid M_1)$ depends on the data only through $\hat{\boldsymbol{\mu}}_1$. The Pearson difference $X^2(M_0) - X^2(M_1)$ can

be negative; the natural analogue is

$$X^2(M_0 \mid M_1) = \sum_i \frac{(\hat{\mu}_{1i} - \hat{\mu}_{0i})^2}{\hat{\mu}_{0i}}. \tag{9.3}$$

Under M_0 both are asymptotically χ^2 with $\mathrm{df} = \mathrm{df}(M_0) - \mathrm{df}(M_1)$, similar even in fairly sparse tables (Haberman). Under M_1 but not M_0, $G^2(M_1)$ stays χ^2 while the others diverge.

9.2.4 Partitioning chi-squared with model comparisons

Applying (9.1) along nested $M_1 \supset M_2 \supset \cdots$ partitions G^2 into asymptotically independent components. For a $2 \times J$ table let M_j: columns $1, \ldots, j$ form an independence table. Then

$$G^2(M_J) = G^2(M_2) + G^2(M_3 \mid M_2) + \cdots + G^2(M_J \mid M_{J-1}),$$

and $G^2(M_j \mid M_{j-1})$ is the independence G^2 for the 2×2 table of combined columns $1, \ldots, j-1$ versus column j, justifying Section 3.3.3.

For s planned nested tests, level $1 - (1-\alpha)^{1/s}$ each keeps the overall asymptotic type I error at most α; for $s = 3$, $\alpha = 0.05$: 0.017.

9.2.5 Identical marginal and conditional tests of independence

If two direct models differ by one pairwise term, $G^2(M_0 \mid M_1)$ equals the independence G^2 in that pair's marginal table (Sundberg, 1975).

Proposition 9.7 (A conditional test that equals a marginal test)

$G^2[(X, Y, Z) \mid (XY, Z)]$ equals $G^2(X, Y)$ for testing independence in the marginal XY table.

Proof. By Theorem 8.20, $\hat{\mu}_{1,ijk} = n_{ij+}n_{++k}/n$ and $\hat{\mu}_{0,ijk} = n_{i++}n_{+j+}n_{++k}/n^2$, so $\hat{\mu}_1/\hat{\mu}_0 = n\, n_{ij+}/(n_{i++}n_{+j+})$ does not depend on k. By (9.2),

$$G^2 = 2\sum_{i,j,k} \frac{n_{ij+}n_{++k}}{n} \log \frac{n\, n_{ij+}}{n_{i++}n_{+j+}} = 2\sum_{i,j} n_{ij+} \log \frac{n_{ij+}}{n_{i++}n_{+j+}/n},$$

after summing n_{++k}/n over k. This is G^2 for independence in the $\{n_{ij+}\}$ table. ■

This is expected: under (XY, Z) the XY association is collapsible over Z.

9.3 Diagnostics for checking models

Motivation. Residuals show cell by cell where a model fails.

9.3.1 Residuals for loglinear models

Definition (Pearson and standardized residuals)

For cell i the *Pearson residual* is

$$e_i = \frac{n_i - \hat{\mu}_i}{\sqrt{\hat{\mu}_i}}, \qquad \sum_i e_i^2 = X^2. \tag{9.4}$$

The *standardized Pearson residual* is $r_i = e_i/\sqrt{1-\hat{h}_i}$, where $\hat{h}_i$ is the leverage (diagonal of the estimated hat matrix).

Pearson residuals have asymptotic variances below 1 (on average residual df/cells); r_i is asymptotically $N(0,1)$ and preferable. Deviance residuals are an alternative.

9.3.2 Student survey example revisited

Five-way survey: with all two-factor terms the only large standardized residual is 3.2 (count 8, fitted 3.1); for model 6, 3.3 (fitted 2.9) in the same cell (nonwhite males smoking cigarettes but using neither alcohol nor marijuana). With so many cells and large n, neither model looks problematic.

9.3.3 Correspondence between loglinear and logit residuals

Proposition 9.8 (Logit versus loglinear Pearson residuals)

For equivalent logit and loglinear models with binary response, the logit Pearson residual of a success count is the loglinear one divided by $\sqrt{1-\hat{\pi}}$, hence larger.

Proof. Let the cell have count y of N trials. Fitted values agree, $\hat{\mu} = N\hat{\pi}$ (Theorem 8.16); the residuals are $(y - N\hat{\pi})/\sqrt{N\hat{\pi}(1-\hat{\pi})}$ and $(y - N\hat{\pi})/\sqrt{N\hat{\pi}}$, with ratio $1/\sqrt{1-\hat{\pi}} > 1$. ■

Standardized, the two residuals coincide, another reason to prefer them.

9.4 Modeling ordinal associations

Motivation. Nominal models ignore order; for ordinal variables a single trend parameter captures departures concentrated in the corners.

Example 9.9 (Premarital sex and teenage birth control)

1991 GSS. Rows: premarital sex always wrong (1), almost always wrong (2), wrong only sometimes (3), not wrong at all (4). Columns: teenage birth control available, strongly disagree (1) to strongly agree (4). Entries: count, (independence fit), [standardized residual], {L×L fit}.

	SD	D	A	SA
1	81 (42.4) [7.6] {80.9}	68 (51.2) [3.1] {67.6}	60 (86.4) [−4.1] {69.4}	38 (67.0) [−4.8] {29.1}
2	24 (16.0) [2.3] {20.8}	26 (19.3) [1.8] {23.1}	29 (32.5) [−0.8] {31.5}	14 (25.2) [−2.8] {17.6}
3	18 (30.1) [−2.7] {24.4}	41 (36.3) [1.0] {36.1}	74 (61.2) [2.2] {65.7}	42 (47.4) [−1.0] {48.8}
4	36 (70.6) [−6.1] {33.0}	57 (85.2) [−4.6] {65.1}	161 (143.8) [2.4] {157.4}	157 (111.4) [6.8] {155.5}

Independence fits poorly, $G^2(I) = 127.6$, df $= 9$; the corner residuals show a positive trend.

9.4.1 Linear-by-linear association in two-way tables

Definition (Linear-by-linear association (L×L) model)

With ordered row scores $u_1 \le \cdots \le u_I$ and column scores $v_1 \le \cdots \le v_J$,

$$\log \mu_{ij} = \lambda + \lambda_i^X + \lambda_j^Y + \beta u_i v_j, \tag{9.5}$$

the special case $\lambda_{ij}^{XY} = \beta u_i v_j$ of the saturated model: one association parameter instead of $(I-1)(J-1)$. Independence is $\beta = 0$. With $u_i = i$, $v_j = j$ it is the *uniform association* model.

Theorem 9.10 (Odds ratios under L×L)

Under (9.5), for rows $a < c$ and columns $b < d$,

$$\log \frac{\mu_{ab}\mu_{cd}}{\mu_{ad}\mu_{cb}} = \beta(u_c - u_a)(v_d - v_b). \tag{9.6}$$

With unit-spaced scores all local odds ratios are e^{β}. If $\beta > 0$, Y tends to increase with X and μ_{ij} exceeds independence where X, Y are both high or both low.

Proof.

Step 1. By Theorem 8.3(b) the log odds ratio equals $\lambda_{ab}^{XY} + \lambda_{cd}^{XY} - \lambda_{ad}^{XY} - \lambda_{cb}^{XY}$ with $\lambda_{ij}^{XY} = \beta u_i v_j$:

$$\beta(u_a v_b + u_c v_d - u_a v_d - u_c v_b) = \beta(u_c - u_a)(v_d - v_b).$$

Step 2. For adjacent rows and columns with unit spacing, $(u_{i+1} - u_i)(v_{j+1} - v_j) = 1$.

Step 3. For $\beta > 0$ all odds ratios with $a < c$, $b < d$ exceed 1 (likelihood-ratio dependence); $|\beta u_i v_j|$ is largest at the corners. ■

$\beta u_i v_j$ is linear in the column scores within a row (slope βu_i) and in the row scores within a column. Scores are often midpoints or standardized (mean 0, variance 1 under the margins); then β is a log odds ratio per SD, and for an underlying bivariate normal $\beta \approx \rho/(1-\rho^2)$ ($\approx \rho$ when weak).

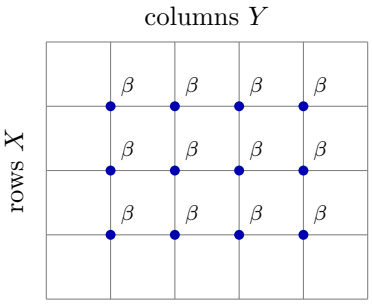


Figure 12: Uniform association: every 2×2 table of adjacent rows and columns has log odds ratio β (book Figure 9.2).

9.4.2 Corresponding logit model for adjacent responses

Proposition 9.11 (Adjacent-categories logit form)

Let $\pi_{j|i} = P(Y = j \mid X = i)$. Under (9.5),

$$\log \frac{\pi_{j+1|i}}{\pi_{j|i}} = (\lambda^Y_{j+1} - \lambda^Y_j) + \beta(v_{j+1} - v_j)u_i,$$

and for unit-spaced $\{v_j\}$ this is $\alpha_j + \beta u_i$: the same linear effect of X on all $J-1$ adjacent-categories logits.

Proof. $\pi_{j+1|i}/\pi_{j|i} = \mu_{i,j+1}/\mu_{ij}$; subtracting (9.5) at j from (9.5) at $j+1$, λ and λ^X_i cancel, leaving the stated expression. ■

9.4.3 Likelihood equations and model fitting

Theorem 9.12 (Likelihood equations for L×L)

The ML fit of (9.5) satisfies

$$\hat\mu_{i+} = n_{i+}, \qquad \hat\mu_{+j} = n_{+j}, \qquad \sum_{i,j} u_i v_j \hat\mu_{ij} = \sum_{i,j} u_i v_j n_{ij}.$$

So fitted and observed distributions share the score means, variances and correlation. Residual df $IJ - I - J$.

Proof.

Step 1. The Poisson log likelihood is

$$L = n\lambda + \sum_i n_{i+}\lambda^X_i + \sum_j n_{+j}\lambda^Y_j + \beta \sum_{i,j} u_i v_j n_{ij} - \sum_{i,j} \exp(\lambda + \lambda^X_i + \lambda^Y_j + \beta u_i v_j).$$

Setting derivatives in λ^X_i, λ^Y_j, β to zero gives the three sets of equations (sufficient statistics: the margins and $\sum u_i v_j n_{ij}$).

Step 2. Dividing by n, the fitted proportions $\hat\pi_{ij}$ and the sample proportions p_{ij} have equal margins, hence equal means and variances of u and of v, and equal $\sum u_i v_j \hat\pi_{ij} = \sum u_i v_j p_{ij}$, hence equal covariance and correlation.

Step 3. The model has $1 + (I-1) + (J-1) + 1$ parameters, so df $= IJ - (I + J)$, unsaturated except for 2×2 tables. ■

The fitted counts therefore reproduce the trend of the data. Fitting uses Newton–Raphson.

9.4.4 Sex opinions example

Example 9.13 (L×L fit to Example 9.9)

Scores $1, 2, 3, 4$: $G^2(\text{L} \times \text{L}) = 11.5$, df $= 8$, versus 127.6 on 9 df, the corners improving most. $\hat\beta = 0.286$ (SE 0.028): local odds ratio 1.33, CI $\exp(0.286 \pm 1.96 \times 0.028) = (1.26, 1.41)$. For the four corner cells, by (9.6),

$$\exp[0.286(4-1)(4-1)] = 13.1 = \frac{80.9 \times 155.5}{29.1 \times 33.0}.$$

Rescaled equal spacing (rows $2, 4, 6, 8$) rescales $\hat{\beta}$ (0.143), same fit. Scores $1, 2, 4, 5$: $G^2 = 8.8$, $\hat{\beta} = 0.146$. Standardized scores: $\hat{\beta} = 0.374$, so $\hat{\rho} = 0.333$ from $\beta = \rho/(1-\rho^2)$.

9.4.5 Directed ordinal test of independence

Within L×L, independence is $H_0 : \beta = 0$, tested by

$$G^2(I \mid \mathrm{L} \times \mathrm{L}) = G^2(I) - G^2(\mathrm{L} \times \mathrm{L}), \qquad \mathrm{df} = 1.$$

For Example 9.9: $127.6 - 11.5 = 116.1$; the Wald statistic $(\hat{\beta}/\mathrm{SE})^2 = 102.5$; the score statistic is the correlation statistic $M^2 = (n-1)r^2 = 112.6$ of Section 3.4.1.

Why the ordinal test is more powerful

Under L×L, $G^2(I \mid \mathrm{L} \times \mathrm{L})$ and $G^2(I)$ have about the same noncentrality but df 1 versus $(I-1)(J-1)$; power falls with df at fixed noncentrality, so the directed test is more powerful, increasingly so with I, J.

9.5 Poisson regression for rates

Motivation. For counts over unequal exposures, an offset models the rate while keeping the Poisson count as the observation.

9.5.1 Analyzing rates using loglinear models with offsets

Proposition 9.14 (Rate model with offset)

For counts n_i with exposure (index) t_i, the model

$$\log(\mu_i/t_i) = \alpha + \beta x_i \tag{9.7}$$

is the Poisson model $\log \mu_i = \log t_i + \alpha + \beta x_i$ with offset $\log t_i$ (coefficient fixed at 1): $\mu_i = t_i \exp(\alpha + \beta x_i)$, and e^{β} is the multiplicative effect on the rate.

Proof. Add $\log t_i$ to both sides of (9.7) and exponentiate. The ratio of rates at $x+1$ and x is $e^{\alpha+\beta(x+1)}/e^{\alpha+\beta x} = e^{\beta}$. ■

The identity link $\mu_i = \alpha t_i + \beta x_i t_i$ needs no offset (predictors t_i, $x_i t_i$, no intercept); effects are additive, and fitting may fail with negative fitted values.

9.5.2 Modeling death rates for heart valve operations

Example 9.15 (Heart valve replacement)

109 patients by valve type and age; time at risk = total follow-up (months) in the cell, deaths and censored alike.

Age		Aortic	Mitral
< 55	Deaths	4	1
	Time at risk	1259	2082
	Death rate	0.0032	0.0005
55+	Deaths	7	9
	Time at risk	1417	1647
	Death rate	0.0049	0.0055

A binary model (died yes/no) would ignore the unequal exposures (3 to 97 months). With dummies a (age 55+) and v (mitral),

$$\log(\mu_{ij}/t_{ij}) = \alpha + \beta_1 a_i + \beta_2 v_j. \tag{9.8}$$

$\hat{\beta}_1 = 1.221$ (SE 0.514), $\hat{\beta}_2 = -0.330$ (SE 0.438). Given valve type the older death rate is $e^{1.221} = 3.4$ times the younger; Wald CI $\exp(1.221 \pm 1.96 \times 0.514) = (1.2, 9.3)$, LR CI $(1.3, 10.4)$. $G^2 = 3.2$, $X^2 = 3.1$, df $= 1$; without valve type $G^2 = 3.8$ (df 2); without age the fit is poor. Fitted rates: < 55: 0.0018 (aortic), 0.0013 (mitral); 55+: 0.0062, 0.0044. Identity link: $G^2 = 1.1$ (df 1), $\hat{\beta}_1 = 0.0040$ (SE 0.0014), an additive rate difference. Only 21 of 109 died, so estimates are imprecise, but all patients contribute through their times at risk.

Chapter 10. Models for Matched Pairs

Matched pairs (repeated measurement, case–control matching, two raters) give a square $I \times I$ table of dependent responses.

10.1 Comparing dependent proportions

Motivation. Matched proportions are correlated; the correct test of equal margins uses only the discordant pairs.

For n pairs let π_{ab} be the probability of outcome a for observation 1 and b for observation 2, n_{ab} the count, $p_{ab} = n_{ab}/n$, and $\{n_{ab}\} \sim \text{multinomial}(n; \{\pi_{ab}\})$.

Definition (Marginal homogeneity)

The square table satisfies **marginal homogeneity** if $\pi_{a+} = \pi_{+a}$ for all a.

For 2×2 tables $\pi_{1+} - \pi_{+1} = \pi_{12} - \pi_{21}$, so marginal homogeneity means $\pi_{12} = \pi_{21}$.

10.1.1 Inference for dependent proportions

Let $\delta = \pi_{+1} - \pi_{1+}$ and $d = p_{+1} - p_{1+} = p_{21} - p_{12}$.

Theorem 10.1 (Variance of a difference of dependent proportions)

Under multinomial sampling,

$$\begin{aligned} n \operatorname{Var}(d) &= \pi_{1+}(1-\pi_{1+}) + \pi_{+1}(1-\pi_{+1}) - 2(\pi_{11}\pi_{22} - \pi_{12}\pi_{21}) \\ &= (\pi_{12} + \pi_{21}) - (\pi_{12} - \pi_{21})^2. \end{aligned}$$

Hence $d \pm z_{\alpha/2}\,\hat{\sigma}(d)$ with $\hat{\sigma}^2(d) = [(p_{12} + p_{21}) - (p_{12} - p_{21})^2]/n$ is a large-sample confidence interval for δ.

Proof.

Step 1. Multinomial moments: $n \operatorname{Var}(p_{ab}) = \pi_{ab}(1 - \pi_{ab})$, $n \operatorname{Cov}(p_{ab}, p_{cd}) = -\pi_{ab}\pi_{cd}$ for distinct cells. Each margin is binomial, and

$$n \operatorname{Cov}(p_{11} + p_{21},\, p_{11} + p_{12}) = \pi_{11}(1 - \pi_{11} - \pi_{12} - \pi_{21}) - \pi_{12}\pi_{21} = \pi_{11}\pi_{22} - \pi_{12}\pi_{21},$$

so $\operatorname{Var}(p_{+1}) + \operatorname{Var}(p_{1+}) - 2\operatorname{Cov}(p_{+1}, p_{1+})$ gives the first line.

Step 2. Directly, $n \operatorname{Var}(p_{21} - p_{12}) = \pi_{21}(1 - \pi_{21}) + \pi_{12}(1 - \pi_{12}) + 2\pi_{12}\pi_{21}$, which is the second line.

Step 3. By the multivariate CLT and consistency of $\hat{\sigma}(d)$, Slutsky's theorem gives the interval. ■

Under $H_0 : \delta = 0$ the variance becomes $(\pi_{12} + \pi_{21})/n$, estimated by $\hat{\sigma}_0^2(d) = (n_{12} + n_{21})/n^2$.

Theorem 10.2 (McNemar's test is the score test)

The score statistic for $H_0 : \pi_{12} = \pi_{21}$ in the multinomial model is z_0^2, where

$$z_0 = \frac{d}{\hat{\sigma}_0(d)} = \frac{n_{21} - n_{12}}{(n_{21} + n_{12})^{1/2}}; \tag{10.1}$$

z_0^2 is asymptotically χ_1^2 under H_0 (**McNemar's test**).

Proof.

Step 1. Reparametrize by $(\pi_{11}, \pi_{22}, \sigma, q)$ with $\sigma = \pi_{12} + \pi_{21}$, $q = \pi_{21}/\sigma$, subject only to $\pi_{11} + \pi_{22} + \sigma = 1$; H_0 is $q = \frac{1}{2}$. With $n^* = n_{12} + n_{21}$,

$$L = n_{11} \log \pi_{11} + n_{22} \log \pi_{22} + n^* \log \sigma + n_{21} \log q + n_{12} \log(1 - q),$$

and q separates from the constrained parameters, so the information is block diagonal and the score test uses only the q-block.

Step 2. $U(q) = n_{21}/q - n_{12}/(1-q)$ and $\mathcal{I}(q) = n\sigma/[q(1-q)]$. At $q = \frac{1}{2}$, $\hat{\sigma} = n^*/n$: $U = 2(n_{21} - n_{12})$, $\mathcal{I} = 4n^*$, and $U^2/\mathcal{I} = (n_{21} - n_{12})^2/n^* = z_0^2$. Also $d/\hat{\sigma}_0(d) = [(n_{21} - n_{12})/n]/[(n^*)^{1/2}/n]$ gives (10.1); the χ_1^2 limit is standard score asymptotics. ■

The diagonal counts do not affect the test, though all pairs enter d and its standard error.

10.1.2 Prime Minister approval rating example

Example 10.3 (Prime Minister approval)

1600 British voters rated the Prime Minister twice, six months apart: $(n_{11}, n_{12}, n_{21}, n_{22}) = (794, 150, 86, 570)$ (1 = approve). $p_{1+} = 0.59$, $p_{+1} = 0.55$, $\hat{\sigma}(d) = 0.0095$, 95% interval for $\pi_{+1} - \pi_{1+}$: $(-0.06, -0.02)$. McNemar $z_0 = (86 - 150)/236^{1/2} = -4.17$: approval dropped.

10.1.3 Increased precision with dependent samples

Independent samples have $n \operatorname{Var}(\hat{\pi}_1 - \hat{\pi}_2) = \pi_1(1 - \pi_1) + \pi_2(1 - \pi_2)$ (no covariance term). Matched data are usually positively dependent ($\pi_{11}\pi_{22} > \pi_{12}\pi_{21}$), reducing $\operatorname{Var}(d)$: SE 0.0095 versus 0.0175 in Example 10.3.

10.1.4 Small-sample test comparing matched proportions

Proposition 10.4 (Exact conditional test)

Conditional on $n^* = n_{12} + n_{21}$, $n_{21} \sim \text{bin}(n^*, \pi_{21}/(\pi_{12} + \pi_{21}))$. Under $H_0 : \pi_{12} = \pi_{21}$ this is $\text{bin}(n^*, \frac{1}{2})$, and the normal approximation to it gives exactly McNemar's z_0.

Proof.

Step 1. A multinomial conditioned on the total of a subset of cells is multinomial on that subset with renormalized probabilities; for the two cells $(1, 2), (2, 1)$ this gives the binomial, with parameter $\frac{1}{2}$ under H_0.

Step 2. With mean $n^*/2$ and variance $n^*/4$, $z = (n_{21} - \frac{1}{2}n^*)/(\frac{1}{4}n^*)^{1/2} = (n_{21} - n_{12})/(n^*)^{1/2} = z_0$. ■

For Example 10.3, $P(\text{bin}(236, \frac{1}{2}) \geq 150) = 0.00002$ (one-sided).

10.1.5 Connection between McNemar and Cochran–Mantel–Haenszel tests

Each pair is its own 2×2 stratum (rows: occasion; columns: outcome; row totals $(1, 1)$): the $2 \times 2 \times n$ **subject-specific table**, versus the **population-averaged table**. Equal success probabilities at both occasions for every subject is conditional independence given subject, implying marginal homogeneity.

Theorem 10.5 (McNemar as a CMH statistic)

The Cochran–Mantel–Haenszel statistic for conditional independence in the $2 \times 2 \times n$ subject-specific table equals $(n_{21} - n_{12})^2/(n_{21} + n_{12})$.

Proof.

Step 1. Let $c_k \in \{0, 1, 2\}$ be the number of successes in stratum k. The hypergeometric moments are $E(n_{11k}) = c_k/2$ and $\mathrm{Var}(n_{11k}) = c_k(2 - c_k)/4$.

Step 2. Concordant strata ($c_k = 0, 2$) have zero variance and $n_{11k} = E(n_{11k})$, so they contribute nothing.

Step 3. Discordant strata have variance $\frac{1}{4}$ and $n_{11k} - \frac{1}{2} = +\frac{1}{2}$ (n_{12} strata) or $-\frac{1}{2}$ (n_{21} strata), so

$$\mathrm{CMH} = \frac{\left[\frac{1}{2}(n_{12} - n_{21})\right]^2}{\frac{1}{4}(n_{12} + n_{21})} = \frac{(n_{21} - n_{12})^2}{n_{21} + n_{12}}.$$

■

With more categories or occasions, generalized CMH tests with one stratum per subject test marginal homogeneity (Section 11.1.4).

10.2 Conditional logistic regression for binary matched pairs

Motivation. Conditioning away one parameter per subject yields n_{21}/n_{12} for the subject-specific odds ratio.

10.2.1 Marginal versus conditional models for matched pairs

Let (Y_1, Y_2) be a subject's pair (1 = success) and $x_1 = 0$, $x_2 = 1$.

Definition (Marginal and conditional models)

A **marginal** (population-averaged) model describes the margins, e.g. logit $P(Y_t = 1) = \alpha + \beta x_t$. A **conditional** (subject-specific) model has link $P(Y_{it} = 1) = \alpha_i + \beta x_t$, $i = 1, \ldots, n$.

In the marginal logit model $\hat{\beta} = \log[p_{+1}p_{2+}/(p_{+2}p_{1+})]$; in the conditional model β is the association in the subject-stratified table.

Proposition 10.6 (Subject-specific versus population-averaged effects)

Let α_i be drawn from a distribution F in the conditional model.
(a) With the identity link, $P(Y_2 = 1) - P(Y_1 = 1) = \beta$; the subject-specific and population-averaged effects coincide.
(b) With the logit link, the population-averaged probabilities are $P(Y_t = 1) = E_F[e^{\alpha+\beta x_t}/(1+e^{\alpha+\beta x_t})]$, which in general is not of the form $e^{a+bx_t}/(1 + e^{a+bx_t})$ with $b = \beta$.

Proof.

(a) Average $(\alpha_i + \beta) - \alpha_i = \beta$ over F.

(b) The first statement is total probability. Take $\alpha = \pm c$ with probability $\frac{1}{2}$ each and $\beta > 0$. Then $E[\mathrm{expit}(\alpha)] = \frac{1}{2}$, so $b = \mathrm{logit}\left(\frac{1}{2}[\mathrm{expit}(c + \beta) + \mathrm{expit}(\beta - c)]\right) \to \mathrm{logit}(\frac{1}{2}) = 0$ as $c \to \infty$, while β is fixed; so $b \neq \beta$ in general. ■

10.2.2 A logit model with subject-specific probabilities

The Cox–Rasch model is

$$\operatorname{logit} P(Y_{it}=1)=\alpha_i+\beta x_t, \qquad x_1=0,\ x_2=1, \tag{10.2}$$

so each subject's odds at occasion 2 are e^{β} times those at occasion 1, responses independent given the parameters.

Proposition 10.7 (Heterogeneity induces positive association)

Suppose $\alpha_1,\dots,\alpha_n$ are i.i.d. from F and, given α_i, Y_{i1},Y_{i2} are independent with the probabilities of (10.2). Then $\operatorname{Cov}(Y_{i1},Y_{i2})\geq 0$ for every β, with equality if and only if F is degenerate.

Proof. [Supplement]

Step 1. With $f(a)=\operatorname{expit}(a)$, $g(a)=\operatorname{expit}(a+\beta)$, conditional independence gives $\operatorname{Cov}(Y_{i1},Y_{i2})=\operatorname{Cov}(f(\alpha),g(\alpha))$, $\alpha\sim F$.

Step 2. For an independent copy α', f,g strictly increasing give $(f(\alpha)-f(\alpha'))(g(\alpha)-g(\alpha'))\geq 0$, whose expectation is $2\operatorname{Cov}(f(\alpha),g(\alpha))$.

Step 3. Equality forces $\alpha=\alpha'$ a.s., possible for independent copies only if F is degenerate. ■

The model captures within-pair dependence, but its many α_i ruin ordinary ML (Proposition 10.9); use *conditional ML*.

10.2.3 Conditional ML inference for binary matched pairs

Theorem 10.8 (Conditional ML estimator of the subject-specific odds ratio)

In model (10.2) let $S_i=y_{i1}+y_{i2}$.
(a) S_i is sufficient for α_i. Given $S_i=0$ or $S_i=2$ the pair is determined; given $S_i=1$,

$$P(Y_{i1}=0,Y_{i2}=1\mid S_i=1)=\frac{e^{\beta}}{1+e^{\beta}}, \qquad P(Y_{i1}=1,Y_{i2}=0\mid S_i=1)=\frac{1}{1+e^{\beta}}.$$

(b) The conditional likelihood is $\ell_c(\beta)=(e^{\beta})^{n_{21}}/(1+e^{\beta})^{n^*}$ with $n^*=n_{12}+n_{21}$, i.e. $n_{21}\mid n^*\sim\operatorname{bin}(n^*,e^{\beta}/(1+e^{\beta}))$.
(c) The conditional ML estimator and its standard error are

$$\hat\beta=\log\frac{n_{21}}{n_{12}}, \qquad \mathrm{SE}=\Big(\frac{1}{n_{21}}+\frac{1}{n_{12}}\Big)^{1/2}.$$

Proof.

Step 1. The joint mass function is proportional to $\exp\big(\sum_i \alpha_i S_i+\beta\sum_i y_{i2}\big)$, an exponential family with S_i sufficient for α_i.

Step 2. Given $S_i\in\{0,2\}$ the pair is fixed. Given $S_i=1$, with $c_i=(1+e^{\alpha_i})(1+e^{\alpha_i+\beta})$, $P(0,1)=e^{\alpha_i+\beta}/c_i$ and $P(1,0)=e^{\alpha_i}/c_i$, so $P(0,1\mid S_i=1)=e^{\beta}/(1+e^{\beta})$.

Step 3. Multiplying over the n^* discordant pairs gives $\ell_c(\beta)=e^{\beta n_{21}}/(1+e^{\beta})^{n^*}$, a binomial likelihood.

Step 4. $\log\ell_c$ is strictly concave with derivative $n_{21}-n^*e^{\beta}/(1+e^{\beta})$, zero at $e^{\hat\beta}=n_{21}/n_{12}$.

Step 5. The information $n^*e^{\beta}/(1+e^{\beta})^2$ equals $n_{21}n_{12}/n^*$ at $\hat\beta$; its inverse is $1/n_{21}+1/n_{12}$. ■

The conditional test of $\beta = 0$ is the binomial test of Proposition 10.4.

Proposition 10.9 (Inconsistency of unconditional ML)

Treating $\alpha_1, \ldots, \alpha_n$ as fixed parameters, the unconditional ML estimator of β in (10.2) is $2\log(n_{21}/n_{12})$, which converges to 2β.

Proof. [Supplement]

Step 1. A concordant pair is fitted perfectly by $\alpha_i \to \pm\infty$ (likelihood 1 for any β) and drops out.

Step 2. A $(1,0)$ pair has likelihood $p_1(1-p_2)$, $p_1 = \operatorname{expit}(\alpha_i)$, $p_2 = \operatorname{expit}(\alpha_i + \beta)$; the α_i-score $(1-p_1) - p_2$ vanishes at $\alpha_i = -\beta/2$, giving $\operatorname{expit}(-\beta/2)^2$. A $(0,1)$ pair profiles to $\operatorname{expit}(\beta/2)^2$.

Step 3. The profile log-likelihood $2n_{21}\log\operatorname{expit}(\beta/2) + 2n_{12}\log\operatorname{expit}(-\beta/2)$ is that of Theorem 10.8 in $\beta/2$, so $\hat\beta/2 = \log(n_{21}/n_{12})$, which tends to β by the law of large numbers. ■

10.2.4 Random effects in binary matched-pairs model

With $\alpha_i \sim N(\mu, \sigma^2)$ integrated out one gets a three-parameter GLMM (Chapter 12), which also gives $\hat\beta = \log(n_{21}/n_{12})$ when the sample log odds ratio is nonnegative.

Proposition 10.10 (Mantel–Haenszel estimator)

Applied to the n subject-specific 2×2 tables, the Mantel–Haenszel estimator of the common odds ratio comparing observation 2 with observation 1 equals n_{21}/n_{12}.

Proof. [Supplement] With rows (observation 2, observation 1) and columns (success, failure), $n_{11k} = y_{k2}$, $n_{12k} = 1 - y_{k2}$, $n_{21k} = y_{k1}$, $n_{22k} = 1 - y_{k1}$, $n_{++k} = 2$, so

$$\hat\theta_{MH} = \frac{\sum_k n_{11k}n_{22k}/n_{++k}}{\sum_k n_{12k}n_{21k}/n_{++k}} = \frac{\sum_k y_{k2}(1-y_{k1})}{\sum_k (1-y_{k2})y_{k1}} = \frac{n_{21}}{n_{12}}.$$

■

	$Y_2 = 1$	$Y_2 = 0$
$Y_1 = 1$	n_{11}	n_{12}
$Y_1 = 0$	n_{21}	n_{22}

grey: $S_i \in \{0, 2\}$, no information on β

orange: $S_i = 1$, $n_{21} \mid n^* \sim \text{bin}\big(n^*, e^\beta/(1+e^\beta)\big)$

Figure 10.1. Only the discordant cells carry conditional information about β.

10.2.5 Logistic regression for matched case–control studies

In a matched case–control study each case $(Y = 1)$ is matched with a control $(Y = 0)$ and both are measured on X. With

$$\operatorname{logit} P(Y_{it} = 1) = \alpha_i + \beta x_{it}, \tag{10.3}$$

e^β is an XY odds ratio and hence estimable from retrospective data.

Example 10.11 (Diabetes and myocardial infarction)

144 Navajo MI cases matched with 144 controls (age, gender). Diabetes: only case 37 pairs, only control 16, both 9, neither 82; $e^{\hat\beta} = 37/16 = 2.3$, $\hat\beta = 0.84$.

10.2.6 Conditional ML for matched pairs with multiple predictors

With p predictors the model is

$$\text{logit}\, P(Y_{it}=1) = \alpha_i + \beta_1 x_{1it} + \cdots + \beta_p x_{pit} = \alpha_i + x_{it}'\boldsymbol{\beta}. \tag{10.4}$$

Theorem 10.12 (Conditional likelihood as no-intercept logistic regression)

In model (10.4), for pairs with $S_i = 1$,

$$P(Y_{i1}=0, Y_{i2}=1 \mid S_i=1) = \frac{e^{x_{i2}'\boldsymbol{\beta}}}{e^{x_{i1}'\boldsymbol{\beta}} + e^{x_{i2}'\boldsymbol{\beta}}} = \frac{e^{x_i^{*\prime}\boldsymbol{\beta}}}{1+e^{x_i^{*\prime}\boldsymbol{\beta}}}, \qquad x_i^* = x_{i2} - x_{i1}.$$

Hence:
(a) the conditional ML estimate of $\boldsymbol{\beta}$ is the no-intercept logistic regression fit to the discordant pairs, with $y_i^* = 1$ for $(0,1)$, $y_i^* = 0$ for $(1,0)$ pairs and predictor x_i^*;
(b) a between-pair predictor ($x_{ji1} = x_{ji2}$ for all i) cancels from the conditional likelihood, so its coefficient is not estimable.

Proof.

Step 1. As in Theorem 10.8, the factors e^{α_i} and $c_i = (1+e^{\alpha_i + x_{i1}'\boldsymbol{\beta}})(1+e^{\alpha_i + x_{i2}'\boldsymbol{\beta}})$ cancel, leaving $e^{x_{i2}'\boldsymbol{\beta}}/(e^{x_{i1}'\boldsymbol{\beta}} + e^{x_{i2}'\boldsymbol{\beta}})$; divide by $e^{x_{i1}'\boldsymbol{\beta}}$.

Step 2. The conditional likelihood $\prod_{S_i=1} \text{expit}(x_i^{*\prime}\boldsymbol{\beta})^{y_i^*}\left[1-\text{expit}(x_i^{*\prime}\boldsymbol{\beta})\right]^{1-y_i^*}$ is the no-intercept logistic likelihood, proving (a).

Step 3. If $x_{ji1} = x_{ji2}$ for all i, every x_i^* has jth component 0 and β_j disappears, proving (b). ■

In Example 10.11 (control $t=1$, case $t=2$) $y_i^* = 1$ and $x_i^* = -1$ (16 pairs) or $+1$ (37 pairs), again giving $\hat{\beta} = \log(37/16)$.

10.2.7 Marginal models and conditional models: extensions

Both kinds of model extend to multinomial responses and clusters; conditional ML estimates only within-cluster effects (Theorem 10.12(b)), random-effects models (Chapter 12) are not so restricted.

10.3 Marginal models for square contingency tables

Motivation. For $I > 2$, model the margins directly: cumulative logits (ordinal) or baseline-category logits (nominal).

For a square table with $\pi_{ab} = P(Y_1 = a, Y_2 = b)$, marginal homogeneity is $P(Y_1 = a) = P(Y_2 = a)$ for all a.

10.3.1 Marginal models for ordinal classifications

The cumulative logit model

$$\text{logit}\, P(Y_t \le j) = \alpha_j + \beta x_t, \qquad t = 1, 2, \quad j = 1, \ldots, I-1, \tag{10.5}$$

($x_1 = 0$, $x_2 = 1$) has proportional odds; $\beta > 0$ means Y_1 tends higher, $\beta = 0$ is marginal homogeneity. ML maximizes the multinomial likelihood in $\{\pi_{ab}\}$ under the margin constraints; df $= 2(I-1) - I = I - 2$.

10.3.2 Premarital and extramarital sex example

Example 10.13 (Premarital and extramarital sex)

1989 GSS: 475 subjects rated premarital and extramarital sex (1 = always wrong, ..., 4 = not wrong at all). Cumulative marginal proportions $(0.307, 0.389, 0.611)$ and $(0.815, 0.918, 0.987)$. $\hat{\beta} = 2.51$ (SE 0.13): far more permissive on premarital sex. Marginal homogeneity $G^2 = 348.1$ (df = 3), (10.5) $G^2 = 35.1$ (df = 2); better models in Section 10.4.7.

10.3.3 Marginal models for nominal classifications

The baseline-category model

$$\log \frac{P(Y_t = j)}{P(Y_t = I)} = \alpha_j + \beta_j x_t, \qquad t = 1, 2, \quad j = 1, \ldots, I-1, \tag{10.6}$$

is saturated for the margins; marginal homogeneity is $\beta_1 = \cdots = \beta_{I-1} = 0$, tested by G^2 or X^2 with df $= I - 1$. A non-iterative alternative uses $d_a = p_{+a} - p_{a+}$ and $\mathbf{d} = (d_1, \ldots, d_{I-1})'$.

Proposition 10.14 (Bhapkar and Stuart statistics)

(a) The covariance matrix $\mathbf{V}$ of $\sqrt{n}\,\mathbf{d}$ has elements

$$v_{ab} = -(\pi_{ab} + \pi_{ba}) - (\pi_{+a} - \pi_{a+})(\pi_{+b} - \pi_{b+}), \qquad a \neq b,$$
$$v_{aa} = \pi_{+a} + \pi_{a+} - 2\pi_{aa} - (\pi_{+a} - \pi_{a+})^2.$$

(b) Under marginal homogeneity the Wald (Bhapkar) statistic $W = n\mathbf{d}'\hat{\mathbf{V}}^{-1}\mathbf{d}$ is asymptotically χ^2_{I-1}. Stuart's statistic $W_0 = n\mathbf{d}'\hat{\mathbf{V}}_0^{-1}\mathbf{d}$ uses the null version $\hat{v}_{ab0} = -(p_{ab} + p_{ba})$, $\hat{v}_{aa0} = p_{+a} + p_{a+} - 2p_{aa}$, and

$$W = \frac{W_0}{1 - W_0/n}.$$

(c) For $I = 2$, W_0 is McNemar's statistic z_0^2.

Proof. [Supplement]

Step 1. $\mathbf{d}$ is the mean of i.i.d. $\mathbf{X}_k$ with $X_{ka} = \mathbb{1}\{Y_{k2} = a\} - \mathbb{1}\{Y_{k1} = a\}$. For $a \neq b$, $E(X_a X_b) = -(\pi_{ab} + \pi_{ba})$, and $E(X_a^2) = \pi_{+a} + \pi_{a+} - 2\pi_{aa}$; subtracting $E(X_a)E(X_b)$ gives (a).

Step 2. By the CLT and consistency of $\hat{\mathbf{V}}$, under H_0 ($E\mathbf{d} = \mathbf{0}$) $W \xrightarrow{d} \chi^2_{I-1}$.

Step 3. $\hat{\mathbf{V}} = \hat{\mathbf{V}}_0 - \mathbf{d}\mathbf{d}'$. By Sherman–Morrison with $q = \mathbf{d}'\hat{\mathbf{V}}_0^{-1}\mathbf{d} = W_0/n$,

$$\mathbf{d}'\hat{\mathbf{V}}^{-1}\mathbf{d} = q + \frac{q^2}{1-q} = \frac{q}{1-q},$$

so $W = W_0/(1 - W_0/n)$.

Step 4. For $I = 2$, $d_1 = p_{21} - p_{12}$ and $\hat{v}_{110} = p_{12} + p_{21}$, so $W_0 = (n_{21} - n_{12})^2/(n_{12} + n_{21})$. ■

For ordered categories a single-df ordinal test is often more powerful.

10.3.4 Migration example

Example 10.15 (Migration 1980–1985)

Region (NE, MW, S, W) in 1980 and 1985 for 55,981 U.S. residents; about 95% on the diagonal; e.g. 124 moved NE→W, 63 W→NE. Marginal homogeneity: $G^2 = 240.8$ (df = 3), Bhapkar $W = 236.5$. Margins $(0.218, 0.260, 0.330, 0.191)$ and $(0.213, 0.253, 0.339, 0.194)$ differ by -0.005, -0.007, 0.009, 0.003 (each ± 0.001, Theorem 10.1): significant only because n is huge.

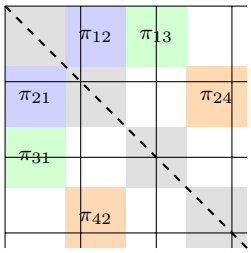
10.4 Symmetry, quasi-symmetry, and quasi-independence

Motivation. Quasi-symmetry fits well, has a subject-specific meaning and gives an LR test of marginal homogeneity.

Definition (Symmetry)

An $I \times I$ distribution satisfies **symmetry** if $\pi_{ab} = \pi_{ba}$ for all $a \neq b$.

Symmetry implies marginal homogeneity ($\pi_{a+} = \sum_b \pi_{ba} = \pi_{+a}$); for $I > 2$ the converse fails.



grey: diagonal, fitted exactly by S, QS, QI

mirror cells $(a, b), (b, a)$: equal under S; under QS, $\pi_{ab}/\pi_{ba} = e^{\tau_b - \tau_a}$

Figure 10.2. Mirror pairs of a square table ($I = 4$).

10.4.1 Symmetry as logit and loglinear models

For $\pi_{ab} > 0$, symmetry is $\log(\pi_{ab}/\pi_{ba}) = 0$ $(a < b)$, or, with $\mu_{ab} = n\pi_{ab}$,

$$\log \mu_{ab} = \lambda + \lambda_a + \lambda_b + \lambda_{ab}, \qquad \lambda_{ab} = \lambda_{ba}. \tag{10.7}$$

Theorem 10.16 (Fitting and testing symmetry)

For Poisson or multinomial counts:
(a) the ML fit of symmetry is $\hat{\mu}_{ab} = (n_{ab} + n_{ba})/2$ (so $\hat{\mu}_{aa} = n_{aa}$);
(b) the residual df is $I(I-1)/2$;
(c) the Pearson statistic is Bowker's statistic

$$X^2 = \sum_{a<b} \frac{(n_{ab} - n_{ba})^2}{n_{ab} + n_{ba}},$$

which for $I = 2$ is McNemar's z_0^2;
(d) the standardized Pearson residuals are $r_{ab} = (n_{ab} - n_{ba})/(n_{ab} + n_{ba})^{1/2} = -r_{ba}$, and $\sum_{a<b} r_{ab}^2 = X^2$.

Proof.

Step 1. (a) With $\log \mu_{ab} = \lambda_{ab}$, $\lambda_{ab} = \lambda_{ba}$, the Poisson log-likelihood splits into $(n_{ab}+n_{ba}) \log \mu - 2\mu$ per pair $a < b$ and $n_{aa} \log \mu_{aa} - \mu_{aa}$ per diagonal cell; maximize each (same multinomial fit).

Step 2. (b) $\text{df} = I^2 - I - \binom{I}{2} = I(I-1)/2$.

Step 3. (c) With $m = (n_{ab}+n_{ba})/2$, both cells of a pair have $(n-m)^2 = (n_{ab}-n_{ba})^2/4$, contributing $2(n_{ab} - n_{ba})^2/(4m)$; diagonal cells contribute 0.

Step 4. (d) Each pair is fitted by one parameter with design $(1,1)'$ and equal weights, so the leverage is $h = \frac{1}{2}$ and $r_{ab} = (n_{ab} - m)/\sqrt{m/2} = (n_{ab} - n_{ba})/(n_{ab} + n_{ba})^{1/2}$; squaring and summing gives (c). ■

10.4.2 Quasi-symmetry

Definition (Quasi-symmetry)

The **quasi-symmetry** (QS) model lets the main effects in (10.7) differ:

$$\log \mu_{ab} = \lambda + \lambda_a^X + \lambda_b^Y + \lambda_{ab}, \qquad \lambda_{ab} = \lambda_{ba}. \tag{10.8}$$

Symmetry is $\lambda_a^X = \lambda_a^Y$; independence is $\lambda_{ab} = 0$.

Since $\lambda_a^X + \lambda_b^Y = (\lambda_a^X + \lambda_b^X) + (\lambda_b^Y - \lambda_b^X)$, QS can be written

$$\log \mu_{ab} = \lambda'_{ab} + \tau_b, \qquad \lambda'_{ab} = \lambda'_{ba}, \quad \tau_b = \lambda_b^Y - \lambda_b^X, \quad \tau_I = 0. \tag{10.9}$$

Proposition 10.17 (Likelihood equations for quasi-symmetry)

The ML fit of QS satisfies

$$\hat{\mu}_{a+} = n_{a+}, \qquad \hat{\mu}_{+b} = n_{+b}, \qquad \hat{\mu}_{ab} + \hat{\mu}_{ba} = n_{ab} + n_{ba} \quad (a \le b);$$

one of the first two sets is implied by the other two. In particular $\hat{\mu}_{aa} = n_{aa}$, and the residual df is $(I-1)(I-2)/2$.

Proof.

Step 1. Poisson likelihood equations equate the sufficient statistics (row totals, column totals, symmetric sums $n_{ab} + n_{ba}$, $a \le b$) to their expectations; $a = b$ gives $\hat{\mu}_{aa} = n_{aa}$.

Step 2. Given the third set and the row equations, $\hat{\mu}_{+a} = \sum_b (n_{ab} + n_{ba} - \hat{\mu}_{ab}) = n_{a+} + n_{+a} - \hat{\mu}_{a+} = n_{+a}$.

Step 3. Form (10.9) has $I(I+1)/2 + (I-1)$ parameters, so $\text{df} = (I-1)(I-2)/2$. ■

QS has the multiplicative form

$$\pi_{ab} = \alpha_a \beta_b \gamma_{ab}, \qquad \gamma_{ab} = \gamma_{ba}, \tag{10.10}$$

with symmetry the case $\alpha_a = \beta_a$.

Theorem 10.18 (Odds-ratio characterization of quasi-symmetry)

Let all $\pi_{ab} > 0$.
(a) QS holds if and only if

$$\frac{\pi_{ab}\pi_{II}}{\pi_{aI}\pi_{Ib}} = \frac{\pi_{ba}\pi_{II}}{\pi_{bI}\pi_{Ia}} \qquad \text{for all } a, b. \tag{10.11}$$

(b) Under QS the local odds ratios satisfy $\theta_{ab} = \theta_{ba}$, where $\theta_{ab} = \pi_{ab}\pi_{a+1,b+1}/(\pi_{a,b+1}\pi_{a+1,b})$.

Proof. [Supplement]

Step 1. ($\Rightarrow$) Under (10.10) the α's and β's cancel, leaving $\gamma_{ab}\gamma_{II}/(\gamma_{aI}\gamma_{Ib})$ and $\gamma_{ba}\gamma_{II}/(\gamma_{bI}\gamma_{Ia})$, equal by symmetry of γ.

Step 2. ($\Leftarrow$) Put $\alpha_a = \pi_{aI}$, $\beta_b = \pi_{Ib}/\pi_{II}$, $\gamma_{ab} = \pi_{ab}\pi_{II}/(\pi_{aI}\pi_{Ib})$; then $\pi_{ab} = \alpha_a\beta_b\gamma_{ab}$ and (10.11) says $\gamma_{ab} = \gamma_{ba}$.

Step 3. (b) Similarly $\theta_{ab} = \gamma_{ab}\gamma_{a+1,b+1}/(\gamma_{a,b+1}\gamma_{a+1,b})$, which is unchanged when a, b are swapped since γ is symmetric. ■

The meaning of QS comes from the subject-specific version of (10.6):

$$\log \frac{P(Y_{it} = j)}{P(Y_{it} = I)} = \alpha_{ij} + \beta_j x_t, \qquad t = 1, 2, \quad j = 1, \ldots, I-1, \tag{10.12}$$

with $x_1 = 0$, $x_2 = 1$, $\alpha_{iI} = \beta_I = 0$, and responses independent given the subject.

Theorem 10.19 (Subject-specific logit models imply quasi-symmetry)

If (10.12) holds and subjects are sampled from any distribution F of $(\alpha_{i1}, \ldots, \alpha_{i,I-1})$, then the population-averaged table satisfies QS with $\tau_j = \lambda_j^Y - \lambda_j^X = \beta_j$ (constraint $\tau_I = 0$).

Proof. [Supplement]

Step 1. With $D_1(\boldsymbol{\alpha}) = \sum_k e^{\alpha_k}$, $D_2(\boldsymbol{\alpha}) = \sum_k e^{\alpha_k+\beta_k}$, conditional independence gives $P(Y_1 = a, Y_2 = b \mid \boldsymbol{\alpha}) = e^{\alpha_a} e^{\alpha_b+\beta_b}/[D_1(\boldsymbol{\alpha})D_2(\boldsymbol{\alpha})]$.

Step 2. Averaging over F, $\pi_{ab} = e^{\beta_b}\gamma_{ab}$ with $\gamma_{ab} = E_F[e^{\alpha_a+\alpha_b}/(D_1D_2)] = \gamma_{ba}$. This is (10.10) with $\alpha_a \equiv 1$, i.e. (10.9) with $\tau_b = \beta_b$. ■

Moreover (Conaway 1989) the conditional ML estimates of $\{\beta_j\}$ equal $\hat\lambda_j^Y - \hat\lambda_j^X$ from the QS fit.

10.4.3 Quasi-independence

Definition (Quasi-independence)

A square table satisfies **quasi-independence** (QI) if the classifications are independent given that the row and column outcomes differ:

$$\log \mu_{ab} = \lambda + \lambda_a^X + \lambda_b^Y + \delta_a\, I(a = b). \tag{10.13}$$

The likelihood equations $\hat\mu_{a+} = n_{a+}$, $\hat\mu_{+a} = n_{+a}$, $\hat\mu_{aa} = n_{aa}$ fit the diagonal exactly; off-diagonal rectangle odds ratios equal 1, and df $= (I-1)^2 - I$ $(I \geq 3)$.

Proposition 10.20 (Quasi-independence within quasi-symmetry)

QI is QS with all γ_{ab}, $a \neq b$, in (10.10) equal; for $I = 3$ they coincide.

Proof. [Supplement]

Step 1. QI is (10.10) with $\gamma_{ab} = 1$ off the diagonal and $\gamma_{aa} = e^{\delta_a}$; conversely a constant off-diagonal γ is absorbed into α, and γ_{aa} into δ_a.

Step 2. For $I = 3$, the positive numbers $c_1 = (\gamma_{12}\gamma_{13}/\gamma_{23})^{1/2}$, $c_2 = (\gamma_{12}\gamma_{23}/\gamma_{13})^{1/2}$, $c_3 = (\gamma_{13}\gamma_{23}/\gamma_{12})^{1/2}$ satisfy $c_ac_b = \gamma_{ab}$ $(a \neq b)$. Then $\pi_{ab} = (\alpha_ac_a)(\beta_bc_b)$ off the diagonal, which is QI; with Step 1 the models coincide. ■

10.4.4 Migration revisited

Example 10.21 (Migration: loglinear models)

Example 10.15: independence ($G^2 = 125{,}923$) and symmetry ($G^2 = 243.6$, df $= 6$) fail; QI ($G^2 = 69.5$, df $= 5$) misfits NE→S and W→S; QS fits ($G^2 = 3.0$, df $= 3$), $\hat{\tau} = (-0.672, -0.623, 0.122)$ (W baseline): odds of living in the South rather than the West were $e^{0.122} = 1.13$ times higher in 1985.

10.4.5 Marginal homogeneity and quasi-symmetry

Theorem 10.22 (Caussinus)

For a table with all $\pi_{ab} > 0$,

$$\text{quasi-symmetry} + \text{marginal homogeneity} \iff \text{symmetry}.$$

Proof.

Step 1. ($\Leftarrow$) Symmetry is QS with $\alpha_a = \beta_a$ and implies marginal homogeneity.

Step 2. ($\Rightarrow$) Write $\pi_{ab} = \kappa_a \gamma^*_{ab}$ with $\kappa_a = \alpha_a/\beta_a$, $\gamma^*_{ab} = \beta_a\beta_b\gamma_{ab} = \gamma^*_{ba} > 0$. Marginal homogeneity $\kappa_j \sum_b \gamma^*_{jb} = \sum_a \kappa_a \gamma^*_{aj}$ reads

$$\kappa_j = \sum_a \kappa_a \, w_{aj}, \qquad w_{aj} = \frac{\gamma^*_{aj}}{\sum_b \gamma^*_{bj}} > 0, \quad \sum_a w_{aj} = 1.$$

Step 3. For j with $\kappa_j = \min_a \kappa_a$, a positive-weight average equal to the minimum forces all $\kappa_a = \kappa$. Then $\pi_{ab} = \kappa\gamma^*_{ab} = \pi_{ba}$. ■

Corollary 10.23 (Conditional test of marginal homogeneity)

Given that QS holds, marginal homogeneity is equivalent to symmetry, i.e. to $\lambda^X_a = \lambda^Y_a$ (all $\tau_a = 0$). It is tested by

$$G^2(S \mid QS) = G^2(S) - G^2(QS), \qquad \text{df} = \tfrac{I(I-1)}{2} - \tfrac{(I-1)(I-2)}{2} = I - 1.$$

Proof. The equivalence is Theorem 10.22. Symmetry is the submodel $\tau_1 = \cdots = \tau_{I-1} = 0$ of (10.9), so the deviance difference is the LR statistic (df from Theorem 10.16(b) and Proposition 10.17). ■

For the migration data $G^2(S \mid QS) = 240.6$ (df $= 3$), close to $G^2 = 240.8$ and $W = 236.5$ of Example 10.15.

10.4.6 Ordinal quasi-symmetry model

With scores $u_1 \le \cdots \le u_I$, the **ordinal quasi-symmetry** model is

$$\log \mu_{ab} = \lambda + \lambda_a + \lambda_b + \beta u_b + \lambda_{ab}, \qquad \lambda_{ab} = \lambda_{ba}, \tag{10.14}$$

QS with $\tau_b = \beta u_b$; symmetry is $\beta = 0$. Equivalently

$$\log \frac{\pi_{ab}}{\pi_{ba}} = \beta(u_b - u_a), \qquad a < b, \tag{10.15}$$

a no-intercept logit model for $n_{ab} \mid (n_{ab} + n_{ba})$ with predictor $u_b - u_a$.

Proposition 10.24 (Ordinal quasi-symmetry and marginal homogeneity)

Let $u_1 < \cdots < u_I$ and all $\pi_{ab} > 0$.
(a) The likelihood equations are $\sum_a u_a \hat{\mu}_{a+} = \sum_a u_a n_{a+}$, $\sum_b u_b \hat{\mu}_{+b} = \sum_b u_b n_{+b}$, $\hat{\mu}_{ab} + \hat{\mu}_{ba} = n_{ab} + n_{ba}$ $(a \le b)$: fitted row and column means, not margins, match.
(b) Under (10.14) the column mean exceeds the row mean iff $\beta > 0$ (smaller iff $\beta < 0$); so ordinal QS + marginal homogeneity = symmetry, and $\beta = 0$ tests marginal homogeneity (df = 1).

Proof. [Supplement]

Step 1. (a) The sufficient statistics of $\log \mu_{ab} = \lambda'_{ab} + \beta u_b$ are the symmetric sums and $\sum_b u_b n_{+b}$, giving the second and third sets. Summing $(u_a + u_b)$ times the third set over $a \le b$ gives $\sum_{a,b}(u_a + u_b)\hat{\mu}_{ab} = \sum_{a,b}(u_a + u_b)n_{ab}$; subtracting the second set gives the first.

Step 2. (b) By (10.15),

$$\sum_{a,b}(u_b - u_a)\pi_{ab} = \sum_{a<b}(u_b - u_a)\pi_{ba}(e^{\beta(u_b - u_a)} - 1),$$

every term having the sign of β. Equal means force $\beta = 0$, i.e. symmetry; the converse is immediate. ■

10.4.7 Premarital and extramarital sex revisited

Example 10.25 (Premarital and extramarital sex: symmetry models)

Example 10.13: symmetry fails ($G^2 = 402.2$, df = 6); QS ($G^2 = 1.4$, df = 3) and ordinal QS with scores $1, \ldots, 4$ ($G^2 = 2.1$, df = 5, $\hat{\beta} = -2.86$) fit. By (10.15), a premarital response x categories more permissive than the extramarital one is $e^{2.86x}$ times as likely as the reverse.

10.4.8 Other ordinal models for square tables

The **conditional symmetry** model (McCullagh 1978)

$$\log(\pi_{ab}/\pi_{ba}) = \tau \qquad \text{for all } a < b \tag{10.16}$$

describes $\pi_{ab} > \pi_{ba}$ for all $a < b$ (or the reverse); symmetry is $\tau = 0$. The **quasi-uniform association** model

$$\log \mu_{ab} = \lambda + \lambda_a^X + \lambda_b^Y + \beta u_a u_b + \delta_a I(a = b) \tag{10.17}$$

adds linear-by-linear association off the diagonal to QI; it is a special case of QS. For Example 10.13, $\hat{\tau} = -4.130$ (SE 0.451) and $\hat{\beta} = 0.632$ (SE 0.106).

10.5 Measuring agreement between observers

Motivation. Agreement is not association; it needs diagonal-oriented models and indices such as kappa.

Example 10.26 (Pathologists' diagnoses)

Pathologists A, B classified 118 slides into four ordered categories (1 = negative, ..., 4 = invasive carcinoma). Diagonal $(22, 7, 36, 10)$; main disagreements: B rated 3 when A rated

2 (14) or 4 (17). Independence fits poorly ($G^2 = 118.0$, df $= 9$), large positive diagonal residuals.

10.5.1 Agreement: departures from independence

$\sum_a \pi_{aa}$ is the agreement probability. Strong agreement needs strong association, not conversely (A might always rate one category higher). Independence is chance agreement.

10.5.2 Using quasi-independence to analyze agreement

For two subjects rated a and b by one observer and b, a or a, b by the other, the odds of agreement are

$$\tau_{ab} = \frac{\pi_{aa}\pi_{bb}}{\pi_{ab}\pi_{ba}} = \frac{\mu_{aa}\mu_{bb}}{\mu_{ab}\mu_{ba}}. \tag{10.18}$$

Proposition 10.27 (Agreement odds under QI and QS)

Under QI, $\tau_{ab} = \exp(\delta_a + \delta_b)$. Under QS, $\tau_{ab} = \exp(\lambda_{aa} + \lambda_{bb} - \lambda_{ab} - \lambda_{ba})$.

Proof. [Supplement] In $\log \mu_{aa} + \log \mu_{bb} - \log \mu_{ab} - \log \mu_{ba}$ the main effects $\lambda_a^X + \lambda_a^Y + \lambda_b^X + \lambda_b^Y$ cancel against those of the off-diagonal cells, leaving $\delta_a + \delta_b$ under (10.13) and $\lambda_{aa} + \lambda_{bb} - \lambda_{ab} - \lambda_{ba}$ under (10.8). ■

For Example 10.26, QI has $G^2 = 13.2$ (df $= 5$) and $\hat{\tau}_{23} = e^{2.5} = 12.3$.

10.5.3 Quasi-symmetry and agreement modeling

QS also allows association given disagreement and fits Example 10.26 well ($G^2 = 1.0$, df $= 2$, $\hat{\tau}_{23} = 10.7$); remove pairs with $n_{ab} + n_{ba} = 0$ (fitted 0) for correct df. Symmetry fits poorly ($G^2 = 39.2$, df $= 5$); $G^2(S \mid QS) = 38.2$ (df $= 3$): strong marginal heterogeneity.

10.5.4 Kappa measure of agreement

Definition (Cohen's kappa)

With $P_o = \sum_a \pi_{aa}$ and $P_e = \sum_a \pi_{a+}\pi_{+a}$ (agreement expected under independence),

$$\kappa = \frac{\sum_a \pi_{aa} - \sum_a \pi_{a+}\pi_{+a}}{1 - \sum_a \pi_{a+}\pi_{+a}} = \frac{P_o - P_e}{1 - P_e}.$$

$\kappa = 0$ at chance agreement, $\kappa = 1$ at perfect agreement.

Theorem 10.28 (Asymptotic variance of sample kappa)

Under multinomial sampling, $\sqrt{n}(\hat{\kappa} - \kappa) \xrightarrow{d} N(0, \sigma_\kappa^2)$, where

$$\sigma_\kappa^2 = \frac{P_o(1-P_o)}{(1-P_e)^2} + \frac{2(1-P_o)[2P_oP_e - \sum_a \pi_{aa}(\pi_{a+} + \pi_{+a})]}{(1-P_e)^3} + \frac{(1-P_o)^2[\sum_a \sum_b \pi_{ab}(\pi_{b+} + \pi_{+a})^2 - 4P_e^2]}{(1-P_e)^4}.$$

The estimated variance of $\hat{\kappa}$ (Fleiss, Cohen and Everitt 1969) substitutes sample proportions

and divides by n.

Proof. [Supplement]

Step 1. By the delta method for the multinomial, $\sigma^2 = \sum_{a,b}\pi_{ab}g_{ab}^2 - \left(\sum_{a,b}\pi_{ab}g_{ab}\right)^2$ with $g_{ab} = \partial\kappa/\partial\pi_{ab}$.

Step 2. With $\kappa = 1-(1-P_o)/(1-P_e)$, $\partial P_o/\partial\pi_{ab} = I(a=b)$ and $\partial P_e/\partial\pi_{ab} = \pi_{+a}+\pi_{b+}$,

$$g_{ab} = \frac{I(a=b)(1-P_e) - (1-P_o)(\pi_{+a}+\pi_{b+})}{(1-P_e)^2}.$$

Step 3. Since $\sum_{a,b}\pi_{ab}(\pi_{+a}+\pi_{b+}) = 2P_e$, $\sum\pi_{ab}g_{ab} = [P_o(1-P_e) - 2P_e(1-P_o)]/(1-P_e)^2$. With $s = \sum_a \pi_{aa}(\pi_{a+}+\pi_{+a})$ and $t = \sum_{a,b}\pi_{ab}(\pi_{+a}+\pi_{b+})^2$,

$$(1-P_e)^4\sum_{a,b}\pi_{ab}g_{ab}^2 = P_o(1-P_e)^2 - 2(1-P_e)(1-P_o)s + (1-P_o)^2 t.$$

Step 4. Subtracting the square of the first moment,

$$(1-P_e)^4\sigma_\kappa^2 = P_o(1-P_o)(1-P_e)^2 + 2(1-P_e)(1-P_o)(2P_oP_e - s) + (1-P_o)^2(t-4P_e^2),$$

which is the stated formula. ■

For Example 10.26, $P_o = 0.636$, $P_e = 0.281$, $\hat\kappa = 0.493$ (SE 0.057).

10.5.5 Weighted kappa: quantifying disagreement

Definition (Weighted kappa)

With weights $0 \le w_{ab} \le 1$, $w_{aa} = 1$, $w_{ab} = w_{ba}$,

$$\kappa_w = \frac{\sum_a\sum_b w_{ab}\pi_{ab} - \sum_a\sum_b w_{ab}\pi_{a+}\pi_{+b}}{1 - \sum_a\sum_b w_{ab}\pi_{a+}\pi_{+b}}.$$

Common choices are $w_{ab} = 1 - |a-b|/(I-1)$ and (Fleiss–Cohen) $w_{ab} = 1-(a-b)^2/(I-1)^2$.

Both kappas depend strongly on the margins; models of the agreement structure say more.

10.5.6 Extensions to multiple observers

With R observers one can model the $\binom{R}{2}$ pairwise marginal tables simultaneously, use generalized kappas, or use latent-class models (Chapter 13).

Chapter 11. Analyzing Repeated Categorical Response Data

Repeated responses arise over time, under several conditions (crossover trials) or within *clusters* (teeth, litters, families), and are dependent within a cluster.

11.1 Comparing marginal distributions: multiple responses

Motivation. The T margins usually matter more than the joint dependence.

11.1.1 Binary marginal models and marginal homogeneity

For binary $(Y_1, \ldots, Y_T)$ the marginal logit model

$$\operatorname{logit} P(Y_t = 1) = \alpha + \beta_t, \qquad t = 1, \ldots, T, \tag{11.1}$$

($\beta_T = 0$) extends Section 10.2.1. Let $\pi_{\mathbf{i}} = P(Y_1 = i_1, \ldots, Y_T = i_T)$ in the 2^T table. (11.1) is saturated; **marginal homogeneity** is $\beta_1 = \cdots = \beta_T$, awkward to fit by ML since the likelihood involves the 2^T joint probabilities. With $p_{\mathbf{i}} = n_{\mathbf{i}}/n$ and $\hat{\pi}^{MH}$ the constrained ML fit,

$$-2\big[L(\hat{\pi}^{MH}) - L(\mathbf{p})\big] = 2\sum_{\mathbf{i}} n_{\mathbf{i}} \log\left(p_{\mathbf{i}}/\hat{\pi}^{MH}_{\mathbf{i}}\right) \tag{11.2}$$

is asymptotically χ^2_{T-1} under H_0.

11.1.2 Crossover drug comparison example

Example 11.1 (Crossover study of three drugs)

46 subjects, drugs A, B, C, each response favorable or not. Proportions favorable $(0.61, 0.61, 0.35)$; (11.2) $= 5.95$ (df $= 2$, $P = 0.05$). Bonferroni interval ($z = 2.39$), A vs C: $0.261 \pm 2.39(0.108) = (0.002, 0.520)$ (Theorem 10.1); same for B vs C.

11.1.3 Modeling margins of a multicategory response

For I categories, the saturated model $\log[P(Y_t = j)/P(Y_t = I)] = \beta_{tj}$ gives a marginal homogeneity test ($\beta_{1j} = \cdots = \beta_{Tj}$) of the form (11.2) with df $= (T-1)(I-1)$. For ordinal responses,

$$\operatorname{logit} P(Y_t \le j) = \alpha_j + \beta_t, \qquad t = 1, \ldots, T, \quad j = 1, \ldots, I-1, \tag{11.3}$$

($\beta_T = 0$) gives a test of $\beta_1 = \cdots = \beta_T$ with df $= T - 1$.

11.1.4 Wald and generalized CMH score tests of marginal homogeneity

Let $p_j(t)$ be the sample proportion in category j at occasion t, $d_j(t) = p_j(t) - \sum_s p_j(s)/T$, and $\mathbf{d}$ the vector of $d_j(t)$, $t \le T-1$, $j \le I-1$. Bhapkar's statistic

$$W = n\,\mathbf{d}'\hat{\mathbf{V}}^{-1}\mathbf{d} \tag{11.4}$$

is asymptotically $\chi^2_{(I-1)(T-1)}$, generalizing Proposition 10.14.

Generalized CMH statistics treat each subject as a $T \times I$ stratum. Conditional independence given subject implies marginal homogeneity; for $I = T = 2$ this is McNemar (Theorem 10.5), for $I = 2$, $T > 2$ Cochran's Q.

Theorem 11.2 (Cochran's Q as a generalized CMH statistic)

For binary responses, let R_k be the number of successes of subject k, C_t the number of successes at occasion t and $N = \sum_t C_t = \sum_k R_k$. The generalized CMH statistic (general association) for the $T \times 2 \times n$ subject-stratified table is

$$Q = \frac{T(T-1)\sum_{t=1}^{T}(C_t - N/T)^2}{TN - \sum_k R_k^2},$$

which is asymptotically χ^2_{T-1} under marginal homogeneity. For $T = 2$, $Q = (n_{21}-n_{12})^2/(n_{21}+n_{12})$.

Proof. [Supplement]

Step 1. Under conditional independence with fixed stratum margins, subject k's R_k successes are allocated to the T occasions completely at random, independently over subjects; the CMH statistic is the quadratic form in $\mathbf{C} = (C_1, \ldots, C_T)'$ about its null mean with a generalized inverse of its null covariance.

Step 2. With $c_k = (R_k/T)(1 - R_k/T)$, sampling without replacement gives $E(Y_{kt}) = R_k/T$, $\mathrm{Var}(Y_{kt}) = c_k$, $\mathrm{Cov}(Y_{kt}, Y_{ks}) = -c_k/(T-1)$, i.e. $\mathrm{Cov}(\mathbf{Y}_k) = \frac{T}{T-1} c_k \mathbf{P}$ with $\mathbf{P} = \mathbf{I} - \frac{1}{T}\mathbf{J}$.

Step 3. Summing over subjects, $E(\mathbf{C}) = (N/T)\mathbf{1}$ and $\mathrm{Cov}(\mathbf{C}) = \kappa \mathbf{P}$ with

$$\kappa = \frac{T}{T-1}\sum_k c_k = \frac{TN - \sum_k R_k^2}{T(T-1)}.$$

Step 4. $\mathbf{P}$ is the projection onto $\{\mathbf{1}'\mathbf{v} = 0\}$, its own generalized inverse, and contains $\mathbf{C} - E\mathbf{C}$ in its range, so the quadratic form is $\kappa^{-1}\sum_t (C_t - N/T)^2 = Q$. Since $\mathrm{rank}\,\mathbf{P} = T - 1$, the CLT gives the χ^2_{T-1} limit.

Step 5. For $T = 2$, concordant subjects contribute 0 to $2N - \sum_k R_k^2$, which equals $n_{12} + n_{21}$, and $\sum_t (C_t - N/2)^2 = (n_{12} - n_{21})^2/2$; hence $Q = (n_{12} - n_{21})^2/(n_{12} + n_{21})$. ■

11.2 Marginal modeling: maximum likelihood approach

Motivation. ML for marginal logit models maximizes the multinomial likelihood under constraints on the margins.

11.2.1 Longitudinal mental depression example

Example 11.3 (Mental depression)

Mild or severe depression ($s = 0, 1$), standard or new drug ($d = 0, 1$), normal or abnormal at weeks 1, 2, 4 ($t = 0, 1, 2$, $\log_2$ week); $n = 340$. The proportion normal rises faster for the new drug (0.18 → 0.83 severe/new, 0.21 → 0.46 severe/standard). Main effects: $G^2 = 34.6$ (df = 8); adding drug × time,

$$\mathrm{logit}\, P(Y_t = 1) = \alpha + \beta_1 s + \beta_2 d + \beta_3 t + \beta_4 dt,$$

gives $G^2 = 4.2$ (df = 7), with $\hat\beta_3 = 0.48$ (SE 0.12), $\hat\beta_4 = 1.01$ (SE 0.18), $\hat\beta_1 = -1.29$ (SE 0.14) and $\hat\beta_2 = -0.06$ (SE 0.22).

11.2.2 Modeling a repeated multinomial response

For I categories, model the $I-1$ marginal logits at each t as $\text{logit}_j(t) = \alpha_j + \boldsymbol{\beta}_j'\mathbf{x}_t$ ($\boldsymbol{\beta}_j = \boldsymbol{\beta}$ under proportional odds), checking whether effects change over t.

11.2.3 Insomnia example

Example 11.4 (Insomnia clinical trial)

Time to fall asleep (4 ordered categories) before ($t=0$) and after ($t=1$) two weeks of active treatment ($x=1$) or placebo ($x=0$). The proportional odds model

$$\text{logit}\, P(Y_t \le j) = \alpha_j + \beta_1 t + \beta_2 x + \beta_3 tx \tag{11.5}$$

has $G^2 = 8.0$ (df $= 6$), $\hat\beta_1 = 1.074$ (SE 0.162), $\hat\beta_2 = 0.046$ (SE 0.236), $\hat\beta_3 = 0.662$ (SE 0.244): active-vs-placebo cumulative odds ratio 1.04 initially, $e^{0.708} = 2.03$ at follow-up.

11.2.4 Comparisons that control for initial response

If initial margins differ between groups, marginal changes may differ even with identical conditional follow-up distributions; then compare follow-up given y_1,

$$\text{logit}\, P(Y_2 \le j) = \alpha_j + \beta_1 x + \beta_2 y_1, \tag{11.6}$$

an ordinal analogue of ANCOVA and a simple transitional model (Section 11.5).

11.3 Marginal modeling: generalized estimating equations (GEE) approach

Motivation. GEE needs only a marginal mean, a variance function and a working correlation, and still estimates $\boldsymbol{\beta}$ consistently.

11.3.1 Generalized estimating equation methodology: basic ideas

For $(Y_1, \ldots, Y_T)$ (T may vary by subject), specify a link for $\mu_t = E(Y_t)$, a variance function $v(\mu_t)$ and a *working correlation*; the estimates solve the **generalized estimating equations** (Liang and Zeger 1986).

Definition (Working correlation structures)

For cluster i with T_i observations, a working correlation matrix $\mathbf{R}(\boldsymbol{\alpha})$ specifies $\text{Corr}(Y_{it}, Y_{is})$:
(i) *independence*: $\mathbf{R} = \mathbf{I}$;
(ii) *exchangeable*: $\text{Corr}(Y_{it}, Y_{is}) = \alpha$ for all $t \neq s$;
(iii) *autoregressive* AR(1): $\text{Corr}(Y_{it}, Y_{is}) = \alpha^{|t-s|}$;
(iv) *unstructured*: a separate α_{ts} for each pair.

The key properties, proved in Section 11.4:

(a) If the mean model is correct, GEE estimators are consistent *even if the working correlation is wrong.*

(b) Naive standard errors are then wrong; *robust* (sandwich) standard errors use the empirical dependence.

Even independence gives good estimates with corrected SEs; a good working correlation improves efficiency. Compare structures for sensitivity; exchangeable is a sensible default, unstructured costs efficiency for large T.

Limitations of GEE

There is no likelihood, so inference uses Wald statistics with the sandwich covariance. With few clusters sandwich SEs are too small and variable; quasi-score tests (Boos 1992; Rotnitzky and Jewell 1990) may be better.

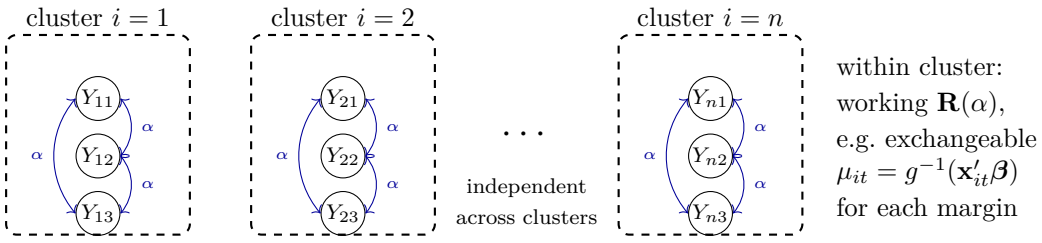


Figure 11.1. n independent clusters with dependent observations; GEE asymptotics are in n.

11.3.2 Longitudinal mental depression example

Example 11.5 (Depression: GEE fit)

Example 11.3: GEE is close to ML for every working correlation; exchangeable gives $\hat{\beta}_3 = 0.48$ (SE 0.12), $\hat{\beta}_4 = 1.02$ (SE 0.19). Independence reproduces ordinary logistic regression on 3×340 observations (intercept -0.028, severe -1.314, new drug -0.060, time 0.482, drug $\times$ time 1.017); naive SEs $(0.164, 0.146, 0.222, 0.115, 0.189)$ and robust $(0.174, 0.146, 0.229, 0.120, 0.188)$ nearly agree as $\hat{\alpha}_{\text{exch}} = -0.003$.

11.3.3 GEE approach for multinomial responses: insomnia example

For multinomial responses (Lipsitz, Kim and Zhao 1994), choose a working correlation for each pair of categories across occasions, with the $(I-1) \times (I-1)$ multinomial covariance at each occasion.

Example 11.6 (Insomnia: GEE fit)

For (11.5) with independence working correlation, $\hat{\beta}_1 = 1.038$ (SE 0.168), $\hat{\beta}_2 = 0.034$ (SE 0.238), $\hat{\beta}_3 = 0.708$ (SE 0.244), similar to ML.

11.4 Quasi-likelihood and its GEE multivariate extension: details

Motivation. Consistency under a wrong correlation and the sandwich both follow from one fact about mean-zero estimating functions (Lemma 11.8).

For subject i let y_i have mean μ_i, variance function $v(\mu_i)$, and $\eta_i = g(\mu_i) = \mathbf{x}_i'\boldsymbol{\beta}$. The **quasi-likelihood** (QL) estimates solve

$$\mathbf{u}(\boldsymbol{\beta}) = \sum_{i=1}^{n} \Big(\frac{\partial \mu_i}{\partial \boldsymbol{\beta}}\Big) \frac{y_i - \mu_i}{v(\mu_i)} = \mathbf{0}, \qquad \mu_i = g^{-1}(\mathbf{x}_i'\boldsymbol{\beta}). \tag{11.7}$$

Proposition 11.7 (Motivations for the quasi-score equations)

(a) If y_i has a natural exponential family distribution with variance function v, (11.7) are the GLM likelihood equations.
(b) With $v(\mu_i)$ replaced by known constants v_i, (11.7) are the normal equations for minimizing $\sum_i (y_i - \mu_i)^2 / v_i$.
(c) Since $\partial \mu_i / \partial \beta_j = (\partial \mu_i / \partial \eta_i) x_{ij}$, (11.7) depend only on the link and the variance function.

Proof. [Supplement] (a) For $f(y;\theta) = \exp\{y\theta - b(\theta) + c(y)\}$, $\mu = b'(\theta)$ and $\partial\mu/\partial\theta = b''(\theta) = v(\mu)$, so

$$\frac{\partial [y_i \theta_i - b(\theta_i)]}{\partial \beta_j} = (y_i - \mu_i) \frac{\partial \theta_i}{\partial \mu_i} \frac{\partial \mu_i}{\partial \beta_j} = \frac{y_i - \mu_i}{v(\mu_i)} \frac{\partial \mu_i}{\partial \beta_j}.$$

(b) $\partial / \partial \beta_j$ of $\sum_i (y_i - \mu_i)^2 / v_i$ is $-2 \sum_i (y_i - \mu_i) v_i^{-1} \partial \mu_i / \partial \beta_j$.

(c) Chain rule. ■

11.4.1 Properties of quasi-likelihood estimators

Definition (Unbiased estimating function)

A function $\mathbf{h}(\mathbf{y}; \boldsymbol{\beta})$ is an **unbiased estimating function** if $E_{\boldsymbol{\beta}}[\mathbf{h}(\mathbf{Y}; \boldsymbol{\beta})] = \mathbf{0}$ for all $\boldsymbol{\beta}$; the equations $\mathbf{h}(\mathbf{y}; \boldsymbol{\beta}) = \mathbf{0}$ that define $\hat{\boldsymbol{\beta}}$ are **estimating equations**.

The quasi-score is unbiased whenever $E(y_i) = \mu_i(\boldsymbol{\beta})$, whatever the true variance.

Lemma 11.8 (Asymptotics of estimating equations)

Let clusters $i = 1, \dots, n$ be independent, $\boldsymbol{\beta} \in \Theta \subseteq \mathbb{R}^p$ open, with true value $\boldsymbol{\beta}_0$, and let $\boldsymbol{\gamma}$ be a nuisance parameter estimated by $\hat{\boldsymbol{\gamma}} \xrightarrow{p} \boldsymbol{\gamma}^*$. Let $\mathbf{U}_n(\boldsymbol{\beta}; \boldsymbol{\gamma}) = \sum_{i=1}^n \boldsymbol{\psi}_i(\boldsymbol{\beta}; \boldsymbol{\gamma})$ and $\mathbf{H}_i = -\partial \boldsymbol{\psi}_i / \partial \boldsymbol{\beta}'$. Assume:
(E1) $E\, \boldsymbol{\psi}_i(\boldsymbol{\beta}_0; \boldsymbol{\gamma}^*) = \mathbf{0}$ for all i;
(E2) $n^{-1} \sum_i \operatorname{Cov}\left(\boldsymbol{\psi}_i(\boldsymbol{\beta}_0; \boldsymbol{\gamma}^*)\right) \to \mathbf{B}$, positive definite, and $\sup_i E \|\boldsymbol{\psi}_i(\boldsymbol{\beta}_0; \boldsymbol{\gamma}^*)\|^{2+\delta} < \infty$ for some $\delta > 0$;
(E3) $\boldsymbol{\psi}_i$ is continuously differentiable in $\boldsymbol{\beta}$, and on neighbourhoods $\mathcal{N}$ of $\boldsymbol{\beta}_0$ and $\mathcal{M}$ of $\boldsymbol{\gamma}^*$, $\sup_{\mathcal{N} \times \mathcal{M}} \| n^{-1} \sum_i \mathbf{H}_i(\boldsymbol{\beta}; \boldsymbol{\gamma}) - \mathbf{A}(\boldsymbol{\beta}, \boldsymbol{\gamma}) \| \xrightarrow{p} 0$ for a continuous $\mathbf{A}(\cdot, \cdot)$ with $\mathbf{A} = \mathbf{A}(\boldsymbol{\beta}_0, \boldsymbol{\gamma}^*)$ non-singular;
(E4) $n^{-1/2} [\mathbf{U}_n(\boldsymbol{\beta}_0; \hat{\boldsymbol{\gamma}}) - \mathbf{U}_n(\boldsymbol{\beta}_0; \boldsymbol{\gamma}^*)] \xrightarrow{p} \mathbf{0}$.
Then:
(i) for every sufficiently small $r > 0$, with probability tending to one, $\mathbf{U}_n(\boldsymbol{\beta}; \hat{\boldsymbol{\gamma}}) = \mathbf{0}$ has exactly one root $\hat{\boldsymbol{\beta}}$ in the ball $\|\boldsymbol{\beta} - \boldsymbol{\beta}_0\| \le r$, and $\hat{\boldsymbol{\beta}} \xrightarrow{p} \boldsymbol{\beta}_0$;
(ii) $\sqrt{n}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0) \xrightarrow{d} N(\mathbf{0}, \mathbf{A}^{-1} \mathbf{B} \mathbf{A}^{-1\prime})$;
(iii) if moreover $n^{-1} \sum_i \boldsymbol{\psi}_i \boldsymbol{\psi}_i'$ converges in probability, uniformly on $\mathcal{N} \times \mathcal{M}$, to a continuous limit equal to $\mathbf{B}$ at $(\boldsymbol{\beta}_0, \boldsymbol{\gamma}^*)$, then with $\hat{\mathbf{A}} = n^{-1} \sum_i \mathbf{H}_i(\hat{\boldsymbol{\beta}}; \hat{\boldsymbol{\gamma}})$ and $\hat{\mathbf{B}} = n^{-1} \sum_i \boldsymbol{\psi}_i \boldsymbol{\psi}_i'(\hat{\boldsymbol{\beta}}; \hat{\boldsymbol{\gamma}})$, the sandwich $\hat{\mathbf{A}}^{-1} \hat{\mathbf{B}} \hat{\mathbf{A}}^{-1\prime} \xrightarrow{p} \mathbf{A}^{-1} \mathbf{B} \mathbf{A}^{-1\prime}$.

Proof. [Supplement]

Step 1. Set-up. Let $c = \|\mathbf{A}^{-1}\|$. Choose $r > 0$ and a neighbourhood $\mathcal{M}' \subseteq \mathcal{M}$ of $\boldsymbol{\gamma}^*$ with $\bar{B}_r = \{\|\boldsymbol{\beta} - \boldsymbol{\beta}_0\| \le r\} \subseteq \mathcal{N}$ and $\|\mathbf{A}(\boldsymbol{\beta}, \boldsymbol{\gamma}) - \mathbf{A}\| \le 1/(4c)$ on $\bar{B}_r \times \mathcal{M}'$. Let ε_n be the supremum in (E3) and

$$E_n = \left\{ \hat{\boldsymbol{\gamma}} \in \mathcal{M}', \quad \varepsilon_n \le \tfrac{1}{4c}, \quad \|n^{-1} \mathbf{U}_n(\boldsymbol{\beta}_0; \hat{\boldsymbol{\gamma}})\| \le \tfrac{r}{2c} \right\}.$$

By (E1)–(E2) and Chebyshev, $n^{-1} \mathbf{U}_n(\boldsymbol{\beta}_0; \boldsymbol{\gamma}^*) \xrightarrow{p} \mathbf{0}$, hence by (E4) also with $\hat{\boldsymbol{\gamma}}$; so $P(E_n) \to 1$.

Step 2. Linearization. For $\boldsymbol{\beta} \in \bar{B}_r$ (convex),

$$n^{-1}\mathbf{U}_n(\boldsymbol{\beta};\hat{\boldsymbol{\gamma}}) = n^{-1}\mathbf{U}_n(\boldsymbol{\beta}_0;\hat{\boldsymbol{\gamma}}) - \mathbf{A}_n(\boldsymbol{\beta})(\boldsymbol{\beta}-\boldsymbol{\beta}_0), \qquad \mathbf{A}_n(\boldsymbol{\beta}) = \int_0^1 n^{-1}\sum_i \mathbf{H}_i(\boldsymbol{\beta}_0 + s(\boldsymbol{\beta}-\boldsymbol{\beta}_0);\hat{\boldsymbol{\gamma}})\, ds,$$

and on E_n, $\|\mathbf{A}_n(\boldsymbol{\beta}) - \mathbf{A}\| \le \varepsilon_n + \frac{1}{4c} \le \frac{1}{2c}$ on $\bar{B}_r$.

Step 3. Existence. The fixed points of $F(\boldsymbol{\beta}) = \boldsymbol{\beta} + \mathbf{A}^{-1}n^{-1}\mathbf{U}_n(\boldsymbol{\beta};\hat{\boldsymbol{\gamma}})$ are roots, and by Step 2

$$F(\boldsymbol{\beta}) - \boldsymbol{\beta}_0 = \mathbf{A}^{-1}n^{-1}\mathbf{U}_n(\boldsymbol{\beta}_0;\hat{\boldsymbol{\gamma}}) + \mathbf{A}^{-1}(\mathbf{A} - \mathbf{A}_n(\boldsymbol{\beta}))(\boldsymbol{\beta}-\boldsymbol{\beta}_0),$$

so on E_n, $\|F(\boldsymbol{\beta}) - \boldsymbol{\beta}_0\| \le c \cdot \frac{r}{2c} + c \cdot \frac{1}{2c} \cdot r = r$. By Brouwer's theorem F has a fixed point $\hat{\boldsymbol{\beta}} \in \bar{B}_r$.

Step 4. Uniqueness. Two roots $\boldsymbol{\beta}_1, \boldsymbol{\beta}_2 \in \bar{B}_r$ give $\mathbf{0} = \tilde{\mathbf{A}}(\boldsymbol{\beta}_1 - \boldsymbol{\beta}_2)$ with $\|\tilde{\mathbf{A}} - \mathbf{A}\| \le \frac{1}{2c}$; since $\|\mathbf{A}^{-1}(\tilde{\mathbf{A}} - \mathbf{A})\| \le \frac{1}{2}$, $\tilde{\mathbf{A}}$ is invertible (Neumann series) and $\boldsymbol{\beta}_1 = \boldsymbol{\beta}_2$.

Step 5. Consistency. The argument holds for every $r' \le r$, and by uniqueness the roots coincide, so $P(\|\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0\| \le r') \to 1$ for all $r' > 0$. This proves (i).

Step 6. Asymptotic normality. On E_n, $\sqrt{n}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0) = \mathbf{A}_n(\hat{\boldsymbol{\beta}})^{-1}\, n^{-1/2}\mathbf{U}_n(\boldsymbol{\beta}_0;\hat{\boldsymbol{\gamma}})$. By (E3), continuity and consistency, $\mathbf{A}_n(\hat{\boldsymbol{\beta}}) \xrightarrow{p} \mathbf{A}$. By (E4), $n^{-1/2}\mathbf{U}_n(\boldsymbol{\beta}_0;\hat{\boldsymbol{\gamma}}) = n^{-1/2}\mathbf{U}_n(\boldsymbol{\beta}_0;\boldsymbol{\gamma}^*) + o_p(1)$, a normalized sum of independent mean-zero vectors; for fixed $\boldsymbol{\lambda} \neq \mathbf{0}$ Lyapunov's condition

$$\frac{\sum_i E|\boldsymbol{\lambda}'\boldsymbol{\psi}_i|^{2+\delta}}{\left(\sum_i \mathrm{Var}(\boldsymbol{\lambda}'\boldsymbol{\psi}_i)\right)^{1+\delta/2}} = \frac{O(n)}{\left(n\,\boldsymbol{\lambda}'\mathbf{B}\boldsymbol{\lambda}\,(1+o(1))\right)^{1+\delta/2}} \to 0$$

holds, so by Cramér–Wold $n^{-1/2}\mathbf{U}_n(\boldsymbol{\beta}_0;\boldsymbol{\gamma}^*) \xrightarrow{d} N(\mathbf{0},\mathbf{B})$, and Slutsky gives (ii).

Step 7. Sandwich. $\hat{\mathbf{A}} \xrightarrow{p} \mathbf{A}$ as in Step 6 and $\hat{\mathbf{B}} \xrightarrow{p} \mathbf{B}$ by the extra assumption; continuity of inversion and multiplication proves (iii). ■

Theorem 11.9 (Consistency and asymptotic normality of QL estimators)

Suppose the mean model $g(\mu_i) = \mathbf{x}_i'\boldsymbol{\beta}$ is correct at $\boldsymbol{\beta}_0$ (the variance function v may be wrong) and that conditions (E2), (E3) and the condition in (iii) of Lemma 11.8 hold for $\boldsymbol{\psi}_i = (\partial\mu_i/\partial\boldsymbol{\beta})(y_i - \mu_i)/v(\mu_i)$ (they hold, e.g., for bounded covariates, smooth g and v, $v > 0$, and bounded fourth moments of y_i). Then the QL estimator is consistent and asymptotically normal, with

$$\mathrm{Cov}(\hat{\boldsymbol{\beta}}) \approx \mathbf{V}\Big[\sum_i \frac{\partial\mu_i}{\partial\boldsymbol{\beta}}\frac{\mathrm{Var}(Y_i)}{v(\mu_i)^2}\frac{\partial\mu_i}{\partial\boldsymbol{\beta}'}\Big]\mathbf{V}, \qquad \mathbf{V} = \Big[\sum_i \frac{\partial\mu_i}{\partial\boldsymbol{\beta}}\frac{1}{v(\mu_i)}\frac{\partial\mu_i}{\partial\boldsymbol{\beta}'}\Big]^{-1}. \tag{11.8}$$

If $\mathrm{Var}(Y_i) = v(\mu_i)$, this reduces to $\mathrm{Cov}(\hat{\boldsymbol{\beta}}) \approx \mathbf{V}$, the GLM formula.

Proof.

Step 1. (E1) holds since $E(y_i) = \mu_i(\boldsymbol{\beta}_0)$ and the coefficient of $y_i - \mu_i$ is nonrandom; (E4) is vacuous.

Step 2. $\mathbf{H}_i = v(\mu_i)^{-1}(\partial\mu_i/\partial\boldsymbol{\beta})(\partial\mu_i/\partial\boldsymbol{\beta}') - (\partial[v(\mu_i)^{-1}\partial\mu_i/\partial\boldsymbol{\beta}]/\partial\boldsymbol{\beta}')(y_i - \mu_i)$; the second term has mean zero at $\boldsymbol{\beta}_0$, giving $\mathbf{A} = \lim n^{-1}\sum_i v(\mu_i)^{-1}(\partial\mu_i/\partial\boldsymbol{\beta})(\partial\mu_i/\partial\boldsymbol{\beta}')$, and $\mathrm{Cov}(\boldsymbol{\psi}_i) = v(\mu_i)^{-2}\,\mathrm{Var}(Y_i)(\partial\mu_i/\partial\boldsymbol{\beta})(\partial\mu_i/\partial\boldsymbol{\beta}')$ gives $\mathbf{B}$.

Step 3. Lemma 11.8(ii) gives $\mathrm{Cov}(\hat{\boldsymbol{\beta}}) \approx n^{-1}\mathbf{A}^{-1}\mathbf{B}\mathbf{A}^{-1}$, which is (11.8); if $\mathrm{Var}(Y_i) = v(\mu_i)$ then $\mathbf{B} = \mathbf{A}$ and it collapses to $\mathbf{V}$. ■

11.4.2 GEE methodology: technical details

For cluster i let $\mathbf{y}_i = (y_{i1},\dots,y_{iT_i})'$, $\boldsymbol{\mu}_i = E(\mathbf{Y}_i)$, $\mathbf{X}_i$ the $T_i \times p$ matrix with rows $\mathbf{x}_{it}'$, and $\eta_{it} = g(\mu_{it}) = \mathbf{x}_{it}'\boldsymbol{\beta}$. Each margin has

$$f(y_{it};\theta_{it},\phi) = \exp\{[y_{it}\theta_{it} - b(\theta_{it})]/\phi + c(y_{it},\phi)\},$$

so $\mu_{it} = b'(\theta_{it})$ and $\operatorname{Var}(Y_{it}) = \phi\, b''(\theta_{it})$.

Definition (Generalized estimating equations)

Let $\mathbf{B}_i = \operatorname{diag}\big(b''(\theta_{i1}), \dots, b''(\theta_{iT_i})\big)$ and let $\mathbf{R}(\boldsymbol{\alpha})$ be a working correlation matrix. The **working covariance** is

$$\mathbf{V}_i = \phi\, \mathbf{B}_i^{1/2} \mathbf{R}(\boldsymbol{\alpha}) \mathbf{B}_i^{1/2} \tag{11.9}$$

($= \operatorname{Cov}(\mathbf{Y}_i)$ if $\mathbf{R}$ is the true correlation). Let $\boldsymbol{\Delta}_i = \operatorname{diag}(\partial\theta_{it}/\partial\eta_{it})$ (the identity for the canonical link) and $\mathbf{D}_i = \partial\boldsymbol{\mu}_i/\partial\boldsymbol{\beta}' = \mathbf{B}_i\boldsymbol{\Delta}_i\mathbf{X}_i$ ($T_i \times p$). The GEE estimator $\hat{\boldsymbol{\beta}}$ solves

$$\mathbf{u}(\boldsymbol{\beta}) = \sum_{i=1}^{n} \mathbf{D}_i' \mathbf{V}_i^{-1} (\mathbf{y}_i - \boldsymbol{\mu}_i) = \mathbf{0}. \tag{11.10}$$

$\mathbf{D}_i = \mathbf{B}_i\boldsymbol{\Delta}_i\mathbf{X}_i$ is the chain rule, and the scalar ϕ cancels from (11.10).

Proposition 11.10 (Independence working correlation gives the GLM fit)

With $\mathbf{R} = \mathbf{I}$, (11.10) becomes $\sum_i \mathbf{X}_i'\boldsymbol{\Delta}_i(\mathbf{y}_i - \boldsymbol{\mu}_i) = \mathbf{0}$ (for the canonical link, $\sum_i \mathbf{X}_i'(\mathbf{y}_i - \boldsymbol{\mu}_i) = \mathbf{0}$): the ordinary GLM likelihood equations treating all y_{it} as independent.

Proof. With $\mathbf{V}_i = \phi\mathbf{B}_i$, $\mathbf{D}_i'\mathbf{V}_i^{-1} = \mathbf{X}_i'\boldsymbol{\Delta}_i\mathbf{B}_i(\phi\mathbf{B}_i)^{-1} = \phi^{-1}\mathbf{X}_i'\boldsymbol{\Delta}_i$; drop $1/\phi$ and compare with Proposition 11.7(a) summed over all (i,t). ■

Definition (Moment estimators of the working parameters)

With Pearson residuals $e_{it} = (y_{it} - \hat{\mu}_{it})/b''(\hat{\theta}_{it})^{1/2}$, $N = \sum_i T_i$ (Liang and Zeger 1986):

$$\hat{\phi} = \frac{1}{N-p} \sum_i \sum_t e_{it}^2, \qquad \hat{\alpha}_{\text{exch}} = \frac{\sum_i \sum_{t<s} e_{it}e_{is}}{\hat{\phi}\big[\sum_i T_i(T_i-1)/2 - p\big]},$$

$$\hat{\alpha}_{\text{AR1}} = \frac{\sum_i \sum_{t \le T_i - 1} e_{it}e_{i,t+1}}{\hat{\phi}\big[\sum_i (T_i - 1) - p\big]}, \qquad \hat{R}_{ts} = \frac{1}{n\hat{\phi}} \sum_i e_{it}e_{is} \ \text{ (unstructured)}.$$

The algorithm alternates a modified Fisher scoring step for $\boldsymbol{\beta}$ given $(\boldsymbol{\alpha}, \phi)$,

$$\boldsymbol{\beta}^{(k+1)} = \boldsymbol{\beta}^{(k)} + \Big[\sum_i \mathbf{D}_i'\mathbf{V}_i^{-1}\mathbf{D}_i\Big]^{-1} \sum_i \mathbf{D}_i'\mathbf{V}_i^{-1}(\mathbf{y}_i - \boldsymbol{\mu}_i),$$

with moment updates of $(\boldsymbol{\alpha}, \phi)$ from the current residuals.

Theorem 11.11 (Liang–Zeger: consistency and asymptotic normality of GEE)

Assume:
(G1) the marginal mean model is correct: $E(\mathbf{Y}_i) = \boldsymbol{\mu}_i(\boldsymbol{\beta}_0)$ for all i (the working correlation may be wrong);
(G2) $\sqrt{n}(\hat{\boldsymbol{\alpha}} - \boldsymbol{\alpha}^*) = O_p(1)$ for some $\boldsymbol{\alpha}^*$ (not necessarily a "true" value);
(G3) conditions (E2), (E3) and the condition in (iii) of Lemma 11.8 hold for $\boldsymbol{\psi}_i(\boldsymbol{\beta}; \boldsymbol{\alpha}) = \mathbf{D}_i'\mathbf{V}_i^{-1}(\boldsymbol{\alpha})(\mathbf{y}_i - \boldsymbol{\mu}_i)$, and $n^{-1}\sum_i \partial\boldsymbol{\psi}_i(\boldsymbol{\beta}_0; \boldsymbol{\alpha})/\partial\boldsymbol{\alpha}' \xrightarrow{p} \mathbf{0}$ uniformly for $\boldsymbol{\alpha}$ near $\boldsymbol{\alpha}^*$.
Then, as the number of clusters $n \to \infty$, $\hat{\boldsymbol{\beta}} \xrightarrow{p} \boldsymbol{\beta}_0$ and $\sqrt{n}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0) \xrightarrow{d} N(\mathbf{0}, \mathbf{V}_G)$, where $\mathbf{V}_G = \lim_n \mathbf{V}_{G,n}$,

$$\mathbf{V}_{G,n} = n\Big[\sum_i \mathbf{D}_i'\mathbf{V}_i^{-1}\mathbf{D}_i\Big]^{-1}\Big[\sum_i \mathbf{D}_i'\mathbf{V}_i^{-1}\operatorname{Cov}(\mathbf{Y}_i)\mathbf{V}_i^{-1}\mathbf{D}_i\Big]\Big[\sum_i \mathbf{D}_i'\mathbf{V}_i^{-1}\mathbf{D}_i\Big]^{-1},$$

evaluated at $(\boldsymbol{\beta}_0, \boldsymbol{\alpha}^*)$. The sandwich estimator $\hat{\mathbf{V}}_{G,n}/n$, which substitutes $\hat{\boldsymbol{\beta}}$, $\hat{\boldsymbol{\alpha}}$, $\hat{\phi}$ and replaces $\mathrm{Cov}(\mathbf{Y}_i)$ by $(\mathbf{y}_i - \hat{\boldsymbol{\mu}}_i)(\mathbf{y}_i - \hat{\boldsymbol{\mu}}_i)'$, is consistent for $\mathrm{Cov}(\hat{\boldsymbol{\beta}})$.

Proof. [Supplement] We verify the hypotheses of Lemma 11.8 with nuisance $\boldsymbol{\gamma} = \boldsymbol{\alpha}$ (ϕ cancels from (11.10)).

Step 1. (E1). At $\boldsymbol{\beta}_0$, $\mathbf{D}_i$ and $\mathbf{V}_i(\boldsymbol{\alpha}^*)$ are nonrandom and $E(\mathbf{y}_i - \boldsymbol{\mu}_i) = \mathbf{0}$ by (G1), so $E\boldsymbol{\psi}_i(\boldsymbol{\beta}_0; \boldsymbol{\alpha}^*) = \mathbf{0}$ *for any* $\boldsymbol{\alpha}^*$: a wrong working correlation does not destroy consistency.

Step 2. $\mathbf{A}$ and $\mathbf{B}$. Differentiating,

$$\mathbf{H}_i = \mathbf{D}_i' \mathbf{V}_i^{-1} \mathbf{D}_i - \sum_j \Big[\frac{\partial (\mathbf{D}_i' \mathbf{V}_i^{-1})}{\partial \beta_j} (\mathbf{y}_i - \boldsymbol{\mu}_i) \Big] \mathbf{e}_j',$$

whose second term has mean zero at $\boldsymbol{\beta}_0$, so $\mathbf{A} = \lim n^{-1} \sum_i \mathbf{D}_i' \mathbf{V}_i^{-1} \mathbf{D}_i$; and $\mathrm{Cov}(\boldsymbol{\psi}_i) = \mathbf{D}_i' \mathbf{V}_i^{-1} \mathrm{Cov}(\mathbf{Y}_i) \mathbf{V}_i^{-1} \mathbf{D}_i$ averages to $\mathbf{B}$. (E2), (E3) hold by (G3).

Step 3. (E4). By the mean value theorem, for some $\bar{\boldsymbol{\alpha}}$ between $\hat{\boldsymbol{\alpha}}$ and $\boldsymbol{\alpha}^*$ (componentwise),

$$n^{-1/2} [\mathbf{U}_n(\boldsymbol{\beta}_0; \hat{\boldsymbol{\alpha}}) - \mathbf{U}_n(\boldsymbol{\beta}_0; \boldsymbol{\alpha}^*)] = \Big[n^{-1} \sum_i \frac{\partial \boldsymbol{\psi}_i(\boldsymbol{\beta}_0; \bar{\boldsymbol{\alpha}})}{\partial \boldsymbol{\alpha}'} \Big] \sqrt{n} (\hat{\boldsymbol{\alpha}} - \boldsymbol{\alpha}^*).$$

Each $\partial \boldsymbol{\psi}_i / \partial \alpha_k = \mathbf{D}_i' (\partial \mathbf{V}_i^{-1} / \partial \alpha_k)(\mathbf{y}_i - \boldsymbol{\mu}_i)$ has mean zero at $\boldsymbol{\beta}_0$; by (G3) the bracket is $o_p(1)$ and by (G2) the last factor is $O_p(1)$, so the product is $o_p(1)$: estimating $\boldsymbol{\alpha}$ has no first-order effect.

Step 4. Lemma 11.8(i)–(ii) gives consistency and $\sqrt{n}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0) \xrightarrow{d} N(\mathbf{0}, \mathbf{A}^{-1} \mathbf{B} \mathbf{A}^{-1})$ with $\mathbf{A}^{-1} \mathbf{B} \mathbf{A}^{-1} = \lim \mathbf{V}_{G,n}$. Lemma 11.8(iii) gives consistency of the sandwich with $\hat{\mathbf{B}} = n^{-1} \sum_i \mathbf{D}_i' \mathbf{V}_i^{-1} (\mathbf{y}_i - \hat{\boldsymbol{\mu}}_i)(\mathbf{y}_i - \hat{\boldsymbol{\mu}}_i)' \mathbf{V}_i^{-1} \mathbf{D}_i$ at $(\hat{\boldsymbol{\beta}}, \hat{\boldsymbol{\alpha}})$; replacing $\hat{\mathbf{A}}$ by $n^{-1} \sum \mathbf{D}_i' \mathbf{V}_i^{-1} \mathbf{D}_i$ at the estimates changes it by $o_p(1)$ (Step 2). ■

Corollary 11.12 (Model-based covariance)

If the working covariance is correct, $\mathrm{Cov}(\mathbf{Y}_i) = \mathbf{V}_i$, then $\mathbf{V}_{G,n}/n = \big(\sum_i \mathbf{D}_i' \mathbf{V}_i^{-1} \mathbf{D}_i \big)^{-1}$.

Proof. Substituting $\mathrm{Cov}(\mathbf{Y}_i) = \mathbf{V}_i$, the middle factor equals $\sum_i \mathbf{D}_i' \mathbf{V}_i^{-1} \mathbf{D}_i$ and cancels one outer inverse. ■

Theorem 11.13 (Optimality of the true covariance as weight)

Consider all estimating equations $\sum_i \mathbf{D}_i' \mathbf{W}_i (\mathbf{y}_i - \boldsymbol{\mu}_i) = \mathbf{0}$ with nonrandom positive definite weights $\mathbf{W}_i$, and let $\boldsymbol{\Sigma}_i = \mathrm{Cov}(\mathbf{Y}_i)$. Their asymptotic covariance satisfies, in the Loewner order,

$$\mathbf{G}^{-1} \mathbf{M} \mathbf{G}^{-1} \;\geq\; \mathbf{F}^{-1}, \qquad \mathbf{G} = \sum_i \mathbf{D}_i' \mathbf{W}_i \mathbf{D}_i, \quad \mathbf{M} = \sum_i \mathbf{D}_i' \mathbf{W}_i \boldsymbol{\Sigma}_i \mathbf{W}_i \mathbf{D}_i, \quad \mathbf{F} = \sum_i \mathbf{D}_i' \boldsymbol{\Sigma}_i^{-1} \mathbf{D}_i,$$

with equality when $\mathbf{W}_i = \boldsymbol{\Sigma}_i^{-1}$. So a correct working covariance gives the most efficient GEE estimator.

Proof. [Supplement]

Step 1. With $\mathbf{K}_i = \begin{pmatrix} \mathbf{W}_i \mathbf{D}_i & \boldsymbol{\Sigma}_i^{-1} \mathbf{D}_i \end{pmatrix}$, the matrix $\mathbf{K}_i' \boldsymbol{\Sigma}_i \mathbf{K}_i = \begin{pmatrix} \mathbf{D}_i' \mathbf{W}_i \boldsymbol{\Sigma}_i \mathbf{W}_i \mathbf{D}_i & \mathbf{D}_i' \mathbf{W}_i \mathbf{D}_i \\ \mathbf{D}_i' \mathbf{W}_i \mathbf{D}_i & \mathbf{D}_i' \boldsymbol{\Sigma}_i^{-1} \mathbf{D}_i \end{pmatrix}$ is positive semidefinite; summing, $\left(\begin{smallmatrix} \mathbf{M} & \mathbf{G} \\ \mathbf{G} & \mathbf{F} \end{smallmatrix}\right) \geq 0$.

Step 2. Since $\mathbf{F} > 0$, the Schur complement $\mathbf{M} - \mathbf{G} \mathbf{F}^{-1} \mathbf{G} \geq 0$.

Step 3. Congruence by the symmetric $\mathbf{G}^{-1}$ gives $\mathbf{G}^{-1}\mathbf{M}\mathbf{G}^{-1} \geq \mathbf{F}^{-1}$; if $\mathbf{W}_i = \boldsymbol{\Sigma}_i^{-1}$ then $\mathbf{G} = \mathbf{M} = \mathbf{F}$. ■

Correlation is an awkward association measure for binary data

The Fréchet bounds $\max(0, p_t + p_s - 1) \leq P(Y_t = Y_s = 1) \leq \min(p_t, p_s)$ restrict $\mathrm{Corr}(Y_t, Y_s)$ by the means (at most 1/3 for $p_t = 0.1$, $p_s = 0.5$). Odds ratios are variation-independent of the means; *alternating logistic regressions* (Carey, Zeger and Diggle 1993) alternate GEE steps for $\boldsymbol{\beta}$ with logistic steps for the log odds ratios.

11.5 Markov chains: transitional modeling

Motivation. Modelling each response given earlier ones factors the joint distribution into transitions, fitted by ordinary multinomial or logistic models.

11.5.1 Transitional models

For a discrete-time, discrete-state process $\{Y_t\}$, **transitional models** describe Y_t given $y_0, \ldots, y_{t-1}$ (and covariates), via

$$f(y_0, \ldots, y_T) = f(y_0) f(y_1 \mid y_0) f(y_2 \mid y_0, y_1) \cdots f(y_T \mid y_0, \ldots, y_{T-1}).$$

11.5.2 First-order Markov chains

Definition (Markov chain)

$\{Y_t\}$ is a (first-order) **Markov chain** if Y_{t+1} given $(Y_0, \ldots, Y_t)$ depends only on Y_t, and **kth-order** if only on $(Y_{t-k+1}, \ldots, Y_t)$. The **transition probabilities** $\pi_{j|i}(t) = P(Y_t = j \mid Y_{t-1} = i)$, $\sum_j \pi_{j|i}(t) = 1$, form the $I \times I$ transition matrix.

Theorem 11.14 (Markov factorization and its loglinear form)

(a) $\{Y_0, \ldots, Y_T\}$ is a Markov chain if and only if

$$f(y_0, \ldots, y_T) = f(y_0) f(y_1 \mid y_0) f(y_2 \mid y_1) \cdots f(y_T \mid y_{T-1}). \tag{11.11}$$

(b) A positive joint distribution is that of a Markov chain if and only if it satisfies the loglinear model $(Y_0Y_1, Y_1Y_2, \ldots, Y_{T-1}Y_T)$, i.e. $\log \pi(y_0, \ldots, y_T) = \sum_{t=1}^{T} h_t(y_{t-1}, y_t)$ for some functions h_t.

Proof. [Supplement]

Step 1. (a) The Markov property turns the general factorization into (11.11). Conversely, summing (11.11) over $y_{t+1}, \ldots, y_T$ gives $f(y_0, \ldots, y_t) = f(y_0) \prod_{s \leq t} f(y_s \mid y_{s-1})$, and the ratio of these for $t+1$ and t is $f(y_{t+1} \mid y_t)$.

Step 2. (b, $\Rightarrow$) Take logs of (11.11), absorbing $\log f(y_0)$ into h_1.

Step 3. (b, $\Leftarrow$) If $\pi(\mathbf{y}) = \prod_t g_t(y_{t-1}, y_t)$, let $G_t(y_t) = \sum_{y_{t+1}, \ldots, y_T} \prod_{s>t} g_s(y_{s-1}, y_s)$ ($G_T \equiv 1$). Then

$$P(y_{t+1} \mid y_0, \ldots, y_t) = \frac{g_{t+1}(y_t, y_{t+1}) G_{t+1}(y_{t+1})}{G_t(y_t)},$$

since the factors $\prod_{s\le t} g_s$ cancel; this depends only on (y_t, y_{t+1}). ■

Adding three-factor terms $(Y_{t-2}Y_{t-1}Y_t)$ gives a second-order chain.

Theorem 11.15 (ML estimation of transition probabilities)

Let $n_{ij}(t)$ count transitions $i \to j$ from time $t-1$ to t, $n_{i+}(t) = \sum_j n_{ij}(t)$, and $\{n_{i0}\}$ the initial counts, multinomial with $\{\pi_{i0}\}$. For n independent subjects the likelihood is proportional to

$$\Big[\prod_{i=1}^{I} \pi_{i0}^{n_{i0}}\Big] \prod_{t=1}^{T}\prod_{i=1}^{I}\prod_{j=1}^{I} \pi_{j|i}(t)^{n_{ij}(t)}, \tag{11.12}$$

and the ML estimates are $\hat\pi_{j|i}(t) = n_{ij}(t)/n_{i+}(t)$ (Anderson and Goodman 1957). For a time-homogeneous chain, $\pi_{j|i}(t) \equiv \pi_{j|i}$, the ML estimate is $\hat\pi_{j|i} = \sum_t n_{ij}(t) / \sum_t n_{i+}(t)$.

Proof.

Step 1. Each subject contributes $\pi_{y_0 0}\prod_t \pi_{y_t|y_{t-1}}(t)$ by (11.11); grouping factors gives (11.12).

Step 2. [Supplement] The likelihood factors into separate multinomial kernels $\prod_j \pi_{j|i}(t)^{n_{ij}(t)}$ with unrelated parameters, each maximized separately.

Step 3. For a kernel with counts n_j, $m = \sum_j n_j > 0$, Jensen's inequality gives

$$\sum_j n_j \log \pi_j - \sum_j n_j \log \frac{n_j}{m} = m \sum_j \frac{n_j}{m} \log \frac{\pi_j}{n_j/m} \le m \log \sum_j \pi_j = 0,$$

with equality iff $\pi_j = n_j/m$.

Step 4. Under time homogeneity the kernels for fixed i merge, with counts $\sum_t n_{ij}(t)$. ■

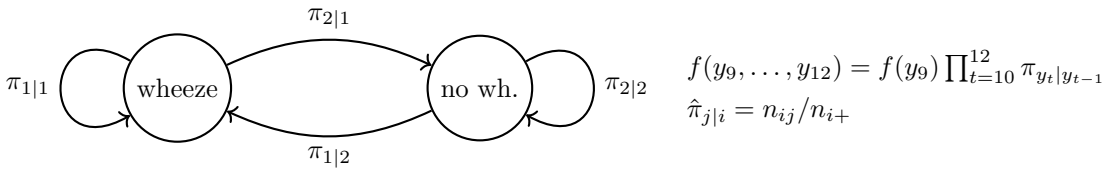


Figure 11.2. A two-state first-order Markov chain.

11.5.3 Respiratory illness example

Example 11.16 (Wheeze at ages 9–12)

Harvard air-pollution study: wheeze at ages 9–12 (2^4 table, $n = 1019$). First-order ($G^2 = 122.9$, df $= 8$) and second-order chains ($G^2 = 23.9$, df $= 4$) fit poorly, perhaps through subject heterogeneity. All pairwise associations fit ($G^2 = 1.5$, df $= 5$), as does one log odds ratio for pairs one year apart (1.75) and another for pairs further apart (1.04) ($G^2 = 2.3$, df $= 9$).

11.5.4 Transitional models with explanatory variables

With covariates, $f(y_1,\ldots,y_T;\mathbf{x}) = f(y_1;\mathbf{x})\prod_{t\ge 2} f(y_t \mid y_1,\ldots,y_{t-1};\mathbf{x})$, and for binary y the **regressive logistic model** (Bonney 1987) is

$$f(y_t \mid y_1,\ldots,y_{t-1};\mathbf{x}_t) = \frac{\exp\left[y_t(\alpha + \beta_1 y_1 + \cdots + \beta_{t-1}y_{t-1} + \boldsymbol{\beta}'\mathbf{x}_t)\right]}{1+\exp(\alpha + \beta_1 y_1 + \cdots + \beta_{t-1}y_{t-1} + \boldsymbol{\beta}'\mathbf{x}_t)}, \qquad y_t = 0, 1.$$

First-order Markov structure drops $y_1, \ldots, y_{t-2}$. Unlike marginal effects, $\boldsymbol{\beta}$ depends on which past responses are conditioned on.

Proposition 11.17 (Fitting transitional models by ordinary logistic regression)

Given the first response(s), the likelihood is a product of Bernoulli likelihoods with the stated logits over subjects and transitions, so conditional ML is ordinary logistic regression with each transition an independent observation and past responses as predictors.

Proof. [Supplement] Subject i contributes $f(y_{i1}; \mathbf{x}) \prod_{t \geq 2} f(y_{it} \mid y_{i1}, \ldots, y_{i,t-1}; \mathbf{x}_{it})$. Conditioning on the first factor, each remaining factor is $\text{expit}(\eta_{it})^{y_{it}}[1 - \text{expit}(\eta_{it})]^{1-y_{it}}$ with η_{it} linear in the parameters and known regressors (past responses, $\mathbf{x}_{it}$). Independence across subjects makes the product over all (i, t) a logistic-regression likelihood with one observation per transition. ■

11.5.5 Child's respiratory illness and maternal smoking

Example 11.18 (Respiratory illness and maternal smoking)

Respiratory illness at ages 7–10; $s = 1$ if the mother smoked. With three transitions per child (12 binomials over (s, t, y_{t-1})), the ML fit is

$$\text{logit}\, \hat{P}(Y_t = 1) = -0.293 + 0.296s - 0.243t + 2.211y_{t-1}, \qquad t = 8, 9, 10,$$

with SEs $(0.846, 0.156, 0.095, 0.158)$ and $G^2 = 3.1$ (df $= 8$). The previous response has a strong effect; the smoking effect is marginal (LR 3.55, $P = 0.06$).